

SN8F5708 Series

Datasheet

8051-based Microcontroller

SN8F5705

SN8F5707

SN8F5708

SN8F570870

SN8F570812

SN8F57082

SN8F570822

SN8F57084

SN8F57085

SN8F57086

SN8F57087

1 Device Overview

1.1 Features

- **Enhanced 8051 microcontroller** with reduced instruction cycle time (up to 12 times 80C51)
- Up to 8 MHz flexible CPU frequency
- Internal 32 MHz Clock Generator (IHRC), 1 MHz to 16 MHz crystal, and external synchronous clock source selections
- Real-time clock with 32.768 kHz crystal
- **Memory configuration**
 - 16 KB on-chip Flash program memory (IROM)
 - 256 bytes on-chip internal RAM (IRAM)
 - 1 KB on-chip external RAM (XRAM)
- **19 interrupt sources with priority levels control and unique interrupt vectors**
- 16 internal interrupts
- 3 external interrupts: INT0, INT1, INT2
- Hardware Multiplication/Division Unit
- 2 set of DPTR
- 2 set 8/16-bit timers with 4 operation modes
- 1 set 16-bit timers with 4 comparison output (PWM) and capture channels
- **3 set 16-bit PWM generators:**
 - each PWM generator has 2 output channels with inverters and dead-band control
- **12-bit SAR ADC** with 12 external and 3 internal channels, and 4 internal reference voltages
- **SPI, UART, I2C** interface with SMBus Support
- **On-Chip Debug Support:**
 - Single-wire debug interface
 - 5 hardware breakpoints
 - Unlimited software breakpoints
 - ROM data security/protection
- Watchdog and programmable external reset
- 1.8/2.4/3.3-V low voltage detectors
- Wide supply voltage (1.8 V – 5.5 V) and temperature (-40 °C to 85 °C) range

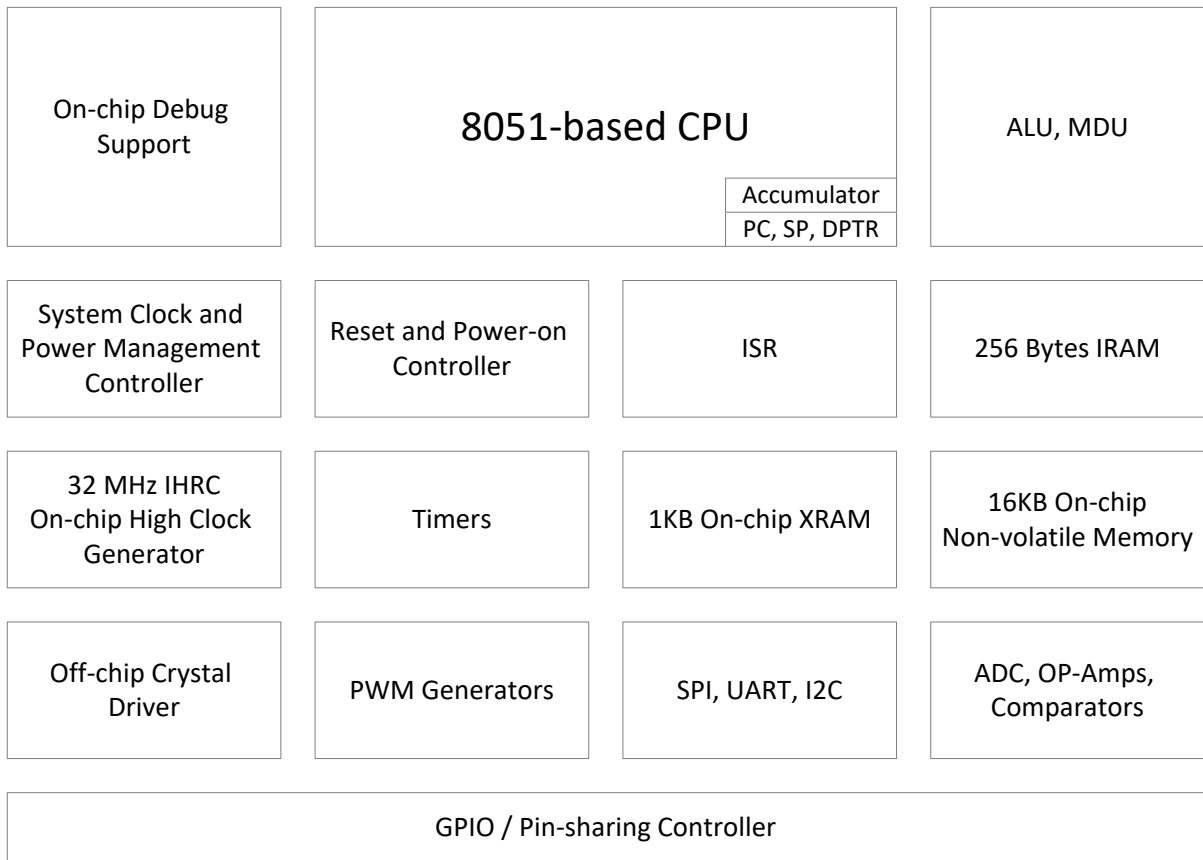
1.2 Applications

- Brushless DC motor
- Home automation
- Household
- Other

1.3 Features Selection Table

	I/O	PWM	I2C	SPI	UART	ADC ext.	OPA	CMP	Ext. INT	Package Types
SN8F5708	46	10	V	V	V	12	2	2	3	LQFP48,QFN48
SN8F570870	44	10	V	V	V	12	2	2	3	QFN46
SN8F5707	42	10	V	V	V	12	2	2	3	LQFP44
SN8F57086	38	10	V	V	V	12	1	2	3	QFN40
SN8F5705	30	9	V	V	V	7	2	2	1	LQFP32,QFN32
SN8F57084	26	9	V	V	V	7	1	1	2	SKDIP28,SOP28,TSSOP28,QFN28
SN8F57082	18	6	V	-	V	7	-	-	2	SOP20,TSSOP20
SN8F570822	18	8	-	-	-	8	-	1	3	TSSOP20
SN8F570812	14	6	-	-	-	9	-	1	3	SOP16
SN8F57085	14	5	V	-	-	6	-	2	1	TSSOP16
SN8F57087	14	6	V	V	V	4	-	-	-	TSSOP16

1.4 Block Diagram



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3 Revision History

Revision	Date	Description
1.0	July 2015	First issue.
1.1	Sep. 2015	Modify LQFP32/QFN32 pin assignment.
1.2	Oct. 2015	1. Modify timer section and electrical characteristic section. 2. Add program memory security section, special function register section and noise filter section. 3. Add SN8F57085/SN8F57086/SN8F57087 part number and SOP20 package type. 4. Modify minimum requirement in debug interface section.
1.3	Oct. 2015	Modify endurance time of electrical characteristic section.
1.4	Nov. 2015	1. Modify SN8F57081 part number to SN8F570870. 2. Modify device nomenclature and electrical characteristic section.
1.5	Dec. 2015	Add QFN28 package type.
1.6	Dec. 2015	Modify electrical characteristic in IHRC section.
1.7	May 2016	1. Add Timer 2 capture function waveform to illustrate operation. 2. Special Function Registers adds Register Declaration section. 3. Add Appendix: Reference Document chapter. 4. Add ROM Programming Pin chapter. 5. Add SN8F570812/SN8F570822 part number and SOP16 package type.
1.8	Sep. 2016	1. Modify the description of OPM register. 2. Add WDT description in watchdog reset section.
1.9	Oct. 2016	1. Add UART Baud Rate Table and modify TB80/RB80. 2. Modify electrical characteristic section. 3. Modify feature section and feature selection table.
2.0	Sep. 2017	1. Modify baud rate control section. 2. Modify pflag initial value. 3. Add pin circuit diagrams section. 4. Modify ordering information. 5. Add package information.
2.1	Jun. 2018	1. Modify external interrupt request flag type. 2. Add SN8F57084XG package type.
2.2	Jul. 2018	1. Repair an error, omission, etc. 2. Add Pin Characteristic section. 3. Modify Internal & External RAM section description. 4. Modify Program Memory section description.

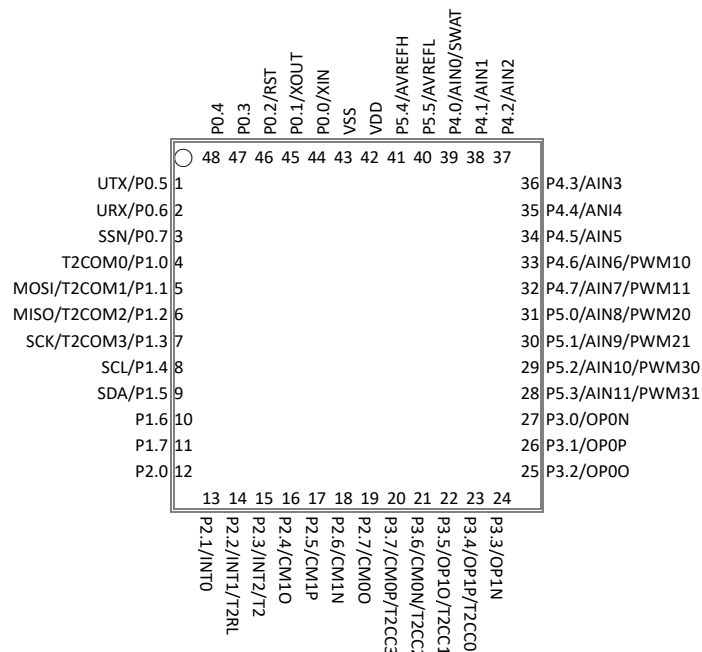
		5. Modify Configuration of Reset and Power-on Controller section description. 6. Modify Power-on Sequence, LVD Reset and External Reset Pin section description. 7. Modify System clock section description. 8. Add High Speed Clock and Real time clock section. 9. Modify Power Management section description. 10. Add System clock timing section. 11. Add System Operating Mode chapter. 12. Modify Interrupt Priority section description. 13. Interrupt chapter adds example section. 14. Modify UART chapter description and baud rate table. 15. I2C chapter adds protocol description diagram and modifies the clock rate table. 16. Debug Interface chapter was renamed Development Environment chapter. Modify Development Environment chapter description. Add Development Tool section. 17. Add SN5708 Starter-kit chapter. 18. Modify ROM Programming Pin chapter description. Add MP5 Hardware Connecting, SN-Link ISP Programming and SN-Link ISP Programming Pin Mapping sections.
2.3	Dec. 2018	1. Modify typing error in pin assignment and T2 comparison output section. 2. Remove SSOP28 and add TSSOP28 package. 3. Modify Pin Circuit Diagrams section.
2.4	Jan. 2019	1. Repair an error, omission, etc. 2. Modify ADC input offset range. 3. Modify power on sequence and system clock timing. 4. Modify Package Information section. 5. Modify SPI operation section description. 6. Modify Timer0/ Timer1 section description.
2.5	May 2022	1. Modify ADC VREFH register description.
2.6	Oct. 2023	1. Added graphics and tables in electrical characteristic section. 2. Modify wake-up timing graph and description in IDLE mode. 3. Modify ordering Information section.
2.7	Apr. 2024	1. Add note for pin assignments and GPIO section about not pin out I/O. 2. Add note for ISP operation.

		3. Modify electrical characteristic section. 4. Modify memory description in feature table 5. Modify typing error.
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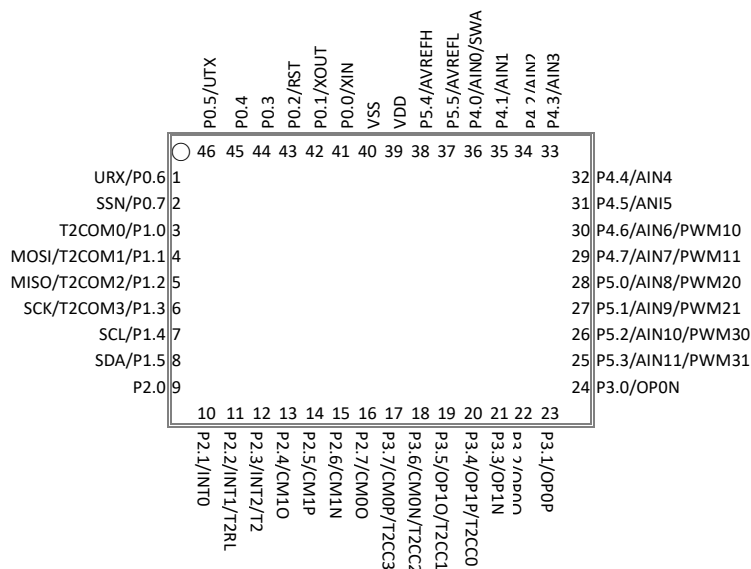
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4 Pin Assignments

4.1 SN8F5708F/J (LQFP48/QFN48)

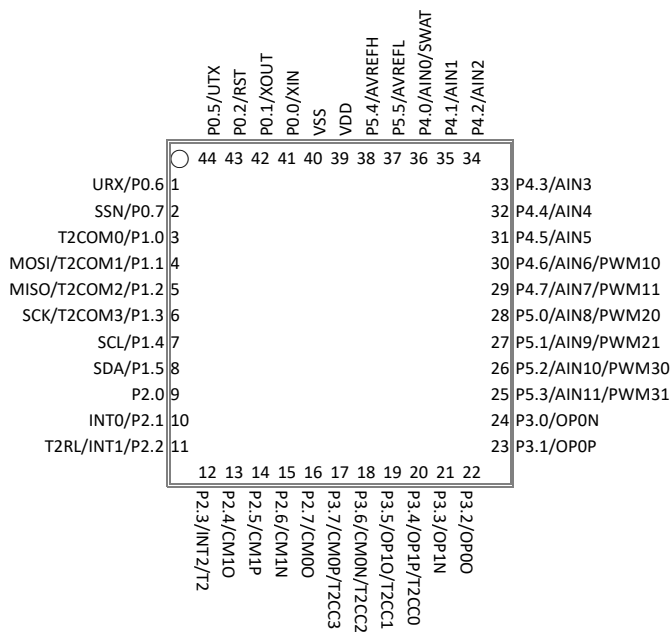


4.2 SN8F570870J (QFN46)



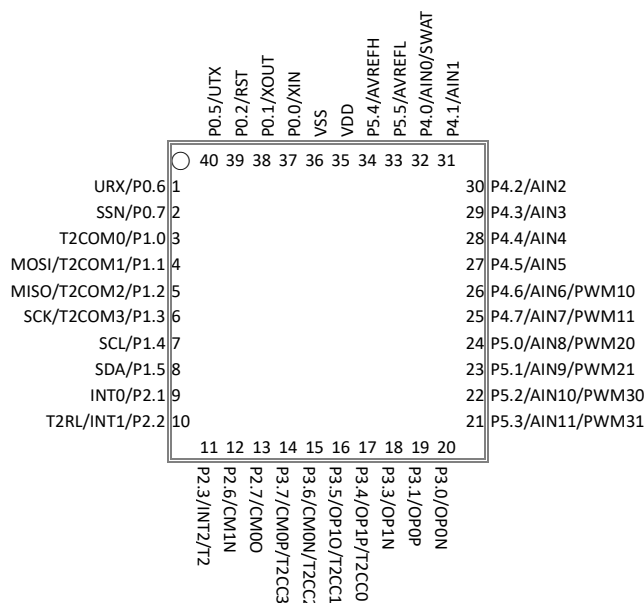
* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.3 SN8F5707F (LQFP44)



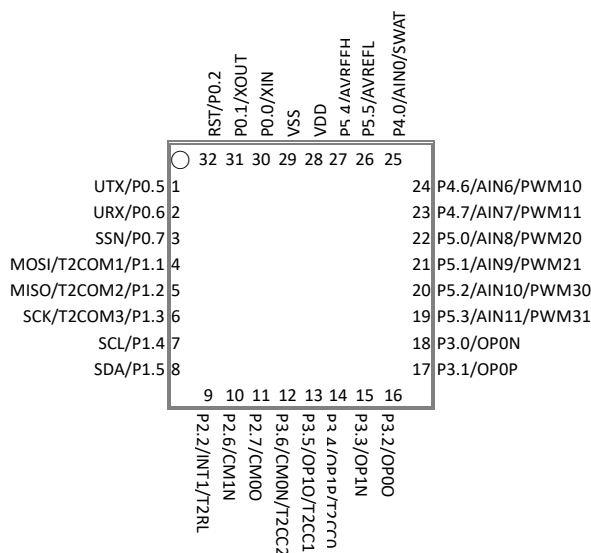
* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.4 SN8F57086J (QFN40)



* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.5 SN8F5705F/J (LQFP32/QFN32)



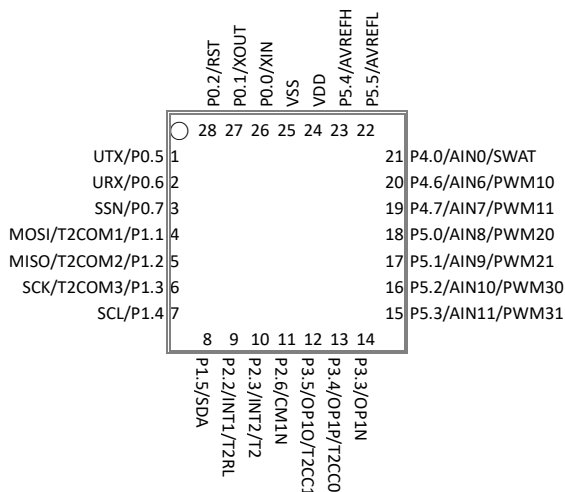
* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.6 SN8F57084K/S/T (SKDIP28/SOP28/TSSOP28)

VSS	1	U	28	VDD
XIN/P0.0	2		27	P5.4/AVREFH
XOUT/P0.1	3		26	P5.5/AVREFL
RST/P0.2	4		25	P4.0/AIN0/SWAT
UTX/P0.5	5		24	P4.6/AIN6/PWM10
URX/P0.6	6		23	P4.7/AIN7/PWM11
SSN/P0.7	7		22	P5.0/AIN8/PWM20
MOSI/T2COM1/P1.1	8		21	P5.1/AIN9/PWM21
MISO/T2COM2/P1.2	9		20	P5.2/AIN10/PWM30
SCK/T2COM3/P1.3	10		19	P5.3/AIN11/PWM31
SCL/P1.4	11		18	P3.3/OP1N
SDA/P1.5	12		17	P3.4/OP1P/T2CC0
T2RL/INT1/P2.2	13		16	P3.5/OP10/T2CC1
T2/INT2/P2.3	14		15	P2.6/CM1N

* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.7 SN8F57084J (QFN28)



* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.8 SN8F57082S/T (SOP20/TSSOP20)

VSS	1	U	20	VDD
XIN/P0.0	2		19	P5.4/AVREFH
XOUT/P0.1	3		18	P5.5/AVREFL
RST/P0.2	4		17	P4.0/AIN0/SWAT
UTX/P0.5	5		16	P4.6/AIN6/PWM10
URX/P0.6	6		15	P4.7/AIN7/PWM11
SCL/P1.4	7		14	P5.0/AIN8/PWM20
SDA/P1.5	8		13	P5.1/AIN9/PWM21
T2RL/INT1/P2.2	9		12	P5.2/AIN10/PWM30
T2/INT2/P2.3	10		11	P5.3/AIN11/PWM31

* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.9 SN8F570822T (TSSOP20)

VDD	1	U	20	P4.0/AIN0/SWAT
VSS	2		19	P4.1/AIN1
XIN/P0.0	3		18	P4.6/AIN6/PWM10
RST/P0.2	4		17	P4.7/AIN7/PWM11
MOSI/T2COM1/P1.1	5		16	P5.0/AIN8/PWM20
SCK/T2COM3/P1.3	6		15	P5.1/AIN9/PWM21
*INT0/P2.1/SCL/P1.4	7		14	P5.2/AIN10/PWM30
T2RL/INT1/P2.2	8		13	P5.3/AIN11/PWM31
T2/INT2/P2.3	9		12	P3.4/OP1P/T2CC0
CM1P/P2.5	10		11	P2.6/CM1N

* **Note:** The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.

4.10 SN8F570812S (SOP16)

INT0/P2.1/SCL/P1.4	1	U	16	P4.4/AIN4/P0.2/RST
T2RL/INT1/P2.2	2		15	P0.0/XIN
T2/INT2/P2.3	3		14	VSS
CM1N/P2.6	4		13	VDD
PWM31/AIN11/P5.3	5		12	P4.0/AIN0/SWAT
PWM30/AIN10/P5.2	6		11	P4.1/AIN1
PWM21/AIN9/P5.1	7		10	P4.6/AIN6/PWM10
PWM20/AIN8/P5.0	8		9	P4.7/AIN7/PWM11

* Pin double bond.

*** Note: The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.**

4.11 SN8F57085T (TSSOP16)

VSS	1	U	16	VDD
XIN/P0.0	2		15	P4.0/AIN0/SWAT
RST/P0.2	3		14	P4.6/AIN6/PWM10
SCL/P1.4	4		13	P4.7/AIN7/PWM11
SDA/P1.5	5		12	P5.0/AIN8/PWM20
INT0/P2.1	6		11	P5.1/AIN9/PWM21
CM1P/P2.5	7		10	P5.3/AIN11/PWM31
CM1N/P2.6	8		9	P3.6/CM0N/T2CC2

*** Note: The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.**

4.12 SN8F57087T (TSSOP16)

XIN/P0.0	1	U	16	VSS
RST/P0.2	2		15	VDD
UTX/P0.5	3		14	P4.0/AIN0/SWAT
URX/P0.6	4		13	P4.6/AIN6/PWM10
MOSI/T2COM1/P1.1	5		12	P4.7/AIN7/PWM11
MISO/T2COM2/P1.2	6		11	P5.0/AIN8/PWM20
SCK/T2COM3/P1.3	7		10	P0.7/SSN
SDA/P1.5	8		9	P1.4/SCL

*** Note: The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.**

4.13 Pin Descriptions

Power Pins

Pin Name	Type	Description
VDD	Power	Power supply
VSS	Power	Ground (0 V)

Port 0

Pin Name	Type	Description
P0.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
XIN	Analog Input	System clock: external clock input
P0.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
XOUT	Analog Output	System clock: drive external crystal/resonator
P0.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
Reset	Digital Input	System reset (active low)
P0.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
P0.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
P0.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
UTX	Digital Output	UART: transmission pin
P0.6	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
URX	Digital Input	UART: reception pin
P0.7	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
SSN	Digital Input	SPI: salve selection pin (slave mode)

Port 1

Pin Name	Type	Description
P1.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
T2COM0	Digital Output	Timer 2: compare 0 output
P1.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
MOSI	Digital I/O	SPI: transmission pin (master) reception pin (slave)
T2COM1	Digital Output	Timer 2: compare 1 output
P1.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
MISO	Digital I/O	SPI: reception pin (master) transmission pin (slave)
T2COM2	Digital Output	Timer 2: compare 2 output
P1.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
SCK	Digital I/O	SPI: clock output (master) clock input (slave)
T2COM3	Digital Output	Timer 2: compare 3 output
P1.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
SCL	Digital I/O	I2C: clock output (master) clock input (slave)
P1.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
SDA	Digital I/O	I2C: data pin
P1.6	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.
P1.7	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors. Level change wake-up.

Port 2

Pin Name	Type	Description
P2.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
P2.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
INT0	Digital Input	INT0: external interrupt 0
P2.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
INT1	Digital Input	INT1: external interrupt 1
T2RL	Digital Input	Timer 2: reload trigger input
P2.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
INT2	Digital Input	INT2: external interrupt 2
T2	Digital Input	Timer 2: event counter input
P2.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM1O	Digital Output	Comparator 1: output
P2.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM1P	Analog Input	Comparator 1: positive input
P2.6	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM1N	Analog Input	Comparator 1: negative input
P2.7	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM0O	Digital Output	Comparator 0: output

Port 3

Pin Name	Type	Description
P3.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP0N	Analog Input	OPA 0: negative input
P3.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP0P	Analog Input	OPA 0: positive input
P3.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP0O	Analog Output	OPA 0: output
P3.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP1N	Analog Input	OPA 1: negative input
P3.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP1P	Analog Input	OPA 1: positive input
T2CC0	Digital Input	Timer 2: capture 0 input
P3.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
OP1O	Analog Output	OPA 1: output
T2CC1	Digital Input	Timer 2: capture 1 input
P3.6	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM0N	Analog Input	Comparator 0: negative input
T2CC2	Digital Input	Timer 2: capture 2 input
P3.7	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
CM0P	Analog Input	Comparator 0: positive input
T2CC3	Digital Input	Timer 2: capture 3 input

Port 4

Pin Name	Type	Description
P4.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
SWAT	Digital I/O	Debug interface
AIN0	Analog Input	ADC: input channel 0
P4.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN1	Analog Input	ADC: input channel 1
P4.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN2	Analog Input	ADC: input channel 2
P4.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN3	Analog Input	ADC: input channel 3
P4.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN4	Analog Input	ADC: input channel 4
P4.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN5	Analog Input	ADC: input channel 5
P4.6	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN6	Analog Input	ADC: input channel 6
PWM10	Digital Output	PWM: programmable PWM output
P4.7	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN7	Analog Input	ADC: input channel 7
PWM11	Digital Output	PWM: programmable PWM output

Port 5

Pin Name	Type	Description
P5.0	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN8	Analog Input	ADC: input channel 8
PWM20	Digital Output	PWM: programmable PWM output
P5.1	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN9	Analog Input	ADC: input channel 9
PWM21	Digital Output	PWM: programmable PWM output
P5.2	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN10	Analog Input	ADC: input channel 10
PWM30	Digital Output	PWM: programmable PWM output
P5.3	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AIN11	Analog Input	ADC: input channel 11
PWM31	Digital Output	PWM: programmable PWM output
P5.4	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AVREFH	Analog Input	ADC: external high reference voltage
P5.5	Digital I/O	GPIO: Bi-direction pin. Schmitt trigger structure as input mode. Built-in pull-up resistors.
AVREFL	Analog Input	ADC: external low reference voltage

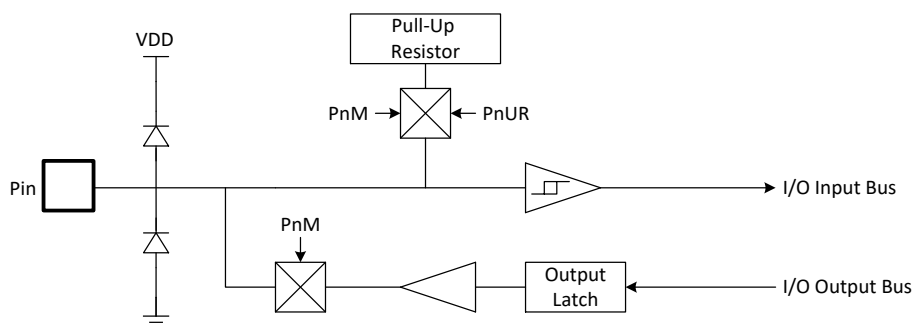
4.14 Pin Characteristic

Port	Open-Drain	Sink Current 100mA VSS+1.5V	Sink Current 20mA VSS+0.5V	External Interrupt	Wakeup (Level change)	Shared Pin
P0.0	-	V	-	-	V	XIN
P0.1	-	V	-	-	V	XOUT
P0.2	-	V	-	-	V	RST
P0.3	-	V	-	-	V	
P0.4	-	V	-	-	V	
P0.5	V	V	-	-	V	UTX
P0.6	V	V	-	-	V	URX
P0.7	-	V	-	-	V	SSN
P1.0	-	V	-	-	V	T2COM0
P1.1	V	V	-	-	V	T2COM1/MOSI
P1.2	V	V	-	-	V	T2COM2/MISO
P1.3	V	V	-	-	V	T2COM3/SCK
P1.4	-	-	V	-	V	SCL
P1.5	-	-	V	-	V	SDA
P1.6	-	-	V	-	V	
P1.7	-	-	V	-	V	
P2.0	-	-	V	-	-	
P2.1	-	-	V	V	-	INT0
P2.2	-	-	V	V	-	INT1/T2RL
P2.3	-	-	V	V	-	INT2/T2
P2.4	-	-	V	-	-	CM1O
P2.5	-	-	V	-	-	CM1P
P2.6	-	-	V	-	-	CM1N
P2.7	-	-	V	-	-	CM0O
P3.0	-	-	V	-	-	OP0N
P3.1	-	-	V	-	-	OP0P
P3.2	-	-	V	-	-	OP0O
P3.3	-	-	V	-	-	OP1N
P3.4	-	-	V	-	-	OP1P/T2CC0
P3.5	-	-	V	-	-	OP1O/T2CC1
P3.6	-	-	V	-	-	CM0N/T2CC2
P3.7	-	-	V	-	-	CM0P/T2CC3
P4.0	-	-	V	-	-	AIN0/SWAT
P4.1	-	-	V	-	-	AIN1
P4.2	-	-	V	-	-	AIN2
P4.3	-	-	V	-	-	AIN3
P4.4	-	-	V	-	-	AIN4
P4.5	-	-	V	-	-	AIN5
P4.6	-	-	V	-	-	AIN6/PWM10
P4.7	-	-	V	-	-	AIN7/PWM11

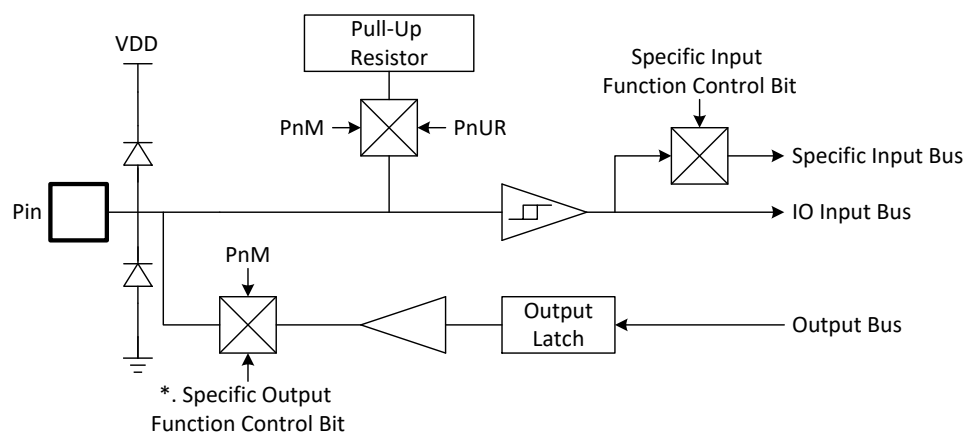
P5.0	-	-	V	-	-	AIN8/PWM20
P5.1	-	-	V	-	-	AIN9/PWM21
P5.2	-	-	V	-	-	AIN10/PWM30
P5.3	-	-	V	-	-	AIN11/PWM31
P5.4	-	-	V	-	-	AVREFH
P5.5	-	-	V	-	-	AVREFL

4.15 Pin Circuit Diagrams

Normal Bi-direction I/O Pin.

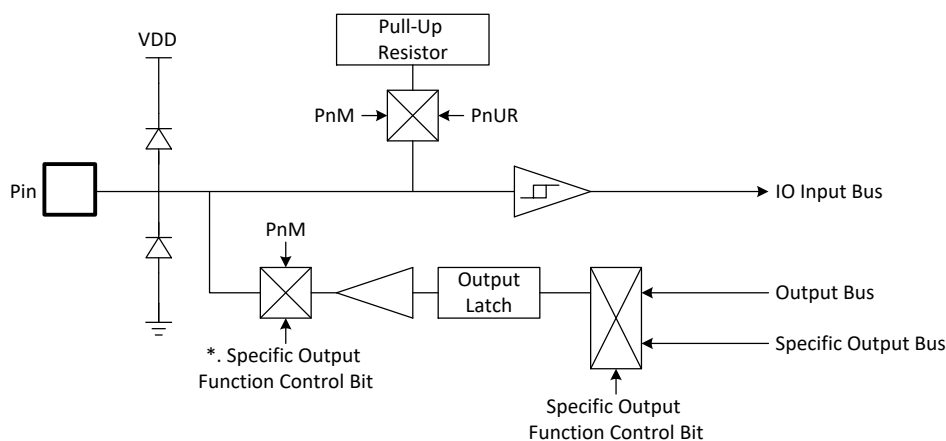


Bi-direction I/O Pin Shared with Specific Digital Input Function, e.g. INT0, Event counter, SIO, UART.



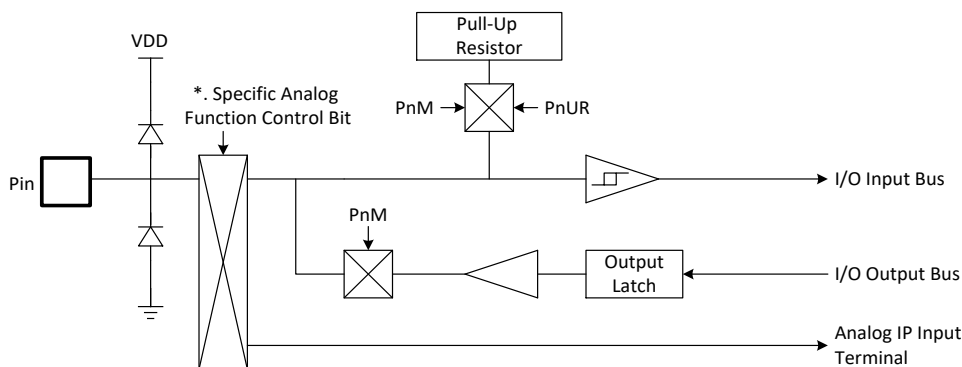
*. Some specific functions switch I/O direction directly, not through PnM register.

Bi-direction I/O Pin Shared with Specific Digital Output Function, e.g. PWM, SIO, UART.



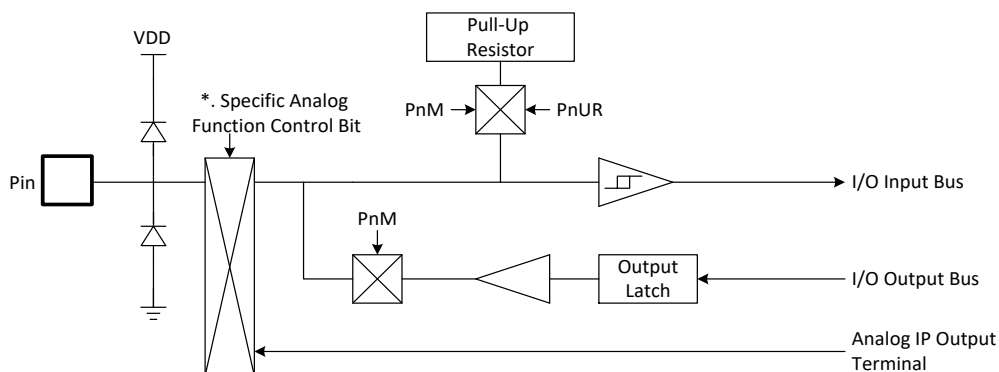
*. Some specific functions switch I/O direction directly, not through PnM register.

Bi-direction I/O Pin Shared with Specific Analog Input Function, e.g. XIN, ADC.



*. Some specific functions switch I/O direction directly, not through PnM register.

Bi-direction I/O Pin Shared with Specific Analog Output Function, e.g. XOUT...



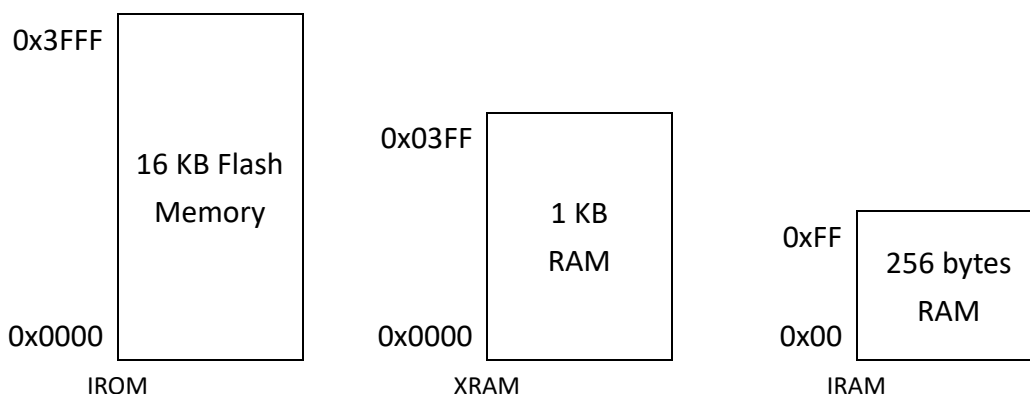
*. Some specific functions switch I/O direction directly, not through PnM register.

5 CPU

SN8F5000 family is an enhanced 8051 microcontroller (MCU). It is fully compatible with MCS-51 instructions, hence the ability to cooperate with modern development environment (e.g. Keil C51). Generally speaking, SN8F5000 CPU has 9.4 to 12.1 times faster than the original 8051 at the same frequency.

5.1 Memory Organization

SN8F5708 builds in three on-chip memories: internal RAM (IRAM), external RAM (XRAM), and program memory (IROM). The internal RAM is a 256-byte RAM which has higher access performance (direct and indirect addressing). By contrast, the external RAM has 1 KB of size, but it requires a longer access period. The program memory is a 16 KB non-volatile memory and has a maximum 8 MHz speed limitation.



5.2 Internal RAM (IRAM)

256 X 8-bit RAM (Internal Data Memory)

Address	RAM Location	
000h	Work Register Area	
01Fh		
020h	Bit Addressable Area	
02Fh		
030h	General Purpose Area	
...		
...		
07Fh		
080h	General Purpose Area (Indirect Access)	Special Function Register (Direct Access)
...		
...		
...		
0FFh		

00h-7Fh of RAM is direct and indirect access RAM

080h-0FFh store special function registers.

End of Bank 0

The 256-byte data RAM in internal data memory is a standard 8051 RAM access configuration. The upper 128-byte RAM is general purpose RAM and can configure by direct addressing access and indirect addressing access. The lower 128-byte can be indirect access RAM in general purpose or direct access RAM in special function register (SFR).

- 0x0000-0x007F: General purpose RAM contains work register area and bit addressable area. In this area, direct or indirect addressing can be used.
- 0x0000-0x001F: Work register area includes 4-bank. Each bank has 8 work registers (R0 - R7) which is selected by RS0/RS1 in PSW register.
- 0x0020-0x002F: Bit addressable area.

In the bit addressable area, user can read or write any single bit in this range by using the unique address for that bit. Supports 16bytes bit addressable RAM area giving 128 addressable bits. Each bit has individual address in the range from 00H to 7FH. Thus, the bit can be addressed directly. Bit0 of the byte 20H has bit address 00H and Bit 7 of the byte 20H has bit address 07H. Bit0 of the byte 2FH has bit address 78H and Bit 7 of the byte 2FH has bit address 7FH. When set "SETB 42H", it means the bit2 of the byte 28H is set.

Bit Addressable Area	Byte Address	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
	0x20	0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07
	0x21	0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F
	0x22	0x10	0x11	0x12	0x13	0x14	0x15	0x16	0x17
	0x23	0x18	0x19	0x1A	0x1B	0x1C	0x1D	0x1E	0x1F
	0x24	0x20	0x21	0x22	0x23	0x24	0x25	0x26	0x27
	0x25	0x28	0x29	0x2A	0x2B	0x2C	0x2D	0x2E	0x2F
	0x26	0x30	0x31	0x32	0x33	0x34	0x35	0x36	0x37
	0x27	0x38	0x39	0x3A	0x3B	0x3C	0x3D	0x3E	0x3F
	0x28	0x40	0x41	0x42	0x43	0x44	0x45	0x46	0x47
	0x29	0x48	0x49	0x4A	0x4B	0x4C	0x4D	0x4E	0x4F
	0x2A	0x50	0x51	0x52	0x53	0x54	0x55	0x56	0x57
	0x2B	0x58	0x59	0x5A	0x5B	0x5C	0x5D	0x5E	0x5F
	0x2C	0x60	0x61	0x62	0x63	0x64	0x65	0x66	0x67
	0x2D	0x68	0x69	0x6A	0x6B	0x6C	0x6D	0x6E	0x6F
	0x2E	0x70	0x71	0x72	0x73	0x74	0x75	0x76	0x77
	0x2F	0x78	0x79	0x7A	0x7B	0x7C	0x7D	0x7E	0x7F

- 0x0080~0x00FF: General purpose area in indirect addressing access or special function register in direct addressing access.

5.3 External RAM (XRAM)

1024 X 8-bit XRAM (Extension Data Memory)

The external RAM enlarges the capacity of variables; it is the lowest access performance in the contrast of internal RAM. Since frequently used variables and local variables are expected to store in internal RAM, the vast majority of external RAM usages are specific. It can be allocated as a variable storage area for lower priority tasks, or look-up table preloaded from ROM to speed up the access period.

Address	RAM Location	
0000h	General Purpose Area	Extension RAM Bank 0
...		
03FFh		End of Bank 0

The upper 1024-byte XRAM is general purpose RAM and can configure by MOVX instruction access.

5.4 Program Memory (IROM)

The program memory is a non-volatile storage area where stores code, look-up ROM table, and other data with occasional modification. It can be updated by debug tools like SN-Link3, and a program can also self-update via in-system program process (refer to In-system Program).

Address	ROM	Comment	
0000H	Reset vector	Reset vector	
0001H	General purpose area	User program	
0002H			
0003H	INT0 Interrupt vector	Interrupt vector	
000BH	TIMER0 Interrupt vector		
0013H	INT1 Interrupt vector		
001BH	TIMER1 Interrupt vector		
0023H	UART Interrupt vector		
002BH	TIMER2 Interrupt vector		
0043H	I2C Interrupt vector		
004BH	SPI Interrupt vector		
0053H	T2COM0 Interrupt vector		
005BH	T2COM1 Interrupt vector		
0063H	T2COM2 Interrupt vector		
006BH	T2COM3 Interrupt vector		
0083H	PWM1 Interrupt vector		
008BH	PWM2 Interrupt vector		
0093H	PWM3 Interrupt vector		
009BH	ADC Interrupt vector		
00A3H	Comparator 0 Interrupt vector	User program	
00ABH	Comparator 1 Interrupt vector		
00EBH	INT2 Interrupt vector		
00ECH	General purpose area		
.			
.			
.			
.	Reserved	End of user program	
3FF6H			
3FF7H			
.			
3FFEh			
3FFFH			

The ROM includes reset vector, Interrupt vector, general purpose area and reserved area. The reset vector is program beginning address. The interrupt vector is the head of interrupt service routine when any interrupt occurring. The general purpose area is main program area including main loop, sub-routines and data table.

- 0x0000 Reset vector: Program counter points to 0x0000 after any reset events (power on reset,

reset pin reset, watchdog reset, LVD reset...).

- 0x0001~0x0002: General purpose area to process system reset operation.
- 0x0003~0x00EB: Multi interrupt vector area. Each of interrupt events has a unique interrupt vector.
- 0x00EC~0x3FDF: General purpose area for user program and ISP (EEPROM function).
- 0x3FE0~0x3FF5: General purpose area for user program. Do not execute ISP.
- 0x3FF6~0x3FFF: Reserved area. Do not execute ISP.

5.5 Program Memory Security

The SN8F5708 provides security options at the disposal of the designer to prevent unauthorized access to information stored in FLASH memory. When enable security option, the ROM code is secured and not dumped complete ROM contents. ROM security rule is all address ROM data protected and outputs 0x00.

5.6 Data Pointer

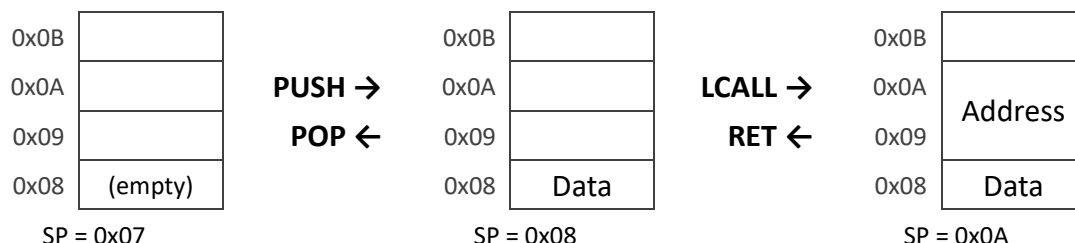
A data pointer helps to specify the XRAM and IROM address while performing MOVX and MOVC instructions. The microcontroller has two set of data pointer (DPH/DPL and DPH1/DPL1) which is selectable by DPS register. The DPC register controls two functions: next DPTR selection and automatically increase/decrease DPTR function.

The next DPTR selection can specify which DPTR is anticipated to use after perform MOVX @DPTR instruction. In other word, the DPS can automatically swap between the two data pointers. To enable this function: write 0 to DPSEL and fill 1 to NDPS firstly, then write 1 to DPSEL and fill 0 to NDPS register.

The automatically increase/decrease DPTR function can make an increment or decrement after perform MOVX @DPTR instruction. As a result, it enables a continuous external RAM access without re-specified DPTR value. Those functions are controlled by the DPC Register, where there are separate DPC register bits for each DPTR, to provide high flexibility in data transfers. The DPC Register address 0x93 points to the window where the actual DPC is selected using the DPS Register, same as for the DPTR.

5.7 Stack

Stack can be assigned to any area of internal RAM (IRAM). However, it requires manual assignment to ensure its area does not overlap other RAM's variables. An overflow and underflow stack could also mistakenly overwrite other RAM's variables; thus, these factors should be considered while arrange the size of stack.



By default, stack pointer (SP register) points to 0x07 which means the stack area begin at IRAM address 0x08. In other word, if a planned stack area is assigned from IRAM address 0xC0, the appropriate SP register is anticipated to set at 0xBF after system reset.

An assembly PUSH instruction costs one byte of stack. LCALL, ACALL instructions and interrupt respectively costs two bytes stack. POP-instruction decreases one count, and a RET/RETI subtract two counts of stack pointer.

* **Note: Stack and IRAM share the same area, Keil C51 compiler will not display "error" or "warning" when overlap condition is occurred so user must pay attention.**

5.8 Stack and Data Pointer Register

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SP	SP7	SP6	SP5	SP4	SP3	SP2	SP1	SP0
DPL	DPL7	DPL6	DPL5	DPL4	DPL3	DPL2	DPL1	DPL0
DPH	DPH7	DPH6	DPH5	DPH4	DPH3	DPH2	DPH1	DPH0
DPL1	DPL17	DPL16	DPL15	DPL14	DPL13	DPL12	DPL11	DPL10
DPH1	DPH17	DPH16	DPH15	DPH14	DPH13	DPH12	DPH11	DPH10
DPS	-	-	-	-	-	-	-	DPSEL
DPC	-	-	-	-	NDPS	ATMS	ATMD	ATME

SP Register (0x81)

Bit	Field	Type	Initial	Description
7..0	SP	R/W	0x07	Stack pointer

DPL Register (0x82)

Bit	Field	Type	Initial	Description
7..0	DPL[7:0]	R/W	0x00	Low byte of DPTR0

DPH Register (0x83)

Bit	Field	Type	Initial	Description
7..0	DPH[7:0]	R/W	0x00	High byte of DPTR0

DPL1 Register (0x84)

Bit	Field	Type	Initial	Description
7..0	DPL1[7:0]	R/W	0x00	Low byte of DPTR1

DPH1 Register (0x85)

Bit	Field	Type	Initial	Description
7..0	DPH1[7:0]	R/W	0x00	High byte of DPTR1

DPS Register (0x92)

Bit	Field	Type	Initial	Description
7..1	Reserved	R	0x00	
0	DPSEL	R/W	0	DPTR selection 0: DPH/DPL (DPTR0) is selected 1: DPH1/DPL1 (DPTR1) is selected

DPC Register (0x93)

Bit	Field	Type	Initial	Description
7..4	Reserved	R	0x0	
3	NDPS	R/W	0	Next DPTR selection The DPSEL loads this bit automatically after perform any

				MOVX @DPTR instruction.
2..1	ATMS/ATMD	R/W	00	Automatically increase/decrease DPTR (if ATME applied) 00: +1 after any MOVX @DPTR instruction 01: -1 after any MOVX @DPTR instruction 10: +2 after any MOVX @DPTR instruction 11: -2 after any MOVX @DPTR instruction
0	ATME	R/W	0	Automatically increase/decrease DPTR function 0: Disable 1: Enable

6 Special Function Registers

6.1 Special Function Register Memory Map

BIN HEX	000	001	010	011	100	101	110	111
F8	P5	P0M	P1M	P2M	P3M	P4M	P5M	PFLAG
F0	B	P0UR	P1UR	P2UR	P3UR	P4UR	P5UR	SRST
E8	P4	MD0	MD1	MD2	MD3	MD4	MD5	ARCON
E0	ACC	SPSTA	SPCON	SPDAT	P1OC	CLKSEL	CLKCMD	TCON0
D8	SOCON2	-	I2CDAT	I2CADR	I2CCON	I2CSTA	SMBSEL	SMBDST
D0	PSW	IEN4	ADM	ADB	ADR	VREFH	P4CON	P5CON
C8	T2CON	-	CRCL	CRCH	TL2	TH2	CMPT	PW3A
C0	IRCON	CCEN	CCL1	CCH1	CCL2	CCH2	CCL3	CCH3
B8	IEN1	IP1	SORELH	PW1DH	PW1DL	PW1A	PW2A	IRCON2
B0	P3	PW3M	PW3YL	PW3YH	PW3BL	PW3BH	PW3DL	PW3DH
A8	IEN0	IP0	SORELL	PW1M	PW1YL	PW1YH	PW1BL	PW1BH
A0	P2	PW2M	PW2YL	PW2YH	PW2BL	PW2BH	PW2DL	PW2DH
98	SOCON	S0BUF	IEN2	OPM	CMP0M	CMP1M	P2CON	P3CON
90	P1	P1W	DPS	DPC	PECMD	PEROML	PEROMH	PERAM
88	TCON	TMOD	TL0	TL1	TH0	TH1	CKCON	PEDGE
80	P0	SP	DPL	DPH	DPL1	DPH1	WDTR	PCON

* **Note: All SFRs in the left-most column are bit-addressable. (Every 0x0/0x8-ending SFR addresses are bit-addressable).**

6.2 Special Function register Description

0x80 - 0x9F Registers Description

Register	Address	Description
P0	080H	Port 0 data buffer.
SP	081H	Stack pointer register.
DPL	082H	Data pointer 0 low byte register.
DPH	083H	Data pointer 0 high byte register.
DPL1	084H	Data pointer 1 low byte register.
DPH1	085H	Data pointer 1 high byte register.
WDTR	086H	Watchdog timer clear register.
PCON	087H	System mode register.
TCON	088H	Timer 0 / 1 controls register.
TMOD	089H	Timer 0 / 1 mode register.
TL0	08AH	Timer 0 counting low byte register.
TL1	08BH	Timer 1 counting low byte register.
TH0	08CH	Timer 0 counting high byte register.
TH1	08DH	Timer 1 counting high byte register.
CKCON	08EH	Extended cycle controls register.
PEDGE	08FH	External interrupt edge controls register.
P1	090H	Port 1 data buffer.
P1W	091H	Port 1 wake-up controls register.
DPS	092H	Data pointer selects register.
DPC	093H	Data pointer controls register.
PECMD	094H	In-System Program command register.
PEROML	095H	In-System Program ROM address low byte
PEROMH	096H	In-System Program ROM address high byte
PERAM	097H	In-System Program RAM mapping address
S0CON	098H	UART control register.
S0BUF	099H	UART data buffer.
IEN2	09AH	Interrupts enable register
OPM	09BH	OP-AMP controls register.
CMP0M	09CH	Comparator 0 controls register.
CMP1M	09DH	Comparator 1 controls register.
P2CON	09EH	Port 2 configuration controls register.
P3CON	09FH	Port 3 configuration controls register.

0xA0 - 0xBF Registers Description

Register	Address	Description
P2	0xA0	Port 2 data buffer
PW2M	0xA1	PW2 controls register.
PW2YL	0xA2	PW2 cycle controls buffer low byte.
PW2YH	0xA3	PW2 cycle controls buffer high byte.
PW2BL	0xA4	PW2 B point dead band controls buffer low byte.
PW2BH	0xA5	PW2 B point dead band controls buffer high byte.
PW2DL	0xA6	PW2 duty controls buffer low byte.
PW2DH	0xA7	PW2 duty controls buffer high byte.
IEN0	0xA8	Interrupts enable register
IPO	0xA9	Interrupts priority register.
SORELL	0xAA	UART reload low byte register.
PW1M	0xAB	PW1 controls register.
PW1YL	0xAC	PW1 cycle controls buffer low byte.
PW1YH	0xAD	PW1 cycle controls buffer high byte.
PW1BL	0xAE	PW1 B point dead band controls buffer low byte.
PW1BH	0xAF	PW1 B point dead band controls buffer high byte.
P3	0xB0	Port 3 data buffer.
PW3M	0xB1	PW3 controls register.
PW3YL	0xB2	PW3 cycle controls buffer low byte.
PW3YH	0xB3	PW3 cycle controls buffer high byte.
PW3BL	0xB4	PW3 B point dead band controls buffer low byte.
PW3BH	0xB5	PW3 B point dead band controls buffer high byte.
PW3DL	0xB6	PW3 duty controls buffer low byte.
PW3DH	0xB7	PW3 duty controls buffer high byte.
IEN1	0xB8	Interrupts enable register
IP1	0xB9	Interrupts priority register.
SORELH	0xBA	UART reload high byte register.
PW1DL	0xBB	PW1 duty controls buffer low byte.
PW1DH	0xBC	PW1 duty controls buffer high byte.
PW1A	0xBD	PW1 A point dead band controls buffer.
PW2A	0xBE	PW2 A point dead band controls buffer.
IRCON2	0xBF	Interrupts request register.

0xC0 - 0xDF Registers Description

Register	Address	Description
IRCON	0xC0	Interrupts request register.
CCEN	0xC1	Timer 2 Compare /capture enable register.
CCL1	0xC2	Timer 2 Compare /capture module 1 low byte register.
CCH1	0xC3	Timer 2 Compare /capture module 1 high byte register.
CCL2	0xC4	Timer 2 Compare /capture module 2 low byte register.
CCH2	0xC5	Timer 2 Compare /capture module 2 high byte register.
CCL3	0xC6	Timer 2 Compare /capture module 3 low byte register.
CCH3	0xC7	Timer 2 Compare /capture module 3 high byte register.
T2CON	0xC8	Timer 2 controls register.
-	0xC9	-
CRCL	0xCA	Timer 2 Compare/capture module 0 & reload function low byte register.
CRCH	0xCB	Timer 2 Compare/capture module 0 & reload function high byte register.
TL2	0xCC	Timer 2 counting low byte register.
TH2	0xCD	Timer 2 counting high byte register.
CMPT	0xCE	Comparator 0 / 1 with PWM triggers select register.
PW3A	0xCF	PW3 A point dead band controls buffer.
PSW	0xD0	System flag register.
IEN4	0xD1	Interrupts enable register
ADM	0xD2	ADC controls register.
ADB	0xD3	ADC data buffer.
ADR	0xD4	ADC resolution selects register.
VREFH	0xD5	ADC reference voltage controls register.
P4CON	0xD6	Port 4 configuration controls register.
P5CON	0xD7	Port 5 configuration controls register.
SOCON2	0xD8	UART baud rate controls register.
-	0xD9	-
I2CDAT	0xDA	I2C data buffer.
I2CADR	0xDB	Own I2C slave address.
I2CCON	0xDC	I2C interface operation control register.
I2CSTA	0xDD	I2C Status Code.
SMBSEL	0xDE	SMBus mode controls register.
SMBDST	0xDF	SMBus internal timeout register.

0xE0 - 0xFF Registers Description

Register	Address	Description
ACC	0xE0	Accumulator register.
SPSTA	0xE1	SPI statuses register.
SPCON	0xE2	SPI control register.
SPDAT	0xE3	SPI data buffer.
P1OC	0xE4	Open drain controls register.
CLKSEL	0xE5	Clock switch selects register.
CLKCMD	0xE6	Clock switch controls Register.
TCON0	0xE7	Timer 0 / 1 clock controls register.
P4	0xE8	Port 4 data buffer.
MD0	0xE9	MDU controls register 0.
MD1	0xEA	MDU controls register 1.
MD2	0xEB	MDU controls register 2.
MD3	0xEC	MDU controls register 3.
MD4	0xED	MDU controls register 4.
MD5	0xEE	MDU controls register 5.
ARCON	0xEF	MDU Arithmetic control register.
B	0xF0	Multiplication/ division instruction data buffer.
P0UR	0xF1	Port 0 pull-up resister controls register.
P1UR	0xF2	Port 1 pull-up resister controls register.
P2UR	0xF3	Port 2 pull-up resister controls register.
P3UR	0xF4	Port 3 pull-up resister controls register.
P4UR	0xF5	Port 4 pull-up resister controls register.
P5UR	0xF6	Port 5 pull-up resister controls register.
SRST	0xF7	Software reset controls register.
P5	0xF8	Port 5 data buffer.
P0M	0xF9	Port 0 input/output mode register.
P1M	0xFA	Port 1 input/output mode register.
P2M	0xFB	Port 2 input/output mode register.
P3M	0xFC	Port 3 input/output mode register.
P4M	0xFD	Port 4 input/output mode register.
P5M	0xFE	Port 5 input/output mode register.
PFLAG	0xFF	Reset flag register.

6.3 System Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
ACC	ACC7	ACC6	ACC5	ACC4	ACC3	ACC2	ACC1	ACC0
B	B7	B6	B5	B4	B3	B2	B1	B0
PSW	CY	AC	F0	RS1	RS0	OV	F1	P

ACC Register (0xE0)

Bit	Field	Type	Initial	Description
7..0	ACC[7:0]	R/W	0x00	The ACC is an 8-bit data register responsible for transferring or manipulating data between ALU and data memory. If the result of operating is overflow (OV) or there is carry (C or AC) and parity (P) occurrence, then these flags will be set to PSW register.

B Register (0xF0)

Bit	Field	Type	Initial	Description
7..0	B[7:0]	R/W	0x00	The B register is used during multiplying and division instructions. It can also be used as a scratch-pad register to hold temporary data.

PSW Register (0xD0)

Bit	Field	Type	Initial	Description
7	CY	R/W	0	Carry flag. 0: Addition without carry, subtraction with borrowing signal, rotation with shifting out logic "0", comparison result < 0. 1: Addition with carry, subtraction without borrowing, rotation with shifting out logic "1", comparison result ≥ 0.
6	AC	R/W	0	Auxiliary carry flag. 0: If there is no a carry-out from 3rd bit of Accumulator in BCD operations. 1: If there is a carry-out from 3rd bit of Accumulator in BCD operations.
5	F0	R/W	0	General purpose flag 0. General purpose flag available for user.
4..3	RS[1:0]	R/W	00	Register bank select control bit, used to select working register bank. 00: 00H – 07H (Bnak0) 01: 08H – 0FH (Bnak1) 10: 10H – 17H (Bnak2) 11: 18H – 1FH (Bnak3)
2	OV	R/W	0	Overflow flag. 0: Non-overflow in Accumulator during arithmetic Operations. 1: overflow in Accumulator during arithmetic Operations.
1	F1	R/W	0	General purpose flag 1. General purpose flag available for user.
0	P	R	0	Parity flag. Reflects the number of '1's in the Accumulator. 0: if Accumulator contains an even number of '1's. 1: Accumulator contains an odd number of '1's.

6.4 Register Declaration

SN8F5708 has many registers to control various functions, but SFR name is not predefined in the C51 / A51 compiler. To make programming easier and therefore need to add header files to declare SFR name.

When using the assembly code programs, please add the following sentence.

```
1 $NOMOD51 ;Do not recognize the 8051-specific predefined special register.
2 #include <SN8F5708.H>
```

When using the C code programs, please add the following sentence.

```
1 #include <SN8F5708.H>
```

After adding the header file, user can use name of registers to program. During compilation, the compiler will register name translate into register position through the header file.

Different devices need to use a different header file to declare, but the option file is to use the same.

Device	Header file	Options file
SN8F5705	SN8F5705.h	OPTIONS_SN8F5708.A51
SN8F5707	SN8F5707.h	
SN8F5708	SN8F5708.h	
SN8F570870	SN8F570870.h	
SN8F57082	SN8F57082.h	
SN8F57084	SN8F57084.h	
SN8F57085	SN8F57085.h	
SN8F57086	SN8F57086.h	
SN8F57087	SN8F57087.h	

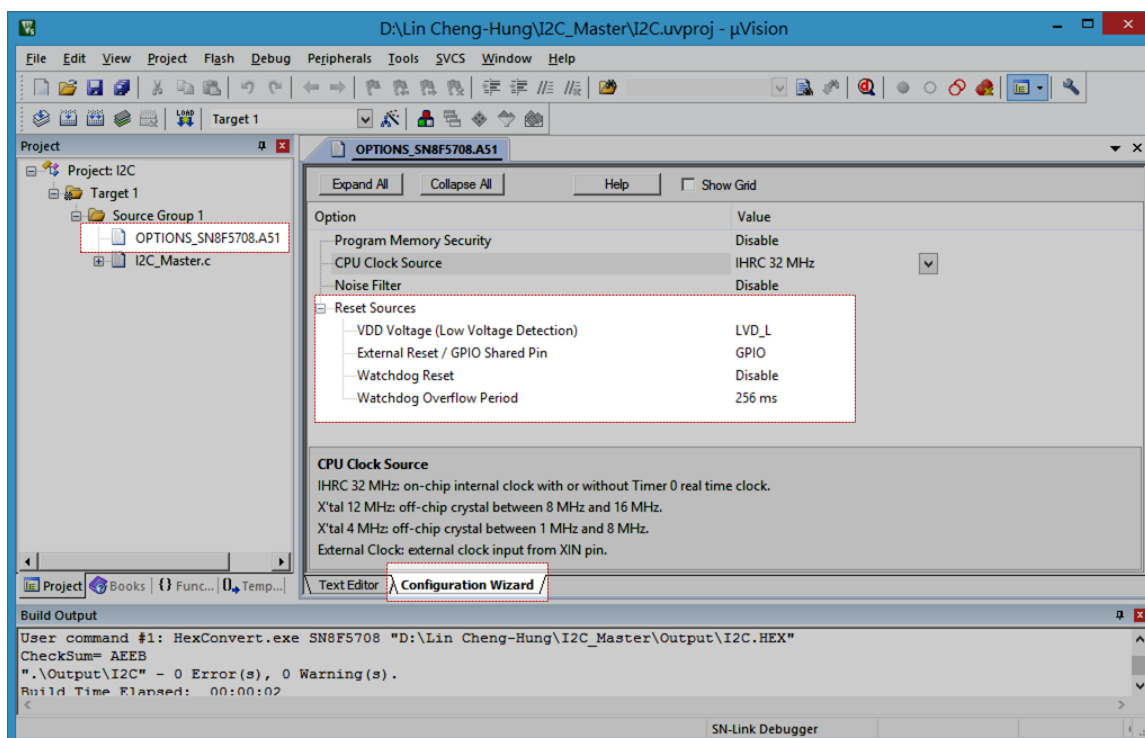
7 Reset and Power-on Controller

The reset and power-on controller has five reset sources: low voltage detectors (LVDs), watchdog, programmable external reset pin, and software reset. The first three sources would trigger an additional power-on sequence. Subsequently, the microcontroller initializes all registers and starts program execution with its reset vector (ROM address 0x0000).

7.1 Configuration of Reset and Power-on Controller

SONiX publishes an *OPTIONS_SN8F5708.A51* file in *SN-Link Driver for Keil C51.exe* (downloadable on cooperative website: www.sonix.com.tw). This *options* file contains appropriate parameters of reset sources and CPU clock source selection, and is strongly recommended to add to Keil project. *SN8F5000 Debug Tool Manual* provides the further detail of this configuration. The option items are as following:

- Program Memory Security
- CPU Clock Source
- Noise Filter
- Reset Source : VDD Voltage (Low Voltage Detection)
- Reset Source : External Reset / GPIO Shared Pin
- Reset Source : Watchdog Reset & Overflow Period



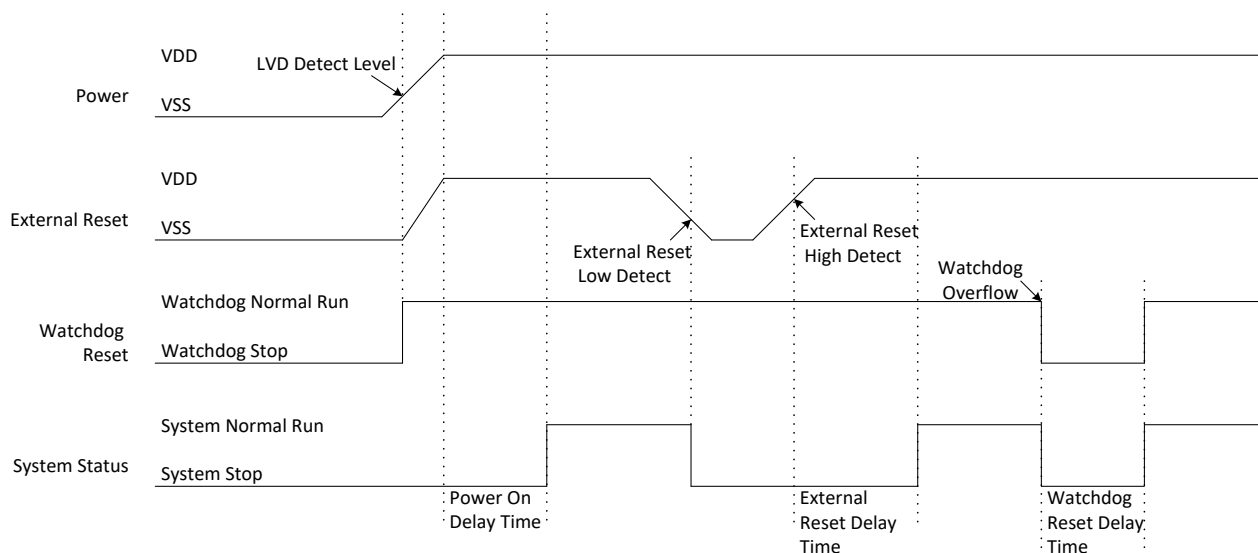
The code option is the system hardware configurations including oscillator type, noise filter option, watchdog timer operation, LVD option, reset pin option and flash ROM security control. The code

option items are as following table:

Code Option	Content	Function Description
Program Memory Security	Security Disable	Disable ROM code Security function
	Security Enable	Enable ROM code Security function
CPU Clock Source	IHRC 32MHz	High speed internal 32MHz RC. XIN/XOUT pins are bi-direction GPIO mode
	IHRC 32MHz with RTC	High speed internal 32MHz RC with low speed crystal/resonator (e.g. 32.768kHz). Low speed crystal/resonator for Timer 0 real time clock.
	X'tal 12MHz	High speed crystal /resonator (e.g. 12MHz) for external high clock oscillator
	X'tal 4MHz	Standard crystal /resonator (e.g. 4M) for external high clock oscillator
	External Clock	XIN pin connect external clock (1M ~32M), XOUT pin is bi-direction GPIO mode
Noise Filter	Disable	Disable Noise Filter
	Enable	Enable Noise Filter
LVD	LVD_L	LVD will reset chip if VDD is below 1.8V.
	LVD_M	LVD will reset chip if VDD is below 1.8V. The LVD24 flag will indicate if VDD is below 2.4V.
	LVD_H	LVD will reset chip if VDD is below 2.4V. The LVD33 flag will indicate if VDD is below 3.3V.
	LVD_Max	LVD will reset chip if VDD is below 3.3V
External Reset	Reset with De-bounce	Enable External reset pin with De-bounce
	Reset without De-bounce	Enable External reset pin without De-bounce
	GPIO with P02	Enable P02
Watchdog Reset	Always	Watchdog timer is always on enable even in STOP mode and IDLE mode
	Enable	Enable watchdog timer. Watchdog timer stops in STOP mode and IDLE mode
	Disable	Disable Watchdog function
Watchdog Overflow Period	64ms	Watchdog timer clock source $F_{ILRC}/4$
	128ms	Watchdog timer clock source $F_{ILRC}/8$
	256ms	Watchdog timer clock source $F_{ILRC}/16$
	512ms	Watchdog timer clock source $F_{ILRC}/32$

7.2 Power-on Sequence

A power-on sequence would be triggered by LVD, watchdog, and external reset pin. It takes place between the end of reset signal and program execution. Overall, it includes two stages: power stabilization period, and clock stabilization period.

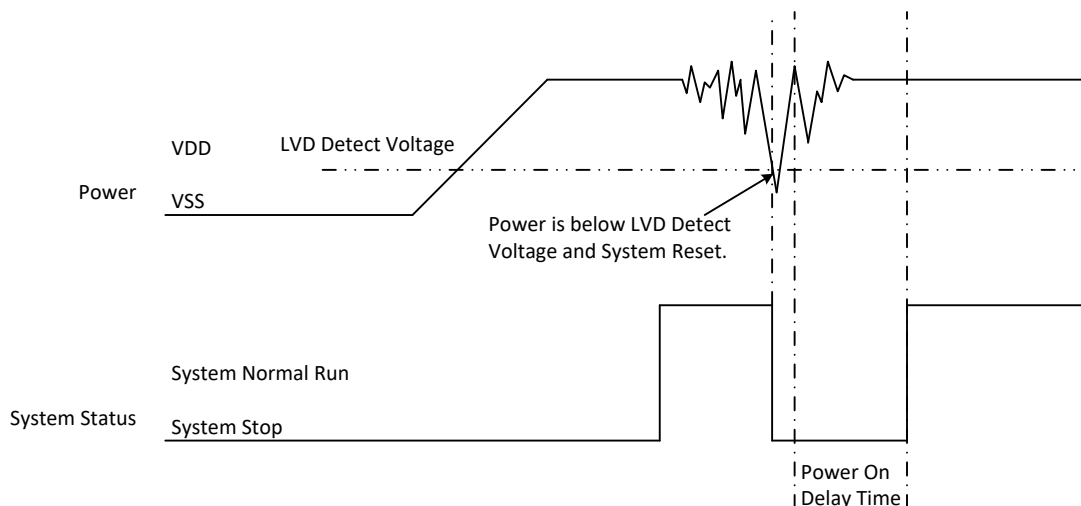


The power stabilization period spends 4.6ms in typical condition. Afterward the microcontroller fetches CPU Clock Source selection automatically. The selected clock source would be driven, and the system counts 2048 times of the clock period and 5 times of the internal low-speed oscillator clocks to ensure its reliability.

* **Note:** In high power noise environment, user can put 10ohm resistor in the front of 0.1uF capacitor & VDD PAD to suppress power noise and avoid IC damage.

7.3 LVD Reset

The low voltage detectors monitor VDD pin's voltage at three levels: 1.8 V, 2.4 V and 3.3 V. Depend on low voltage detection configuration, the comparison result can be seen as a system reset signal, or simply as LVD24/LVD33 register flags. The table below lists four different low voltage detection configurations, from LVD_Max to LVD_L, and the respectively results of VDD pin's condition.



Condition	LVD_Max	LVD_H	LVD_M	LVD_L
$VDD \leq 3.3\text{ V}$	Reset	LVD33 = 1	-	-
$VDD \leq 2.4\text{ V}$	Reset	Reset	LVD24 = 1	-
$VDD \leq 1.8\text{ V}$	Reset	Reset	Reset	Reset

7.4 Watchdog Reset

Watchdog is a periodic reset signal generator for the purpose of monitoring the execution flow. Its internal timer is expected to be cleared in a check point of program flow; therefore, the actual reset signal would be generated only after a software problem occurs. Writing 0x5A to WDTR is the proper method to place a check point in program.

```
1 WDTR = 0x5A;
```

Watchdog timer interval time = $256 * 1 / (\text{Internal Low-Speed oscillator frequency} / \text{WDT Pre-scaler})$
 $= 256 / (F_{ILRC} / \text{WDT Pre-scaler}) \dots \text{sec}$

Internal low-speed oscillator	WDT pre-scaler	Watchdog interval time
$F_{ILRC} = 16 \text{ kHz}$	$F_{ILRC} / 4$	$256 / (16000 / 4) = 64 \text{ ms}$
	$F_{ILRC} / 8$	$256 / (16000 / 8) = 128 \text{ ms}$
	$F_{ILRC} / 16$	$256 / (16000 / 16) = 256 \text{ ms}$
	$F_{ILRC} / 32$	$256 / (16000 / 32) = 512 \text{ ms}$

The operation mode of watchdog is configurable in options file:

Always mode counts its internal timer in all CPU operation modes (normal, IDLE, SLEEP);

Enable mode counts its internal timer during CPU stays in normal mode, and it would not trigger watchdog reset in IDLE and STOP modes;

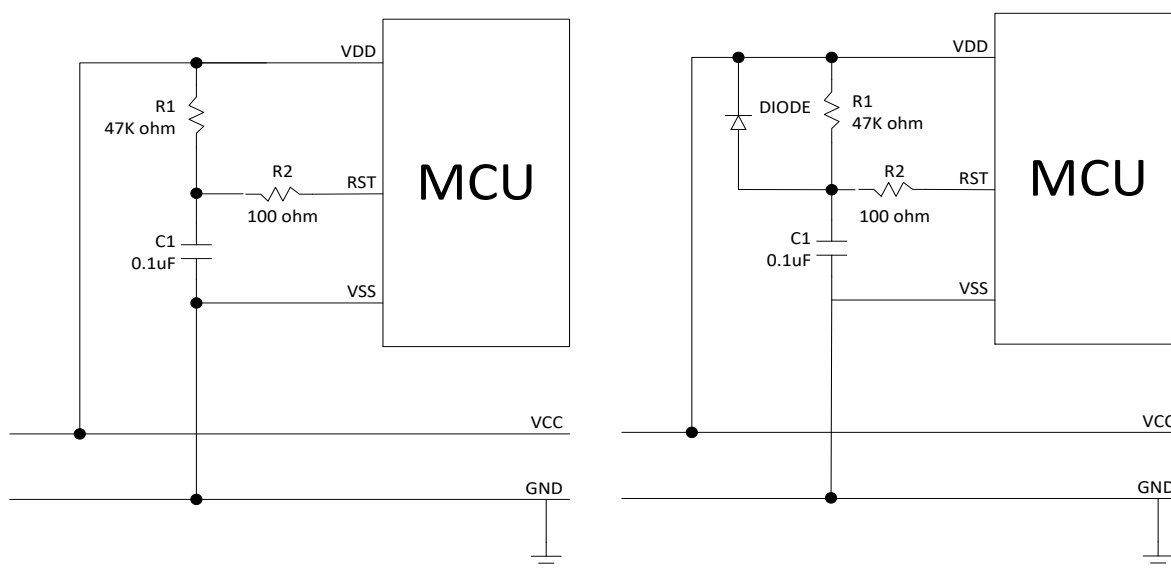
Disable mode suspends its internal timer at all CPU modes, and the watchdog would not trigger in this condition.

When watchdog is operating in always mode, the system will consume additional power.

7.5 External Reset Pin

Programmable external reset pin is configurable in *options file*. Once it is enabled, it monitors its shared pin's logic level. A logical low (lower than 30% of VDD) would immediately trigger system reset until the input is recovered to high (larger than 70% of VDD).

An optional de-bounce period can improve reset signal's stability. Instead of immediate reset, the system reset requires an 8-ms-long logic low to avoid bouncing from a button key. Any signal lower than de-bounce period would not affect the CPU's execution.



✱ **Note:**

1. The reset circuit is no any protection against unusual power or brown out reset on the left side of the figure.
2. The R2 100 ohm resistor of "Simply reset circuit" and "Diode & RC reset circuit" is necessary to limit any current flowing into reset pin from external capacitor C in the event of reset pin breakdown due to Electrostatic Discharge (ESD) or Electrical Over-stress (EOS) on the right side of the figure.

7.6 Software Reset

A software reset would be generated after consecutively set SRSTREQ register. As a result, this procedure enables firmware's ability to reset microcontroller (e.g. reset after firmware update). The following sample C code repeatedly set the least bit of SRST register to perform software reset.

```
1  SRST = 0x01;
2  SRST = 0x01;
```

7.7 Reset and Power-on Controller Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PFLAG	POR	WDT	RST	-	-	LVD24	LVD33	-
SRST	-	-	-	-	-	-	-	SRSTREQ
WDTR	WDTR7	WDTR6	WDTR5	WDTR4	WDTR3	WDTR2	WDTR1	WDTR0

PFLAG Register

Bit	Field	Type	Initial	Description
7	POR	R	-	This bit is automatically set if the microcontroller has been reset by LVD.
6	WDT	R	-	This bit is automatically set if the microcontroller has been reset by watchdog.
5	RST	R	-	This bit is automatically set if the microcontroller has been reset by external reset pin.
4..3	Reserved	R	0	
2	LVD24	R	-	This bit is automatically set if the currently VDD pin is lower than 2.4 V
1	LVD33	R	-	This bit is automatically set if the currently VDD pin is lower than 3.3 V
0	Reserved	R	0	

SRST Register

Bit	Field	Type	Initial	Description
7..1	Reserved	R	0	
0	SRSTREQ	R/W	0	Consecutively set this bit for two times to trigger software reset.

WDTR Register (0x86)

Bit	Field	Type	Initial	Description
7..0	WDTR[7:0]	W	-	Watchdog clear is controlled by WDTR register. Moving 0x5A data into WDTR is to reset watchdog timer.

8 System Clock and Power Management

For power saving purpose, the microcontroller built in three different operation modes: normal, IDLE, and STOP mode.

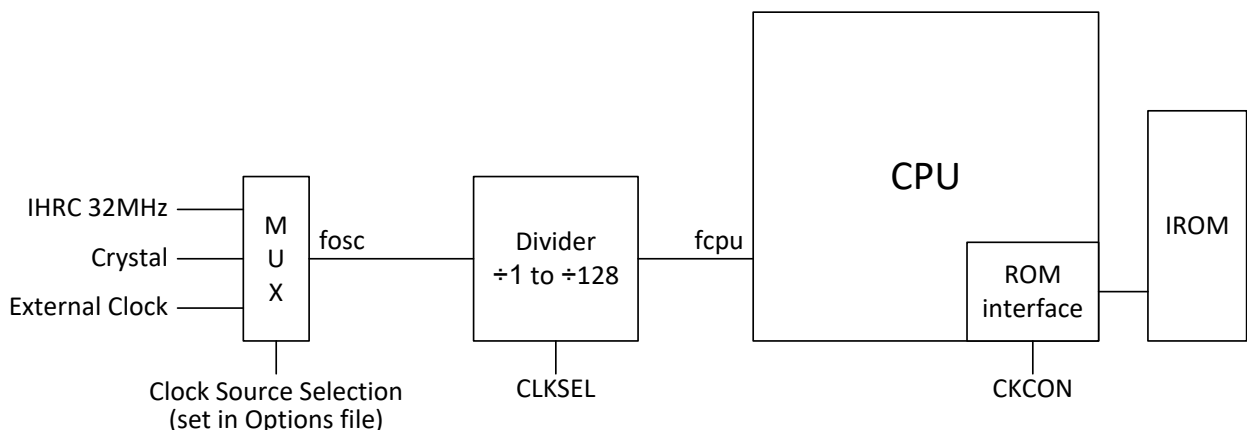
The normal mode means that CPU and peripheral functions are under normally execution. The system clock is based on the combination of source selection, clock divider, and program memory wait state. IDLE mode is the situation that temporarily suspends CPU clock and its execution, yet it remains peripherals' functionality (e.g. timers, PWM, SPI, UART, and I2C). By contrast, STOP mode disables all functions and clock generator until a wakeup signal to return normal mode.

8.1 System Clock

The microcontroller includes an on-chip clock generator (IHRC 32MHz), crystal/resonator driver, and an external clock input. The reset and power-on controller automatically loads clock source selection during power-on sequence. Therefore, the selected clock source is seen as 'fosc' domain which is a fixed frequency at any time.

Subsequently, the selected clock source (fosc) is divided by 1 to 128 times which is controlled by CLKSEL register. The CPU input the divided clock as its operation base (named fcpu). Applying CLKSEL's setting when CLKCMD register be written 0x69.

```
1  CLKSEL = 0x05;    // set fcpu = fosc / 4
2  CLKCMD = 0x69;    // Apply CLKSEL's setting
3  CKCON = 0x00;    // IROM fetch = fcpu / 1
```



ROM interface is built in between CPU and IROM (program memory). It optionally extends the data fetching cycle in order to support lower speed program memory.

$$\text{IROM fetching cycle (Instruction cycle)} \leq 8\text{MHz}$$

* **Note:** For user develop program in C language or assembly, the first line of the program “must be set” CLKSEL= 0x07~0x00, CLKMD= 0x69 and then set CKCON= 0x00~0x70, this priority cannot be modified.

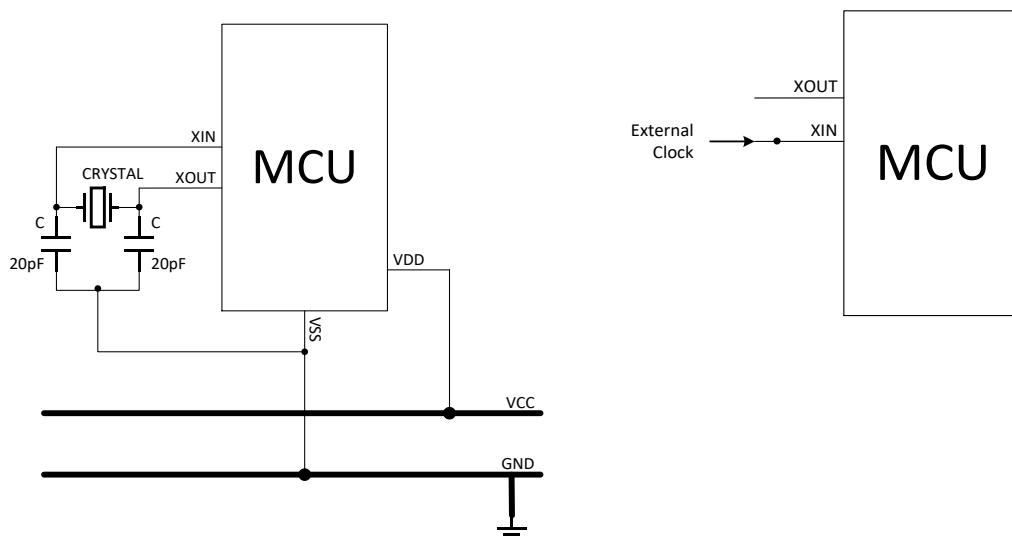
System clock rate and program memory extended cycle limitation as follows.

Code Option CPU Clock Source	Fcpu = CLKSEL[2:0]	IROM Fetch = CKCON[6:4]
IHRC 32M IHRC 32M with RTC External Clock (16-32MHz)	Only Support 000 = fosc / 128 001 = fosc / 64 010 = fosc / 32 011 = fosc / 16 100 = fosc / 8 101 = fosc / 4	Support 000 = fcpu / 1 => Recommend! 001 = fcpu / 2 010 = fcpu / 3 011 = fcpu / 4 100 = fcpu / 5 101 = fcpu / 6 110 = fcpu / 7 111 = fcpu / 8
X'tal 12M (Crystal 8-16MHz) External Clock (8-16MHz)	Only Support 000 = fosc / 128 001 = fosc / 64 010 = fosc / 32 011 = fosc / 16 100 = fosc / 8 101 = fosc / 4 110 = fosc / 2	
X'tal 12M (Crystal 4-8MHz) X'tal 4M (Crystal 1-4MHz) External Clock (1-8MHz)	Support 000 = fosc / 128 001 = fosc / 64 010 = fosc / 32 011 = fosc / 16 100 = fosc / 8 101 = fosc / 4 110 = fosc / 2 111 = fosc / 1	

8.2 High Speed Clock and Real time clock

High-speed clock has internal and external two-type. The external high-speed clock includes 4MHz, 12MHz crystal/ceramic and external clock input mode. The internal high-speed oscillator is 32MHz RC type. These high-speed oscillators are selected by *SN8F5708_OPTIONS.A51*.

- **IHRC 32M:** The system high-speed clock source is internal high-speed 32MHz RC type oscillator. In the mode, XIN and XOUT pins are bi-direction GPIO mode, and not to connect any external oscillator device.
- **IHRC 32M with RTC:** The system high-speed clock source is internal high-speed 32MHz RC type oscillator. In the mode, the XIN and XOUT pins switch to crystal mode to drive an off-chip 32.768 kHz crystal.
- **X'tal 12M:** The system high-speed clock source is external high-speed crystal/ceramic. The oscillator bandwidth is 4MHz~16MHz and connected to XIN/XOUT pins with 20pF capacitors to ground.
- **X'tal 4M:** The system high-speed clock source is external high-speed crystal/resonator. The oscillator bandwidth is 1MHz~4MHz and connected to XIN/XOUT pins with 20pF capacitors to ground.
- **External Clock:** The system high-speed clock source is external clock input mode. The input signal only connects to XIN pin, and the XOUT pin is bi-direction GPIO mode.



SN8F5708 supplies external low-speed clock (f_{RTC}) for the real time clock of Timer 0. In IHRC 32M with RTC mode, the XIN and XOUT pins switch to crystal mode to drive an off-chip 32.768 kHz crystal. The crystal is connected to XIN/XOUT pins with 20pF capacitors to ground.

8.3 Noise Filter

The Noise Filter controlled by Noise Filter option is a low pass filter and supports crystal mode. The purpose is to filter high rate noise coupling on high clock signal from external oscillator. In high noisy environment, enable Noise Filter option is the strongly recommendation to reduce noise effect.

8.4 Power Management

After the end of reset signal and power-on sequence, the CPU starts program execution at the speed of fcpu. Overall, the CPU and all peripherals are functional in this situation (categorized as normal mode).

The least two bits of PCON register (IDLE at bit 0 and STOP at bit 1) control the microcontroller's power management unit.

If IDLE bit is set by program, only CPU clock source would be gated. Consequently, peripheral functions (such as timers, PWM, and I2C) and clock generator (IHRC 32 MHz/crystal driver) remain execution in this status. Any change from P0/P1 input and interrupt events can make the microcontroller turns back to normal mode, and the IDLE bit would be cleared automatically.

- Any function can work in IDLE mode. Only CPU is suspended
- The IDLE mode wake-up sources are P0/P1 level change trigger and any interrupt event.

If STOP bit is set, by contrast, CPU, peripheral functions, and clock generator are suspended. Data storage in registers and RAM would be kept in this mode. Any change from P0/P1 can wake up the microcontroller and resume system's execution. STOP bit would be cleared automatically.

- CPU, peripheral functions, and clock generator are suspended.
- The STOP mode wake-up source is P0/P1 level change trigger.

For user who is develop program in C language, IDLE and STOP macros is strongly recommended to control the microcontroller's system mode, instead of set IDLE and STOP bits directly.

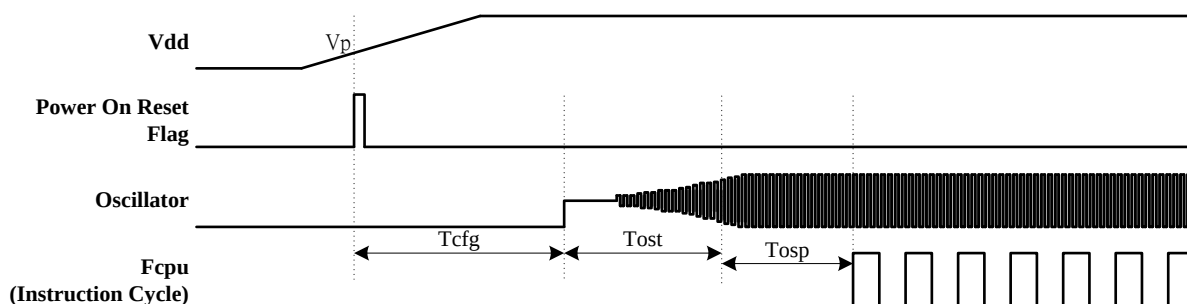
```
1  IDLE();
2  STOP();
```

* ***Note: Into IDLE mode or STOP mode by "Assembly Language" must be using MOV instruction.***

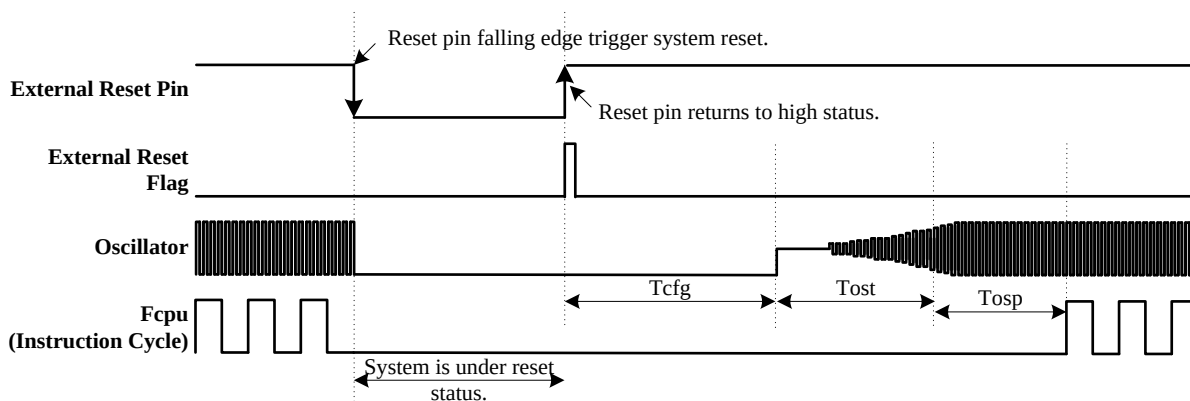
8.5 System Clock Timing

Parameter	Symbol	Description	Typical
Hardware configuration time	Tcfg	$8 \cdot F_{ILRC} + 2^{17} \cdot F_{IHRC}$	4.6ms @ $F_{ILRC} = 16\text{kHz}$ & $F_{IHRC} = 32\text{MHz}$
Oscillator start up time	Tost	The start-up time is depended on oscillator's material, factory and architecture. Normally, the low-speed oscillator's start-up time is lower than high-speed oscillator. The RC type oscillator's start-up time is faster than crystal type oscillator.	-
Oscillator warm-up time	Tosp	Oscillator warm-up time of reset condition. $2048 \cdot F_{hosc} + 5 \cdot F_{ILRC}$ (Power on reset, LVD reset, watchdog reset, external reset pin active.)	825us @ $F_{hosc} = 4\text{MHz}$ 441us @ $F_{hosc} = 16\text{MHz}$ 377us @ $F_{hosc} = 32\text{MHz}$
		Oscillator warm-up time of power down mode wake-up condition. $2048 \cdot F_{hosc} + 5 \cdot F_{ILRC}$Crystal/resonator type oscillator, e.g. 32768Hz crystal, 4MHz crystal, 16MHz crystal... $64 \cdot F_{hosc} + 5 \cdot F_{ILRC}$RC type oscillator, e.g. internal high-speed RC type oscillator.	X'tal: 825us @ $F_{hosc} = 4\text{MHz}$ 441us @ $F_{hosc} = 16\text{MHz}$ RC: 315us @ $F_{hosc} = 32\text{MHz}$

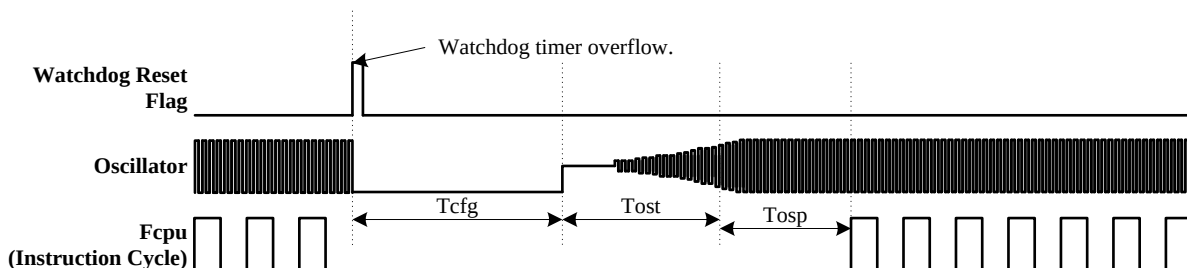
● Power On Reset Timing



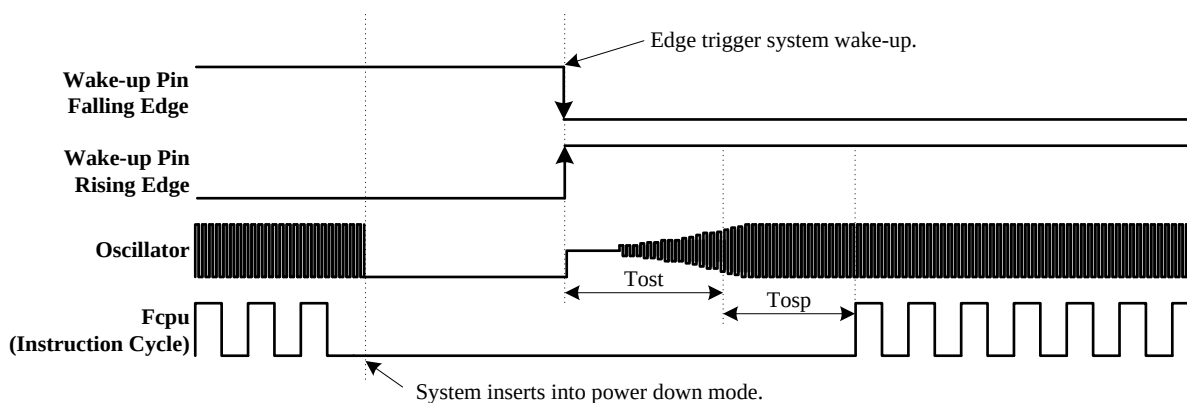
● External Reset Pin Reset Timing



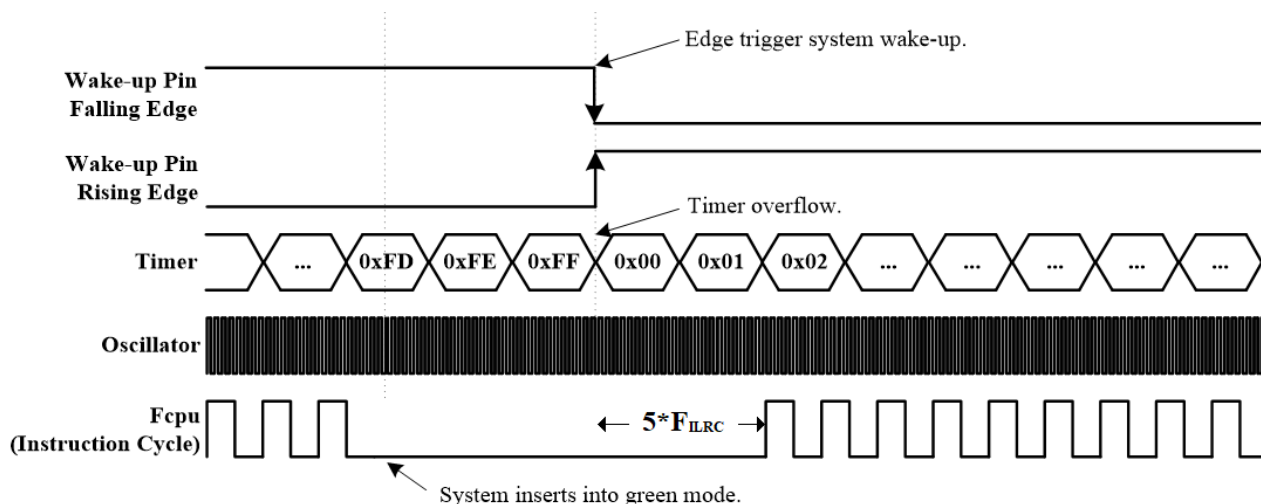
● Watchdog Reset Timing



● STOP Mode Wake-up Timing

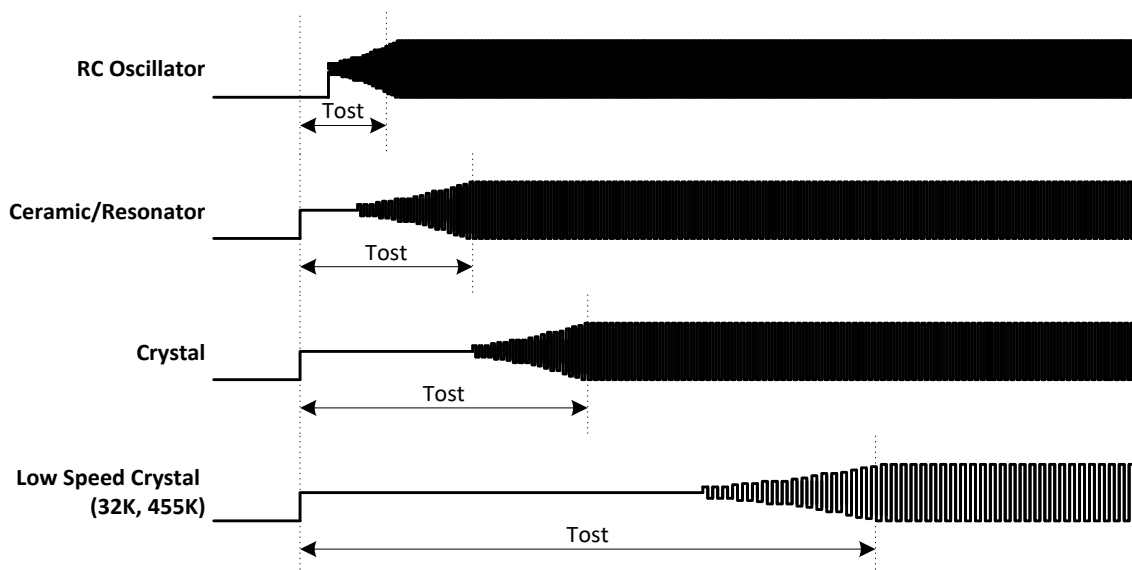


● IDLE Mode Wake-up Timing



● Oscillator Start-up Time

The start-up time is depended on oscillator's material, factory and architecture. Normally, the low-speed oscillator's start-up time is lower than high-speed oscillator.



8.6 System Clock and Power Management Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CKCON	-	PWSC2	PWSC1	PWSC0	ESYN	EWSC2	EWSC1	EWSC0
CLKSEL	-	-	-	-	-	CLKSEL2	CLKSEL1	CLKSEL0
CLKCMD	CMD7	CMD6	CMD5	CMD4	CMD3	CMD2	CMD1	CMD0
PCON	SMOD	-	-	-	P2SEL	GF0	STOP	IDLE
P1W	P17W	P16W	P15W	P14W	P13W	P12W	P11W	P10W

CKCON Register (0x8E)

Bit	Field	Type	Initial	Description
7	Reserved	R	0	
6..4	PWSC[2:0]	R/W	111	Extended cycle(s) applied to reading program memory 000: non 001: 1 cycle 010: 2 cycles 011: 3 cycles 100: 4 cycles 101: 5 cycles 110: 6 cycles 111: 7 cycles
3	ESYN	R/W	0	Extended extra cycles to write XRAM
2..0	EWSC[2:0]	R/W	001	Extended cycle(s) applied to reading XRAM 000: non 001: 1 cycle 010: 2 cycles 011: 3 cycles 100: 4 cycles 101: 5 cycles 110: 6 cycles 111: 7 cycles

CLKSEL Register (0xE5)

Bit	Field	Type	Initial	Description
7..3	Reserved	R	0x00	
2..0	CLKSEL[2:0]	R/W	111	CLKSEL would be applied by writing CLKCMD. 000: fcpu = fosc / 128 001: fcpu = fosc / 64 010: fcpu = fosc / 32 011: fcpu = fosc / 16 100: fcpu = fosc / 8 101: fcpu = fosc / 4 110: fcpu = fosc / 2 111: fcpu = fosc / 1

CLKCMD Register (0xE6)

Bit	Field	Type	Initial	Description
7..0	CMD[7:0]	W	0x00	Writing 0x69 to apply CLKSEL's setting.

PCON Register (0x87)

Bit	Field	Type	Initial	Description
7				Refer to other chapter(s)
6..4	Reserved	R	0x00	
3	P2SEL	R/W	1	High-order address byte configuration bit. Chooses the higher byte of address ("XRAM[15:8]") during MOVX @Ri operations 0: The "XRAM[15:8]" = "P2REG". The "P2REG" is the contents of Port2 output register. 1: The "XRAM[15:8]" = 0x00.
2	GF0	R/W	0	General Purpose Flag
1	STOP	R/W	0	1: Microcontroller switch to STOP mode
0	IDLE	R/W	0	1: Microcontroller switch to IDLE mode

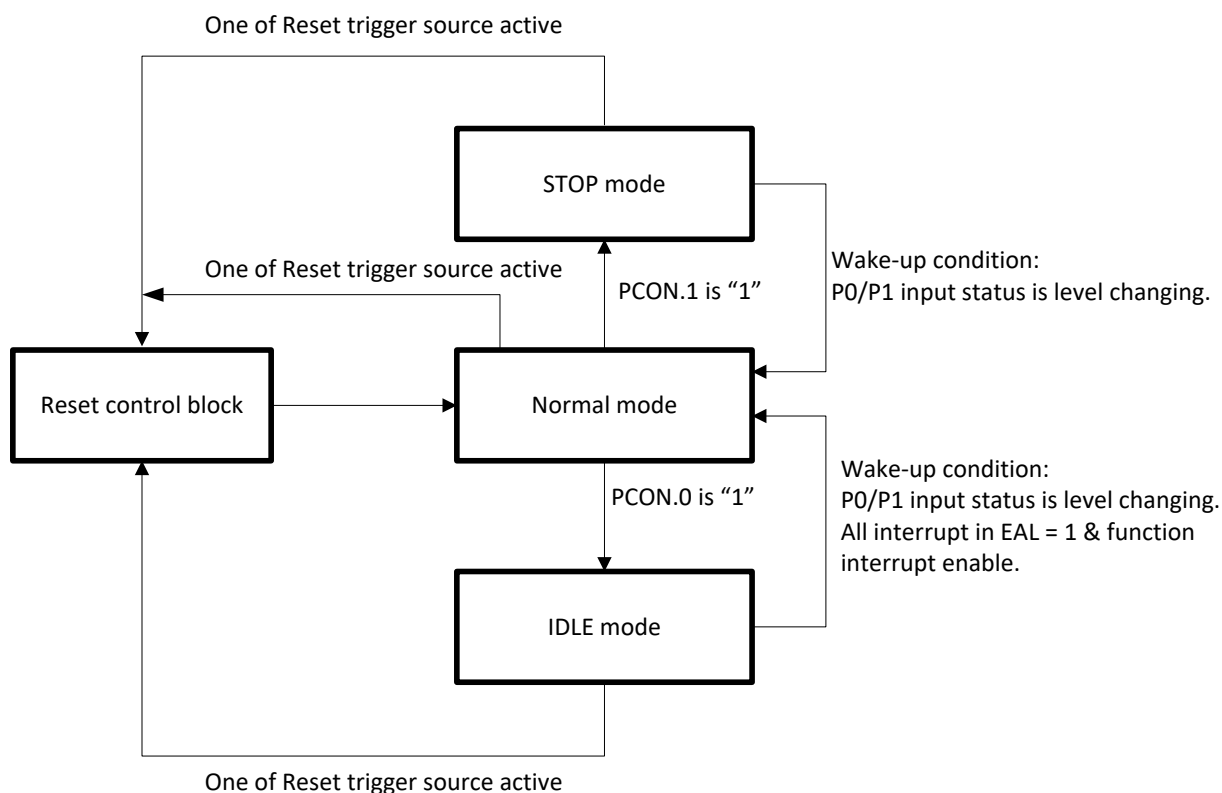
P1W Register (0x91)

Bit	Field	Type	Initial	Description
7..0	P1nW	R/W	0	0: Disable P1.n wakeup functionality 1: Enable P1.n wakeup functionality

9 System Operating Mode

The chip builds in three operating mode for difference clock rate and power saving reason. These modes control oscillators, op-code operation and analog peripheral devices' operation.

- Normal mode: System high-speed operating mode
- IDLE mode: System idle mode (Green mode)
- STOP mode: System power saving mode (Sleep mode)



The operating mode clock control as following table:

Operating Mode	Normal Mode	IDLE Mode	STOP Mode
IHRC	IHRC: Running Ext. OSC: Disable	IHRC: Running Ext. OSC: Disable	Stop
ILRC	Running	Running	Watchdog always: Running Other : stop
Ext. OSC	IHRC: Disable Ext. OSC : Running	IHRC: Disable Ext. OSC : Running	Stop
CPU instruction	Executing	Stop	Stop
Timer 0 (Timer, Event counter)	Active by TR0	Active by TR0	Inactive
Timer 1 (Timer, Event counter)	Active by TR1	Active by TR1	Inactive
Timer 2 (Timer, capture, T2COM)	Active as enable	Active as enable	Inactive
PWM1	Active as enable	Active as enable	Inactive
PWM2	Active as enable	Active as enable	Inactive
PWM3	Active as enable	Active as enable	Inactive
UART	Active as enable	Active as enable	Inactive
SPI	Active as enable	Active as enable	Inactive
I2C	Active as enable	Active as enable	Inactive
ADC	Active as enable	Active as enable	Inactive
Comparator 0	Active as enable	Active as enable	Active as enable
Comparator 1	Active as enable	Active as enable	Active as enable
OPA	Active as enable	Active as enable	Active as enable
Watchdog timer	By Watchdog Code option	By Watchdog Code option	By Watchdog Code option
Internal interrupt	All active	All active	All inactive
External interrupt	All active	All active	All inactive
Wakeup source	-	P0, P1, Reset, All interrupt in EAL = 1 & function interrupt enable	P0, P1, Reset

- Ext. OSC: External high/low-speed oscillator (XIN/XOUT).
- IHRC: Internal high-speed oscillator RC type.
- ILRC: Internal low-speed oscillator RC type.

9.1 Normal Mode

The Normal Mode is system high clock operating mode. The system clock source is from high speed oscillator. The program is executed. After power on and any reset trigger released, the system inserts into normal mode to execute program. When the system is wake-up from STOP/IDLE mode, the system also inserts into normal mode. In normal mode, the high speed oscillator is active, and the power consumption is largest of all operating modes.

- The program is executed, and full functions are controllable.
- The system rate is high speed.
- The high speed oscillator and internal low speed RC type oscillator are active.
- Normal mode can be switched to other operating modes through PCON register.
- STOP/IDLE mode is wake-up to normal mode.

9.2 STOP Mode

The STOP mode is the system ideal status. No program execution and oscillator operation. Only internal regulator is active to keep all control gates status, register status and SRAM contents. The STOP mode is waked up by P0/P1 hardware level change trigger. P0 wake-up function is always enables. The STOP mode is wake-up to normal mode. Inserting STOP mode is controlled by stop bit of PCON register. When stop = 1, the system inserts into STOP Mode. After system wake-up from STOP mode, the stop bit is disabled (zero status) automatically.

- The program stops executing, and full functions are disabled.
- All oscillators including external high/low speed oscillator, internal high speed oscillator and internal low speed oscillator stop.
- Only internal regulator is active to keep all control gates status, register status and SRAM contents.
- The system inserts into normal mode after wake-up from STOP mode.
- The STOP mode wake-up source is P0/P1 level change trigger.

9.3 IDLE Mode

The IDLE mode is another system ideal status not like STOP mode. In STOP mode, all functions and hardware devices are disabled. But in IDLE mode, the system clock source keeps running, so the power consumption of IDLE mode is larger than STOP mode. In IDLE mode, the program isn't executed, but the timer with wake-up function is active as enabled, and the timer clock source is the non-stop system clock. The IDLE mode has 2 wake-up sources. One is the P0/P1 level change trigger wake-up. The other one is any interrupt in EAL = 1 & function interrupt enable. That's mean users can setup any function with interrupt enable, and the system is waked up until the interrupt issue. Inserting IDLE mode is controlled by idle bit of PCON register. When idle = 1, the system inserts into IDLE mode. After system wake-up from IDLE mode, the idle bit is disabled (zero status) automatically.

- The program stops executing, and full functions are disabled.
- Only the timer with wake-up function is active.
- The oscillator to be the system clock source keeps running, and the other oscillators operation depend on system operation mode configuration.
- If inserting IDLE mode from normal mode, the system insets to normal mode after wake-up.
- The IDLE mode wake-up sources are P0/P1 level change trigger.
- If the function clock source is system clock, the functions are workable as enabled and under IDLE mode, e.g. Timer, PWM, event counter...
- All interrupt in EAL = 1 & function interrupt enable can wake-up in IDLE mode.

9.4 Wake up

Under STOP mode (sleep mode) or idle mode, program doesn't execute. The wakeup trigger can wake the system up to normal mode. The wakeup trigger sources are external trigger (P0/P1 level change) and internal trigger (any interrupt in EAL = 1 & function interrupt enable). The wakeup function builds in interrupt operation issued request flag and trigger system executing interrupt service routine as system wakeup occurrence.

When the system is in STOP mode the high clock oscillator stops. When waked up from STOP mode, MCU waits for 2048 external high-speed oscillator clocks + 5 internal low-speed oscillator clocks and 64 internal high-speed oscillator clocks + 5 internal low-speed oscillator clocks as the wakeup time to stable the oscillator circuit. When waked up from IDLE mode, MCU waits for 5 internal low-speed oscillator clocks. After the wakeup time, the system goes into the normal mode.

The value of the external high clock oscillator wakeup time is as the following.

The Wakeup time = $1/F_{osc} * 2048$ (sec) + $1/F_{osc} * 5$ + high clock start-up time (STOP mode)

The Wakeup time = $1/F_{osc} * 5$ (IDLE mode)

Example: In STOP mode (sleep mode), the system is waked up. After the wakeup time, the system goes into normal mode. The wakeup time is as the following.

The wakeup time = $1/F_{osc} * 2048 + 1/F_{osc} * 5 = 0.825$ ms ($F_{osc} = 4\text{MHz}$)

The total wakeup time = 0.825 ms + oscillator start-up time

The value of the internal high clock oscillator RC type wakeup time is as the following.

The Wakeup time = $1/F_{osc} * 64$ (sec) + $1/F_{osc} * 5$ + high clock start-up time

Example: In STOP mode (sleep mode), the system is waked up. After the wakeup time, the system goes into normal mode. The wakeup time is as the following.

The wakeup time = $1/F_{osc} * 64 + 1/F_{osc} * 5 = 315$ us ($F_{osc} = 32\text{MHz}$)

* ***Note: The high clock start-up time is depended on the VDD and oscillator type of high clock.***

Under STOP mode and green mode, the I/O ports with wakeup function are able to wake the system up to normal mode. The wake-up trigger edge is level changing in rising edge or falling edge. The Port 0 and Port 1 have wakeup function. Port 0 wakeup functions always enables, but the Port 1 is controlled by the P1W register.

P1W Register (0x91)

Bit	Field	Type	Initial	Description
7..0	P1nW	R/W	0	0: Disable P1.n wakeup functionality 1: Enable P1.n wakeup functionality

10 Interrupt

The MCU provides 19 interrupt sources (3 external and 16 interrupt) with 4 priority levels. Each interrupt source includes one or more interrupt request flag(s). When interrupt event occurs, the associated interrupt flag is set to logic 1. If both interrupt enable bit and global interrupt (EAL=1) are enabled, the interrupt request is generated and interrupt service routine (ISR) will be started. Most interrupt request flags must be cleared by software. However, some interrupt request flags can be cleared by hardware automatically. In the end, ISR is finished after complete the RETI instruction. The summary of interrupt source, interrupt vector, priority order and control bit are shown as the table below.

Interrupt	Enable Interrupt	Request (IRQ)	IRQ Clearance	Priority / Vector
System Reset	-	-	-	0 / 0x0000
INT0	EX0	IE0	Automatically	1 / 0x0003
PWM1	EPWM1	PWM1F	By firmware	2 / 0x0083
I2C	EI2C	SI	By firmware	3 / 0x0043
Timer 0	ET0	TF0	Automatically	4 / 0x000B
PWM2	EPWM2	PWM2F	By firmware	5 / 0x008B
SPI	ESPI	SPIF / MODF	By firmware	6 / 0x004B
INT1	EX1	IE1	Automatically	7 / 0x0013
PWM3	EPWM3	PWM3F	By firmware	8 / 0x0093
T2COM0	ET2C0	TF2C0	Automatically	9 / 0x0053
Timer 1	ET1	TF1	Automatically	10 / 0x001B
ADC	EADC	ADCF	By firmware	11 / 0x009B
T2COM1	ET2C1	TF2C1	Automatically	12 / 0x005B
UART	ES0	TIO / RIO	By firmware	13 / 0x0023
Comparator 0	ECMP0	CMP0F	By firmware	14 / 0x00A3
T2COM2	ET2C1	TF2C2	Automatically	15 / 0x0063
Timer 2	ET2 / ET2RL	TF2 / TF2RL	By firmware	16 / 0x002B
Comparator 1	ECMP1	CMP1F	By firmware	17 / 0x00AB
INT2	EX2	IE2	Automatically	18 / 0x00EB
T2COM3	ET2C3	TF2C3	Automatically	19 / 0x006B

* **Note: Don't clear Interrupt request flags by firmware when Interrupt request flags can be cleared by hardware automatically.**

10.1 Interrupt Operation

Interrupt operation is controlled by interrupt request flag and interrupt enable bits. Interrupt request flag is interrupt source event indicator, no matter what interrupt function status (enable or disable). Both interrupt enable bit and global interrupt (EAL=1) are enabled, the system executes interrupt operation when each of interrupt request flags activates. The program counter points to interrupt vector (0x03 – 0xEB) and execute ISR.

10.2 Interrupt Priority

Each interrupt source has its specific default priority order. If two interrupts occurs simultaneously, the higher priority ISR will be service first. The lower priority ISR will be serviced after the higher priority ISR completes. The next ISR will be service after the previous ISR complete, no matter the priority order.

For special priority needs, 4-level priority levels (Level 0 – Level 3) are used. All interrupt sources are classified into 6 priority groups (Group0 – Group5). Each group can be set one specific priority level. Priority level is selected by IP0/IP1 registers. Level 3 is the highest priority and Level 0 is the lowest. The interrupt sources inside the same group will share the same priority level. With the same priority level, the priority rule follows default priority.

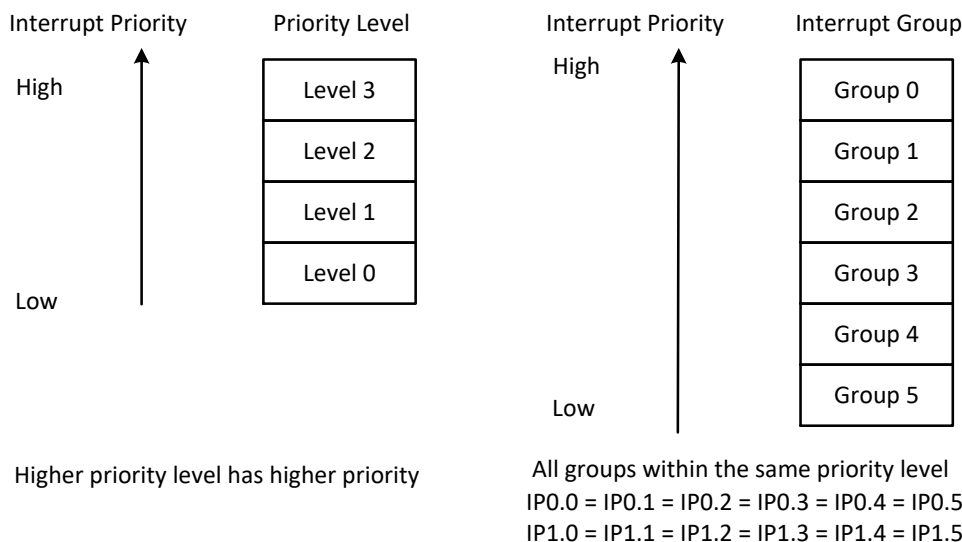
Priority Level	IP1.x	IP0.x
Level 0	0	0
Level 1	0	1
Level 2	1	0
Level 3	1	1

The ISR with the higher priority level can be serviced first; even can break the on-going ISR with the lower priority level. The ISR with the lower priority level will be pending until the ISR with the higher priority level completes.

Group	Interrupt Source			
Group 0	INT0	PWM1	I2C	
Group 1	Timer 0	PWM2	SPI	
Group 2	INT1	PWM3	T2COM0	
Group 3	Timer 1	ADC	T2COM1	
Group 4	UART	Comparator 0	T2COM2	
Group 5	Timer 2	Comparator 1	INT2	T2COM3

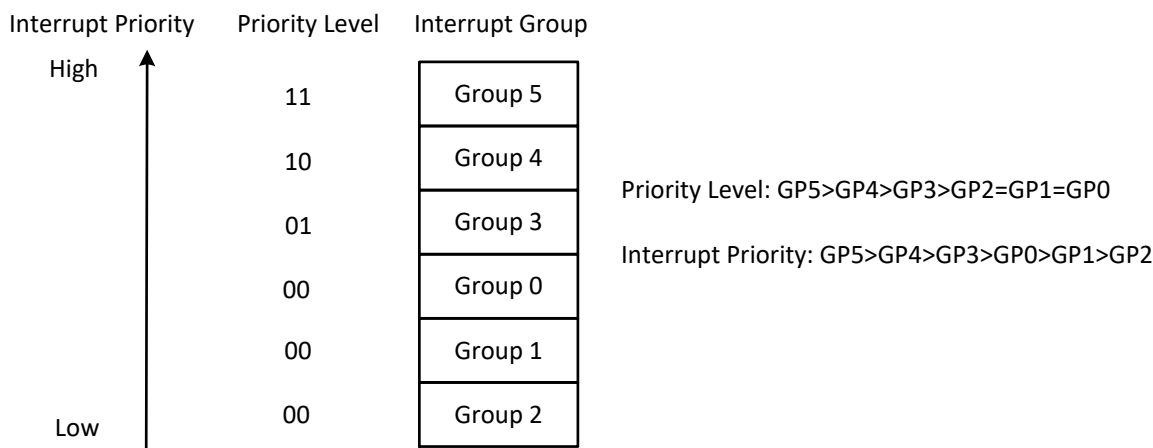
When more than one interrupt request occur, the highest priority request must be executed first. Choose the highest priority request according natural priority and priority level. The steps are as the following:

1. Choose the groups which have the highest priority level between all groups.
2. Choose the group which is the highest nature priority between the groups with the highest priority level.
3. Choose the ISR which has the highest nature priority inside the group with the highest priority.



As the example, group5 has the highest priority level and group0~group2 have the lowest priority level. It means the interrupt vector in group5 has the highest interrupt priority, the 2nd interrupt priority in group4 and the 3rd interrupt priority in group3. Group0~ group2 have the same priority level thus the nature priority rule will be followed. Therefore, interrupt priority will be group5> group4> group3> group0> group1> group2.

```
MOV    IP0, #00101000B    ; Set group0 - group5 in different priority level.
MOV    IP1, #00110000B
```



IP0, IP1 Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
IP0	-	-	IP05	IP04	IP03	IP02	IP01	IP00
IP1	-	-	IP15	IP14	IP13	IP12	IP11	IP10

IP0 Register (0XA9)

Bit	Field	Type	Initial	Description
5..0	IP0[5:0]	R/W	0	Interrupt priority. Each bit together with corresponding bit from IP1 register specifies the priority level of the respective interrupt priority group.
Else	Reserved	R	0	

IP1 Register (0XB9)

Bit	Field	Type	Initial	Description
5..0	IP1[5:0]	R/W	0	Interrupt priority. Each bit together with corresponding bit from IP0 register specifies the priority level of the respective interrupt priority group.
Else	Reserved	R	0	

10.3 Interrupt Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
IEN1	ET2RL	-	ET2C3	ET2C2	ET2C1	ET2C0	ESPI	EI2C
IEN2	-	-	ECMP1	ECMP0	EADC	EPWM3	EPWM2	-
IEN4	EPWM1	EX2	-	-	PWM1F	IE2	-	-
IRCON	TF2RL	TF2	TF2C3	TF2C2	TF2C1	TF2C0	-	-
IRCON2	-	-	-	CMP1F	CMP0F	ADCF	PWM3F	PWM2F
TCON	TF1	TR1	TF0	TR0	IE1	-	IE0	-
SOCON	SM0	SM1	SM20	REN0	TB80	RB80	TIO	RIO
SPSTA	SPIF	WCOL	SSERR	MODF	-	-	-	-
I2CCON	CR2	ENS1	STA	STO	SI	AA	CR1	CRO

IEN0 Register (0XA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Enable all interrupt control bit. 0: Disable all interrupt function. 1: Enable all interrupt function.
5	ET2	R/W	0	T2 timer interrupt control bit 0: Disable T2 interrupt function. 1: Enable T2 interrupt function.
4	ES0	R/W	0	UART interrupt control bit. 0: Disable UART interrupt function. 1: Enable UART interrupt function.
3	ET1	R/W	0	T1 timer interrupt control bit. 0: Disable T1 interrupt function. 1: Enable T1 interrupt function.
2	EX1	R/W	0	External P2.2 interrupt (INT1) control bit. 0: Disable INT1 interrupt function. 1: Enable INT1 interrupt function.
1	ET0	R/W	0	T0 timer interrupt control bit. 0: Disable T0 interrupt function. 1: Enable T0 interrupt function
0	EX0	R/W	0	External P2.1 interrupt (INT0) control bit. 0: Disable INT0 interrupt function. 1: Enable INT0 interrupt function.
Else	Reserved	R	0	

IEN1 Register (0XB8)

Bit	Field	Type	Initial	Description
7	ET2RL	R/W	0	T2 Timer external reload interrupt control bit. 0: Disable T2 external reload interrupt function. 1: Enable T2 external reload interrupt function.
5	ET2C3	R/W	0	T2 Timer COM3 interrupt control bit. 0: Disable T2COM3 interrupt function. 1: Enable T2COM3 interrupt function.
4	ET2C2	R/W	0	T2 Timer COM2 interrupt control bit. 0: Disable T2COM2 interrupt function. 1: Enable T2COM2 interrupt function.
3	ET2C1	R/W	0	T2 Timer COM1 interrupt control bit. 0: Disable T2COM1 interrupt function. 1: Enable T2COM1 interrupt function.
2	ET2C0	R/W	0	T2 Timer COM0 interrupt control bit. 0: Disable T2COM0 interrupt function. 1: Enable T2COM0 interrupt function.
1	ESPI	R/W	0	SPI interrupt control bit 0: Disable SPI interrupt function. 1: Enable SPI interrupt function.
0	EI2C	R/W	0	I2C interrupt control bit. 0: Disable I2C interrupt function. 1: Enable I2C interrupt function.
Else	Reserved	R	0	

IEN2 Register (0X9A)

Bit	Field	Type	Initial	Description
5	ECMP1	R/W	0	Comparator 1 interrupt control bit. 0: Disable CMP1 interrupt function. 1: Enable CMP1 interrupt function.
4	ECMP0	R/W	0	Comparator 0 interrupt control bit. 0: Disable CMP0 interrupt function. 1: Enable CMP0 interrupt function.
3	EADC	R/W	0	ADC interrupt control bit. 0: Disable ADC interrupt function. 1: Enable ADC interrupt function.
2	EPWM3	R/W	0	PWM3 interrupt control bit. 0: Disable PWM3 interrupt function. 1: Enable PWM3 interrupt function.
1	EPWM2	R/W	0	PWM2 interrupt control bit 0: Disable PWM2 interrupt function. 1: Enable PWM2 interrupt function.
Else	Reserved	R	0	

IEN4 Register (0XD1)

Bit	Field	Type	Initial	Description
7	EPWM1	R/W	0	PWM1 interrupt control bit. 0 = Disable PWM1 interrupt function. 1 = Enable PWM1 interrupt function.
6	EX2	R/W	0	External P2.3 interrupt (INT2) control bit. 0: Disable INT2 interrupt function. 1: Enable INT2 interrupt function.
3	PWM1F	R/W	0	PWM1 interrupt request flag. 0: None PWM1 interrupt request 1: PWM1 interrupt request.
2	IE2	R	0	External P2.3 interrupt (INT2) request flag 0: None INT2 interrupt request. 1: INT2 interrupt request.
Else	Reserved	R	0	

IRCON Register (0xC0)

Bit	Field	Type	Initial	Description
7	TF2RL	R/W	0	T2 timer external reload interrupt request flag. 0: None TF2RL interrupt request 1: TF2RL interrupt request.
6	TF2	R/W	0	T2 timer interrupt request flag. 0: None T2 interrupt request. 1: T2 interrupt request.
5	TF2C3	R/W	0	T2 Timer COM3 interrupt request flag. 0: None T2COM3 interrupt request. 1: T2COM3 interrupt request.
4	TF2C2	R/W	0	T2 Timer COM2 interrupt request flag. 0: None T2COM2 interrupt request. 1: T2COM2 interrupt request.
3	TF2C1	R/W	0	T2 Timer COM1 interrupt request flag. 0: None T2COM1 interrupt request. 1: T2COM1 interrupt request.
2	TF2C0	R/W	0	T2 Timer COM0 interrupt request flag. 0: None T2COM0 interrupt request. 1: T2COM0 interrupt request.
Else	Reserved	R	0	

IRCON2 Register (0XBF)

Bit	Field	Type	Initial	Description
4	CMP1F	R/W	0	Comparator 1 interrupt request flag. 0: None CMP1 interrupt request 1: CMP1 interrupt request
3	CMP0F	R/W	0	Comparator 0 interrupt request flag. 0: None CMP0 interrupt request. 1: CMP0 interrupt request.
2	ADCF	R/W	0	ADC interrupt request flag. 0: None ADC interrupt request. 1: ADC interrupt request.
1	PWM3F	R/W	0	PWM3 interrupt request flag. 0: None PWM3 interrupt request 1: PWM3 interrupt request.
0	PWM2F	R/W	0	PWM2 interrupt request flag. 0: None PWM2 interrupt request 1: PWM2 interrupt request.
Else	Reserved	R	0	

TCON Register (0X88)

Bit	Field	Type	Initial	Description
7	TF1	R/W	0	T1 timer external reload interrupt request flag. 0: None T1 interrupt request 1: T1 interrupt request.
5	TF0	R/W	0	T0 timer external reload interrupt request flag. 0: None T0 interrupt request 1: T0 interrupt request.
3	IE1	R	0	External P2.2 interrupt (INT1) request flag 0: None INT1 interrupt request. 1: INT1 interrupt request.
1	IE0	R	0	External P2.1 interrupt (INT0) request flag 0: None INT0 interrupt request. 1: INT0 interrupt request.
Else				Refer to other chapter(s)

S0CON Register (0X98)

Bit	Field	Type	Initial	Description
1	TIO	R/W	0	UART transmit interrupt request flag. It indicates completion of a serial transmission at UART. It is set by hardware at the end of bit 8 in mode 0 or at the beginning of a stop bit in other modes. It must be cleared by software. 0: None UART transmit interrupt request. 1: UART transmit interrupt request.
0	RIO	R/W	0	UART receive interrupt request flag. It is set by hardware after completion of a serial reception at UART. It is set by hardware at the end of bit 8 in mode 0 or in the middle of a stop bit in other modes. It must be cleared by software. 0: None UART receive interrupt request. 1: UART receive interrupt request.
Else				Refer to other chapter(s)

SPSTA Register (0XE1)

Bit	Field	Type	Initial	Description
7	SPIF	R	0	SPI complete communication flag Set automatically at the end of communication Cleared automatically by reading SPSTA, SPDAT registers
4	MODF	R	0	Mode fault flag
Else				Refer to other chapter(s)

I2CCON Register (0XDC)

Bit	Field	Type	Initial	Description
7	SI	R/W	0	Serial interrupt flag The SI is set by hardware when one of 25 out of 26 possible I2C states is entered. The only state that does not set the SI is state F8h, which indicates that no relevant state information is available. The SI flag must be cleared by software. In order to clear the SI bit, '0' must be written to this bit. Writing a '1' to SI bit does not change value of the SI.
Else				Refer to other chapter(s)

10.4 Example

Defining Interrupt Vector. The interrupt service routine is following user assembly code program.

```

                ORG      0          ; 0000H
                JMP      START      ; Jump to user program address.
                ...
                ORG      0X000B     ; Jump to interrupt service routine address.
                JMP      ISR_T0
                ORG      0X0013
                JMP      ISR_INT1
                ...
                ORG      0X009B
                JMP      ISR_ADC
                ...
                ORG      0X00EC
START:          ; 00ECH, The head of user program.
                ... ; User program.
                ...
                JMP      START      ; End of user program.
                ...
ISR_T0:         ; The head of interrupt service routine.
                PUSH     ACC        ; Save ACC to stack buffer.
                PUSH     PSW        ; Save PSW to stack buffer.
                ...
                POP      PSW        ; Load PSW from stack buffer.
                POP      ACC        ; Load ACC from stack buffer.
                RETI              ; End of interrupt service routine.
ISR_ADC:        ;
                PUSH     ACC        ; Save ACC to stack buffer.
                PUSH     PSW        ; Save PSW to stack buffer.
                ...
                POP      PSW        ; Load PSW from stack buffer.
                POP      ACC        ; Load ACC from stack buffer.
                RETI              ; End of interrupt service routine.
                ...
ISR_INT1        ;
                PUSH     ACC        ; Save ACC to stack buffer.
                PUSH     PSW        ; Save PSW to stack buffer.
                ...
                POP      PSW        ; Load PSW from stack buffer.
                POP      ACC        ; Load ACC from stack buffer.
                RETI              ; End of interrupt service routine.
                ...
                END              ; End of program.

```

11 MDU

The multiplication division unit is an on-chip arithmetic co-processor which enables the microcontroller to perform additional extended arithmetic operations. This unit provides 32-bit unsigned division, 16-bit unsigned multiplication, shift and normalize operations. These operations are identified by the different sequences of writing MD0 to MD5 registers.

11.1 Multiplication (16-bit x 16-bit)

The elements of a multiplication include three parts: multiplicand, multiplier and product. To start a multiplication requires following writing sequence: MD0 (low byte of multiplicand), MD4 (low byte of multiplier), MD1 (high byte of multiplicand), and MD5 (high byte of multiplier).

By the end of writing MD5 register, the multiplication is automatically started and takes 11 CPU cycles for its operation. The product of this term operation would be available to read by a specific sequence: MD0 (LSB), MD1, MD2, and MD3 (MSB) registers.

11.2 Division (32-bit/16-bit and 16-bit/16-bit)

The MDU supports two kind of division: 32-bit by 16-bit, and 16-bit by 16-bit. The first operation takes 17 CPU cycles to compute, whereas the second one takes 9 cycles only.

A 32-bit division started by a specific sequence of writing registers: MD0, MD1, MD2, MD3, MD4, and MD5. In this case, the 32-bit dividend is expected to store in MD3 (most significant bit) to MD0 registers, and 16-bit divisor is stored in MD5 and MD4 registers (MSB in MD5 register).

A 16-bit division operation cooperates with four registers only. The 16-bit dividend is stored in MD1 and MD0 registers, and the 16-bit divisor is stored in MD5 and MD4 registers (MD1 and MD5 for most signification bit). The appropriate performing sequence is 'MD0, MD1, MD4, and MD5.'

The MDU starts computing from MD5 register is written. It spends 9 or 17 CPU cycles, depends on the length of dividend, before the outcome is generated. The quotient is stored in MD3 to MD0 registers for 32-bit division, and MD1 to MD0 registers for 16-bit division (LSB in MD0 register). The reminder would be placed in MD5 (MSB) and MD4 registers no matter which division is performed. However, reading MD5 register must be the last operation to indicate the full division is completed.

11.3 Shifting and Normalizing

The shifting and normalizing operations rotate the 32-bit registers (MD3 to MD0, MSB in MD3) for a certain or uncertain time.

In shift operation, the 32-bit unsigned integer is shifted left or right by a specified number of bits. The direction and shifting number is specified in ARCON register. A shift operation takes 3 to 18 CPU

cycles depends on the shift time.

In normalizing operation, the 32-bit unsigned integer would be shifted left repeatedly until the most significant bit (7th bit of MD3 register) is 1. A normalizing operation takes 4 to 19 CPU cycles depends on the actual shift time.

Both shifting and normalizing operations are started by proper sequence of writing registers: MD0, MD1, MD2, MD3, and finally ARCON register. The result would be placed in MD0 to MD3 registers which should be read in the sequence of MD0, MD1, MD2, and MD3.

11.4 Cooperate with Keil C51

Because Keil C51 supports both of hardware and software multiplication/division operators, a command line '#pragma mdu_r515' is required in C to enable the hardware MDU functionality for higher performance. Subsequently, Keil C51 would compile mathematic operators with MDU support.

```
1 #include <SN8F5708.H>
2 #pragma mdu_r515           //Keil C51 MDU command line
```

11.5 The Error Flag (MDEF)

The "MDEF" error flag indicates an improperly performed operation (when one of the arithmetic operations has been restarted or interrupted by a new operation). The error flag mechanism is automatically enabled with the first write operation to "MD0" and disabled with the final read instruction from "MD3" (multiplication or shift/normalize) or "MD5" (division) in phase three.

The error flag is set when:

There is a write access to 'MDx' registers (any of 'MD0' to 'MD5' and ARCON) during phase two of MDU operation (restart or calculations interrupting)

There is a read access to one of MDx registers during phase two of MDU operation when the error flag mechanism is enabled. In such condition error flag is set but the calculation is not interrupted.

The error flag is reset only after read access to "ARCON" register. The error flag is read only.

11.6 The Overflow Flag (MDOV)

The MDOV overflow flag is set when one of the following conditions occurs:

Division by zero

Multiplication with a result greater than 0000 FFFFh

Start of normalizing if the most significant bit of MD3 is set (MD3.7= 1)

Any operation of the MDU that does not match the above conditions clears the overflow flag. Note that the overflow flag is exclusively controlled by hardware. It cannot be written.

11.7 MDU Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
MD0	MD07	MD06	MD05	MD04	MD03	MD02	MD01	MD00
MD1	MD17	MD16	MD15	MD14	MD13	MD12	MD11	MD10
MD2	MD27	MD26	MD25	MD24	MD23	MD22	MD21	MD20
MD3	MD37	MD36	MD35	MD34	MD33	MD32	MD31	MD30
MD4	MD47	MD46	MD45	MD44	MD43	MD42	MD41	MD40
MD5	MD57	MD56	MD55	MD54	MD53	MD52	MD51	MD50
ARCON	MDEF	MDOV	SLR	SC4	SC3	SC2	SC1	SC0

MD Registers (MD0 – MD5: 0xE9 – 0xEE)

Bit	Field	Type	Initial	Description
7..0	MD[7:0]	R/W	0x00	Multiplication/Division Registers

ARCON Register (0xEF)

Bit	Field	Type	Initial	Description
7	MDEF	R/W	0	MDU error flag MDEF Indicates an improperly performed operation (when one of the arithmetic operations has been restarted or interrupted by a new operation).
6	MDOV	R/W	0	MDU overflow flag Overflow occurrence in the MDU operation.
5	SLR	R/W	0	Shift direction 0: Shift left operation 1: Shift right operation
4..0	SC[4:0]	R/W	0x00	Shift counter Write 0x00: Perform normalizing. The actual shift time would be readable after operation. Write else values: Specify the times of shift operation.

11.8 Sample Code

The following sample code demonstrates how to perform MDU 32 bit / 16 bit.

```
1  #include <SN8F5708.H>
2  #pragma mdu_r515 //Keil C51 MDU command line
3
4  void main(void)
5  {
6      unsigned int Divisor; // 16-bit divisor
7      unsigned long Dividend; // 32-bit dividend
8      unsigned long Quotient; // 32-bit Quotient
9      unsigned int Remainder; // 16-bit Remainder
10
11     Divisor = 0x1234;
12     Dividend = 0x56789ABC;
13     Quotient = Dividend / Divisor; //0x0004C016
14     Remainder = Dividend % Divisor; //0x0A44
15
16     while(1);
17 }
18
```

12 GPIO

The microcontroller has up to 46 bidirectional general purpose I/O pin (GPIO). Unlike the original 8051 only has open-drain output, SN8F5708 builds in push-pull output structure to improve its driving performance.

12.1 Input and Output Control

The input and output direction control is configurable through P0M to P5M registers. These bits specify each pin that is either input mode or output mode.

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P0M	P07M	P06M	P05M	P04M	P03M	P02M	P01M	P00M
P1M	P17M	P16M	P15M	P14M	P13M	P12M	P11M	P10M
P2M	P27M	P26M	P25M	P24M	P23M	P22M	P21M	P20M
P3M	P37M	P36M	P35M	P34M	P33M	P32M	P31M	P30M
P4M	P47M	P46M	P45M	P44M	P43M	P42M	P41M	P40M
P5M	-	-	P55M	P54M	P53M	P52M	P51M	P50M
P1OC	PW3EN	PW2EN	PW1EN	P06OC	P05OC	P13OC	P12OC	P11OC

P0M: 0xF9, P1M: 0xFA, P2M: 0xFB, P3M: 0xFC, P4M: 0xFD, P5M: 0xFE

Bit	Field	Type	Initial	Description
7	P07M	R/W	0	Mode selection of P0.7 0: Input mode 1: Output mode
6	P06M	R/W	0	Mode selection of P0.6 0: Input mode 1: Output mode
5	P05M	R/W	0	Mode selection of P0.5 0: Input mode 1: Output mode
4..0				et cetera

*** Note: The pins which are not pin-out shall be set correctly to decrease power consumption in low-power modes. Strongly recommended to set these pins as input pull-up.**

P1OC Register (0xE4)

Bit	Field	Type	Initial	Description
7..5		R/W	000	Refer to PWM chapter
4	P06OC	R/W	0	P0.6 open-drain output mode 0: Disable 1: Enable, output high status becomes to input mode
3	P05OC	R/W	0	P0.5 open-drain output mode 0: Disable 1: Enable, output high status becomes to input mode
2	P13OC	R/W	0	P1.3 open-drain output mode 0: Disable 1: Enable, output high status becomes to input mode
1	P12OC	R/W	0	P1.2 open-drain output mode 0: Disable 1: Enable, output high status becomes to input mode
0	P11OC	R/W	0	P1.1 open-drain output mode 0: Disable 1: Enable, output high status becomes to input mode

12.2 Input Data and Output Data

By a read operation from any registers of P0 to P5, the current pin's logic level would be fetch to represent its external status. This operation remains functional even the pin is shared with other function like UART and I2C which can monitor the bus condition in some case.

A write P0 to P5 register value would be latched immediately, yet the value would be outputted until the mapped P0M – P5M is set to output mode. If the pin is currently in output mode, any value set to P0 to P5 register would be presented on the pin immediately.

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P0	P07	P06	P05	P04	P03	P02	P01	P00
P1	P17	P16	P15	P14	P13	P12	P11	P10
P2	P27	P26	P25	P24	P23	P22	P21	P20
P3	P37	P36	P35	P34	P33	P32	P31	P30
P4	P47	P46	P45	P44	P43	P42	P41	P40
P5	-	-	P55	P54	P53	P52	P51	P50

P0: 0x80, P1: 0x90, P2: 0xA0, P3: 0xFC, P4: 0xE8, P5: 0xF8

Bit	Field	Type	Initial	Description
7	P07	R/W	1	Read: P0.7 pin's logic level Write 1/0: Output logic high or low (applied if P07M = 1)
6	P06	R/W	1	Read: P0.6 pin's logic level Write 1/0: Output logic high or low (applied if P06M = 1)
5	P05	R/W	1	Read: P0.5 pin's logic level Write 1/0: Output logic high or low (applied if P05M = 1)
4..0				et cetera

12.3 On-chip Pull-up Resistors

The P0UR to P5UR registers are mapped to each pins' internal 100 kΩ (in typical value) pull-up resistor.

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P0UR	P07UR	P06UR	P05UR	P04UR	P03UR	P02UR	P01UR	P00UR
P1UR	P17UR	P16UR	P15UR	P14UR	P13UR	P12UR	P11UR	P10UR
P2UR	P27UR	P26UR	P25UR	P24UR	P23UR	P22UR	P21UR	P20UR
P3UR	P37UR	P36UR	P35UR	P34UR	P33UR	P32UR	P31UR	P30UR
P4UR	P47UR	P46UR	P45UR	P44UR	P43UR	P42UR	P41UR	P40UR
P5UR	-	-	P55UR	P54UR	P53UR	P52UR	P51UR	P50UR

P0UR: 0xF1, P1UR: 0xF2, P2UR: 0xF3, P3UR: 0xF4, P4UR: 0xF5, P5UR: 0xF6

Bit	Field	Type	Initial	Description
7	P07UR	R/W	0	On-chip pull-up resistor control of P0.7 0: Disable* 1: Enable
6	P06UR	R/W	0	On-chip pull-up resistor control of P0.6 0: Disable* 1: Enable
5	P05UR	R/W	0	On-chip pull-up resistor control of P0.5 0: Disable* 1: Enable
4..0				et cetera

* Recommended disable pull-up resistor if the pin is output mode or analog function

12.4 Pin Shared with Analog Function

The microcontroller builds in analog functions, such as ADC, OPA and comparator. The Schmitt trigger of input channel is strongly recommended to switch off if the pin's shared analog function is enabled.

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P2CON	P2CON7	P2CON6	P2CON5	P2CON4	-	-	-	-
P3CON	P3CON7	P3CON6	P3CON5	P3CON4	P3CON3	P3CON2	P3CON1	P3CON0
P4CON	P4CON7	P4CON6	P4CON5	P4CON4	P4CON3	P4CON2	P4CON1	P4CON0
P5CON	-	-	P5CON5	P5CON4	P5CON3	P5CON2	P5CON1	P5CON0

P2CON: 0x9E, P3CON: 0x9F, P4CON: 0xD6, P5CON: 0xD7

Bit	Field	Type	Initial	Description
7	P2CON7	R/W	0	Schmitt trigger control of P2.7 0: Enable 1: Disable
6	P2CON6	R/W	0	Schmitt trigger control of P2.6 0: Enable 1: Disable
5	P2CON5	R/W	0	Schmitt trigger control of P2.5 0: Enable 1: Disable
4..0				et cetera

13 External Interrupt

INT0, INT1 and INT2 are external interrupt trigger sources. Build in edge trigger configuration function and edge direction is selected by PEDGE register. When both external interrupt (EX0/EX1/EX2) and global interrupt (EAL) are enabled, the external interrupt request flag (IE0/IE1/IE2) will be set to “1” as edge trigger event occurs. The program counter will jump to the interrupt vector (ORG 0x0003/0x0013/0x00EB) and execute interrupt service routine. Interrupt request flag will be cleared by hardware before ISR is executed.

13.1 External Interrupt Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PEDGE	-	-	EX2G1	EX2G0	EX1G1	EX1G0	EX0G1	EX0G0
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
TCON	TF1	TR1	TF0	TR0	IE1	-	IE0	-
IEN4	EPWM1	EX2	-	-	PWM1F	IE2	-	-

PEDGE Register (0x8F)

Bit	Field	Type	Initial	Description
5..4	EX2G[1:0]	R/W	10	External interrupt 2 trigger edge control register. 00: Reserved. 01: Rising edge trigger. 10: Falling edge trigger (default) 11: Both rising and falling edge trigger
3..2	EX1G[1:0]	R/W	10	External interrupt 1 trigger edge control register. 00: Reserved. 01: Rising edge trigger. 10: Falling edge trigger (default) 11: Both rising and falling edge trigger
1..0	EX0G[1:0]	R/W	10	External interrupt 0 trigger edge control register. 00: Reserved. 01: Rising edge trigger. 10: Falling edge trigger (default) 11: Both rising and falling edge trigger
Else	Reserved	R	0	

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Enable all interrupt control bit. 0: Disable all interrupt function. 1: Enable all interrupt function.
2	EX1	R/W	0	External P2.2 interrupt (INT1) control bit. 0: Disable INT1 interrupt function. 1: Enable INT1 interrupt function.
0	EX0	R/W	0	External P2.1 interrupt (INT0) control bit. 0: Disable INT0 interrupt function. 1: Enable INT0 interrupt function.
Else				Refer to other chapter(s)

TCON Register (0x88)

Bit	Field	Type	Initial	Description
3	IE1	R/W	0	External P2.2 interrupt (INT1) request flag 0: None INT1 interrupt request. 1: INT1 interrupt request.
1	IE0	R/W	0	External P2.1 interrupt (INT0) request flag 0: None INT0 interrupt request. 1: INT0 interrupt request.
Else				Refer to other chapter(s)

IEN4 Register (0xD1)

Bit	Field	Type	Initial	Description
6	EX2	R/W	0	External P2.3 interrupt (INT2) control bit. 0: Disable INT2 interrupt function. 1: Enable INT2 interrupt function.
0	IE2	R/W	0	External P2.3 interrupt (INT2) request flag 0: None INT2 interrupt request. 1: INT2 interrupt request.
Else				Refer to other chapter(s)

*** Note: Before clear one of TF0, TF1, IE0 or IE1 flag manually by firmware, user must be made sure others request flag in TCON register doesn't active.**

13.2 Sample Code

The following sample code demonstrates how to perform INT0/INT1/INT2 with interrupt.

```

1  #define INT0Rsing      (1 << 0) //INT0 trigger edge is rising edge
2  #define INT0Falling   (2 << 0) //INT0 trigger edge is falling edge
3  #define INT0LeChge    (3 << 0) //INT0 trigger edge is level chagne
4  #define EINT0         (1 << 0) //INT0 interrupt enable
5
6  #define INT1Rsing      (1 << 2) //INT1 trigger edge is rising edge
7  #define INT1Falling   (2 << 2) //INT1 trigger edge is falling edge
8  #define INT1LeChge    (3 << 2) //INT1 trigger edge is level chagne
9  #define EINT1         (1 << 2) //INT1 interrupt enable
10
11 #define INT2Rsing      (1 << 4) //INT2 trigger edge is rising edge
12 #define INT2Falling   (2 << 4) //INT2 trigger edge is falling edge
13 #define INT2LeChge    (3 << 4) //INT2 trigger edge is level chagne
14 #define EINT2         (1 << 6) //INT2 interrupt enable
15
16 void EnableINT(void)
17 {
18     // INT0 rising edge, INT1 falling edge, INT2 level change
19     PEDGE = INT0Rising | INT1Falling | INT2LeChge;
20
21     // Enable INT0/INT1 interrupt
22     IEN0 |= EINT0 | EINT1;
23     // Enable INT2 interrupt
24     IEN4 |= EINT2;
25     // Enable total interrupt
26     IEN0 |= 0x80;
27
28     P0 = 0x00;
29     POM = 0x07;
30 }
31
32 void INT0Interrupt(void) interrupt ISRInt0 //0x03
33 { //IE0 clear by hardware
34     P00 = ~P00;
35 }
36
37 void INT1Interrupt(void) interrupt ISRInt1 //0x13
38 { //IE1 clear by hardware
39     P01 = ~P01;
40 }
41
42 void INT2Interrupt(void) interrupt ISRInt2 //0xEB
43 { //IE2 clear by hardware
44     P02 = ~P02;
45 }

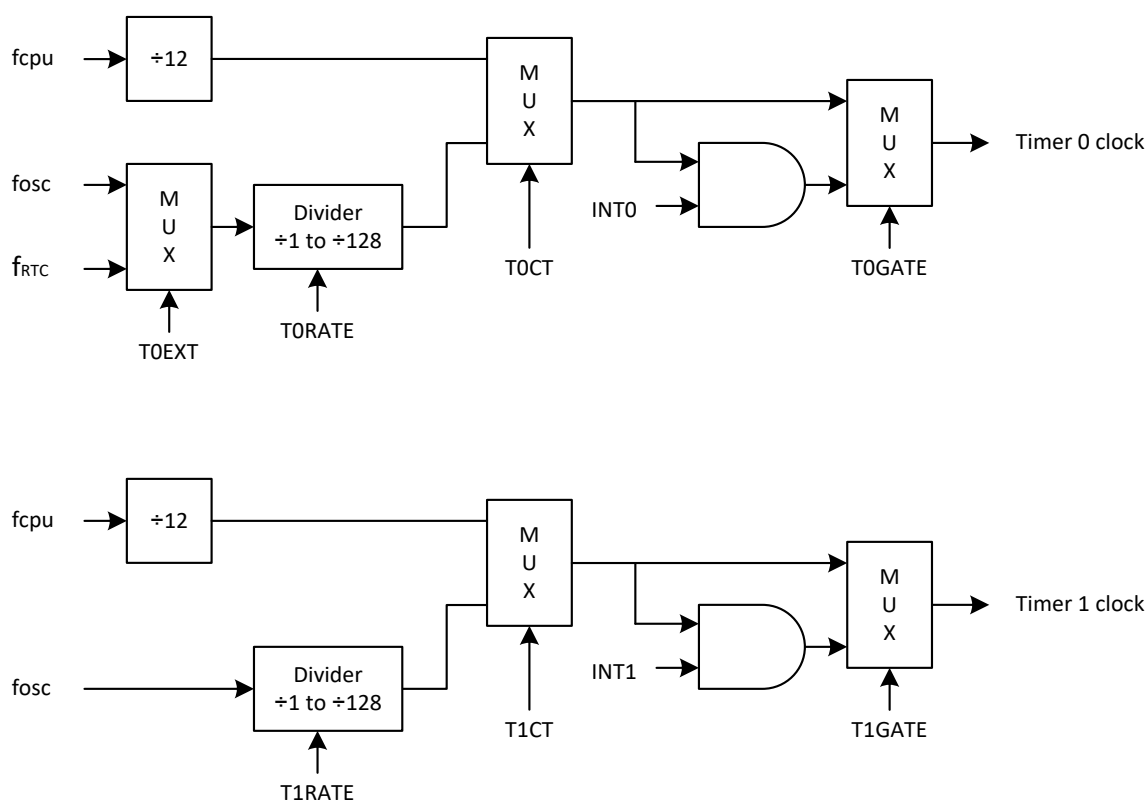
```

14 Timer 0 and Timer 1

Timer 0 and Timer 1 are two independent binary up timers. Timer 0 has four different operation modes: (1) 13-bit up counting timer, (2) 16-bit up counting timer, (3) 8-bit up counting timer with specified reload value support, and (4) separated two 8-bit up counting timer. By contrast, Timer 1 has only mode 0 to mode 2 which are same as Timer 0. Timer 0 and Timer 1 respectively support ET0 and ET1 interrupt function.

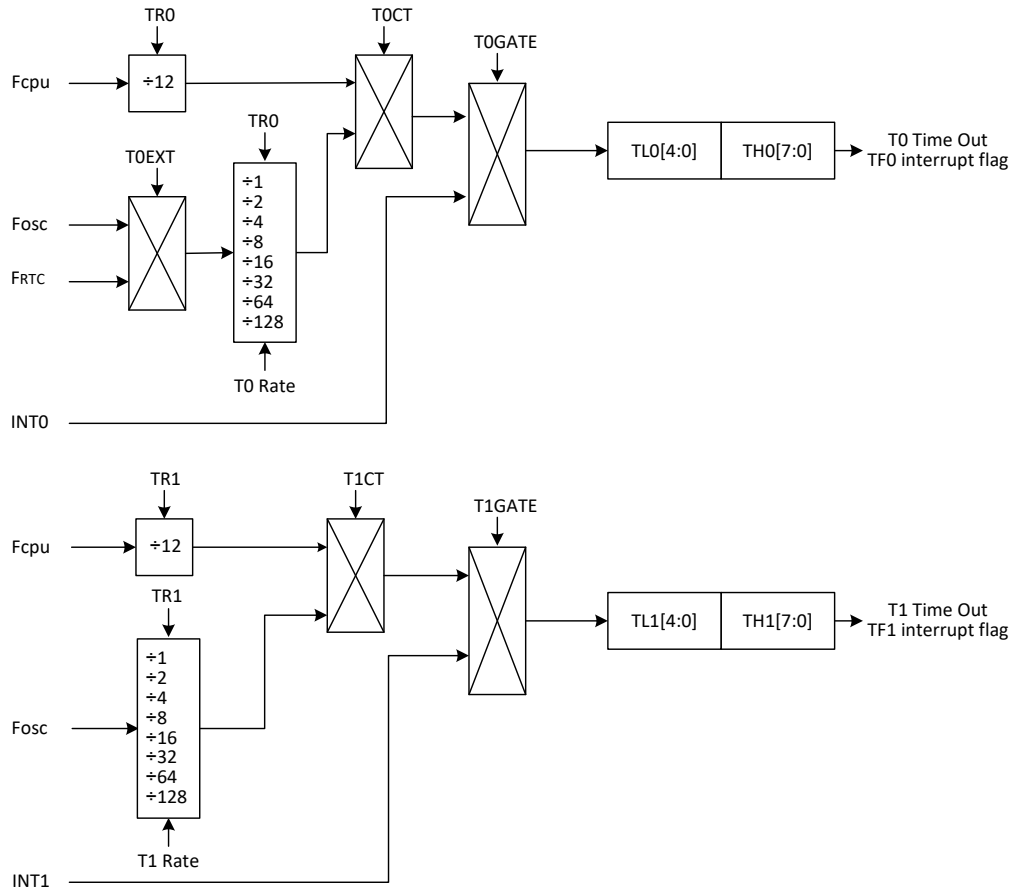
14.1 Timer 0 and Timer 1 Clock Selection

The figures below illustrate the clock selection circuit of Timer 0 and Timer 1. Timer 0 has three clock sources selection: f_{cpu} , f_{osc} , and f_{RTC} . All clock sources can be gated (pause) by INT0 pin if T0GATE is applied. Timer 1 clock sources selection: f_{cpu} and f_{osc} . All clock sources can be gated (pause) by INT1 pin if T1GATE is applied. Overall, the major difference between the two timers is that Timer 0 additionally supports f_{RTC} clock source (real time counter, RTC) which is functional if the microcontroller's CPU clock is 'IHRC 32 MHz with RTC' (refer to *Reset and Power-on Controller*) and an off-chip 32 kHz crystal is connected.



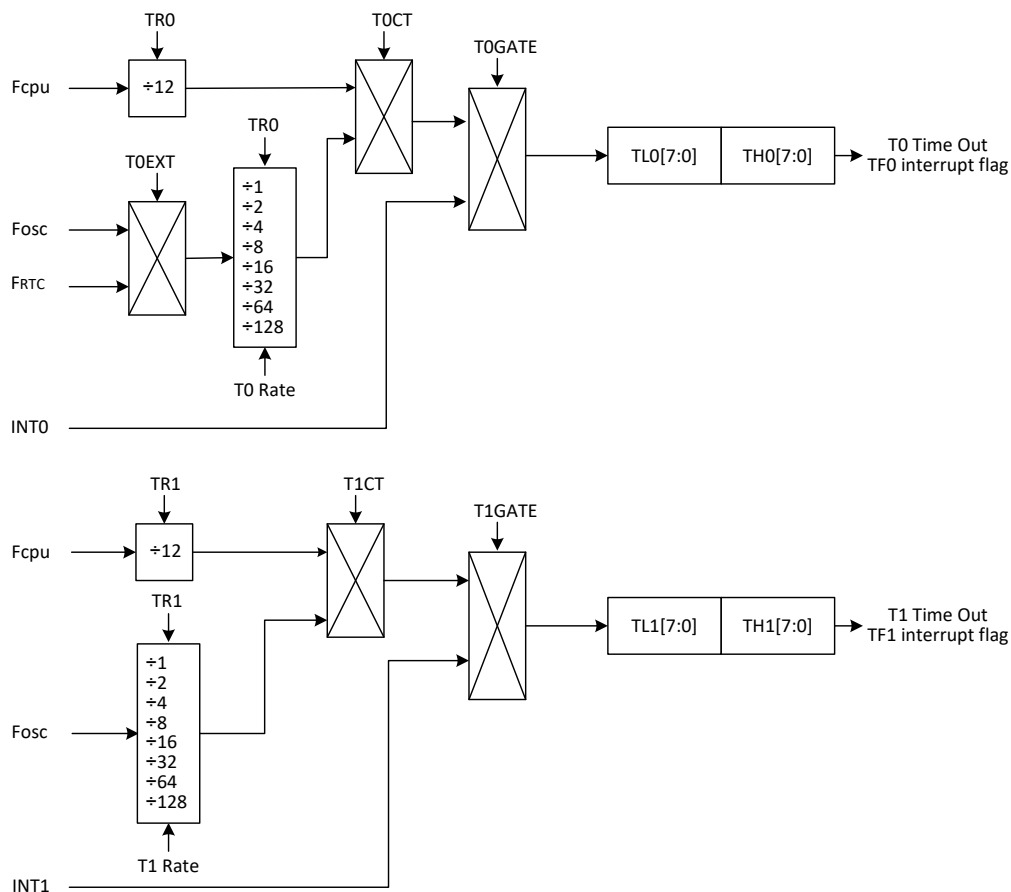
14.2 Mode 0: 13-bit Up Counting Timer

Mode 0 is a 13-bit up counting timer (the upper 3 bits of TL0 is suspended). Once the timer's counter is overflow (counts from 0xFF1F to 0x0000), TF0/TF1 flag would be issued immediately. This flag is readable by firmware if ET0/ET1 does not apply, or can be handled by interrupt controller if ET0/ET1 is applied.



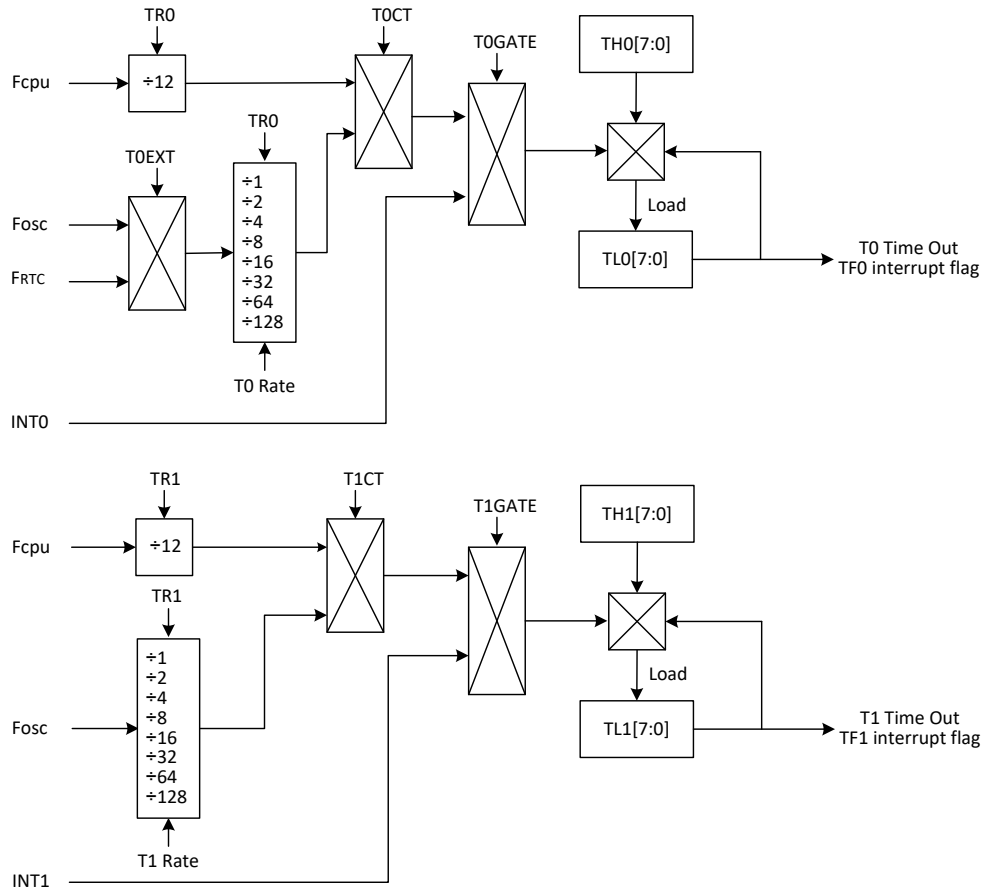
14.3 Mode 1: 16-bit Up Counting Timer

Mode 1 is a 16-bit up counting timer. Once the timer's counter overflow is occurred (from 0xFFFF to 0x0000), TF0/TF1 would be issued which is readable by firmware or can be handled by interrupt controller (if ET0/ET1 applied).



14.4 Mode 2: 8-bit Up Counting Timer with Specified Reload Value Support

Mode 2 is an 8-bit up counting timer (TL0/TL1) with a specifiable reload value. An overflow event (TL0/TL1 counts from 0xFF to 0x00) issues its TF0/TF1 flag for firmware or interrupt controller; meanwhile, the timer duplicates TH0/TH1 value to TL0/TL1 register in the same time. As a result, the timer is actually counts from 0xFF to the value of TH0/TH1.

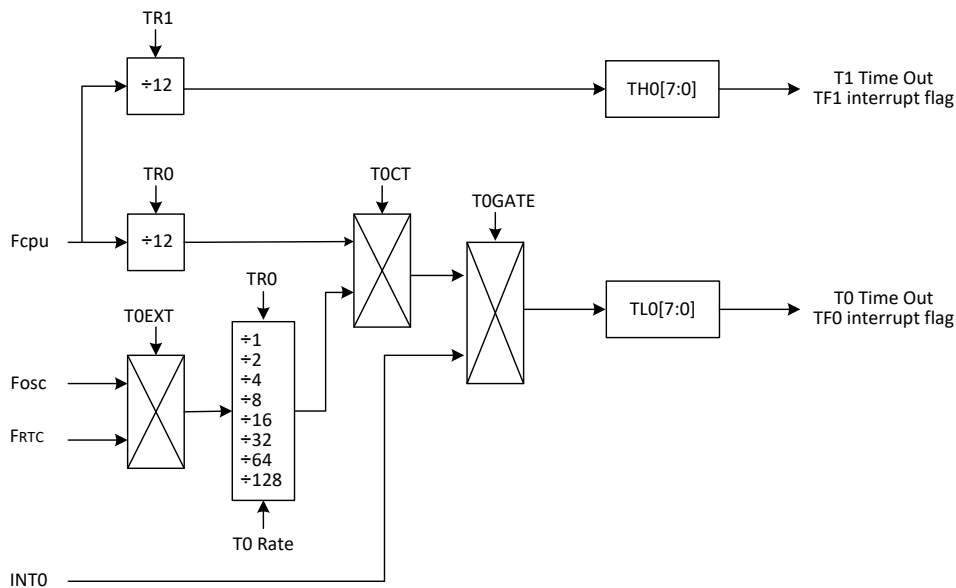


14.5 Mode 3 (Timer 0 only): Separated Two 8-bit Up Counting Timer

Mode 3 treats TH0 and TL0 as two separated 8-bit timers. TL0 is an 8-bit up counting timer with RTC support or two clock sources selection (fcpu and fosc), whereas TH0 clock source is fixed at fcpu/12. Only TL0 clock source can be gated (pause) by INTO pin if TOGATE is applied.

In this mode TL0 counter is enabled by TR0, and its overflow signal is reflected in TF0 flag. TH0 counter is controlled by TR1, and TF1 flag is also occupied by TH0 overflow signal.

Timer 1 cannot issue any overflow event in this situation, and it can be seen as a self-counting timer without flag support.



14.6 Timer 0 and Timer 1 Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
TCON	TF1	TR1	TF0	TR0	IE1	-	IE0	-
TCON0	TOEXT	TORATE2	TORATE1	TORATE0	-	T1RATE2	T1RATE1	T1RATE0
TMOD	T1GATE	T1CT	T1M1	T1M0	TOGATE	TOCT	T0M1	T0M0
TH0	TH07	TH06	TH05	TH04	TH03	TH02	TH01	TH00
TL0	TL07	TL06	TL05	TL04	TL03	TL02	TL01	TL00
TH1	TH17	TH16	TH15	TH14	TH13	TH12	TH11	TH10
TL1	TL17	TL16	TL15	TL14	TL13	TL12	TL11	TL10

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
3	ET1	R/W	0	Timer 1 interrupt 0: Disable 1: Enable
1	ET0	R/W	0	Timer 0 interrupt 0: Disable 1: Enable

TCON Register (0x88)

Bit	Field	Type	Initial	Description
7	TF1	R/W	0	Timer 1 overflow event 0: Timer 1 does not have any overflow event 1: Timer 1 has overflowed This bit can be cleared automatically by interrupt handler, or manually by firmware
6	TR1	R/W	0	Timer 1 function 0: Disable 1: Enable
5	TF0	R/W	0	Timer 0 overflow event 0: Timer 0 does not have any overflow event 1: Timer 0 has overflowed This bit can be cleared automatically by interrupt handler, or manually by firmware

4	TR0	R/W	0	Timer 0 function 0: Disable 1: Enable
3	IE1	R	0	Refer to INT1
2	Reserved	R	0	
1	IE0	R	0	Refer to INT0
0	Reserved	R	0	

*** Note: Before clear one of TF0, TF1, IE0 or IE1 flag manually by firmware, user must be made sure others request flag in TCON register doesn't active.**

TCON0 Register (0xE7)

Bit	Field	Type	Initial	Description
7	TOEXT	R/W	0	Timer 0 real time counter 0: Disable 1: Enable*
6..4	TORATE[2:0]	R/W	000	Clock divider of Timer 0 external clock source 000: $f_{EXT0} / 128$ 001: $f_{EXT0} / 64$ 010: $f_{EXT0} / 32$ 011: $f_{EXT0} / 16$ 100: $f_{EXT0} / 8$ 101: $f_{EXT0} / 4$ 110: $f_{EXT0} / 2$ 111: $f_{EXT0} / 1$
3	Reserved	R	0	
2..0	T1RATE[2:0]	R/W	000	Clock divider of Timer 0 external clock source 000: $f_{EXT1} / 128$ 001: $f_{EXT1} / 64$ 010: $f_{EXT1} / 32$ 011: $f_{EXT1} / 16$ 100: $f_{EXT1} / 8$ 101: $f_{EXT1} / 4$ 110: $f_{EXT1} / 2$ 111: $f_{EXT1} / 1$

* TOEXT = 1 is reserved for 'IHRC 32 MHz with RTC' CPU clock source only; remaining '0' if other source is chosen.

TMOD Register (0x89)

Bit	Field	Type	Initial	Description
7	T1GATE	R/W	0	Timer 1 gate control mode 0: Disable 1: Enable, Timer 1 clock source is gated by INT1
6	T1CT	R/W	0	Timer 1 clock source selection 0: $f_{\text{Timer1}} = f_{\text{cpu}} / 12$ 1: $f_{\text{Timer1}} = f_{\text{EXT1}} / \text{T1RATE}$ (refer to T1RATE) ^{*(1)}
5..4	T1M[1:0]	R/W	00	Timer 1 operation mode 00: 13-bit up counting timer 01: 16-bit up counting timer 10: 8-bit up counting timer with reload support 11: Reserved
3	TOGATE	R/W	0	Timer 0 gate control mode 0: Disable 1: Enable, Timer 0 clock source is gated by INTO
2	TOCT	R/W	0	Timer 0 clock source selection 0: $f_{\text{Timer0}} = f_{\text{cpu}} / 12$ 1: $f_{\text{Timer0}} = f_{\text{EXT0}} / \text{TORATE}$ (refer to TORATE) ^{*(2)}
1..0	T0M[1:0]	R/W	00	Timer 0 operation mode 00: 13-bit up counting timer 01: 16-bit up counting timer 10: 8-bit up counting timer with reload support 11: Separated two 8-bit up counting timer

*(1) $f_{\text{EXT1}} = f_{\text{osc}}$.

*(2) $f_{\text{EXT0}} = f_{\text{osc}}$ or f_{RTC} .

TH0 / TH1 Registers (TH0: 0x8C, TH1: 0x8D)

Bit	Field	Type	Initial	Description
7..0	TH0/TH1	R/W	0x00	High byte of Timer 0 and Timer 1 counter

TL0 / TL1 Register (TL0: 0x8A, TL1: 0x8B)

Bit	Field	Type	Initial	Description
7..0	TL0/TL1	R/W	0x00	Low byte of Timer 0 and Timer 1 counter

14.7 Sample Code

The following sample code demonstrates how to perform T0/T1 with interrupt.

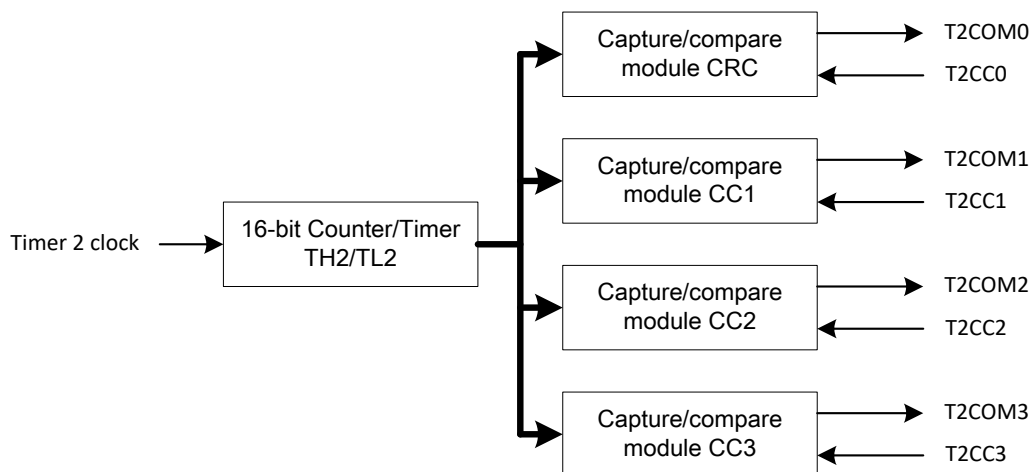
```

1  #define T0Mode0      (0 << 0) //T0 mode0, 13-bit counter
2  #define T0Model1     (1 << 0) //T0 model, 16-bit counter
3  #define T0Mode2      (2 << 0) //T0 mode2, 8-bit auto-reload counter
4  #define T0Mode3      (3 << 0) //T0 mode3, T0 two 8-bit counter/T1 no flag
5  #define T0GATE       (8 << 0) //T0 gating clock by INT0
6  #define T0ClkFcpu    (0 << 0) //T0 clock source from Fcpu/12
7  #define T0ClkExt     (4 << 0) //T0 clock source from Fosc or FRTC
8  #define T0ExtFosc    (0 << 4) //T0 clock source from Fosc
9  #define T0ExtFRTC    (8 << 4) //T0 clock source from FRTC
10
11 #define T1Mode0      (0 << 4) //T1 mode0, 13-bit counter
12 #define T1Model1     (1 << 4) //T1 model, 16-bit counter
13 #define T1Mode2      (2 << 4) //T1 mode2, 8-bit auto-reload counter
14 #define T1Mode3      (3 << 4) //T1 mode3, T1 stop
15 #define T1GATE       (8 << 4) //T1 gating clock by INT1
16 #define T1ClkFcpu    (0 << 4) //T1 clock source from Fcpu/12
17 #define T1ClkExt     (4 << 4) //T1 clock source from Fosc
18
19 void InitT0T1(void)
20 {
21     // T0/T1_initial
22     TH0 = 0x00;
23     TL0 = 0x00;
24     TH1 = 0x00;
25     TL1 = 0x00;
26     // T0 mode0 with gating clock by INT0, clock source from Fosc or FRTC
27     TMOD |= T0Mode0 | T0GATE | T0ClkExt;
28     // T0 clock source = FRTC/1
29     TCON0 |= T0ExtFRTC | 0x70;
30     // T1 model, clock source from Fcpu/12
31     TMOD |= T1Model1 | T1ClkFcpu;
32     // Timer 0/1 enable. Clear TF0/TF1
33     TCON |= 0x50;
34     // Enable T0/T1 interrupt
35     IEN0 |= 0x0A;
36     // Enable total interrupt
37     IEN0 |= 0x80;
38
39     P0 = 0x00;
40     POM = 0x03;
41 }
42
43 void T0Interrupt(void) interrupt ISRTimer0 //0x0B
44 { //TF0 clear by hardware
45     P00 = ~P00;
46 }
47 void T1Interrupt(void) interrupt ISRTimer1 //0x1B
48 { //TF1 clear by hardware
49     P01 = ~P01;
50 }

```

15 Timer 2

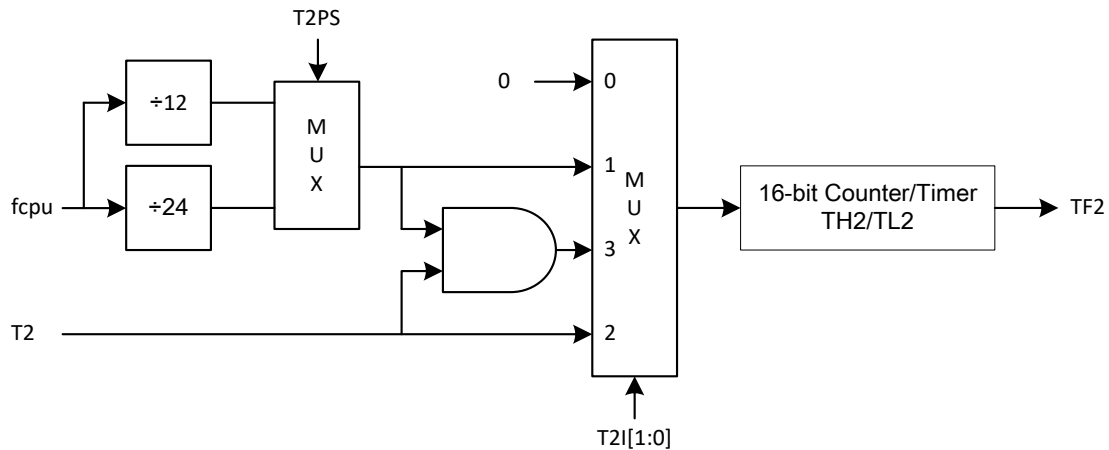
Timer 2 is a 16-bit up counting timer which has several optional extensions: specified reload value, comparison output (PWM) and capture function. Timer 2 consists of a dedicated 16-bit counter/timer and four 16-bit capture/compare modules. Each capture/compare module has its own associated I/O when enabled. Each capture/compare module may be configured to operate independently in one of 3 modes: compare, capture with rising edge, or capture with register be written.



15.1 Timer 2 Up-counting Control

Timer 2 has three operation modes by its clock source: specify fcpu clocks (fcpu/12 and fcpu/24), specify fcpu clocks with a stop control, and external clock input. The table below categorizes these three operation modes and its related registers (T2I1, T2I0 and T2PS). Once the timer's counter is overflow (counts from 0xFFFF to 0x0000), TF2 would be issued immediately which can read/write by firmware. Timer 2 interrupt function is controlled by ET2.

T2I1	T2I0	T2PS	Timer 2 Clock Source
0	0	X	Disable Timer 2
0	1	0	fcpu/12
0	1	1	fcpu/24
1	1	0	fcpu/12 (stop counting when T2 pin is low, resume when T2 is high)
1	1	1	fcpu/24 (stop counting when T2 pin is low, resume when T2 is high)
1	0	X	T2 pin rising edge (T2 pin is shared with P2.3, clock rate $\leq 0.5 * fcpu$)

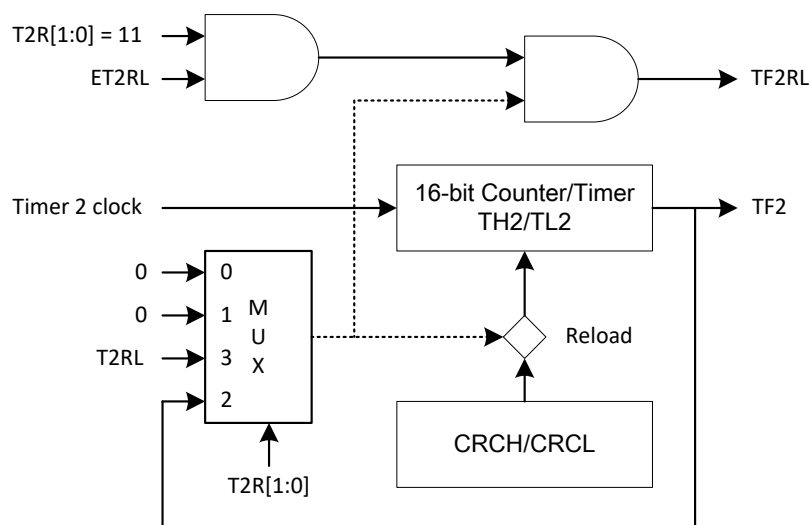


15.2 Specified Timer 2 Reload Value

The specified reload value is an optional function which can reload Timer 2 counter by overflow or external control pin.

If overflow-to-reload is selected, Timer 2 duplicates CRCH/CRCL value to its counter (TH2/TL2) automatically by overflow signal. As a result, Timer 2 would repeatedly counts from CRCH/CRCL value to 0xFFFF.

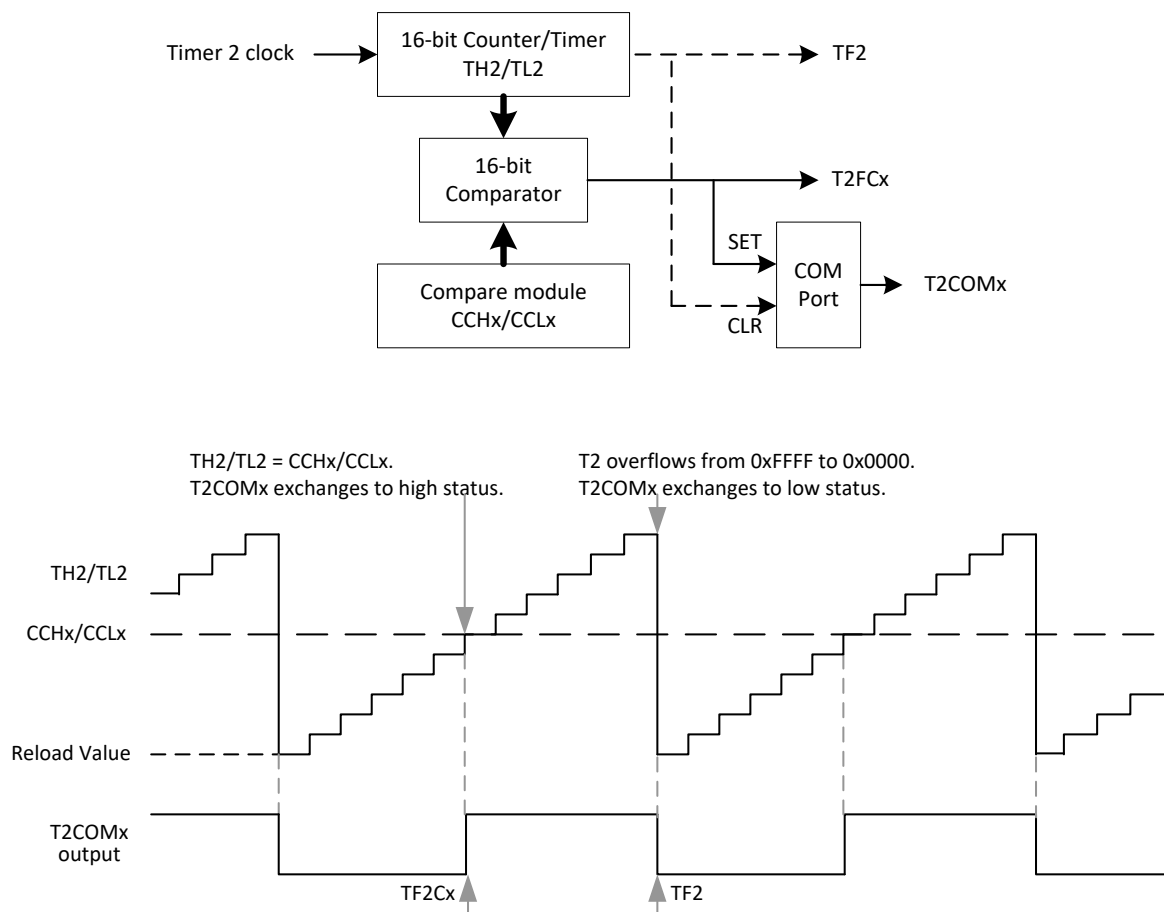
On the other hand, a falling edge of external pin T2RL (shared with P2.2) can also be chosen as a reload signal. In this situation, Timer 2 normally counts its counter from 0x0000 to 0xFFFF if T2RL pin remains stable, yet the counter value would be replaced at any time by CRCH/CRCL value as long as T2RL pin has a falling signal. Subsequently, Timer 2 continues its counting routine from CRCH/CRCL value, and external reload flag (TF2RL) would be issued if interrupt function is enabled (both ET2RL and ET2 are set). External reload interrupt vector is shared with Timer 2 interrupt vector and identify event by firmware.



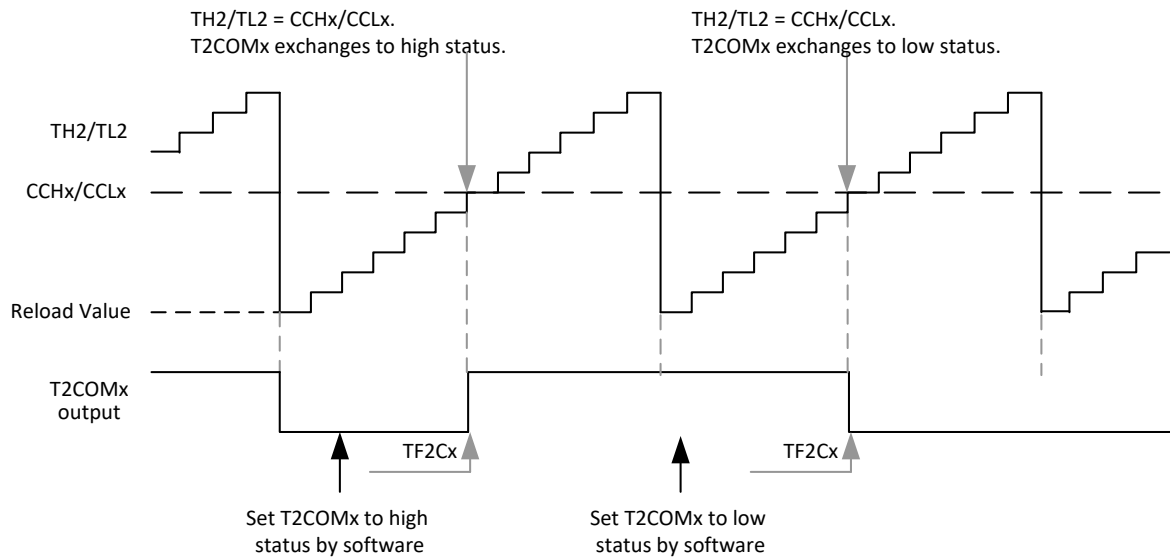
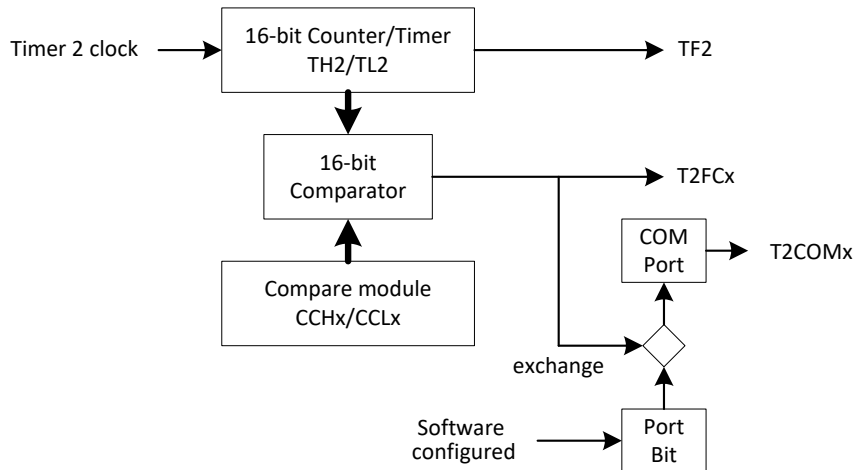
15.3 Comparison Output (PWM)

Timer 2 has up to four set of comparison output. Each set (CRC/CC1/CC2/CC3) independently compares its value to Timer 2 counter (TH2/TL2) and outputs the comparison result on T2COM0 to T2COM3 pins (shared with P1.0 to P1.3). The comparison result has two output methods: directly output and indirectly output.

The directly method is that the mapped pin outputs low status if Timer 2 counter is lower than CRC/CC1/CC2/CC3 register, whereas it outputs high status if its register value is equal/larger than Timer 2 counter. Thus, the output status is changed twice at crossover points. As CRC/CC1/CC2/CC3 register is equal to Timer 2 counter, a TF2C0/TF2C1/TF2C2/TF2C3 flag is issued which can read/write by firmware. Compare interrupt function is controlled by ET2C0/ET2C1/ET2C2/ET2C3.

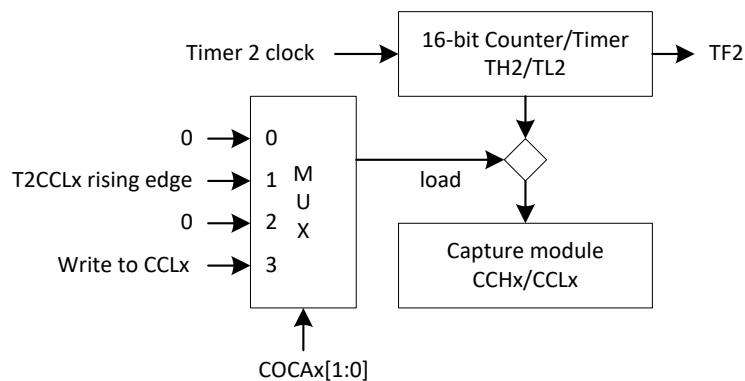


By contrast, the indirectly output method is an event which keep the mapped pin's previous output setting until Timer 2 counter overtakes CRC/CC1/CC2/CC3 register value. In this mode, the transition of the output signal can be configured by software. In other word, the P1.0 register bit would be affect T2COM0/P1.0 pin when TH2/TL2 equal to CRC registers. A Timer 2 overflow causes no output change.

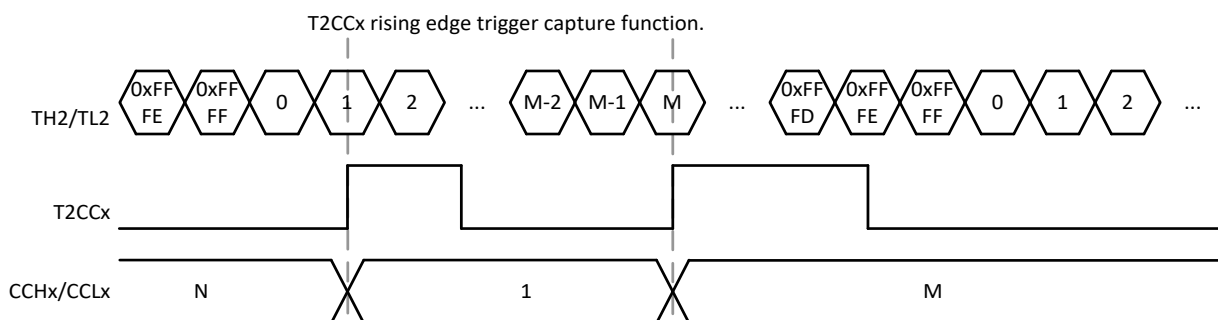


15.4 Capture Function

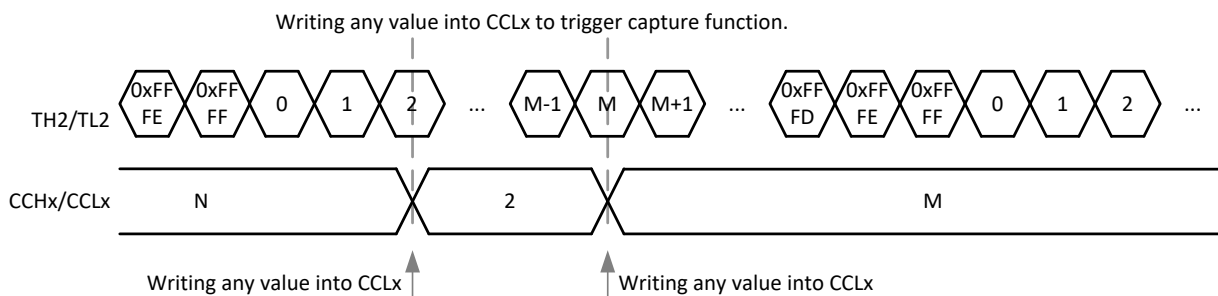
The capture function is similar to split/lap button of a stopwatch. While Timer 2 counter (TH2/TL2) routinely count up, a split event records counter value in CRC/CC1/CC2/CC3 register(s).



The split event can from hardware or software. The T2CC0 pin (shared with P3.4) can trigger a hardware split event that duplicates TH2/TL2 value to CRCH/CRCL registers, whereas T2CC1 (P3.5), T2CC2 (P3.6) and T2CC3 (P3.7) respectively control CC1 to CC3 registers.



A software split event is triggered by writing any value into CRCL/CCL1/CCL2/CCL3 register. While perform a writing instruction to these registers, the present TH2/TL2 value would be record in the paired registers instead.



15.5 Timer 2 Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
T2CON	T2PS	I3FR	-	T2R1	T2R0	T2CM	T2I1	T2I0
CCEN	COCA31	COCA30	COCA21	COCA20	COCA11	COCA10	COCA01	COCA00
TH2	TH27	TH26	TH25	TH24	TH23	TH22	TH21	TH20
TL2	TL27	TL26	TL25	TL24	TL23	TL22	TL21	TL20
CRCH	CRCH7	CRCH6	CRCH5	CRCH4	CRCH3	CRCH2	CRCH1	CRCH0
CRCL	CRCL7	CRCL6	CRCL5	CRCL4	CRCL3	CRCL2	CRCL1	CRCL0
CCH3	CCH37	CCH36	CCH35	CCH34	CCH33	CCH32	CCH31	CCH30
CCL3	CCL37	CCL36	CCL35	CCL34	CCL33	CCL32	CCL31	CCL30
CCH2	CCH27	CCH26	CCH25	CCH24	CCH23	CCH22	CCH21	CCH20
CCL2	CCL27	CCL26	CCL25	CCL24	CCL23	CCL22	CCL21	CCL20
CCH1	CCH17	CCH16	CCH15	CCH14	CCH13	CCH12	CCH11	CCH10
CCL1	CCL17	CCL16	CCL15	CCL14	CCL13	CCL12	CCL11	CCL10
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0

IEN1	ET2RL	-	ET2C3	ET2C2	ET2C1	ET2C0	ESPI	EI2C
IRCON	TF2RL	TF2	TF2C3	TF2C2	TF2C1	TF2C0	-	-

T2CON Register (0xC8)

Bit	Field	Type	Initial	Description
7	T2PS	R/W	0	Timer 2 pre-scalar 0: fcpu/12 1: fcpu/24
6	I3FR	R/W	0	In compare mode: 0: The COM0 interrupt would be generated when the TH2/TL2 becomes not equal to the CRC register (e.g. Timer 2: 0x8081, CRC: 0x8080). 1: The COM0 interrupt would be generated when the TH2/TL2 becomes equal to the CRC register. In capture mode 0: 0: The timer 2 content would be latched into CRC register by T2CC0 is falling edge. 1: The timer 2 content would be latched into CRC register by T2CC0 is rising edge.
5	Reserved	R/W	0	
4..3	T2R[1:0]	R/W	00	Specified Timer 2 reload value 00: Disable 01: Disable 10: Load CRCH/CRCL to TH2/TL2 by counter overflow 11: Load CRCH/CRCL to TH2/TL2 by T2RL pin
2	T2CM	R/W	0	Timer 2 comparison output 0: Directly output method 1: Indirectly output, next output status can be specified
1..0	T2I[1:0]	R/W	00	Timer 2 up counting control 00: Disable 01: Clock rate is defined by T2PS 10: Clock source is T2 pin 11: Clock rate is defined by T2PS with T2 pin gate control

CCEN Register (0xC1)

Bit	Field	Type	Initial	Description
7..6	COCA3[1:0]	R/W	00	Comparison and capture function of CC3 00: Disable 01: Capture by T2CC3 pin rising edge 10: Comparison function 11: Capture by writing CCL3 register
5..4	COCA2[1:0]	R/W	00	Comparison and capture function of CC2 00: Disable 01: Capture by T2CC2 pin rising edge 10: Comparison function 11: Capture by writing CCL2 register
3..2	COCA1[1:0]	R/W	00	Comparison and capture function of CC1 00: Disable 01: Capture by T2CC1 pin rising edge 10: Comparison function 11: Capture by writing CCL1 register
1..0	COCA0[1:0]	R/W	00	Comparison and capture function of CRC 00: Disable 01: Capture by T2CC0 pin rising edge 10: Comparison function 11: Capture by writing CRCL register

TH2/TL2 Registers (TH2: 0xCD, TL2: 0xCC)

Bit	Field	Type	Initial	Description
7..0	TH2/TL2	R/W	0x00	Timer 2 16-bit counter registers.

CRC Registers (CRCH: 0xCB, CRCL: 0xCA)

Bit	Field	Type	Initial	Description
7..6	CRCH[15:0]	R/W	0x00	16-bit compare/capture registers.

CCH3/CCL3 Registers (CCH3: 0xC7, CCL3: 0xC6)

Bit	Field	Type	Initial	Description
7..6	CCH3/CCL3	R/W	0x00	16-bit compare/capture registers.

CCH2/CCL2 Registers (CCH2: 0xC5, CCL2: 0xC4)

Bit	Field	Type	Initial	Description
7..6	CCH2 /CCL2	R/W	0x00	16-bit compare/capture registers.

CCH1/CCL1 Registers (CCH1: 0xC3, CCL1: 0xC2)

Bit	Field	Type	Initial	Description
7..6	CCH1/CCL1	R/W	0x00	16-bit compare/capture registers.

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
5	ET2	R/W	0	Enable Timer 2 interrupt
Else				Refer to other chapter(s)

IEN1 Register (0xB8)

Bit	Field	Type	Initial	Description
7	ET2RL	R/W	0	T2 Timer external reload interrupt control bit 0: Disable 1: Enable
6	Reserved	R	0	
5	ET2C3	R/W	0	T2 Timer COM3 interrupt control bit 0: Disable 1: Enable
4	ET2C2	R/W	0	T2 Timer COM2 interrupt control bit 0: Disable 1: Enable
3	ET2C1	R/W	0	T2 Timer COM1 interrupt control bit 0: Disable 1: Enable
2	ET2C0	R/W	0	T2 Timer COM0 interrupt control bit 0: Disable 1: Enable
Else				Refer to other chapter(s)

IRCON Register (0xC0)

Bit	Field	Type	Initial	Description
7	TF2RL	R/W	0	T2 timer external reload interrupt request flag. 0: None TF2RL interrupt request 1: TF2RL interrupt request.
6	TF2	R/W	0	T2 timer interrupt request flag. 0: None T2 interrupt request. 1: T2 interrupt request.
5	TF2C3	R/W	0	T2 Timer COM3 interrupt request flag. 0: None T2COM3 interrupt request. 1: T2COM3 interrupt request.
4	TF2C2	R/W	0	T2 Timer COM2 interrupt request flag. 0: None T2COM2 interrupt request. 1: T2COM2 interrupt request.
3	TF2C1	R/W	0	T2 Timer COM1 interrupt request flag. 0: None T2COM1 interrupt request. 1: T2COM1 interrupt request.
2	TF2C0	R/W	0	T2 Timer COM0 interrupt request flag. 0: None T2COM0 interrupt request. 1: T2COM0 interrupt request.
Else	Reserved	R	0	

15.6 Sample Code

The following sample code demonstrates how to perform T2 compare function with interrupt.

```

1  #define T2ClkFcpu      (1 << 0) //T2 clock from Fcpu
2  #define T2ClkPin      (2 << 0) //T2 clock from T2 pin
3  #define T2ClkGate     (3 << 0) //T2 clock from Fcpu with T2 pin gating
4  #define T2Fcpu12      (0 << 7) //T2 clock = Fcpu/12
5  #define T2Fcpu24      (1 << 7) //T2 clock = Fcpu/24
6  #define T2RLMode0     (2 << 3) //T2 reload mode0 = auto-reload
7  #define T2RLModel1    (3 << 3) //T2 reload model1 = T2RL falling edge
8  trigger
9  #define ComMode0       (0 << 2) //Compare mode = directly method
10 #define ComModel1      (1 << 2) //Compare mode = indirectly output method
11 #define T2COM0EdNE     (0 << 6) //T2COM0 interrupt edge = no eque CRC
12 #define T2COM0EdE     (1 << 6) //T2COM0 interrupt edge = eque CRC
13 #define T2COM0En       (2 << 0) //T2COM0 compare function enable
14 #define T2COM1En       (2 << 2) //T2COM1 compare function enable
15 #define T2COM2En       (2 << 4) //T2COM2 compare function enable
16 #define T2COM3En       (2 << 6) //T2COM3 compare function enable
17
18 void InitT2(void)
19 {
20     // T2_Initial
21     TH2 = 0x00;
22     TL2 = 0x00;
23     CRCH = 0x80;
24     CRCL = 0x00;
25     CCH1 = 0xC0;
26     CCL1 = 0x00;
27     CCH2 = 0xE0;
28     CCL2 = 0x00;
29     CCH3 = 0xF0;
30     CCL3 = 0x00;
31
32     // T2 clock from Fcpu/24 with T2 pin gating
33     // Reload model = T2RL falling edge trigger
34     // Compare mode = directly method
35     // T2COM0 interrupt trigger = eque CRC
36     T2CON |= T2ClkGate | T2Fcpu24 | T2RLModel1 | ComMode0 | T2COM0EdE;
37
38     // Compare function T2COM0/1/2/3 enable
39     CCEN |= T2COM0En | T2COM1En | T2COM2En | T2COM3En;
40
41     // P23(T2)/P22(T2RL) is input mode with pull-high resister
42     P2M &= 0xF3;
43     P2UR &= 0x0C;
44
45     // Enable T2RL/T2COM0/1/2/3 interrupt
46     IEN1 |= 0xBC;
47
48     // Enable total/Timer2 interrupt
49     IEN0 |= 0xA0;
50
51     P0 = 0x00;
52     POM = 0x3F;
53 }

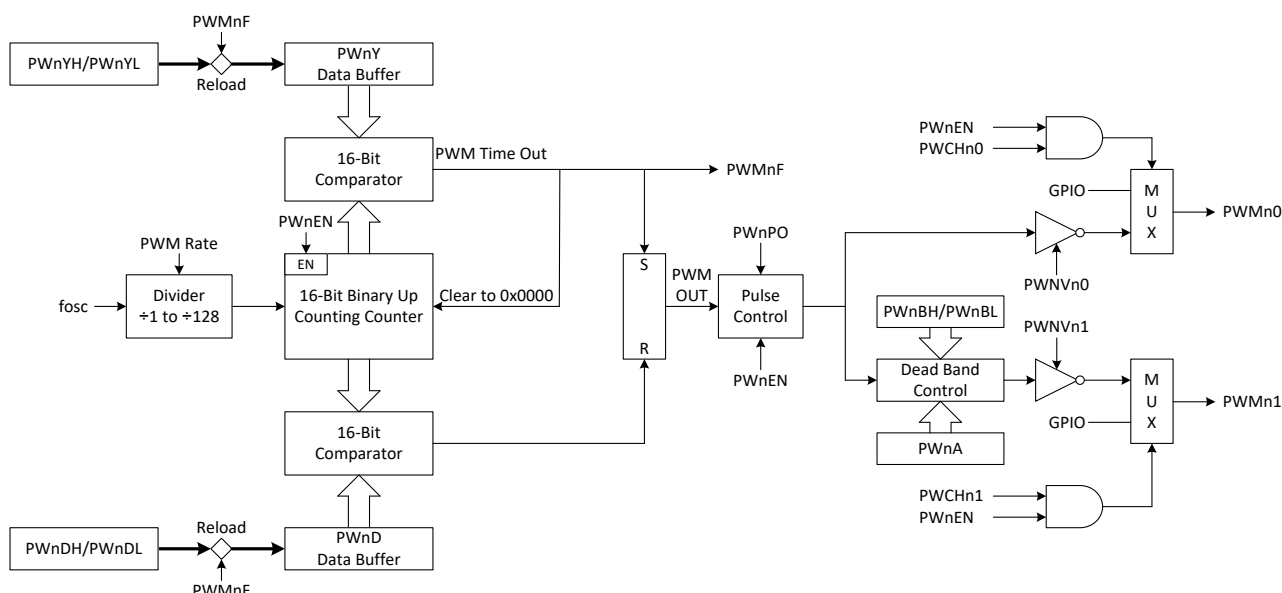
```

```
54
55 void T2Interrupt(void) interrupt ISRTimer2 //0x2B
56 { //TF2/TF2RL clear by software
57     if ((IRCON & 0x40) == 0x40) {
58         IRCON &= 0xBF; //Clear TF2
59         P00 = ~P00;
60     }
61     if ((IRCON & 0x80) == 0x80) {
62         IRCON &= 0x7F; //Clear TF2RL
63         P01 = ~P01;
64     }
65 }
66
67 void T2COM0Interrupt(void) interrupt ISRCom1 //0x53
68 { //TF2C0 clear by hardware
69     P02 = ~P02;
70 }
71
72 void T2COM1Interrupt(void) interrupt ISRCom2 //0x5B
73 { //TF2C1 clear by hardware
74     P03 = ~P03;
75 }
76
77 void T2COM2Interrupt(void) interrupt ISRCom3 //0x63
78 { //TF2C2 clear by hardware
79     P04 = ~P04;
80 }
81
82 void T2COM3Interrupt(void) interrupt ISRCom4 //0x6B
83 { //TF2C3 clear by hardware
84     P05 = ~P05;
85 }
```

16 PWM

The PW1/PW2/PW3 timer respectively includes a 16-bit binary up 2-channel PWM, and one pulse PWM function functions. By the counter reaches the up-boundary value (PW1Y/PW2Y/PW3Y), it clears its counter and triggers an interrupt signal. Each PWM's duty cycle is controlled by PW1D/PW2D/PW3D register.

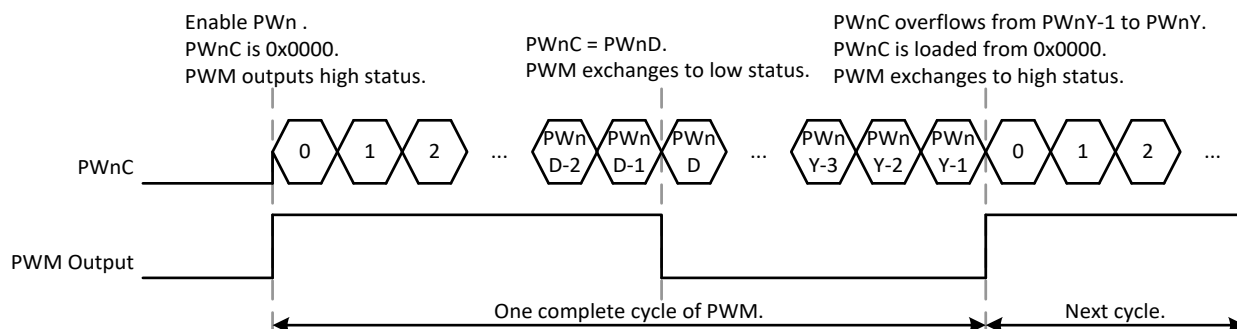
The PWMs also support one pulse output signal which can disables itself by the end of first PWM cycle. Thus, only one pulse would be generated in this condition. The PWM has two programmable channels shared with GPIO pins and controlled by PWCH[1:0] bit. The output operation must be through enabled each bit/channel of PWCH[1:0] bits. The enabled PWM channel exchanges from GPIO to PWM output. When the PWCH[1:0] bits disables, the PWM channel returns to last status of GPIO mode. The PWM build in IDLE Mode wake-up function if interrupt enable. When PWM timer overflow occurs (counts from PWN_Y-1 to PWN_Y), PWM_nF would be issued immediately which can read/write by firmware. PWM_n interrupt function is controlled by EPWM_n.



16.1 General PWM

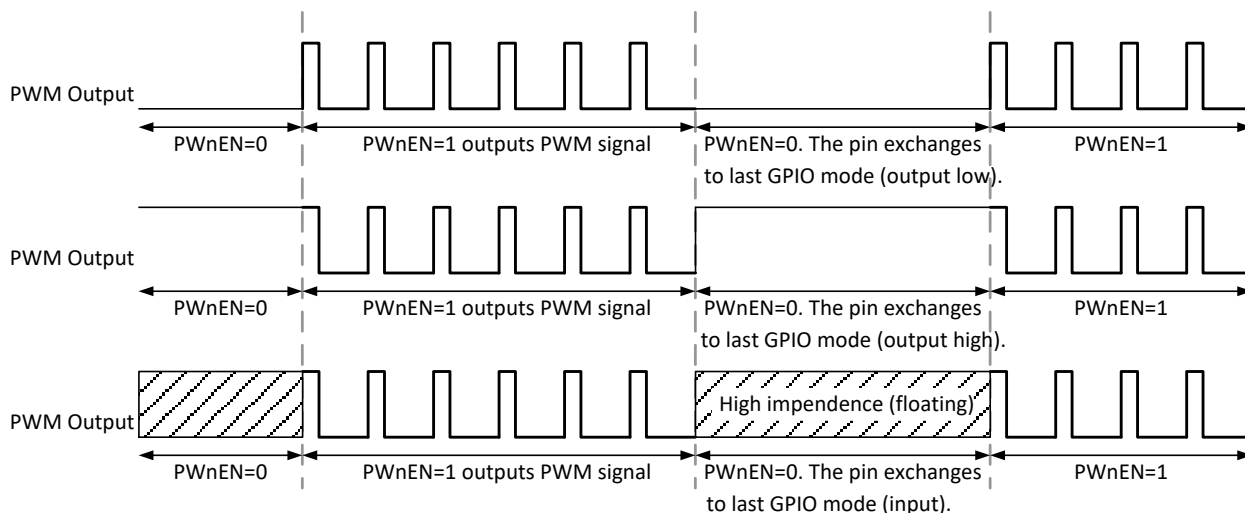
PW_n timer builds in PWM function controlled by PW_nEN and PWCH[1:0] bits. PWM_n0, PWM_n1 are output pins. The PWM_n0, PWM_n1 output pins are shared with GPIO pin controlled by PWCH[1:0] bits. When output PWM function, we must be set PW_nEN = 1. When PWM output signal synchronize finishes, the PWM channel exchanges from GPIO to PWM output. When PW_nEN = 0, the PWM channel returns to GPIO mode and last status. PWM signal is generated from the result of PW_nY and PW_nD comparison combination. When PW_nC counts from 0x0000, the PWM outputs high status which is the PWM initial status. PW_nC is loaded new data from PW_nY register to decide PWM cycle and resolution. PW_nC keeps counting, and the system compares PW_nC and PW_nD. When PW_nC =

PWnD, the PWM output status exchanges to low PWnC keeps counting. When PWM timer overflow occurs (PWnY-1 to PWnY), and one cycle of PWM signal finishes. PWnC is reloaded from 0x0000 automatically, and PWM output status exchanges to high for next cycle. PWnD decides the high duty duration, and PWnY decides the resolution and cycle of PWM. PWnD can't be larger than PWnY, or the PWM signal is error. PWM clock source is fosc, PWnRATE[2:0] bits: 000 = fosc/128, 001 = fosc/64, 010 = fosc/32, 011 = fosc/16, 100 = fosc/8, 101 = fosc/4, 110 = fosc/2, 111 = fosc/1.



PWM Period = PWnY

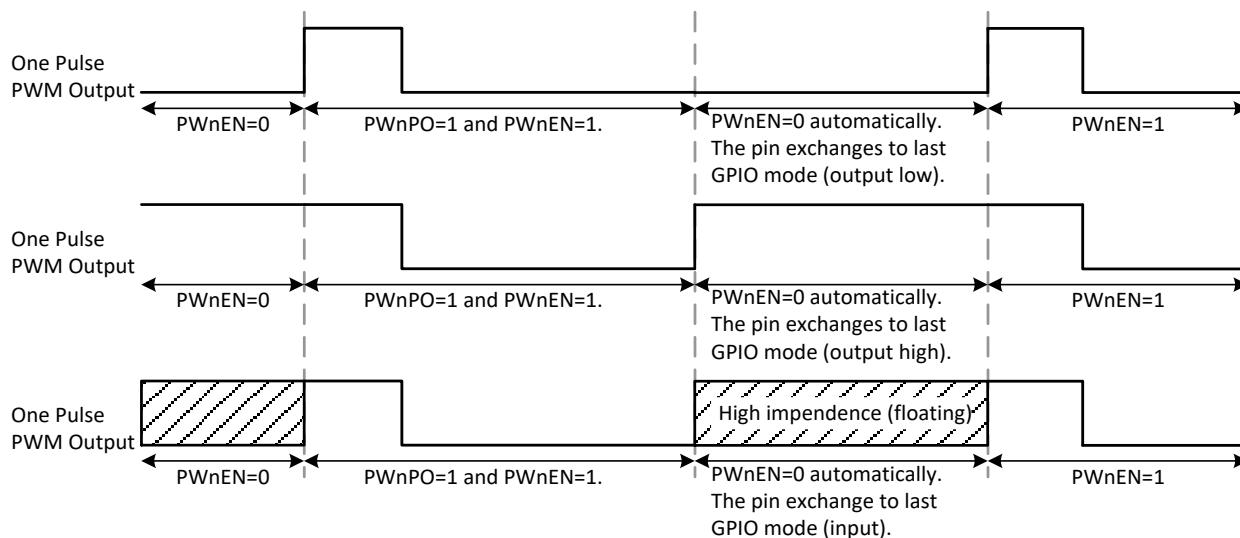
PWM duty = (PWnD): (PWnY-PWnD)



16.2 One Pulse PWM

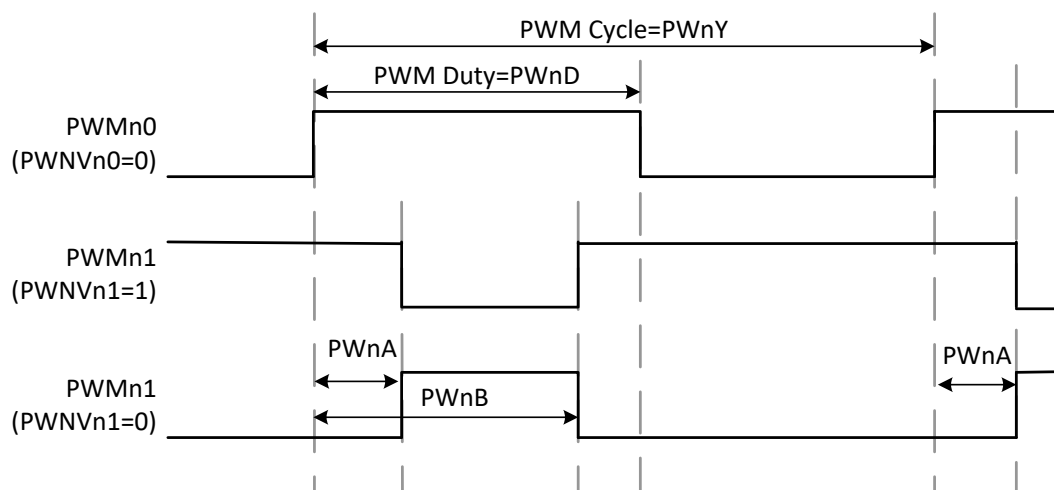
When PWnPO = 0, PWn is PWM function mode. When PWnPO = 1 and PWnEN = 1, PWn will output one pulse PWM function and the PWMnF is issued as PWn counter overflow. PWnEN bit is cleared automatically and pulse output pin returns to idle status. To output next pulse is to set PWnEN bit by program again. One pulse PWM channels selected by PWCH[1:0] bits. PWMn0, PWMn1 are output pins. The PWMn0, PWMn1 output pins are shared with GPIO pin controlled by PWCH[1:0] bits. When output one pulse PWM function, we must be set PWnPO = PWnEN=1. When one pulse PWM output signal synchronize finishes, the PWM channel exchanges from GPIO to PWM output.

When one pulse PWM output finishes, $PWnEN = 0$, the PWM channel returns to GPIO mode and last status.



16.3 Inverse and Dead Band

The PWM builds in inverse output function. The PWM has one inverse PWM signal as $PWNV = 1$. When $PWNV = 1$, the PWn outputs the inverse PWM signal of $PWMn$. When $PWNV = 0$, the PWn outputs the non-inverse PWM signal of $PWMn$. The inverse PWM output waveform is below diagram.



The PWM dead band occurs in PWM high pulse width, and the dead band period is programmable from $PWnA$ and $PWnD - PWnB$ registers. The dead band period is symmetrical at left-right terminal of PWM pulse width or not. If the dead band period is longer than PWM duty, the PWM is no output.

16.4 PWM Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PW1M	PW1R2	PW1R1	PW1R0	PWNV11	PWNV10	PWCH11	PWCH10	PW1PO
PW2M	PW2R2	PW2R1	PW2R0	PWNV21	PWNV20	PWCH21	PWCH20	PW2PO
PW3M	PW3R2	PW3R1	PW3R0	PWNV31	PWNV30	PWCH31	PWCH30	PW3PO
P1OC	PW3EN	PW2EN	PW1EN	P06OC	P05OC	P13OC	P12OC	P11OC
PW1YH	PW1Y15	PW1Y14	PW1Y13	PW1Y12	PW1Y11	PW1Y10	PW1Y9	PW1Y8
PW1YL	PW1Y7	PW1Y6	PW1Y5	PW1Y4	PW1Y3	PW1Y2	PW1Y1	PW1Y0
PW2YH	PW2Y15	PW2Y14	PW2Y13	PW2Y12	PW2Y11	PW2Y10	PW2Y9	PW2Y8
PW2YL	PW2Y7	PW2Y6	PW2Y5	PW2Y4	PW2Y3	PW2Y2	PW2Y1	PW2Y0
PW3YH	PW3Y15	PW3Y14	PW3Y13	PW3Y12	PW3Y11	PW3Y10	PW3Y9	PW3Y8
PW3YL	PW3Y7	PW3Y6	PW3Y5	PW3Y4	PW3Y3	PW3Y2	PW3Y1	PW3Y0
PW1DH	PW1D15	PW1D14	PW1D13	PW1D12	PW1D11	PW1D10	PW1D9	PW1D8
PW1DL	PW1D7	PW1D6	PW1D5	PW1D4	PW1D3	PW1D2	PW1D1	PW1D0
PW2DH	PW2D15	PW2D14	PW2D13	PW2D12	PW2D11	PW2D10	PW2D9	PW2D8
PW2DL	PW2D7	PW2D6	PW2D5	PW2D4	PW2D3	PW2D2	PW2D1	PW2D0
PW3DH	PW3D15	PW3D14	PW3D13	PW3D12	PW3D11	PW3D10	PW3D9	PW3D8
PW3DL	PW3D7	PW3D6	PW3D5	PW3D4	PW3D3	PW3D2	PW3D1	PW3D0
PW1BH	PW1B15	PW1B14	PW1B13	PW1B12	PW1B11	PW1B10	PW1B9	PW1B8
PW1BL	PW1B7	PW1B6	PW1B5	PW1B4	PW1B3	PW1B2	PW1B1	PW1B0
PW2BH	PW2B15	PW2B14	PW2B13	PW2B12	PW2B11	PW2B10	PW2B9	PW2B8
PW2BL	PW2B7	PW2B6	PW2B5	PW2B4	PW2B3	PW2B2	PW2B1	PW2B0
PW3BH	PW3B15	PW3B14	PW3B13	PW3B12	PW3B11	PW3B10	PW3B9	PW3B8
PW3BL	PW3B7	PW3B6	PW3B5	PW3B4	PW3B3	PW3B2	PW3B1	PW3B0
PW1A	PW1A7	PW1A6	PW1A5	PW1A4	PW1A3	PW1A2	PW1A1	PW1A0
PW2A	PW2A7	PW2A6	PW2A5	PW2A4	PW2A3	PW2A2	PW2A1	PW2A0
PW3A	PW3A7	PW3A6	PW3A5	PW3A4	PW3A3	PW3A2	PW3A1	PW3A0
OPM	PW3CM1	PW3CM0	PW2CM1	PW2CM0	PW1CM1	PW1CM0	OP1EN	OP0EN
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
IEN2	-	-	ECMP1	ECMP0	EADC	EPWM3	EPWM2	-
IEN4	EPWM1	EX2	-	-	PWM1F	IE2	-	-
IRCON2	-	-	-	CMP1F	CMP0F	ADCF	PWM3F	PWM2F

PWnM Registers (PW1M: 0xAB, PW2M: 0xA1, PW3M: 0xB1)

Bit	Field	Type	Initial	Description
7..5	PWnRATE	R/W	000	PWM timer clock source 000: fosc / 128 001: fosc / 64 010 fosc / 32 011: fosc / 16 100: fosc / 8 101: fosc / 4 110: fosc / 2 111: fosc / 1
4	PWNVn1	R/W	0	PWMn1 pin output control 0: Non-inverse 1: Inverse
3	PWNVn0	R/W	0	PWMn0 pin output control 0: Non-inverse 1: Inverse
2	PWCHn1	R/W	0	PWMn1 shared-pin control 0: GPIO 1: PWM output
1	PWCHn0	R/W	0	PWMn0 shared-pin control 0: GPIO 1: PWM output
0	PWnPO	R/W	0	One pulse function 0: Disable 1: Enable

P1OC Register (0xE4)

Bit	Field	Type	Initial	Description
7	PW3EN	R/W	0	PW3 function 0: Disable 1: Enable
6	PW2EN	R/W	0	PW2 function 0: Disable 1: Enable
5	PW1EN	R/W	0	PW1 function 0: Disable 1: Enable
Else				Refer to other chapter(s)

PW1YH/PW1YL Registers (PW1YH: 0xAD, PW1YL: 0xAC)

Bit	Field	Type	Initial	Description
7..0	PW1YH/L	R/W	0x00	16-bit PWM1 period control

PW2YH/PW2YL Registers (PW2YH: 0xA3, PW2YL: 0xA2)

Bit	Field	Type	Initial	Description
7..0	PW2YH/L	R/W	0x00	16-bit PWM2 period control

PW3YH/PW3YL Registers (PW3YH: 0xB3, PW3YL: 0xB2)

Bit	Field	Type	Initial	Description
7..0	PW3YH/L	R/W	0x00	16-bit PWM3 period control

PW1DH/PW1DL Registers (PW1DH: 0xBC, PW1DL: 0xBB)

Bit	Field	Type	Initial	Description
7..0	PW1DH/L	R/W	0x00	16-bit PWM1 duty control

PW2DH/PW2DL Registers (PW2DH: 0xA7, PW2DL: 0xA6)

Bit	Field	Type	Initial	Description
7..0	PW2DH/L	R/W	0x00	16-bit PWM2 duty control

PW3DH/PW3DL Registers (PW3DH: 0x87, PW3DL: 0x86)

Bit	Field	Type	Initial	Description
7..0	PW3DH/L	R/W	0x00	16-bit PWM3 duty control

PW1BH/PW1BL Registers (PW1BH: 0xAF, PW1BL: 0xAE)

Bit	Field	Type	Initial	Description
7..0	PW1BH/L	R/W	0x00	16-bit PWM1 dead band control

PW2BH/PW2BL Registers (PW2BH: 0xA5, PW2BL: 0xA4)

Bit	Field	Type	Initial	Description
7..0	PW2BH/L	R/W	0x00	16-bit PWM2 dead band control

PW3BH/PW3BL Registers (PW3BH: 0xB5, PW3BL: 0xB4)

Bit	Field	Type	Initial	Description
7..0	PW3BH/L	R/W	0x00	16-bit PWM3 dead band control

PW1A Register (PW1A: 0xBD)

Bit	Field	Type	Initial	Description
7..0	PW1A	R/W	0x00	8-bit PWM1 dead band control

PW2A Register (PW2A: 0xBE)

Bit	Field	Type	Initial	Description
7..0	PW2A	R/W	0x00	8-bit PWM2 dead band control

PW3A Registers (PW3A: 0xCF)

Bit	Field	Type	Initial	Description
7..0	PW3A	R/W	0x00	8-bit PWM3 dead band control

OPM Register (0x9B)

Bit	Field	Type	Initial	Description
7	PW3CM1	R/W	0	PWM3 output and CMP1 trigger synchronous control bit 0: Disable 1: Enable
6	PW3CM0	R/W	0	PWM3 output and CMP0 trigger synchronous control bit 0: Disable 1: Enable
5	PW2CM1	R/W	0	PWM2 output and CMP1 trigger synchronous control bit 0: Disable 1: Enable
4	PW2CM0	R/W	0	PWM2 output and CMP0 trigger synchronous control bit 0: Disable 1: Enable
3	PW1CM1	R/W	0	PWM1 output and CMP1 trigger synchronous control bit 0: Disable 1: Enable
2	PW1CM0	R/W	0	PWM1 output and CMP0 trigger synchronous control bit 0: Disable 1: Enable
Else				Refer to other chapter(s)

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
Else				Refer to other chapter(s)

IEN2 Register (0X9A)

Bit	Field	Type	Initial	Description
2	EPWM3	R/W	0	PWM3 interrupt control bit. 0: Disable PWM3 interrupt function. 1: Enable PWM3 interrupt function.
1	EPWM2	R/W	0	PWM2 interrupt control bit 0: Disable PWM2 interrupt function. 1: Enable PWM2 interrupt function.
Else	Reserved	R	0	

IEN4 Register (0XD1)

Bit	Field	Type	Initial	Description
7	EPWM1	R/W	0	PWM1 interrupt control bit. 0 = Disable PWM1 interrupt function. 1 = Enable PWM1 interrupt function.
3	PWM1F	R/W	0	PWM1 interrupt request flag. 0: None PWM1 interrupt request 1: PWM1 interrupt request.
Else	Reserved	R	0	

IRCON2 Register (0XBF)

Bit	Field	Type	Initial	Description
1	PWM3F	R/W	0	PWM3 interrupt request flag. 0: None PWM3 interrupt request 1: PWM3 interrupt request.
0	PWM2F	R/W	0	PWM2 interrupt request flag. 0: None PWM2 interrupt request 1: PWM2 interrupt request.
Else	Reserved	R	0	

16.5 Sample Code

The following sample code demonstrates how to perform PW1/PW2/PW3 with interrupt.

```

1  #define PW1Inv0      (1 << 3) //PWM10 output inverse
2  #define PW1Inv1      (1 << 4) //PWM11 output inverse
3  #define PW1CMP0Tri    (1 << 2) //PW1 output Control by Comparator 0 trigger
4  #define PW1CMP1Tri    (1 << 3) //PW1 output Control by Comparator 1 trigger
5  #define PW1OnePu      (1 << 0) //Enable PW1 pulse output function
6  #define PWM10En       (1 << 1) //Enable PWM10 output function
7  #define PWM11En       (1 << 2) //Enable PWM11 output function
8  #define PW1En         (1 << 5) //Enable PWM1 function
9
10 #define PW2Inv0      (1 << 3) //PWM20 output inverse
11 #define PW2Inv1      (1 << 4) //PWM21 output inverse
12 #define PW2CMP0Tri    (1 << 4) //PW2 output Control by Comparator 0 trigger
13 #define PW2CMP1Tri    (1 << 5) //PW2 output Control by Comparator 1 trigger
14 #define PW2OnePu      (1 << 0) //Enable PW2 pulse output function
15 #define PWM20En       (1 << 1) //Enable PWM20 output function
16 #define PWM21En       (1 << 2) //Enable PWM21 output function
17 #define PW2En         (1 << 6) //Enable PWM2 function
18
19 #define PW3Inv0      (1 << 3) //PWM30 output inverse
20 #define PW3Inv1      (1 << 4) //PWM31 output inverse
21 #define PW3CMP0Tri    (1 << 6) //PW3 output Control by Comparator 0 trigger
22 #define PW3CMP1Tri    (1 << 7) //PW3 output Control by Comparator 1 trigger
23 #define PW3OnePu      (1 << 0) //Enable PW3 pulse output function
24 #define PWM30En       (1 << 1) //Enable PWM30 output function
25 #define PWM31En       (1 << 2) //Enable PWM31 output function
26 #define PW3En         (1 << 7) //Enable PWM3 function
27
28
29 void InitPWM(void)
30 {
31     // PWM1_Initial
32     PW1YH = 0x80;
33     PW1YL = 0x00;
34     PW1DH = 0x40;
35     PW1DL = 0x00;
36     PW1BH = 0x60;
37     PW1BL = 0x00;
38     PW1A = 0x80;
39
40     // PWM1 clock = Fosc/32, PW10/11 channel enable
41     PW1M = 0x40 | PWM10En | PWM11En;
42
43     // PWM2_Initial
44     PW2YH = 0x00;
45     PW2YL = 0x80;
46     PW2DH = 0x00;
47     PW2DL = 0x40;
48     PW2BH = 0x00;
49     PW2BL = 0x60;
50     PW2A = 0x20;
51
52     // PWM2 clock = Fosc/128, PW20/21 channel enable with inverse
53     PW2M = 0x00 | PW2Inv0 | PW2Inv1 | PWM20En | PWM21En;

```



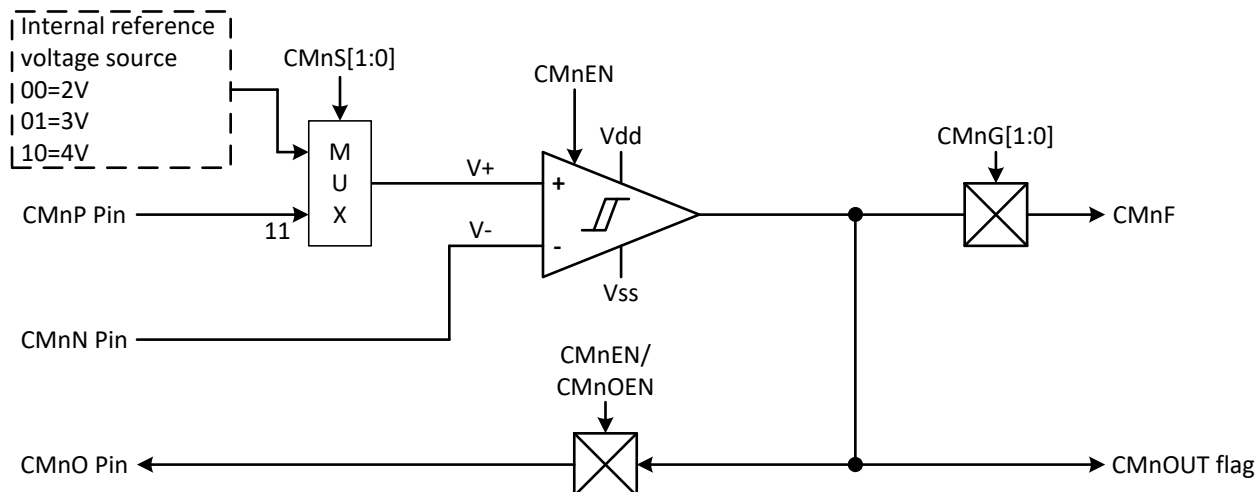
```

54
55 // PWM3_Initial
56 PW3YH = 0xF0;
57 PW3YL = 0xF0;
58 PW3DH = 0x80;
59 PW3DL = 0x80;
60 PW3BH = 0x40;
61 PW3BL = 0x40;
62 PW3A = 0xF0;
63
64 // PWM3 clock = Fosc/1, PW30/31 channel enable with one pulse
65 PW3M = 0xE0 | PWM30En | PWM31En | PW3OnePu;
66
67 // PW1/PW2/PW3 enable
68 P1OC |= PW3En | PW2En | PW1En;
69
70 // Enable PWM1 interrupt
71 IEN4 |= 0x80;
72
73 // Enable PWM2/PWM3 interrupt
74 IEN2 |= 0x06;
75
76 // Enable total interrupt
77 IEN0 |= 0x80;
78
79 P0 = 0x00;
80 POM |= 0x07;
81 }
82
83 void PW1Interrupt(void) interrupt ISRPwm1 //0x83
84 { //PWM1F clear by software
85     if ((IEN4 & 0x08) == 0x08) {
86         IEN4 &= 0xF7; //Clear PWM1F
87         P00 = ~P00;
88     }
89 }
90
91 void PW2Interrupt(void) interrupt ISRPwm2 //0x8B
92 { //PWM2F clear by software
93     if ((IRCON2 & 0x01) == 0x01) {
94         IRCON2 &= 0xFE; //Clear PWM2F
95         P01 = ~P01;
96     }
97 }
98
99 void PW3Interrupt(void) interrupt ISRPwm3 //0x93
100 { //PWM3F clear by software
101     if ((IRCON2 & 0x02) == 0x02) {
102         IRCON2 &= 0xFD; //Clear PWM3F
103         P02 = ~P02;
104     }
105 }

```

17 Comparator

The microcontroller builds in two comparator functions. When the positive input voltage is greater than the negative input voltage, the comparator output is high. When the positive input voltage is smaller than the negative input voltage, the comparator output is low. Comparator positive voltage is from internal 2V/3V/4V or CMnP. There is a programmable direction function to decide comparator trigger edge for indicator function. The comparator has flag indicator, interrupt function and IDLE Mode weak-up function for different application.



17.1 Configurations of Operation

The Comparator pins are shared with GPIO controlled by CMnEN bit. When CMnEN=1, CMnN/CMnP pin is enabled connected to Comparator negative terminal. CMnOEN controls Comparator output connected to GPIO or not. When CMnOEN=1, Comparator output terminal is connected to GPIO pins and isolate GPIO function.

The internal reference has three steps including 2V/3V/4V controlled by CMnS[1:0] bits. Comparator pins configuration table is as following.

Comparator NO.	CMnEN	Comparator Negative Pin	Comparator Positive Pin (CMnS[1:0])				Comparator Output Pin (CMnOEN)	
			00	01	10	11	0	1
CMP0	CM0EN=0	All pins are GPIO mode. Comparator is disabled.						
	CM0EN=1	CM0N	2V	3V	4V	CM0P	GPIO	CM0O
CMP1	CM1EN=0	All pins are GPIO mode. Comparator is disabled.						
	CM1EN=1	CM1N	2V	3V	4V	CM1P	GPIO	CM1O

17.2 Comparator Output Function

The comparator output signal is the source of comparator output function. The comparator output function includes:

CMnOUT output flag: The comparator output signal is connected to CMnOUT flag directly. CMnOUT bit responses comparator status immediately. Program reads comparator status from CMnOUT bit.

Comparator extern pin output function: The comparator output status can output to CMnO pin controlled by CMnOEN bit. When CMnOEN=0, the comparator output pin is GPIO mode. If CMnOEN=1, CMnO pin outputs comparator output status and isolates GPIO mode.

Comparator edge trigger and interrupt function: The comparator builds in interrupt function, and the trigger edge is programmable. CMnG[1:0] bit controls comparator trigger edge. When the edge trigger condition occurs, CMPnF will be set automatically. To clear CMPnF bit must be through program. If CMnEN=1, program counter will be pointed to interrupt vector to execute interrupt service routine as CMPnF is setting. When the trigger edge direction is equal to interrupt trigger condition, the system will execute interrupt operation immediately.

Comparator IDLE Mode Wake-up function: The comparator's wake-up function only support IDLE Mode (interrupt needs enable), not STOP Mode. If the trigger edge condition (comparator output status exchanging) is found, the system will be wake-up from IDLE Mode. Because in interrupt trigger condition, the CMPnF is set as "1". Of course the interrupt routine is executed if the interrupt function enabled.

17.3 PWM Output Control

The results of comparator can be used to control PWM outputs. User can select the appropriate control mode through CMPT register. The following table lists these types of control methods.

CMnT[1:0]	PWM Synchronous Trigger Operation
00	CMPn with PWM output is not related.
01	CMPnF = 1 → PWM stop
10	CMnP > CMnN (Rising edge trigger) → PWM output CMnP < CMnN (Falling edge trigger) → PWM stop
11	CMnP < CMnN (Falling edge trigger) → PWM output CMnP > CMnN (rising edge trigger) → PWM stop

The comparator can control those PWM outputs, depending PWnCM* bits set. See detailed description of PWM section. OPM register bit 7 – 2: PW3CM1, PW3CM0, PW2CM0, PW2CM1, PW1CM1, PW1CM0.

17.4 Comparator Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CMP0M	CM0EN	-	CM0S1	CM0S0	CM0OEN	CM0OUT	CM0G1	CM0G0
CMP1M	CM1EN	-	CM1S1	CM1S0	CM1OEN	CM1OUT	CM1G1	CM1G0
CMPT	-	-	-	-	CM1T1	CM1T0	CM0T1	CM0T0
P3CON	P3CON7	P3CON6	P3CON5	P3CON4	P3CON3	P3CON2	P3CON1	P3CON0
P2CON	P2CON7	P2CON6	P2CON5	P2CON4	-	-	-	-
IEN2	-	-	ECMP1	ECMP0	EADC	EPWM3	EPWM2	-
IRCON2	-	-	-	CMP1F	CMP0F	ADCF	PWM3F	PWM2F

CMP0M Register (0x9C)

Bit	Field	Type	Initial	Description
7	CM0EN	R/W	0	Comparator 0 control bit. 0: Disable. CM0P/CM0N pins are GPIO mode. 1: Enable. CM0P/CM0N pins are CMP input pins.
5..4	CM0S[1:0]	R/W	00	CMP0 positive input voltage control bit. 00: 2.0V 01: 3.0V 10: 4.0V 11: CM0P
3	CM0OEN	R/W	0	Comparator 0 output pin control bit. 0: Disable. CM0O is GPIO mode. 1: Enable. CM0O is comparator output pin and isolate GPIO function.
2	CM0OUT	R	0	Comparator 0 output flag bit. 0: CM0P voltage is less than CM0N voltage. 1: CM0P voltage is larger than CM0N voltage.
1..0	CM0G[1:0]	R/W	00	Comparator interrupt trigger direction control bit. 00: Reserved. 01: Rising edge trigger. CM0P > CM0N. 10: Falling edge trigger. CM0P < CM0N. 11: Both rising and falling edge trigger

CMP1M Register (0x9D)

Bit	Field	Type	Initial	Description
7	CM1EN	R/W	0	Comparator 1 control bit. 0: Disable. CM1P/CM1N pins are GPIO mode. 1: Enable. CM1P/CM1N pins are CMP input pins.
5..4	CM1S[1:0]	R/W	00	CMP1 positive input voltage control bit. 00: 2.0V 01: 3.0V 10: 4.0V 11: CM1P
3	CM1OEN	R/W	0	Comparator 1 output pin control bit. 0: Disable. CM1O is GPIO mode. 1: Enable. CM1O is comparator output pin and isolate GPIO function.
2	CM1OUT	R	0	Comparator 1 output flag bit. 0: CM1P voltage is less than CM1N voltage. 1: CM1P voltage is larger than CM1N voltage.
1..0	CM1G[1:0]	R/W	00	Comparator interrupt trigger direction control bit. 00: Reserved. 01: Rising edge trigger. CM1P > CM1N. 10: Falling edge trigger. CM1P < CM1N. 11: Both rising and falling edge trigger (Level change trigger).

CMPT Register (0xCE)

Bit	Field	Type	Initial	Description
3..2	CM1T[1:0]	R/W	00	CMP1 with PWM trigger select bits. 00: CMP1 with PWM output is not related. 01: CMP1F = 1 → PWM stop 10: CM1P > CM1N → PWM output; CM1P < CM1N → PWM stop 11: CM1P < CM1N → PWM output; CM1P > CM1N → PWM stop
1..0	CM0T[1:0]	R/W	00	CMP0 with PWM trigger select bits. 00: CMP0 with PWM output is not related. 01: CM0F = 1 → PWM stop 10: CM0P > CM0N → PWM output; CM0P < CM0N → PWM stop 11: CM0P < CM0N → PWM output; CM0P > CM0N → PWM stop

P3CON Register (0x9F)

Bit	Field	Type	Initial	Description
7..6	P3CON[7:6]	R/W	0x00	P3 configuration control bit*. 0: P3 can be analog input pin (CMP input pin) or digital GPIO pin. 1: P3 is pure analog input pin and can't be a digital GPIO pin.
5..0				Refer to other chapter(s)

* P3CON [7:6] will configure related Port3 pin as pure analog input pin to avoid current leakage.

P2CON Register (0x9E)

Bit	Field	Type	Initial	Description
6..5	P2CON[6:5]	R/W	0x0	P2 configuration control bit*. 0: P2 can be analog input pin (CMP input pin) or digital GPIO pin. 1: P2 is pure analog input pin and can't be a digital GPIO pin.
Else				Refer to other chapter(s)

* P2CON [6:5] will configure related Port2 pin as pure analog input pin to avoid current leakage.

IEN2 Register (0x9A)

Bit	Field	Type	Initial	Description
5	ECMP1	R/W	0	Comparator 1 interrupt control bit. 0: Disable CMP1 interrupt function. 1: Enable CMP1 interrupt function.
4	ECMP0	R/W	0	Comparator 0 interrupt control bit. 0: Disable CMP0 interrupt function. 1: Enable CMP0 interrupt function.
Else				Refer to other chapter(s)

IRCON2 Register (0xBF)

Bit	Field	Type	Initial	Description
4	CMP1F	R/W	0	Comparator 1 interrupt request flag. 0: None CMP1 interrupt request 1: CMP1 interrupt request.
3	CMP0F	R/W	0	Comparator 0 interrupt request flag. 0: None CMP0 interrupt request. 1: CMP0 interrupt request.
Else				Refer to other chapter(s)

17.5 Sample Code

The following sample code demonstrates how to perform CMP0 with interrupt.

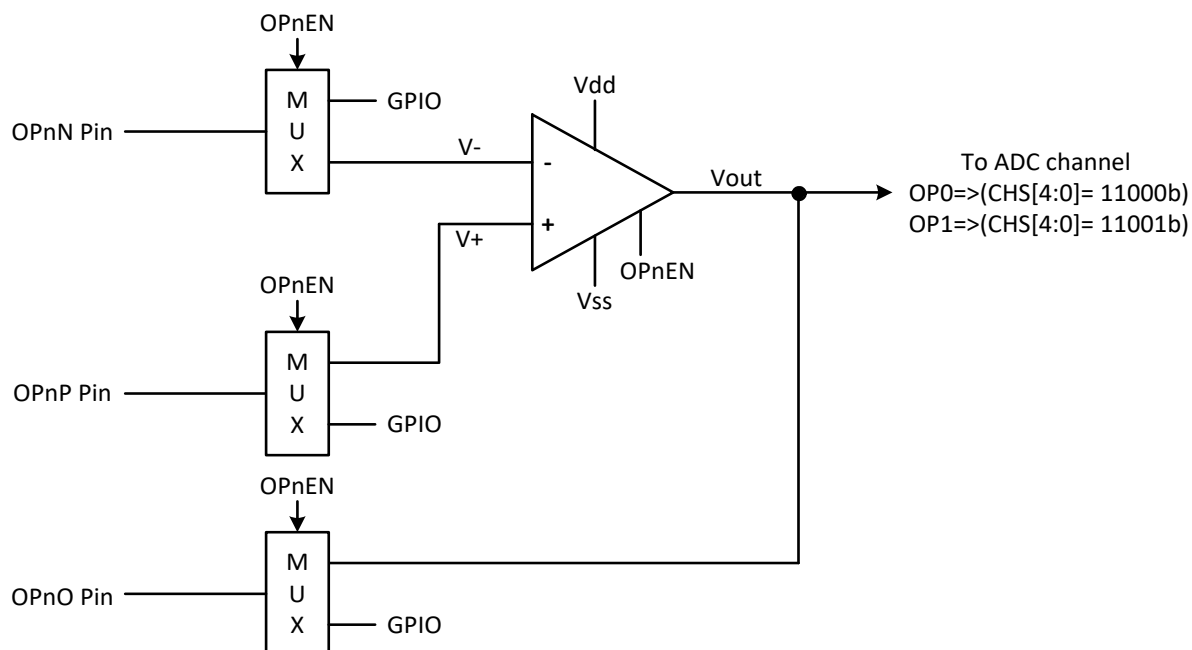
```

1  #define CM0LeChge    (3 << 0) // CM0P > CM0N or CM0P < CM0N
2  #define CM0NGreCM0P (2 << 0) // CM0P < CM0N
3  #define CM0PGreCM0N (1 << 0) // CM0P > CM0N
4  #define CMP0OEN      (1 << 3) // CMP0 output pin enable
5  #define CMP0Vin2V    (0 << 4) // CMP0 positive Vin connect 2.0V
6  #define CMP0Vin3V    (1 << 4) // CMP0 positive Vin connect 3.0V
7  #define CMP0Vin4V    (2 << 4) // CMP0 positive Vin connect 4.0V
8  #define CMP0VINP      (3 << 4) // CMP0 positive Vin connect CM0P
9  #define CMP0EN        (1 << 7) //enable CMP0
10 #define CM1LeChge    (3 << 0) // CM1P > CM1N or CM1P < CM1N
11 #define CM1NGreCM1P (2 << 0) // CM1P < CM1N
12 #define CM1PGreCM1N (1 << 0) // CM1P > CM1N
13 #define CMP1OEN      (1 << 3) // CMP1 output pin enable
14 #define CMP1Vin2V    (0 << 4) // CMP1 positive Vin connect 2.0V
15 #define CMP1Vin3V    (1 << 4) // CMP1 positive Vin connect 3.0V
16 #define CMP1Vin4V    (2 << 4) // CMP1 positive Vin connect 4.0V
17 #define CMP1VINP      (3 << 4) // CMP1 positive Vin connect CM1P
18 #define CMP1EN        (1 << 7) //enable CMP1
19
20 void CMPInit(void)
21 {
22     P1 = 0x00;
23     P1M = 0xC0;
24     // set CMP0 & CMP1 pins' mode at pure analog pin
25     P3CON |= 0xC0; //P36/P37
26     P2CON |= 0x60; //P26/P25
27     // configure CMP0 positive Vin = 4.0V and interrupt trigger = CM0P > CM0N
28     // enable CMP0 and output pin
29     CMP0M = CMP0EN | CMP0Vin4V | CMP0OEN | CM0PGreCM0N;
30     // configure CMP1 positive Vin = CM1P and interrupt trigger = CM1P < CM1N
31     // enable CMP1 and output pin
32     CMP1M = CMP1EN | CMP1VINP | CMP1OEN | CM1NGreCM1P;
33
34     // enable CMP0 & CMP1 interrupt
35     IEN2 |= 0x30;
36     IEN0 |= 0x80; //enable global interrupt
37 }
38
39 void CMP0Interrupt(void) interrupt ISRCmp0 //0xA3
40 {
41     if ((IRCON2 & 0x08) == 0x08) {
42         P16 = ~P16;
43         IRCON2 &= 0xF7; //Clear CMP0F
44     }
45 }
46 void CMP1Interrupt(void) interrupt ISRCmp1 //0xAB
47 {
48     if ((IRCON2 & 0x10) == 0x10) {
49         P17 = ~P17;
50         IRCON2 &= 0xEF; //Clear CMP1F
51     }
52 }

```


18 OPA

The microcontroller builds in two operational amplifiers (OP0 and OP1). The OP-Amp power range is VSS – VDD. OP-Amp input signal and output voltage are within the voltage range. The OP-Amp output pin is programmable to connect with ADC input channel for voltage measurement.



18.1 Configurations of Operation

The OP-AMP pins are shared with GPIO controlled by OP0EN/OP1EN bit. When OPEN=0, OP AMP pins are GPIO mode. When OPEN=1, GPIO pins switch to OP-AMP and isolate GPIO path. OP-AMP pins selection table is as following.

OP-AMP NO.	OPnEN	OP Positive Pin	OP Negative Pin	OP Output Pin
OP0	OP0EN = 0	All pins are GPIO mode.		
	OP0EN = 1	OP0P (Vin+)	OP0N (Vin-)	OP0O (Vout)
OP1	OP1EN = 0	All pins are GPIO mode.		
	OP1EN = 1	OP0P (Vin+)	OP0N (Vin-)	OP0O (Vout)

OP0/OP1 output pins are also connected to ADC internal AIN12 and AIN13 channel => CHS[4:0]. See detailed description of ADC section.

18.2 OPA Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
OPM	PW3CM1	PW3CM0	PW2CM1	PW2CM0	PW1CM1	PW1CM0	OP1EN	OP0EN
P3CON	P3CON7	P3CON6	P3CON5	P3CON4	P3CON3	P3CON2	P3CON1	P3CON0

OPM Register (0x9B)

Bit	Field	Type	Initial	Description
7..2				Refer to other chapter(s)
1	OP1EN	R/W	0	OP-Amp 1 enable bit. 0: Disable. OP-Amp 1 disable, OP1O/OP1P/OP1N pins are GPIO mode. 1: Enable. OP-Amp 1 enables, OP1O/OP1P/OP1N pins are OP-Amp 1 input and output pins.
0	OP0EN	R/W	0	OP-Amp 0 enable bit. 0: Disable. OP-Amp 0 disable, OP0O/OP0P/OP0N pins are GPIO mode. 1: Enable. OP-Amp 0 enables, OP0O/OP0P/OP0N pins are OP-Amp 0 input and output pins.

P3CON Register (0x9F)

Bit	Field	Type	Initial	Description
7..6				Refer to other chapter(s)
5..0	P3CON[5:0]	R/W	0x00	P3 configuration control bit*. 0: P3 can be analog input pin (OP input pin) or digital GPIO pin. 1: P3 is pure analog input pin and can't be a digital GPIO pin.

* P3CON [7:0] will configure related Port3 pin as pure analog input pin to avoid current leakage.

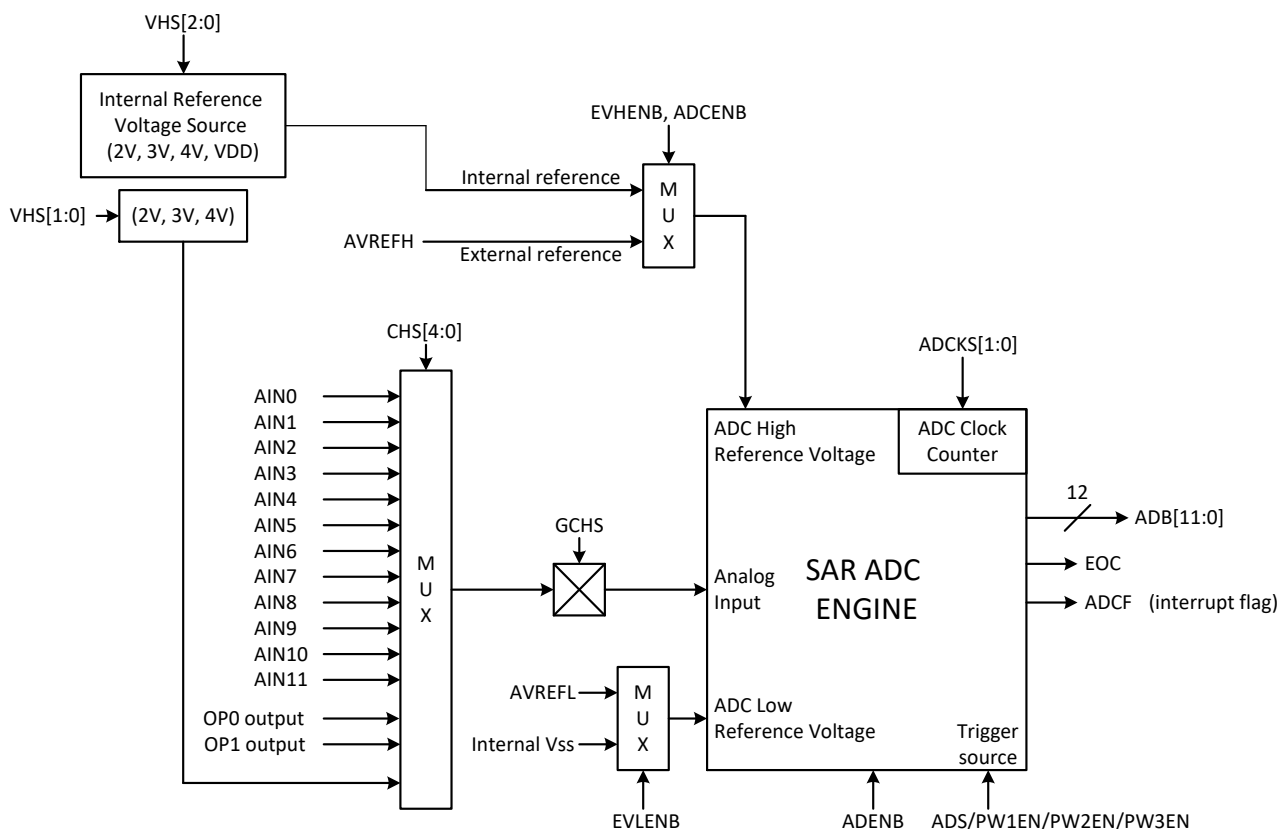
18.3 Sample Code

The following sample code demonstrates how to perform OP0.

```
1 #define OP0EN          (1 << 0)
2 #define OP1EN          (1 << 1)
3
4 void OP0En(void)
5 {
6     // set OP0 pins' mode at pure analog pin
7     P3CON = 0x07;
8
9     // enable OP0
10    OPM |= OP0EN;
11 }
12
13 void OP1En(void)
14 {
15     // set OP1 pins' mode at pure analog pin
16     P3CON = 0x38;
17
18     // enable OP1
19     OPM |= OP1EN;
20 }
```

19 ADC

The analog to digital converter (ADC) is SAR structure with 12-input sources and up to 4096-step resolution to transfer analog signal into 12-bits digital buffers. The ADC builds in 12-channel input source to measure 12 different analog signal sources. The ADC resolution is 12-bit. The ADC has four clock rates to decide ADC converting rate. The ADC reference high voltage includes 5 sources. Four internal power source including VDD, 4V, 3V and 2V. The other one is external reference voltage input pin from AVREFH pin. The ADC reference low voltage includes 2 sources controlled by VREFH register. One is internal VSS and the other one is external reference voltage input pin from AVREFL. The ADC builds in P2CON/P3CON/P4CON/P5CON registers to set pure analog input pin. After setup ADENB and ADS bits, the ADC starts to convert analog signal to digital data. Besides ADS bit can start to convert analog signal, PW1EN/ PW2EN/ PW3EN also have convert analog signal ADC function. ADC can work in idle mode. After ADC operating, the system would be waked up from green mode to normal mode if interrupt enable.



19.1 Configurations of Operation

These configurations must be setup completely before starting ADC converting. ADC is configured using the following steps:

1. Choose and enable the start of conversion ADC input channel. (By CHS[4:0] bits and GCHS bit)
2. The GPIO mode of ADC input channel must be set as input mode. (By PnM register)
3. The internal pull-up resistor of ADC input channel must be disabled. (By PnUR register)
4. The configuration control bit of ADC input channel must be set. (By PnCON register)
5. Choose ADC high reference voltage and low reference Voltage. (By VREFH register)
6. Choose ADC Clock Rate. (By ADCKS[1:0] bits)
7. After setup ADENB bits, the ADC ready to convert analog signal to digital data.

When ADC IP is enabled by ADENB bit, it is necessary to make an ADC start-up by program. Writing a 1 to the ADS bit of register ADM. After setup ADENB and ADS bits, the ADC starts to convert analog signal to digital data. The ADS bit is reset to logic 0 when the conversion is complete. When the conversion is complete, the ADC circuit will set EOC and ADCF bits to “1” and the digital data outputs in ADB and ADR registers. If ADC interrupt function is enabled (EADC = 1), the ADC interrupt request occurs and executes interrupt service routine when ADCF is “1” after ADC converting. Clear ADCF by hardware automatically in interrupt procedure. Note that when ADPWS bit is “1”, if PWM enable trigger be used as the conversion source, the ADC will continuous conversions until PWM is disabled.

19.2 ADC input channel

The ADC builds in 12-channel input source (AIN0 – AIN11) to measure 12 different analog signal sources controlled by CHS[4:0] and GCHS bits. AIN12, AIN13 channels are OP0 and OP1-Amp output terminal. The AIN14 is internal 2V or 3V or 4V input channel. There is no any input pin from outside. In this time ADC reference voltage must be internal VDD and External voltage, not internal 2V or 3V or 4V. AIN14 can be a good battery detector for battery system. To select appropriate internal AVREFH level and compare value, a high performance and cheaper low battery detector is built in the system.

CHS[4:0]	Channel	Pin name	Remark
00000	AIN0	P4.0	-
00001	AIN1	P4.1	-
00010	AIN2	P4.2	-
00011	AIN3	P4.3	-
00100	AIN4	P4.4	-
00101	AIN5	P4.5	-
00110	AIN6	P4.6	-
00111	AIN7	P4.7	-
01000	AIN8	P5.0	-
01001	AIN9	P5.1	-
01010	AIN10	P5.2	-
01011	AIN11	P5.3	-
01100 – 10111	-	-	Reserved
11000	AIN12	OP0	OP0-AMP output terminal
11001	AIN13	OP1	OP1-AMP output terminal
11010	AIN14	Internal 2V or 3V or 4V	Battery detector channel
11011 – 11111	-	-	No connection

19.2.1 Pin Configuration

ADC input channels are shared with Port2, Port3, Port4 and Port5. ADC channel selection is through CHS[4:0] bit. Only one pin of Port2, Port3, Port4 and Port5 can be configured as ADC input in the same time. The pins of Port2, Port3, Port4 and Port5 configured as ADC input channel must be set input mode, disable internal pull-up and enable P2CON/P3CON/P4CON/P5CON first by program. After selecting ADC input channel through CHS[4:0], set GCHS bit as “1” to enable ADC channel function.

ADC input pins are shared with digital I/O pins. Connect an analog signal to CMOS digital input pin, especially, the analog signal level is about 1/2 VDD will cause extra current leakage. In the power down mode, the above leakage current will be a big problem. Unfortunately, if users connect more than one analog input signal to Port2, Port3, Port4 or Port5 will encounter above current leakage situation. Write “1” into PnCON register will configure related pin as pure analog input pin to avoid current leakage.

Note that When ADC pin is general I/O mode, the bit of P2CON, P3CON, P4CON and P5CON must be set to “0”, or the digital I/O signal would be isolated.

19.3 Reference Voltage

The ADC builds in five high reference voltage source controlled through VREFH register. There are one external voltage source and four internal voltage source (VDD, 4V, 3V, 2V). When EVHENB bit is “1”, ADC reference voltage is external voltage source from AVREFH/P5.4. In the condition, P5.4 GPIO mode must be set as input mode and disable internal pull-up resistor. The ADC reference low voltage includes two sources controlled by EVLENB bit. One is internal VSS (EVLENB=0), and the other one is external reference voltage input pin from AVREFL/P5.5 (EVLENB=1). When the P5.5 be ADC external low reference voltage input pin, P5.5 GPIO mode must be set as input mode and disable internal pull-up resistor.

If EVHENB bit is “0”, ADC reference high voltage is from internal voltage source selected by VHS[1:0] bits. If VHS[1:0] is “11”, ADC reference high voltage is VDD. If VHS[1:0] is “10”, ADC reference high voltage is 4V. If VHS[1:0] is “01”, ADC reference high voltage is 3V. If VHS[1:0] is “00”, ADC reference high voltage is 2V. The limitation of internal high reference voltage application is VDD can’t below each of internal high voltage level, or the level is equal to VDD. If AIN14 channel is selected as internal 2V or 3V or 4V input channel. There is no any input pin from outside. In this time ADC high reference voltage must be internal VDD or External voltage, not internal 2V/3V/4V.

19.3.1 Signal Format

ADC sampling voltage range is limited by high/low reference voltage. The ADC low reference voltage includes internal VSS and external reference voltage source from P5.5/AVREFL controlled by EVLENB bit. The ADC high reference voltage includes internal VDD/4V/3V/2V and external reference voltage source from P5.4/AVREFH pin controlled by EVHENB bit. ADC reference voltage range limitation is “(ADC high reference voltage - low reference voltage) $\geq$ 2V”. So ADC high reference voltage range is AVREFL+2V to VDD. The range is ADC external high reference voltage range. ADC low reference voltage range is VSS to AVREFH-2V. The range is ADC external high reference voltage range.

- ADC Internal Low Reference Voltage = 0V. (EVLENB=0)
- ADC External Low Reference Voltage = VSS to AVREFH-2V. (EVLENB=1)
- ADC Internal High Reference Voltage = VDD/4V/3V/2V. (EVHENB=0)
- ADC External High Reference Voltage = AVREFL+2V to VDD. (EVHENB=1)

ADC sampled input signal voltage must be from ADC low reference voltage to ADC high reference. If the ADC input signal voltage is over the range, the ADC converting result is error (full scale or zero).

- ADC Low Reference Voltage $\leq$ ADC Sampled Input Voltage $\leq$ ADC High Reference Voltage

19.4 Converting Time

The ADC converting time is from ADS=1 (Start to ADC convert) to EOC=1 (End of ADC convert). The converting time duration is depend on ADC clock rate. 12-bit ADC’s converting time is

1/ (ADC clock /4)*16 sec. ADC clock source is fosc and includes fosc/1, fosc/2, fosc/8 and fosc/16 controlled by ADCKS[1:0] bits.

The ADC converting time affects ADC performance. If input high rate analog signal, it is necessary to select a high ADC converting rate. If the ADC converting time is slower than analog signal variation rate, the ADC result would be error. So to select a correct ADC clock rate to decide a right ADC converting rate is very important.

$$12 \text{ bits ADC conversion time} = \frac{16}{\text{ADC clock rate}/4}$$

ADCKS[1:0]	ADC clock rate	fosc = 16MHz		fosc = 32MHz	
		Converting time	Converting rate	Converting time	Converting rate
00	fosc/16	$1/(16\text{MHz}/16/4)*16$ = 64us	15.625kHz	$1/(32\text{MHz}/16/4)*16$ = 32us	31.25kHz
01	fosc/8	$1/(16\text{MHz}/8/4)*16$ = 32us	31.25kHz	$1/(32\text{MHz}/8/4)*16$ = 16us	62.5kHz
10	fosc	$1/(16\text{MHz}/4)*16$ = 4us	250kHz	$1/(32\text{MHz}/4)*16$ = 2us	500kHz
11	fosc/2	$1/(16\text{MHz}/2/4)*16$ = 8us	125kHz	$1/(32\text{MHz}/2/4)*16$ = 4us	250kHz

19.5 Data Buffer

ADC data buffer is 12-bit length to store ADC converter result. The high byte is ADB register, and the low-nibble is ADR[3:0] bits. The ADB register is only 8-bit register including bit 4 – bit 11 ADC data. To combine ADB register and the low-nibble of ADR will get full 12-bit ADC data buffer. The ADC data buffer is a read-only register and the initial status is unknown after system reset.

Table 19-1 The AIN input voltage vs. ADB output data

AIN n	ADB11	ADB10	ADB9	ADB8	ADB7	ADB6	ADB5	ADB4	ADB3	ADB2	ADB1	ADB0
0/4096*VREFH	0	0	0	0	0	0	0	0	0	0	0	0
1/4096*VREFH	0	0	0	0	0	0	0	0	0	0	0	1
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
4094/4096*VREFH	1	1	1	1	1	1	1	1	1	1	1	0
4095/4096*VREFH	1	1	1	1	1	1	1	1	1	1	1	1

19.6 ADC Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
ADM	ADENB	ADS	EOC	CHS4	CHS3	CHS2	CHS1	CHS0
ADB	ADB11	ADB10	ADB9	ADB8	ADB7	ADB6	ADB5	ADB4
ADR	-	GCHS	ADCKS1	ADCKS0	ADB3	ADB2	ADB1	ADB0
VREFH	EVHENB	EVLENB	-	ADPWS	-	VHS2	VHS1	VHS0
P5CON	-	-	P5CON5	P5CON4	P5CON3	P5CON2	P5CON1	P5CON0
P4CON	P4CON7	P4CON6	P4CON5	P4CON4	P4CON3	P4CON2	P4CON1	P4CON0
P3CON	P3CON7	P3CON6	P3CON5	P3CON4	P3CON3	P3CON2	P3CON1	P3CON0
P2CON	P2CON7	P2CON6	P2CON5	P2CON4	-	-	-	-
IEN2	-	-	ECMP1	ECMP0	EADC	EPWM3	EPWM2	-
IRCON2	-	-	-	CMP1F	CMP0F	ADCF	PWM3F	PWM2F

ADM Register (0xD2)

Bit	Field	Type	Initial	Description
7	ADENB	R/W	0	ADC control bit. In stop mode, disable ADC to reduce power consumption. 0: Disable 1: Enable
6	ADS	R/W	0	ADC conversion control Write 1: Start ADC conversion (automatically cleared by the end of conversion)
5	EOC	R/W	0	ADC status bit. 0: ADC progressing 1: End of conversion (automatically set by hardware; manually cleared by firmware)
4..0	CHS[4:0]	R/W	0x00	ADC input channel select bit. 00000: AIN0, 00001: AIN1, 00010: AIN2, 00011: AIN3, 00100: AIN4, 00101: AIN5, 00110: AIN6, 00111: AIN7, 01000: AIN8, 01001: AIN9, 01010: AIN10, 01011: AIN11, 11000: AIN12 ^{*(1)} , 11001: AIN13 ^{*(2)} , 11010: AIN14 ^{*(3)} , others: Reserved.

*(1) AIN12 channel is OP0-Amp output terminal.

*(2) AIN13 channel is OP1-Amp output terminal.

*(3) The AIN14 is internal 2V or 3V or 4V input channel. There is no any input pin from outside. In this time ADC reference voltage must be internal VDD and External voltage, not internal 2V or 3V or 4V.

ADB Register (0xD3)

Bit	Field	Type	Initial	Description
7..0	ADB[11:4]	R	-	ADC Result Bit [11:4]* in 12-bit ADC resolution mode.

* ADC data buffer is 12-bit length to store ADC converter result. The high byte is ADB register, and the low-nibble is ADR[3:0] bits.

ADR Register (0xD4)

Bit	Field	Type	Initial	Description
6	GCHS	R/W	0	ADC global channel select bit. 0: Disable AIN channel. 1: Enable AIN channel.
5..4	ADCKS[1:0]	R/W	00	ADC's clock source select bit. 00 = $f_{osc}/16$, 01 = $f_{osc}/8$, 10 = $f_{osc}/1$, 11 = $f_{osc}/2$
3..0	ADB[3:0]	R	-	ADC Result Bit [3:0]* in 12-bit ADC resolution mode.

* ADC data buffer is 12-bit length to store ADC converter result. The high byte is ADB register, and the low-nibble is ADR[3:0] bits.

VREFH Register (0xD5)

Bit	Field	Type	Initial	Description
7	EVHENB	R/W	0	ADC internal reference high voltage control bit. 0: Enable ADC internal VREFH function. AVREFH/P5.4 pin is GPIO. 1: Disable ADC internal VREFH function. AVREFH/P5.4 pin is external AVREFH ^{*(1)} input pin.
6	EVLENB	R/W	0	ADC internal reference low voltage control bit. 0: Enable ADC internal VREFL function. AVREFL/P5.5 pin is GPIO. 1: Disable ADC internal VREFL function. AVREFL/P5.5 pin is external AVREFL ^{*(2)} input pin.
4	ADPWS	R/W	0	PWM trigger ADC start control bit. 0: Disable PWM trigger ADC start. 1: Enable PWM trigger ADC start.
2..0	VHS[2:0]	R/W	00	ADC internal reference high voltage selects bits. ^{*(3)} 000: VREFH = 2.0V. 001: VREFH = 3.0V. 010: VREFH = 4.0V. 011: VREFH = VDD. 100: VREFH = VDD and AIN14 = 2.0V. 101: VREFH = VDD and AIN14 = 3.0V. 110: VREFH = VDD and AIN14 = 4.0V. Others: Reserved.

^{*(1)} The AVREFH level must be between the VDD and AVREFL + 2.0V.

^{*(2)} The AVREFL level must be between the VSS and AVREFH - 2.0V.

^{*(3)} If AIN14 channel is selected as internal 2V or 3V or 4V input channel. There is no any input pin from outside. In this time ADC reference high voltage must be internal VDD or External voltage, not internal 2V/3V/4V.

P5CON Register (0xD7)

Bit	Field	Type	Initial	Description
5..0	P5CON[5:0]	R/W	0x00	P5 configuration control bit*. 0: P5 can be analog input pin (ADC input pin) or digital GPIO pin. 1: P5 is pure analog input pin and can't be a digital GPIO pin.

* P5CON [5:0] will configure related Port5 pin as pure analog input pin to avoid current leakage.

P4CON Register (0xD6)

Bit	Field	Type	Initial	Description
7..0	P4CON[7:0]	R/W	0x00	P4 configuration control bit*. 0: P4 can be analog input pin (ADC input pin) or digital GPIO pin. 1: P4 is pure analog input pin and can't be a digital GPIO pin.

* P4CON [7:0] will configure related Port4 pin as pure analog input pin to avoid current leakage.

P3CON Register (0x9F)

Bit	Field	Type	Initial	Description
7..0	P3CON[7:0]	R/W	0x00	P3 configuration control bit*. 0: P3 can be analog input pin (ADC input pin) or digital GPIO pin. 1: P3 is pure analog input pin and can't be a digital GPIO pin.

* P3CON [7:0] will configure related Port3 pin as pure analog input pin to avoid current leakage.

P2CON Register (0x9E)

Bit	Field	Type	Initial	Description
7..4	P2CON[7:4]	R/W	0x0	P2 configuration control bit*. 0: P2 can be analog input pin (ADC input pin) or digital GPIO pin. 1: P2 is pure analog input pin and can't be a digital GPIO pin.

* P2CON [7:4] will configure related Port2 pin as pure analog input pin to avoid current leakage.

IEN2 Register (0x9A)

Bit	Field	Type	Initial	Description
3	EADC	R/W	0	ADC interrupt control bit. 0: Disable ADC interrupt function. 1: Enable ADC interrupt function.
Else				Refer to other chapter(s)

IRCON2 Register (0xBF)

Bit	Field	Type	Initial	Description
2	ADCF	R/W	0	ADC interrupt request flag. 0 = None ADC interrupt request. 1 = ADC interrupt request.
Else				Refer to other chapter(s)

19.7 Sample Code

The following sample code demonstrates how to perform ADC to convert AIN5 with interrupt.

```

1  #define ADCAIN14_VDD    (3 << 0) //AIN14 = VDD
2  #define ADCAIN14_4V    (2 << 0) //AIN14 = 4.0V
3  #define ADCAIN14_3V    (1 << 0) //AIN14 = 3.0V
4  #define ADCAIN14_2V    (0 << 0) //AIN14 = 2.0V
5  #define ADCInRefVDD    (1 << 2) //internal reference from VDD
6  #define ADCExLowRef    (1 << 6) //low reference from AVREFL/P5.5
7  #define ADCExHighRef   (1 << 7) //high reference from AVREFH/P5.4
8  #define ADCSpeedDiv16  (0 << 4) //ADC clock = fosc/16
9  #define ADCSpeedDiv8   (1 << 4) //ADC clock = fosc/8
10 #define ADCSpeedDiv1   (2 << 4) //ADC clock = fosc/1
11 #define ADCSpeedDiv2   (3 << 4) //ADC clock = fosc/2
12 #define ADCChannelEn   (1 << 6) //enable ADC channel
13 #define SelAIN5        (5 << 0) //select ADC channel 5
14 #define ADCStart       (1 << 6) //start ADC conversion
15 #define ADCISREN       (1 << 7) //enable ADC
16 #define EADC           (1 << 3) //enable ADC interrupt
17 #define ClearEOC       0xDF;
18
19 unsigned int  ADCBuffer; // data buffer
20
21 void ADCInit(void)
22 {
23     P1  = 0x00;
24     P1M = 0x80;
25
26     // set AIN5 pin's mode at pure analog pin
27     P4CON |= 0x20; //AIN5/P45
28     P4M  &= 0xDF; //input mode
29     P4UR  &= 0xDF; //disable pull-high
30
31     // configure ADC channel and enable ADC.
32     ADM = ADCEn | SelAIN5;
33
34     // enable channel and select conversion speed
35     ADR = ADCChannelEn | ADCSpeedDiv1;
36
37     // configure reference voltage
38     VREFH = ADCExLowRef | ADCInRefVDD;
39
40     // set external low reference pin's mode at pure analog pin
41     P5CON |= 0x20; //AVREFL/P55
42     P5M  &= 0xDF; //input mode
43     P5UR  &= 0xDF; //disable pull-high
44
45     // enable global interrupt
46     IEN0 |= 0x80;
47
48     //enable ADC interrupt
49     IEN2 |= EADC;
50
51     // start ADC conversion
52     ADM |= ADCStart;
53 }

```

```
54
55 void ADCInterrupt(void) interrupt ISRAdc
56 {
57     if ((IRCON2 & 0x04) == 0x04) {
58         P17 = ~P17;
59         IRCON2 &= 0xFB; //Clear ADCF
60         ADCBuffer = (ADB << 4) + (ADR & 0x0F);
61         ADM &= ClearEOC;
62         ADM |= ADCStart;
63     }
64 }
```


20 UART

The UART provides a flexible full-duplex synchronous/asynchronous receiver/transmitter. The serial interface provides an up to 0.25MHz flexible full-duplex transmission. It can operate in four modes (one synchronous and three asynchronous). Mode0 is a shift register mode and operates as synchronous transmitter/receiver. In Mode1-Mode3 the UART operates as asynchronous transmitter/receiver with 8-bit or 9-bit data. The transfer format has start bit, 8-bit/ 9-bit data and stop bit. Transmission is started by writing to the SOBUF register. After reception, input data are available after completion of the reception in the SOBUF register. TB80/RB80 bit can be used as the 9th bit for transmission and reception in 9-bit UART mode. Programmable baud rate supports different speed peripheral devices.

The UART features include the following:

- Full-duplex, 2-wire synchronous/asynchronous data transfer.
- Programmable baud rate.
- 8-bit shift register: operates as synchronous transmitter/receiver
- 8-bit / 9-bit UART: operates as asynchronous transmitter/receiver with 8 or 9-bit data bits and programmable baud rate.

20.1 UART Operation

The UART UTX and URX pins are shared with GPIO. In synchronous mode, the UTX/URX shared pins must set output high by software. In asynchronous mode (8-bit/9-bit UART), the UTX shared pins must set output high and URX set input high by software. Thus, URX/UTX pins will transfers to UART purpose. When UART disables, the UART pins returns to GPIO last status.

The UTX/URX pins also support open-drain structure. The open-drain option is controlled by PnOC bit. When PnOC=0, disable UTX/URX open-drain structure. When PnOC=1, enable UTX/URX open-drain structure. If enable open-drain structure, UTX/URX pin must set high level (IO mode control will be ignored) and need external pull-up resistor.

The UART supports interrupt function. ES0 is UART0 transfer interrupt function control bit. UART transmitter and receiver interrupt function is controlled by ES0. When ES0 = 0, disable transmitter/receiver interrupt function. When ES0 = 1, enable UART transmitter/ receiver interrupt function. UART transmitter and receiver interrupt function are share interrupt vector 0x0023. When UART interrupt function enable, the program counter points to interrupt vector to do UART interrupt service routine after UART operating. TI0/RI0 is UART0 interrupt request flag, and also to be the UART operating status indicator when interrupt is disabled. TI0 and RI0 must clear by software.

UART provides four operating mode (one synchronous and three asynchronous) controlled by SOCON register. These modes can be support in different baud rate and communication protocols.

SM0	SM1	Mode	Synchronization	Clock Rate	Start Bit	Data Bits	Stop Bit	UART pins' mode and data
0	0	0	Synchronous	Fcpu/12	X	8	X	UTX pin: P00M=1 and P00=1 URX pin: Transmitter: P01M=1 and P01=1 Receiver: P01M=0 and P01=1
0	1	1	Asynchronous	Baud rate generator or T1 overflow rate	1	8	1	UTX pin: P00M=1 and P00=1 URX pin: P01M=0
1	0	2	Asynchronous	Fcpu/64 or Fcpu/32	1	9	1	
1	1	3	Asynchronous	Baud rate generator or T1 overflow rate	1	9	1	

20.2 Mode 0: Synchronous 8-bit Receiver/Transmitter

Mode0 is a shift register mode. It operates as synchronous transmitter/receiver. The UTX pin output shift clock for both transmit and receive condition. The URX pin is used to transmit and receive data. 8-bit data will be transmit and receive with LSB first. The baud rate is fcpu/12. Data transmission is started by writing data to SOBUF register. In the end of the 8th bit transmission, the TIO flag is set. Data reception is controlled by REN0 bit and clearing RIO bits. When REN0=1 and RIO is from 1 to 0, data transmission starts and the RIO flag is set at the end of the 8th bit reception.

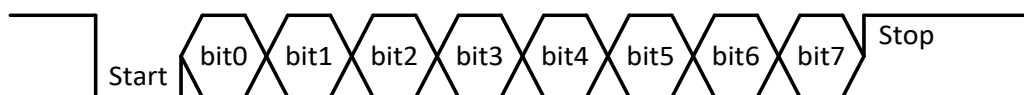
20.3 Mode 1: 8-bit Receiver/Transmitter with Variable Baud Rate

Mode1 supports an asynchronous 8-bit UART with variable baud rate. The transfer format includes 1 start bit, 8 data bits (LSB first) and 1 stop bit. Data is transmitted by UTX pin and received by URX pin. The baud rate clock source can be baud rate generator or T1 overflow controlled by BD bit. When BD=0, the baud rate clock source is from T1 overflow. When BD=1, the baud rate clock source is from baud rate generator controlled by SORELH and SORELL. Additionally, the baud rate can be doubled by SMOD bit.

Data transmission is controlled by REN0 bit. After transmission configuration, load transmitted data

into S0BUF, and then UART starts to transmit the packet. The TIO flag is set at the beginning of the stop bit.

Data reception is controlled by REN0 bit. When REN0=1, data reception function is enabled. Data reception starts by receiving the start bit for master terminal, URX detects the falling edge of start bit, and then the R10 flag is set in the middle of a stop bit. Until reception completion, input data is stored in S0BUF register and the stop bit is stored in RB80.



20.4 Mode 2: 9-bit Receiver/Transmitter with Fixed Baud Rate

Mode2 supports an asynchronous 9-bit UART with fixed baud rate. The transfer format includes 1 start bit, 9 data bits (LSB first) and 1 stop bit. Data is transmitted by UTX pin and received by URX pin. The baud rate clock source is fixed to $fcpu/64$ or $fcpu/32$ and is controlled by SMOD bit. When SMOD=0, baud rate is $fcpu/64$. When SMOD=1, baud rate is $fcpu/32$.

Data transmission is controlled by REN0 bit. After transmission configuration, load transmitted data into S0BUF, and then UART starts to transmit the packet. The 9th data bit is taken from TB80. The TIO flag is set at the beginning of the stop bit.

Data reception is controlled by REN0 bit. When REN0=1, data reception function is enabled. Data reception starts by receiving the start bit for master terminal, URX detects the falling edge of start bit, and then the R10 flag is set in the middle of a stop bit. Until reception completion, lower 8-bit input data is stored in S0BUF register and the 9th bit is stored in RB80.



20.5 Mode 3: 9-bit Receiver/Transmitter with Variable Baud Rate

Mode3 supports an asynchronous 9-bit UART with variable baud rate. The transfer format includes 1 start bit, 9 data bits (LSB first) and 1 stop bit. Data is transmitted by UTX pin and received by URX pin. The different between Mode2 and Mode3 is baud rate selection. In the Mode3, the baud rate clock source can be baud rate generator or T1 overflow controlled by BD bit. When BD=0, the baud rate clock source is from T1 overflow. When BD=1, the baud rate clock source is from baud rate generator controlled by SORELH and SORELL. Additionally, the baud rate can be doubled by SMOD bit.

Data transmission is controlled by REN0 bit. After transmission configuration, load transmitted data into S0BUF, and then UART starts to transmit the packet. The 9th data bit is taken from TB80. The TIO flag is set at the beginning of the stop bit.

Data reception is controlled by REN0 bit. When REN0=1, data reception function is enabled. Data reception starts by receiving the start bit for master terminal, URX0 detects the falling edge of start bit, and then the RIO flag is set in the middle of a stop bit. Until reception completion, lower 8-bit input data is stored in S0BUF register and the 9th bit is stored in RB80.



20.6 Multiprocessor Communication

UART supports multiprocessor communication between a master device and one or more slaver device in Mode2 and Mode3 (9-bit UART). The master identifies correct slavers by using the 9th data bit. When the communication starts, the master transmits a specific address byte with the 9th bit is set "1" to selected slavers, and then transmits a data byte with the 9th bit is set "0" in the following transmission.

Multiprocessor communication is controlled by SM20 bit. When SM20=0, disable multiprocessor communication. When SM20=1, enable multiprocessor communication. If SM20 is set, the UART reception interrupt is only generated when the 9th received bit is "1" (RB80). The slavers will compare received data with its own address data by software. If address byte is match, the slavers clear SM20 bit to enable interrupt function in the following data transmission. The slavers with unmatched address, their SM20 keep in "1" and will not generate interrupt in the following data transmission.

20.7 Baud Rate Control

The UART mode 0 has a fixed baud rate at $f_{cpu}/12$, and the mode 2 has two baud rate selection which is chosen by SMOD register: $f_{cpu}/64$ (SMOD = 0) and $f_{cpu}/32$ (SMOD = 1).

The baud rate of UART mode 1 and mode 3 is generated by either S0RELH/S0RELL registers (BD = 1) or Timer 1 overflow period (BD = 0). The SMOD bit doubles the frequency from the generator.

If the S0RELH/S0RELL is selected (BD = 1) in mode 1 and 3, the baud rate is generated as following equation.

$$\text{Baud Rate} = 2^{\text{SMOD}} \times \frac{f_{cpu}}{64 \times (1024 - \text{S0REL})} \text{bps}$$

Table 20-1 Recommended Setting for Common UART Baud Rates (fcpu = 8 MHz)

Baud Rate	SMOD	SORELH	SORELL	Accuracy
4800	0	0x03	0xE6	0.16 %
9600	0	0x03	0xF3	0.16 %
19200	1	0x03	0xF3	0.16 %
38400	1	0x03	0xF9	-6.99 %
56000	1	0x03	0xFB	-10.71 %
57600	1	0x03	0xFC	8.51 %
115200	1	0x03	0xFE	8.51 %
128000	1	0x03	0xFE	-2.34 %
250000	1	0x03	0xFF	0 %

If the Timer 1 overflow period is selected (BD = 0) in mode 1 and 3, the baud rate is generated as following equation. The Timer 1 must be in 8-bit auto-reload mode which can generate periodically overflow signals.

$$\text{Baud Rate} = 2^{\text{SMOD}} \times \frac{1}{32 \times \text{Timer 1 period}} \text{ bps}$$

Table 20-2 Recommended Setting T1 overflow period (T1 clock=32M) for Common UART Baud Rates (fcpu = 8 MHz)

Baud Rate	SMOD	Timer Period	TH1/TL1	Accuracy
4800	0	6.510 us	0x30	0.16 %
9600	1	6.510 us	0x30	0.16 %
19200	1	3.255 us	0x98	0.16 %
38400	1	1.628 us	0xCC	0.16 %
56000	1	1.116 us	0xDC	-0.80 %
57600	1	1.085 us	0xDD	-0.80 %
115200	1	0.543 us	0xEF	2.08 %
128000	1	0.488 us	0xF0	-2.40 %

*** Note:**

- 1. When baud rate generator source is T1 overflow rate, the max counter value is 0xFB. (Only supports 0x00~0xFB).**
- 2. When baud rate generator source is T1 overflow rate, the T1 overflow rate must be greater four times to system clock fcpu.**

20.8 UART Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SOCON	SM0	SM1	SM20	REN0	TB80	RB80	TIO	RI0
SOCON2	BD	-	-	-	-	-	-	-
S0BUF	S0BUF7	S0BUF6	S0BUF5	S0BUF4	S0BUF3	S0BUF2	S0BUF1	S0BUF0
PCON	SMOD	-	-	-	P2SEL	GF0	STOP	IDLE
SORELH	-	-	-	-	-	-	SOREL9	SOREL8
SORELL	SOREL7	SOREL6	SOREL5	SOREL4	SOREL3	SOREL2	SOREL1	RORELO
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
P1OC	PW3EN	PW2EN	PW1EN	P06OC	P05OC	P13OC	P12OC	P11OC
P0M	P07M	P06M	P05M	P04M	P03M	P02M	P01M	P00M
P0	P07	P06	P05	P04	P03	P02	P01	P00

SOCON Register (0x98)

Bit	Field	Type	Initial	Description
7..6	SM[0:1]	R/W	00	UART mode selection 00: Mode 0 01: Mode 1 10: Mode 2 11: Mode 3
5	SM20	R/W	0	Multiprocessor communication (mode 2, 3) 0: Disable 1: Enable
4	REN0	R/W	0	UART reception function 0: Disable 1: Enable
3	TB80	R/W	0	The 9 th bit transmission data (mode 2, 3)
2	RB80	R/W	0	The 9 th bit data from reception
1	TIO	R/W	0	UART interrupt flag of transmission
0	RI0	R/W	0	UART interrupt flag of reception

*** Note: TIO and RI0 are clear by software when interrupt is enabled.**

S0CON2 Register (0xD8)

Bit	Field	Type	Initial	Description
7	BD	R/W	0	Baud rate generators selection (mode 1, 3) 0: Timer 1 overflow period 1: Controlled by S0RELH, S0RELL registers
6..0	Reserved	R	0x00	

S0BUF Register (0x99)

Bit	Field	Type	Initial	Description
7..0	S0BUF	R/W	0x00	Action of writing data triggers UART communication (LSB first). Reception data is available to read by the end of packages.

PCON Register (0x87)

Bit	Field	Type	Initial	Description
7	SMOD	R/W	0	UART baud rate control In UART mode 0: Unused. In UART mode 1, 3: The baud rate is generated as the equation in section 20.7 (Baud Rate Control). In UART mode 2: 0: fcpu/64 1: fcpu/32
6..0				Refer to other chapter(s)

S0RELH/S0RELL Registers (S0RELH: 0xBA, S0RELL: 0xAA)

Bit	Field	Type	Initial	Description
15..10	Reserved	R	0x00	
9..0	S0REL[9:0]	R/W	0x00	S0RELH[1:0] & S0RELL[7:0]. UART Reload Register is used for UART baud rate generation.

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
4	ES0	R/W	0	Enable UART interrupt

Else	Refer to other chapter(s)
------	---------------------------

P1OC Register (0xE4)

Bit	Field	Type	Initial	Description
4	P06OC	R/W	0	0: Switch P0.6 (URX) to input mode (required) 1: Switch P0.6 (URX) to open-drain mode*
3	P05OC	R/W	0	0: Switch P0.5 (UTX) to push-pull mode 1: Switch P0.5 (UTX) to open-drain mode
Else	Refer to other chapter(s)			

* Setting P06OC as high causes URX cannot receive data.

P0M Register (0xF9)

Bit	Field	Type	Initial	Description
6	P06M	R/W	0	0: Set P0.6 (URX) as input mode (required) 1: Set P0.6 (URX) as output mode*
5	P05M	R/W	0	0: Set P0.5 (UTX) as input mode* 1: Set P0.5 (UTX) as output mode (required)
Else	Refer to other chapter(s)			

* The URX and UTX respectively require input and output mode selection to receive/transmit data appropriately.

P0 Register (0x80)

Bit	Field	Type	Initial	Description
6	P06	R/W	0	This bit is available to read at any time for monitoring the bus statue.
5	P05	R/W	0	0: Set P0.5 (UTX) always low* 1: Make P0.5 (UTX) can output UART data (required)
Else	Refer to other chapter(s)			

* Setting P05 initially high because UART block drive the shared pin low signal only.

20.9 Sample Code

The following sample code demonstrates how to perform UART mode 1 with interrupt.

```

1  #define SYSUartSM0    (0 << 6)
2  #define SYSUartSM1    (1 << 6)
3  #define SYSUartSM2    (2 << 6)
4  #define SYSUartSM3    (3 << 6)
5  #define SYSUartREN    (1 << 4)
6  #define SYSUartSMOD    (1 << 7)
7  #define SYSUartES0    (1 << 4)
8
9  Unsigned int uartTxBuf;
10 Unsigned int uartRxBuf;
11
12
13 void SYSUartInit(void)
14 {
15     // set UTX, URX pins' mode at here or at GPIO initialization
16     P05 = 1;
17     P0M = P0M | 0x20 & ~0x40;
18
19     // configure UART mode between SM0 and SM3, enable URX
20     S0CON = SYSUartSM1 | SYSUartREN;
21
22     // configure UART baud rate
23     PCON = SYSUartSMODE1;
24     S0CON2 = SYSUartBD1;
25     S0RELH = 0x03;
26     S0RELL = 0xFE;
27
28     // enable UART interrupt
29     IEN0 |= SYSUartES0;
30
31     // send first UTX data
32     S0BUF = uartTxBuf;
33 }
34
35 void SYSUartInterrupt(void) interrupt ISRUart
36 {
37     if (TI0 == 1) {
38         S0BUF = uartTxBuf;
39         TI0 = 0;
40     } else if (RI0 == 1) {
41         uartRxBuf = S0BUF;
42         RI0 = 0;
43     }
44 }

```

21 SPI

The SPI is a serial communication interface for data exchanging from one MCU to one MCU or other hardware peripherals. It is a simple 8-bit interface without a major definition of protocol, packet or control bits. The SPI transceiver includes three pins, clock (SCK), data input and data output (MISO/MOSI) to send data between master and slave terminals. An optional slave select pin (SSN) can be enabled by register in slave mode. The SPI interface builds in 4-mode which are the clock idle status and the clock phases.

- Full-duplex, 3-wire synchronous data transfer.
- Master (SCK is clock output) or Slave (SCK is clock input) operation.
- Seven SPI Master baud rates.
- Slave Clock rate up to $f_{cpu}/8$.
- 8-bit data transmitted MSB first, LSB last.
- Serial clock with programmable polarity and phase.
- Master Mode fault error flag with MCU interrupt capability.
- Write collision flag protection.

21.1 SPI Operation

The SPCON register can control SPI operating function, such as: transmit/receive, clock rate, data transfer direction, SPI clock idle status and clock control phase and enable this circuit. This SPI circuit will transmit or receive 8-bit data automatically by setting SPEN in SPCON register and write or read SPDAT register.

CPOL bit is designed to control SPI clock idle status. CPHA bit is designed to control the clock edge direction of data receive. CPOL and CPHA bits decide the SPI format. The SPI data transfer direction is MSB bit to LSB bit.

The SPI supports 4-mode format controlled by CPOL and CPHA bits. The edge direction is “Data Transfer Edge”. When setting rising edge that means to receive and transmit one bit data at SCK rising edge, and data transition is at SCK falling edge. When setting falling edge, that means to receive and transmit one bit data at SCK falling edge, and data transition is at SCK rising edge.

“CPHA” is the clock phase bit controls the phase of the clock on which data is sampled. When CPHA=1, the SCK first edge is for data transition, and receive and transmit data is at SCK 2nd edge. When CPHA=0, the 1st bit is fixed already, and the SCK first edge is to receive and transmit data. The SPI data transfer timing as following figure:

C P O L	C P H A	Diagrams	Description
0	1		SCK idle status = Low. The transfer first bit = MSB. SCK data transfer edge = Falling edge.
1	1		SCK idle status = High. The transfer first bit = MSB. SCK data transfer edge = Rising edge.
0	0		SCK idle status = Low. The transfer first bit = MSB. SCK data transfer edge = Rising edge.
1	0		SCK idle status = High. The transfer first bit = MSB. SCK data transfer edge = Falling edge.

The SPI supports interrupt function. ESPI is SPI interrupt function control bit. ESPI=0, disable SPI interrupt function. ESPI=1, enable SPI interrupt function. When SPI interrupt function enable, the program counter points to interrupt vector to do SPI interrupt service routine after SPI operating. SPIF is SPI interrupt request flag, and also to be the SPI operating status indicator when ESPI= 0, but cleared by reading the SPSTA, SPDAT registers.

SPI builds in chip selection function to implement SPI multi-device mode. One master communicating with several slave devices in SPI bus, and the chip selection decides the pointed device. The chip selection pin is SSN pin.

The SPI pins also support open-drain structure. The open-drain option is controlled by PnOC bits. When PnOC=0, disable SPI open-drain structure. When PnOC=1, enable SPI open-drain structure. If enable open-drain structure, SPI pins must be set input mode and need external pull-up resistor.

21.2 SPI Master

The SPI master mode has seven types of clock generator from $f_{cpu}/2$ to $f_{cpu}/128$. Generated clock is outputted through SCK pin (shared with P1.3) and its idle status is controlled by CPOL.

The phase of data input and output is automatically specified by CPHA register. In master mode MOSI pin (shared with P1.1) plays the role of data output, and MISO pin (shared with P1.2) fetches data from slave device. A SPI communication is started by writing SPDAT register; the received data from MISO is available to read after the end of data transmission.

The master mode has two status flags with interrupt function:

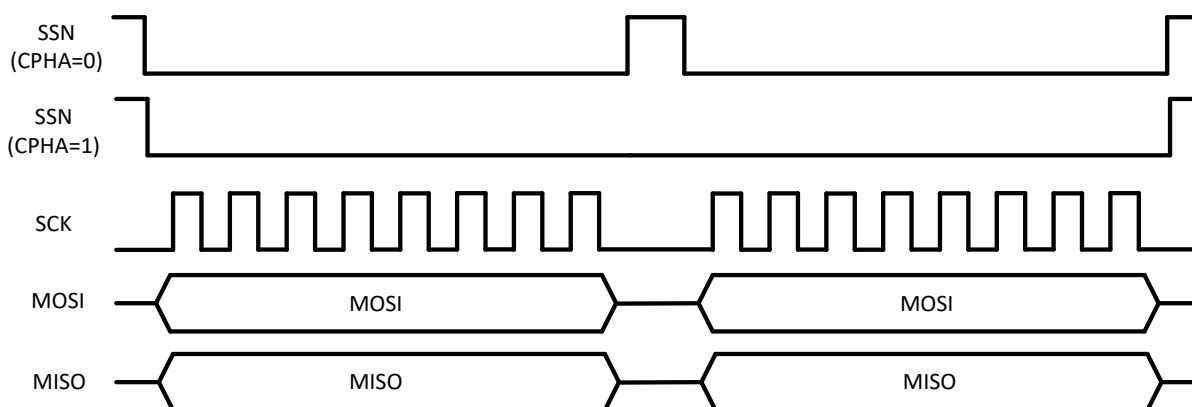
SPIF register indicates the end of one byte data communication. An interrupt would be issued at the same time if ESPI bit is enabled.

MODF is issued by SSN (shared with P0.7) low status while transmission. This interrupt source can be masked by setting SSDIS bit.

21.3 SPI Slave

The SPI slave mode monitors SCK pin to control its MISO and MOSI communication. However, the maximum clock rate is limited at $f_{cpu}/8$. Slave device(s) are expected to specify its CPOL and CPHA setting as the same configuration of the connected SPI bus.

The slave mode treats MOSI pin as its data input, and MISO pin as its data transmission. By default, the SSDIS register is low which means the slave select pin (SSN) is functional. A SPI communication would be processed if the SSN is low status. Thus, a slave device is suspended if its SSN is high status. But in CPHA = 0, Strictly SSN must follow each 8-bit data needs to be included with falling edge and rising edge, CPHA=1 is not limitation.



The slave mode has two status flags with interrupt function:

SPIF indicates the end of one byte data communication. The original SPDAT's value has been transmitted, and the received data from MOSI is ready to be read on SPDAT.

MODF indicates that the slave select pin (SSN) has turned high before a completion of one byte communication. In other word, the last time of SPI communication is broken.

21.4 SPI Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SPCON	SPR2	SPEN	SSDIS	MATR	CPOL	CPHA	SPR1	SPR0
SPSTA	SPIF	WCOL	SSERR	MODF	-	-	-	-
SPDAT	SPDAT7	SPDAT6	SPDAT5	SPDAT4	SPDAT3	SPDAT2	SPDAT1	SPDAT0
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
IEN1	ET2RL	-	ET2C3	ET2C2	ET2C1	ET2C0	ESPI	EI2C
P1OC	PW3EN	PW2EN	PW1EN	P06OC	P05OC	P13OC	P12OC	P11OC
P0M	P07M	P06M	P05M	P04M	P03M	P02M	P01M	P00M
P1M	P17M	P16M	P15M	P14M	P13M	P12M	P11M	P10M

SPCON Register (0xE2)

Bit	Field	Type	Initial	Description
7,1,0	SM[2:0]	R/W	000	SPI baud rate generator (master mode only) 000: fcpu/2 001: fcpu/4 010: fcpu/8 011: fcpu/16 100: fcpu/32 101: fcpu/64 110: fcpu/128 111: reserved
6	SPEN	R/W	0	SPI communication function 0: Disable 1: Enable
5	SSDIS	R/W	0	Slave select pin function (MSTR = 0, CPHA = 0 only) 0: Enable slave selection pin (SSN) function 1: Disable slave select pin (SSN) function
4	MSTR	R/W	1	SPI mode 0: Slave mode 1: Master mode
3	CPOL	R/W	0	SCK pin idle status 0: SCK idle low 1: SCK idle high
2	CPHA	R/W	1	Clock phase of data latch control 0: Data latched by the first of clock edge 1: Data latched by the second of clock edge

SPSTA Register (0xE1)

Bit	Field	Type	Initial	Description
7	SPIF	R	0	SPI complete communication flag Set automatically at the end of communication Cleared automatically by reading SPSTA, SPDAT registers
6	WCOL	R	0	Write collision flag Set automatically if write SPDAT during communication Cleared automatically by reading SPSTA, SPDAT registers
5	SSERR	R	0	Synchronous slave select pin error Set automatically if SSN error controlling Cleared automatically by clear SPEN
4	MODF	R	0	Mode fault flag
3..0	Reserved	R	0x00	

SPDAT Register (0xE3)

Bit	Field	Type	Initial	Description
7..0	SPDAT	R/W	0x00	Master mode: action of writing data triggers SPI communication; reception data is readable after the end of one byte communication (SPIF automatically set). Slave mode: written data would be transmitted by SCK input; reception data is available to read after the end of one byte communication (SPIF automatically set).

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
Else				Refer to other chapter(s)

IEN1 Register (0xB8)

Bit	Field	Type	Initial	Description
1	ESPI	R/W	0	Enable SPI interrupt
Else				Refer to other chapter(s)

P1OC Register (0xE4)

Bit	Field	Type	Initial	Description
2	P13OC	R/W	0	0: Switch P1.3 (SCK) to input or output mode 1: Switch P1.3 (SCK) to open-drain mod
1	P12OC	R/W	0	0: Switch P1.2 (MISO) to input or output mode 1: Switch P1.2 (MISO) to open-drain mod
0	P11OC	R/W	0	0: Switch P1.1 (MOSI) to input or output mode 1: Switch P1.1 (MOSI) to open-drain mode
Else				Refer to other chapter(s)

P0M Register (0xF9)

Bit	Field	Type	Initial	Description
7	P07M	R/W	0	0: Set P0.7 (SSN) as input mode* 1: Set P0.7 (SSN) as output mode*
Else				Refer to other chapter(s)

*If slave mode with SSN function: essentially to set SSN as input mode.

P1M Register

Bit	Field	Type	Initial	Description
3	P13M	R/W	0	0: Set P1.3 (SCK) as input mode ^{slave mode} 1: Set P1.3 (SCK) as output mode ^{master mode}
2	P12M	R/W	0	0: Set P1.2 (MISO) as input mode ^{master mode} 1: Set P1.2 (MISO) as output mode ^{slave mode}
1	P11M	R/W	0	0: Set P1.1 (MOSI) as input mode ^{slave mode} 1: Set P1.1 (MOSI) as output mode ^{master mode}
Else				Refer to other chapter(s)

¹Setting SCK as input mode is essential in slave mode; setting as output mode is recommended in master mode.

²Setting MISO as input mode is essential in master mode; setting as output mode is recommended in slave mode.

³Setting MOSI as input mode is essential in slave mode; setting as output mode is recommended in master mode.

21.5 Sample Code

The following sample code demonstrates how to perform SPI Master with interrupt.

```

1  #define SpiMaster      (1 << 4) //SPI = Master mode
2  #define SpiSlave      (1 << 4) //SPI = Slave mode
3  #define SpiMode0      (0 << 2) //SCK idle low, data latch at rising edge
4  #define SpiMode1      (1 << 2) //SCK idle low, data latch at falling edge
5  #define SpiMode2      (2 << 2) //SCK idle high, data latch at falling edge
6  #define SpiMode3      (3 << 2) //SCK idle high, data latch at rising edge
7  #define SpiEn         (1 << 6) //Enable SPI
8  #define SpiSSNEn      (0 << 5) //SSN pin function enable
9  #define SpiSSNDis     (1 << 5) //SSN pin function disable
10
11 unsigned char u8SpiData = 0; // data buffer
12 unsigned char u8TxCompleted = 0;
13
14 void SpiMaster(void)
15 {
16     unsigned char u8RcvData = 0;
17
18     //SCK & MOSI = output, MISO = input
19     P1M |= 0x0A;
20     //Enable Spi, Master mode, SSN pin disable, Fclk/128
21     //SCK idle low, data latch at falling edge
22     SPCON = SpiEn | SpiMaster | SpiMode1 | SpiSSNDis | 0x82;
23     //Enable Global/SPI interrupt
24     IEN1 |= 0x02;
25     IEN0 |= 0x80; //enable global interrupt
26
27     while (1) {
28         SPDAT = 0x55;
29         while(!u8TxCompleted); // wait end of transmittion
30         u8TxCompleted = 0; // clear sw flag
31         u8RcvData = u8SpiData; // receive 0x66
32
33         SPDAT = 0x99;
34         while(!u8TxCompleted); // wait end of transmittion
35         u8TxCompleted = 0; // clear sw flag
36         u8RcvData = u8SpiData; // receive 0xAA
37     }
38 }
39
40 void SpiInterrupt(void) interrupt ISRSpi //0x4B
41 {
42     switch ( SPSTA ) // Clear SPI flag (SPIF) by reading
43     {
44         case 0x80:
45             u8SpiData = SPDAT;
46             u8TxCompleted = 1;
47             break;
48         case 0x10:
49             // Mode Fault
50             break;
51     }
52 }

```

22 I2C

The I2C is a serial communication interface for data exchanging from one MCU to one MCU or other hardware peripherals. The device can transmit data as a master or a slave with two bi-directional IO, SDA (Serial data output) and SCL (Serial clock input).

When a master transmit data to a slave, it's called "WRITE" operation; when a slave transmit data to a master, it's called "READ" operation. It also supports multi-master communication and keeps data transmission correctly by an arbitration method to decide one master has the control on bus and transmit its data.

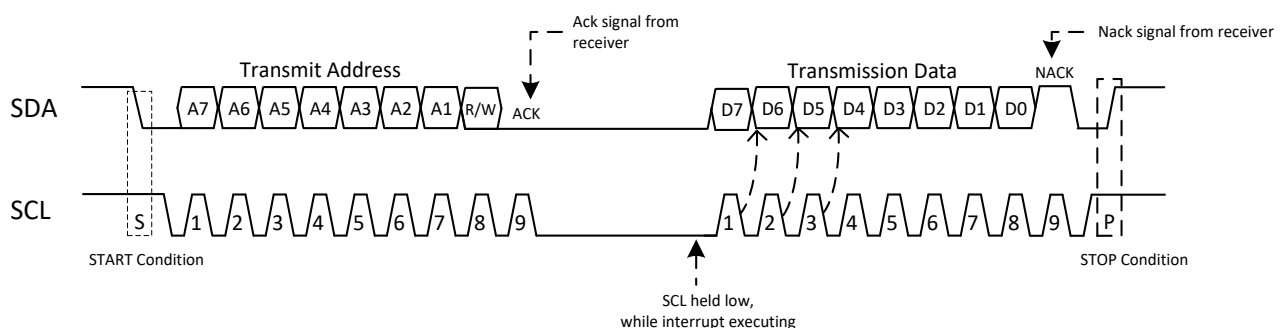
- Master Tx, Rx Mode
- Slave Tx, Rx mode (with general address call) for multiplex slave in single master situation.
- 2-wire synchronous data transfer/receiver.
- Support 100K/400K clock rate.

22.1 I2C Protocol

I2C transmission structure includes a START(S) condition, 8-bit address byte, one or more data byte and a STOP (P) condition. START condition is generated by master to initial any transmission.

Data is transmitted with the Most Significant Bit (MSB) first. In address byte, the higher 7-bit is address bit and the lowest bit is data direction (R/W) bit. When R/W=0, it assigns a "WRITE" operation. When R/W=1, it assigns a "READ" operation.

After each byte is received, the receiver (a master or a slave) must send an acknowledge (ACK). If transmitter can't receive an ACK, it will recognize a not acknowledge (NACK). In WRITE operation, the master will transmit data to the slave and then waits for ACK from slave. In READ operation, the slave will transmit data to the master and then waits for ACK from master. In the end, the master will generate a STOP condition to finish transmission.

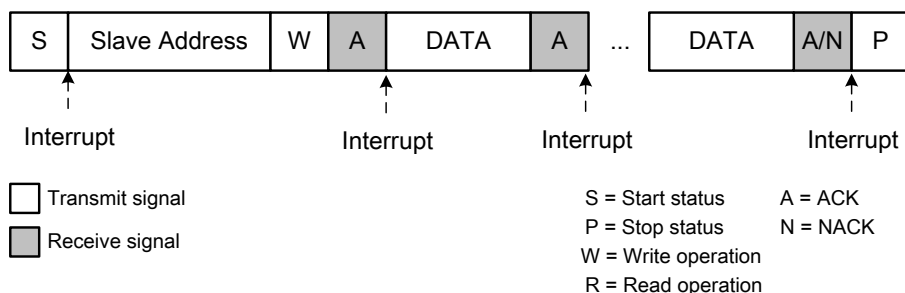


22.2 I2C Transfer Modes

The I2C can operate as a master/slave to execute the 8-bit serial data transmission/reception operation. Thus, the module can operate in one of four modes: Master Transmitter, Master Receiver, Slave Transmitter and Slave Receiver.

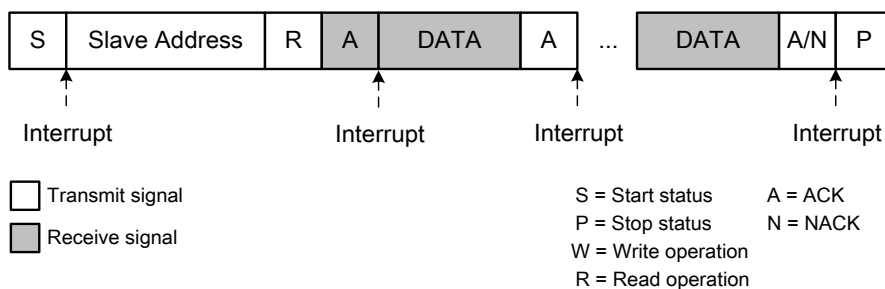
22.2.1 Master Transmitter Mode

The master transmits information to the slave. The serial data is output via SDA while the serial clock is output on SCL. Data transmission starts via generate a START(S) signal. After the START signal, the specific address byte of slave device is sent. The address byte includes 7-bit address bit and an 8th data direction (R/W) bit. The R/W is set "0" to enable the master transmission. In the following, the master transmits one or more data byte to the slaver. After each data is transmitted, the master waits for the acknowledge (ACK) from the slave. In the end, the master generates a STOP (P) signal to terminate the data transmission.



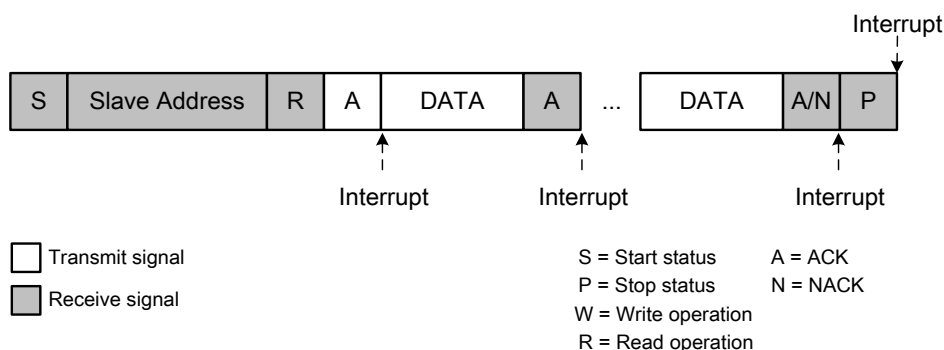
22.2.2 Master Receiver Mode

The master receives the information from the slave. The serial data input via SDA while the serial clock output on SCL. Data reception starts via generate a START(S) signal. After the START signal, the specific address byte of slave device is sent. The address byte includes 7-bit address bit and an 8th data direction (R/W) bit. The R/W is set "1" to enable the master reception. In the following, the master receives one or more data byte from the slaver. After each data is received, the master generates the acknowledge (ACK) or not acknowledge (NACK) to the slave via the status of AA bit. In the end, the master generates a STOP (P) signal to terminate the data transmission.



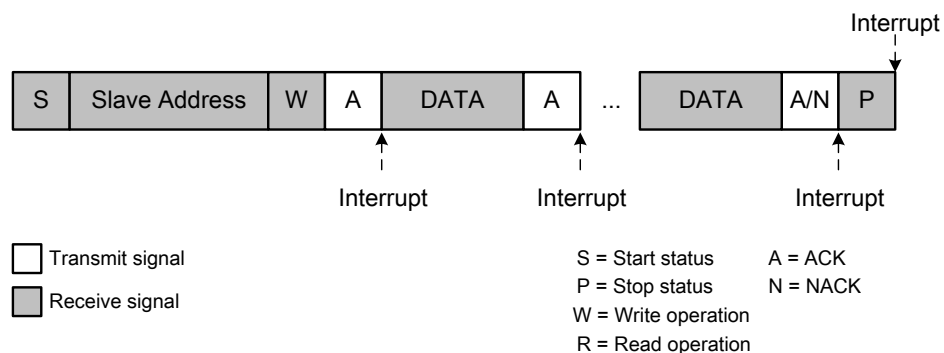
22.2.3 Slave Transmitter Mode

The slave transmits information to the master. The serial data output via SDA while the serial clock input on SCL. Data transmission starts via receive a START(S) signal from the master. After the START signal, the specific address byte of slave device is received. The address byte includes 7-bit address bit and an 8th data direction (R/W) bit. The R/W is set "1" to enable the slave transmission. If the received address byte match the address in I2CADDR register, the slave generate an acknowledge (ACK). Otherwise, if general call address condition is set (GC=1), the slave also generate an acknowledge (ACK) after general call address (0x00) is received. In the following, the slave transmits one or more data byte to the master. After each data is transmitted, the slave waits for the acknowledge (ACK) from the master. In the end, the slave receives a STOP (P) signal from the master to terminate the data transmission.



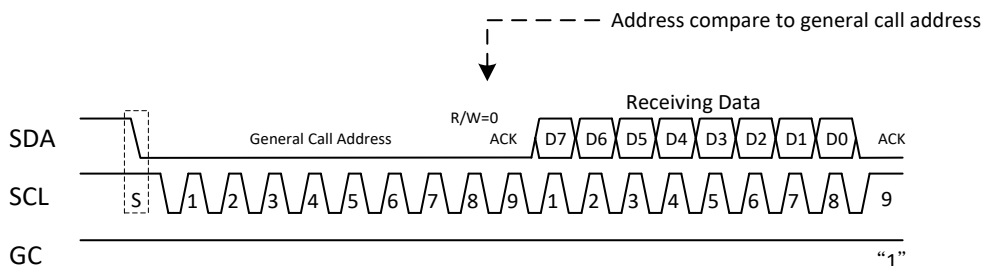
22.2.4 Slave Receiver Mode

The slave receives information from the master. Both the serial data and the serial clock are input on SDA and SCL. Data reception starts via receive a START(S) signal from the master. After the START signal, the specific address byte of slave device is received. The address byte includes 7-bit address bit and an 8th data direction (R/W) bit. The R/W is set "0" to enable the slave reception. If the received address byte match the address in I2CADDR register, the slave generate an acknowledge (ACK). Otherwise, if general call address condition is set (GC=1), the slave also generate an acknowledge (ACK) after general call address (0x00) is received. In the following, the slave receives one or more data byte from the master. After each data is receives, the slave generates the acknowledge (ACK) or not acknowledge (NACK) to the master via the status of AA bit. In the end, the slave receives a STOP (P) signal from the master to terminate the data transmission.



22.3 General Call Address

In I2C bus, the first 7-bit is the slave address. Only the address matches slave address, the slave will response an ACK. The exception is the general call address which can address all slave devices. When this address occur, all devices should response an acknowledge (ACK). The general call address is a special address which is reserved as all "0" of 7-bit address. The general call address function is control by GC bit. Set this bit will enable general call address and clear it will disable. When GC=1, the general call address will be recognized. When GC=0, the general call address will be ignored.



22.4 Serial Clock Generator

In master mode, the SCL clock rate generator's is controlled by CR[2:0] bit of I2CCON register.

When CR[2:0]=000~110, SCL clock rate is from internal clock generator.

$$\text{SCL Clock Rate} = \frac{F_{\text{cpu}}}{\text{Prescaler}} (\text{Prescaler} = 256 \sim 60)$$

When CR[2:0]=111, SCL clock rate is from Timer 1 overflow rate .

$$\text{SCL Clock Rate} = \frac{\text{Timer 1 Overflow}}{8}$$

The table below shows the clock rate under different setting.

CR2	CR1	CR0	I2C	Bit Frequency (kHz)	
			Prescaler	6MHz	8MHz
0	0	0	256	23	31
0	0	1	224	27	36
0	1	0	192	31	42
0	1	1	160	37	50
1	0	0	960	6.25	8
1	0	1	120	50	67
1	1	0	60	100	133
1	1	1	(Timer 1 overflow rate)/8		

*

Note:

- 1. The first step of I2C operation is to setup the I2C pins' mode. Must be set "input mode" in SDA/SCL pins.**
- 2. When clock generator source is T1 overflow rate, the max counter value is 0xFB. (Only supports 0x00~0xFB). And in this time if T1 clock rate is IHRC_32MHz, SCL maximum clock rate is 800kHz.**
- 3. If user wants to generate SCL clock rate is 100kHz/400kHz, you can set T1 counter value is 0xD8/0xF6 easily.**

22.5 Synchronization and Arbitration

In multi-master condition, more than one master may transmit on bus in the same time. It must be decided which master has the control of bus and complete its transmission. Clock synchronization and arbitration are used to configure multi-master transmission. Clock synchronization is executed by synchronizing the SCL signal with another devices.

When two masters want to transmit data in the same, the clock synchronization will start by the High to Low transition on the SCL. If master 1 clock set LOW first, it holds the SCL in LOW status until the clock transit to HIGH status. However, if another master clock still keep LOW status, the Low to High transition of master 1 may not change SCL status (SCL keep LOW). In the other word, SCL keep LOW by the master with the longest clock time in LOW status. The SCL will transit from LOW to HIGH when the all devices clock transit to HIGH status. In the duration, the master1 will keep in HIGH status and wait for SCL transition (from LOW to HIGH), then continue its transmission. After clock synchronization, all devices clock and SCL clock are the same. Arbitration is used to decide which master can complete its transmission by SDA signal. Two masters may send out a START condition and transmit data on bus in the same time. They may influence by each other. Arbitration will force one master to lose the control on bus. Data transmission will keep until master output different data signal. If one master transmits HIGH status and another master transmits LOW status, the SDA will be pull low. The master output High will detect the different with SDA and lose the control on bus. The master with LOW status wins the bus control and continues its transmission. There is no data miss during arbitration.

22.6 System Management Bus Extension

The optional System Management Bus (SMBus) protocol hardware supports 3 types timeout detection: (1) Tmext Timeout Detection: The cumulative stretch clock cycles within one byte. (2)Tsext Timeout Detection: The cumulative stretch clock cycles between start and stop condition. (3)Timeout Detection: The clock low measurement.

Timeout detection is controlled by SMBSEL and SMBDST registers. The SMBEXE bit of SMBSEL is SMBus extension function enable bit. When SMBEXE=1, SMBus extension function is enabled. Otherwise, Disable SMBus extension function. Timeout type and period setting is controlled by SMBTOP[2:0] and SMBDST. The period of SMBus timeout is controlled by three 16-bit buffers of Tmext, Tsext and Tout. The equation is as following.

$$T_{mext}/T_{sext}/T_{out} = \frac{\text{Timeout Period(sec)} \times F_{cpu}(\text{Hz})}{1024}$$

Tmext is support by two 8-bit register of Tmext_L and Tmext_H . Tmext_L hold the low byte and Tmext_H hold high byte. Tsext is support by two 8-bit register of Tsext_L and Tsext_H . Tsext_L hold the low byte and Tsext_H hold high byte. Tout is support by two 8-bit register of Tout_L and Tout_H . Tout_L hold the low byte and Tout_H hold high byte.

Type	Time out period	Fcpu=8MHz	
		DEC	HEX
Tmext	5ms	39	27
Tsext	25ms	195	C3
Tout	35ms	273	111

By the setting of SMBTOP[2:0] to choose register type (as the table below), and write to register by write data to SMBDST register.

SMBTOP[2:0]	SMBDST	Description
000	Tmext_L	Select the low byte of Tmext register.
001	Tmext_H	Select the high byte of Tmext register.
010	Tsext_L	Select the low byte of Tsext register.
011	Tsext_H	Select the high byte of Tsext register.
100	Tout_L	Select the low byte of Tout register.
101	Tout_H	Select the high byte of Tout register.

When the SMBus extension function is enabled the lower 3-bit of I2CSTA hold the information about time out as the table below.

I2CSTA	Description
XXXX X000	No timeout errors.
XXXX XXX1	Tout timeout error.
XXXX XX1X	Tsxt timeout error.
XXXX X1XX	Tmext timeout error.

22.7 I2C Registers

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
I2CDAT	I2CDAT7	I2CDAT6	I2CDAT5	I2CDAT4	I2CDAT3	I2CDAT2	I2CDAT1	I2CDAT0
I2CADR	ADR6	ADR5	ADR4	ADR3	ADR2	ADR1	ADR0	GC
I2CCON	CR2	ENS1	STA	STO	SI	AA	CR1	CR0
I2CSTA	I2CSTA7	I2CSTA6	I2CSTA5	I2CSTA4	I2CSTA3	I2CSTA2	I2CSTA1	I2CSTA0
SMBSEL	SMBEXE	-	-	-	-	SMBTOP2	SMBTOP1	SMBTOP0
SMBDST	SMBD7	SMBD6	SMBD5	SMBD4	SMBD3	SMBD2	SMBD1	SMBD0
IEN0	EAL	-	ET2	ES0	ET1	EX1	ET0	EX0
IEN1	ET2RL	-	ET2C3	ET2C2	ET2C1	ET2C0	ESPI	EI2C
P1M	P17M	P16M	P15M	P14M	P13M	P12M	P11M	P10M
P1	P17	P16	P15	P14	P13	P12	P11	P10

I2CDAT Register (0xDA)

Bit	Field	Type	Initial	Description
7:0	I2CDAT[7:0]	R/W	0x00	The I2CDAT register contains a byte to be transmitted through I2C bus or a byte which has just been received through I2C bus. The CPU can read from and write to this 8-bit, directly addressable SFR while it is not in the process of byte shifting. The I2CDAT register is not shadowed or double buffered so the user should only read I2CDAT when an I2C interrupt occurs.

I2CADR Register (0xDB)

Bit	Field	Type	Initial	Description
7:1	I2CADR[6:0]	R/W	0x00	I2C slave address
0	GC	R/W	0	General call address (0X00) acknowledgment 0: ignored 1: recognized

I2CCON Register (0xDC)

Bit	Field	Type	Initial	Description
7,1,0	CR[2:0]	R/W	0	I2C clock rate 000: fcpu/256 001: fcpu/224 010: fcpu/192 011: fcpu/160 100: fcpu/960 101: fcpu/120 110: fcpu/60 111: Timer 1 overflow-period/8
6	ENS1	R/W	0	I2C functionality 0: Disable 1: Enable
5	STA	R/W	0	START flag 0: No START condition is transmitted. 1: A START condition is transmitted if the bus is free.
4	STO	R/W	0	STOP flag 0: No STOP condition is transmitted. 1: A STOP condition is transmitted to the I2C bus in master mode.
3	SI	R/W	0	Serial interrupt flag The SI is set by hardware when one of 25 out of 26 possible I2C states is entered. The only state that does not set the SI is state F8h, which indicates that no relevant state information is available. The SI flag must be cleared by software. In order to clear the SI bit, '0' must be written to this bit. Writing a '1' to SI bit does not change value of the SI.
2	AA	R/W	0	Assert acknowledge flag 0: A NACK will be returned when a byte has received 1: An ACK will be returned when a byte has received

I2CSTA Register (0xDD)

Bit	Field	Type	Initial	Description
7:3	I2CSTA[7:3]	R	11111	I2C Status Code
2..0	I2CSTA[2:0]	R	000	SMBus Status Code

I2C status code and status

Mode	Status Code	Status of the I2C	Application software response					Next action taken by I2C hardware	
			To/from I2CDAT	TO I2CCON					
				STA	STO	SI	AA		
Master Transmitter/ Receiver	08H	A START condition has been transmitted	Load SLA+R/W	X	0	0	X	SLA+R/W will be transmitted; ACK will be received	
	10H	A repeated START condition has been transmitted.	Load SLA+R/W	X	0	0	X	SLA+R/W will be transmitted; ACK will be received	
			Load SLA+R/W					SLA+W will be transmitted; I2C will be switched to MST/TRX mode.	
Master Transmitter	18H	SLA+W has been transmitted; ACK has been received	Load data byte	0	0	0	X	Data byte will be transmitted; ACK will be received.	
			No action	1	0	0	X	Repeated START will be transmitted.	
			No action	0	1	0	X	STOP condition will be transmitted; STO flag will be reset.	
			No action	1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset.	
	20H	SLA+W has been transmitted; not ACK has been received	Load data byte*	0	0	0	X	Data byte will be transmitted; ACK will be received.	
			No action	1	0	0	X	Repeated START will be transmitted.	
			No action	0	1	0	X	STOP condition will be transmitted; STO flag will be reset.	
			No action	1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset.	
	28H	Data byte in I2CDAT has been transmitted; ACK has been received	Load data byte	0	0	0	X	Data byte will be transmitted; ACK bit will be received.	
			No action	1	0	0	X	Repeated START will be transmitted.	
			No action	0	1	0	X	STOP condition will be transmitted; STO flag will be reset.	
			No action	1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset.	
	30H	Data byte in I2CDAT has been transmitted; not ACK has been received	Load data byte*	0	0	0	X	Data byte will be transmitted; ACK will be received.	
			No action	1	0	0	X	Repeated START will be transmitted.	
			No action	0	1	0	X	STOP condition will be transmitted; STO flag will be reset.	
			No action	1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset.	
	Master Receiver	40H	SLA+R has been transmitted; ACK has been received	No action	0	0	0	0	Data byte will be received; not ACK will be returned
				No action	0	0	0	1	Data byte will be received; ACK will be returned
		48H	SLA+R has been transmitted; not ACK has been received	No action	1	0	0	X	Repeated START condition will be transmitted
				No action	0	1	0	X	STOP condition will be transmitted; STO flag will be reset
No action				1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset	
50H		Data byte has been received; ACK has been returned	Read data byte	0	0	0	0	Data byte will be received; not ACK will be returned	
			Read data byte	0	0	0	1	Data byte will be received; ACK will be returned	
58H		Data byte has been received; not ACK has been returned	Read data byte	1	0	0	X	Repeated START condition will be transmitted	
			Read data byte	0	1	0	X	STOP condition will be transmitted; STO flag will be reset	
			Read data byte	1	1	0	X	STOP condition followed by a START condition will be transmitted; STO flag will be reset	

Mode	Status Code	Status of the I2C	Application software response					Next action taken by I2C hardware
			To/from I2CDAT	TO I2CCON				
				STA	STO	SI	AA	
Slave Receiver	60H	Own SLA+W has been received; ACK has been returned	No action	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	68H	Arbitration lost in SLA+R/W as master; own SLA+W has been received, ACK returned	No action	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	70H	General call address (00H) has been received; ACK has been returned	No action	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	78H	Arbitration lost in SLA+R/W as master; general call address has been received, ACK returned	No action	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	80H	Previously addressed with own SLV address; DATA has been received; ACK returned	Read data byte	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	88H	Previously addressed with own SLA; DATA byte has been received; not ACK returned	Read data byte	0	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address
			Read data byte	0	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized
			Read data byte	1	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address; START condition will be transmitted when the bus becomes free

Slave Transmitter			Read data byte	1	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized; START condition will be transmitted when the bus becomes free
	90H	Previously addressed with general call address; DATA has been received; ACK returned	Read data byte	X	0	0	0/1	Data byte will be received and not ACK/ACK will be returned
	98H	Previously addressed with general call address; DATA has been received; not ACK returned	Read data byte	0	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address
			Read data byte	0	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized
			Read data byte	1	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address; START condition will be transmitted when the bus becomes free
			Read data byte	1	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized; START condition will be transmitted when the bus becomes free
	A0H	A STOP condition or repeated START condition has been received while still addressed as SLV/REC or SLV/TRX	No action	0	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address
			No action	0	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized
			No action	1	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address; START condition will be transmitted when the bus becomes free
			No action	1	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized; START condition will be transmitted when the bus becomes free
	A8H	Own SLA+R has been received; ACK has been returned	Load data byte	X	0	0	0	Last data byte will be transmitted and ACK will be received
			Load data byte	X	0	0	1	Data byte will be transmitted; ACK will be received.
	B0H	Arbitration lost in SLA+R/W as master; own SLA+R has been received, ACK has been returned.	Load data byte	X	0	0	0	Last data byte will be transmitted and ACK will be received
			Load data byte	X	0	0	1	Data byte will be transmitted; ACK will be received.
	B8H	Data byte has been transmitted; ACK will be received.	Load data byte	X	0	0	0	Last data byte will be transmitted and ACK will be received
			Load data byte	X	0	0	1	Data byte will be transmitted; ACK will be received.
	C0H	Data byte has been transmitted; not ACK has been received.	No action	0	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address.
			No action	0	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized.
			No action	1	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address; START condition will be transmitted when the bus becomes free.
			No action	1	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized; START condition will be transmitted when the bus becomes free.
	C8H	Last data byte has been transmitted; ACK has been received.	No action	0	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address.
			No action	0	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized.
			No action	1	0	0	0	Switched to not addressed SLV mode; no recognition of own SLA or general call address; START condition will be transmitted when the bus becomes free.
			No action	1	0	0	1	Switched to not addressed SLV mode; own SLA or general call address will be recognized; START condition will be transmitted when the bus becomes free.
Miscellaneous	F8H	No relevant state information available; SI=0	No action	No action				Wait or proceed current transfer
	38H	Arbitration lost	No action	0	0	0	X	I2C will be released; A start condition will be transmitted.
			No action	1	0	0	X	When the bus becomes free. (enter to a master mode)
	00H	Bus error during MST or selected slave modes	No action	0	1	0	X	Only the internal hardware is affected in the MST or addressed SLV modes. In all cases, the bus is released and I2C is switched to the not addressed SLV mode. STO flag is reset.

“SLA” means slave address, “R” means R/W=1, “W” means R/W=0

*For applications where NACK doesn't mean the end of communication.

SMBSEL Register (0xDE)

Bit	Field	Type	Initial	Description
7	SMBEXE	R/W	0	SMBus extension functionality 0: Disable 1: Enable
2..0	SMBTOP[2:0]	R/W	000	SMBus timeout register

SMBDST Register (0xDF)

Bit	Field	Type	Initial	Description
7..0	SMBD[7:0]	R/W	0x00	This register is used to provide a read/write access port to the SMBus timeout registers. Data read or written to that register is actually read or written to the Timeout Register which is pointed by the SMBSEL register.

IEN0 Register (0xA8)

Bit	Field	Type	Initial	Description
7	EAL	R/W	0	Interrupts enable. Refer to Chapter Interrupt
Else				Refer to other chapter(s)

IEN1 Register (0xB8)

Bit	Field	Type	Initial	Description
0	EI2C	R/W	0	Interrupts enable. Refer to Chapter Interrupt
Else				Refer to other chapter(s)

P1M Register (0xFA)

Bit	Field	Type	Initial	Description
5	P15M	R/W	0	0: Set P1.5 (SDA) as input mode (required) 1: Set P1.5 (SDA) as output mode*
4	P14M	R/W	0	0: Set P1.4 (SCL) as input mode (required) 1: Set P1.4 (SCL) as output mode*
Else				Refer to other chapter(s)

* The P15M and P14M respectively require be set input mode.

22.8 Sample Code

The following sample code demonstrates how to perform I2C with interrupt.

```

1 unsigned int  I2CAddr;
2 unsigned int  I2C_TXData0;
3 unsigned int  I2C_TXDatan;
4 unsigned int  I2C_RXData0;
5 unsigned int  I2C_RXDatan;
6
7 void I2CInit(void)
8 {
9     P1M &= 0xCF;        // P14 & P15 as input
10
11     // configure I2C clock(T1)and enable I2C.
12     I2CCON |= 0xC3;
13     TMOD = 0x60;        // auto reload
14     TCON0 = 0x07;        // Fosc/1
15     TH1 = 0xF6;         //400kHz
16     TL1 = 0xF6;         //400kHz or
17     TH1 = 0xD8;         //100kHz
18     TL1 = 0xD8;         //100kHz
19     TR1 = 1;
20
21     // enable I2C interrupt
22     EI2C = 1;
23     //enable global interrupt
24     EAL = 1;
25
26     I2CCON |= 0x20;        // START (STA) = 1
27 }
28 void I2cInterrupt(void) interrupt ISRI2c //0x43
29 {
30     switch (I2CSTA)
31     {
32         // tx mode
33         case 0x08:
34             I2CCON &= 0xDF;        // START (STA) = 0
35             I2CDAT = I2CAddr;        // Tx/Rx addr
36             break;
37         case 0x18:                // write first byte
38             I2CDAT = I2C_TXData0;
39             break;
40         case 0x28:                // write n byte
41             I2CDAT = I2C_TXDatan;
42             break;
43         case 0x30:                // STOP (STO)
44             I2CCON |= 0x10;
45             break;
46         // rx mode
47         case 0x40:                // get slave addr
48             I2CCON |= 0x04;        // AA = 1
49             break;
50         case 0x50:                // read n byte
51             I2C_RXData0 = I2CDAT;
52             I2CCON &= 0xFB;        // AA = 0
53             break;

```

```
54         case 0x58:                                // read last byte & stop
55             I2C_RXDatan = I2CDAT;
56             I2CCON |= 0x10;                        // STOP (STO)
57             break;
58         default:
59             I2CCON |= 0x10;                        // STOP (STO)
60     }
61
62     I2CCON &= 0xF7;                                // Clear I2C flag (SI)
63 }
```

23 In-System Program

SN8F5708 builds in an on-chip 16 KB program memory, aka IROM, which is equally divided to 512 pages (32 bytes per page). The in-system program is a procedure that enables a firmware to freely modify every page's data; in other word, it is the channel to store value(s) into the non-volatile memory and/or live update firmware.

0x3FFF	Page 511
0x3FE0	
0x3FDF	
0x3FC0	Page 510
	...
0x003F	Page 1
0x0020	
0x001F	
0x0000	Page 0

Program memory (IROM)

23.1 Page Program

Because each page of the program memory has 32 bytes in length, a page program procedure requires 32 bytes IRAM as its data buffer.

ISP ROM MAP		ROM address bit0~bit4 (hex) =0
ROM address bit5~bit15 (hex)	0000	These pages include reset vector and interrupt sector. We strongly recommend to reserve the area not to do ISP erase.
	0020	
	0040	
	...	
	00C0	
	00E0	
	0100	One ISP Program Page
	0120	One ISP Program Page
	...	One ISP Program Page
	1000	One ISP Program Page
	1020	One ISP Program Page
	...	One ISP Program Page
	3700	One ISP Program Page
	3720	One ISP Program Page
	...	One ISP Program Page
	3FE0	This page includes ROM reserved area. We strongly recommend to reserve the area not to do ISP erase.

These configurations must be setup completely before starting Page Program. ISP is configured using the following steps:

1. Save program data into IRAM. The data continues for 32 bytes.
2. Set the start address of the content location to PERAM.
3. Set the start address of the anticipated update area to PEROM [15:5]. (By PEROMH/PRROML registers)
4. Write '0xA5A' into PECMD [11:0] to trigger ISP function. Before writing '0x5A' into PECMD[7:0], PECMD[11:8] must be written '0xA'.
5. Write 'NOP' instruction twice.

As an example, assume the 510th page of program memory (IROM, 0x3FC0 – 0x3FDF) is the anticipated update area; the content is already stored in IRAM address 0x60 – 0x7F. To perform the in-system program, simply write starting IROM address 0x3FC0 to EPROMH/EPROML registers, and then specify buffer starting address 0x60 to EPRAM register. Subsequently, write '0xA5A' into PECMD [11:0] registers to duplicate the buffer's data to 510th page of IROM.

In general, every page has the capability to be modified by in-system program procedure. However, since the first and least pages (page 0 and 511) respectively stores reset vector and information for power-on controller, incorrectly perform page program (such as turn off power while programming) may cause faulty power-on sequence / reset.

* **Note:**

- 1. Watch dog timer should be clear before the Flash write (program) operation, or watchdog timer would overflow and reset system during ISP operating.**
- 2. Don't execute ISP flash ROM program operation for the first page and the last page, or affect program operation.**
- 3. ISP operation (page program) actually perform Flash erase and program procedures in the background. Don't execute ISP flash ROM program operation in low-voltage condition (ex. VDD < 2.5V), or ISP operation maybe not complete before power-off.**

23.2 In-system Program Register

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PERAM	PERAM7	PERAM6	PERAM5	PERAM4	PERAM3	PERAM2	PERAM1	PERAM0
PEROMH	PEROM15	PEROM14	PEROM13	PEROM12	PEROM11	PEROM10	PEROM9	PEROM8
PEROML	PEROM7	PEROM6	PEROM5	-	PECMD11	PECMD10	PECMD9	PECMD8
PECMD	PECMD7	PECMD6	PECMD5	PECMD4	PECMD3	PECMD2	PECMD1	PECMD0

PERAM Register (0x97)

Bit	Field	Type	Initial	Description
7..0	PERAM[7:0]	R/W	0x00	The first address of data buffer (IRAM)

PEROMH Register (0x96)

Bit	Field	Type	Initial	Description
7..0	PEROM[15:8]	R/W	0x00	The first address (15 th – 8 th bit) of program page (IROM)

PEROML Register (0x95)

Bit	Field	Type	Initial	Description
7..5	PEROM[7:5]	R/W	000	The first address (7 th – 5 th) of program page (IROM)
4	Reserved	R	0	
3..0	PECMD[11:8]	W	-	0xA: Enable in-system program Else values: Disable in-system program*

* Disabling in-system program can avoid mistakenly trigger ISP function.

PECMD Register (0x94)

Bit	Field	Type	Initial	Description
7..0	PECMD[7:0]	W	-	0x5A: Start page program procedure ^{*(1)} Else values: Reserved ^{*(2)}

*(1) Before writing '0x5A' into PECMD[7:0], PECMD[11:8] must be written '0xA'.

*(2) Not permitted to write any other to PECMD register.

23.3 Sample Code

```
1 unsigned cahr idata dataBuffer[32] _at_ 0xE0; // IRAM 0xE0 to 0xFF
2
3 void SYSIspSetDataBuffer(unsigned char address, unsigned char data)
4 {
5     dataBuffer[address & 0x1F] = data;
6 }
7
8 void SYSIspStart(unsigned int pageAddress)
9 {
10     ISP(pageAddress, 0xE0);
11 }
```

24 Electrical Characteristics

24.1 Absolute Maximum Ratings

Voltage applied at VDD to VSS	- 0.3V to 6.0V
Voltage applied at any pin to VSS.....	- 0.3V to VDD+0.3V
Operating ambient temperature.....	-40°C to 85°C
Storage ambient temperature	-40°C to 125°C

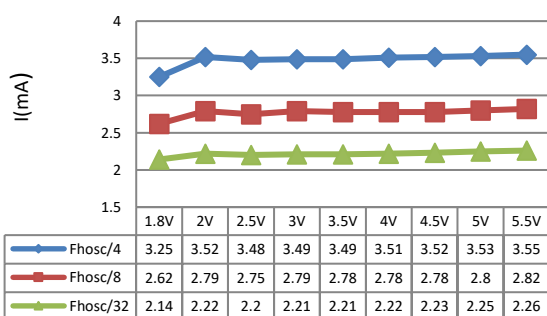
24.2 System Operation Characteristics

	Parameter	Test Condition	MIN	TYP	MAX	UNIT
VDD	Operating voltage	fcpu = 1MHz	1.8		5.5	V
V _{DR}	RAM data retention Voltage		1.5			V
V _{POR}	VDD rising rate*		0.05			V/ms
I _{DD1}	Normal mode supply current	VDD = 3V, fcpu = 0.25MHz		2.05		mA
		VDD = 5V, fcpu = 0.25MHz		2.06		mA
		VDD = 3V, fcpu = 1MHz		2.25		mA
		VDD = 5V, fcpu = 1MHz		2.30		mA
		VDD = 3V, fcpu = 8MHz		3.50		mA
		VDD = 5V, fcpu = 8MHz		3.55		mA
I _{DD2}	STOP mode supply current	VDD = 3V		3.5	8.5	μA
		VDD = 5V		4.0	9.0	μA
I _{DD3}	IDLE mode supply current (fcpu = 1MHz)	VDD = 3V, 32MHz IHRC		0.63		mA
		VDD = 5V, 32MHz IHRC		0.65		mA
		VDD = 3V, 16MHz Crystal		0.65		mA
		VDD = 5V, 16MHz Crystal		1.25		mA
		VDD = 3V, 4MHz Crystal		0.60		mA
		VDD = 5V, 4MHz Crystal		0.75		mA
F _{IHRC}	Internal high clock generator	VDD = 1.8V to 5.5V, 25°C	31.84	32	32.16	MHz
		VDD = 1.8V to 5.5V, 25°C to 85°C	31.68		31.99	MHz
		VDD = 1.8V to 5.5V, -40°C to 25°C	32.31		32.64	MHz
F _{ILRC}	Internal low clock generator	VDD = 5V, 25°C	12	16	24	kHz
V _{LVD18}	LVD18 detect voltage	25°C	1.6	1.7	1.8	V
		-40°C to 85°C	1.5	1.7	1.9	V
V _{LVD24}	LVD24 detect voltage	25°C	2.3	2.4	2.5	V

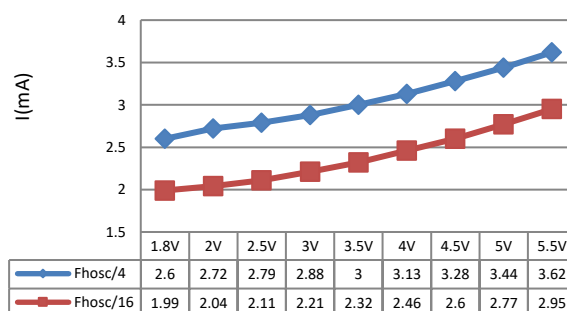
		-40°C to 85°C	2.2	2.4	2.6	V
V_{LVD33}	LVD33 detect voltage	25°C	3.2	3.3	3.4	V
		-40°C to 85°C	3.0	3.3	3.6	V
V_{ESD_HBM}	ESD human body mode		3000			V
V_{ESD_MM}	ESD machine mode		300			V

* Parameter(s) with star mark are non-verified design reference. Ambient temperature is 25°C.

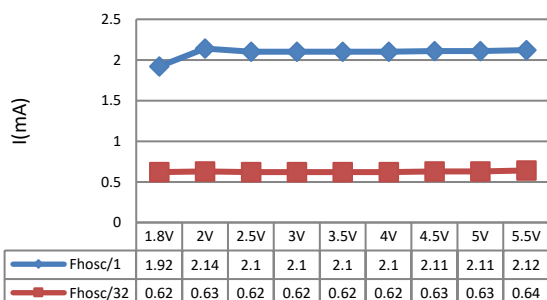
IHRC32M Operating Current



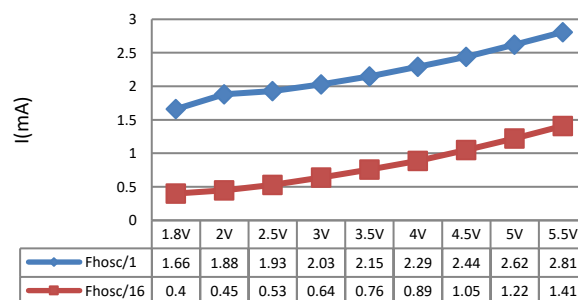
X'tal16M Operating Current



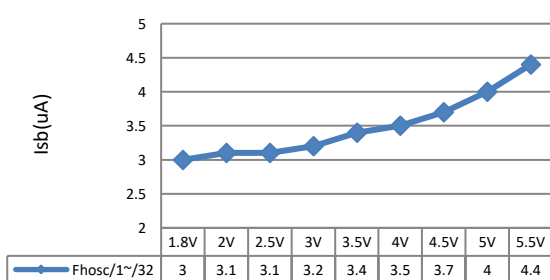
IHRC32M Green Current



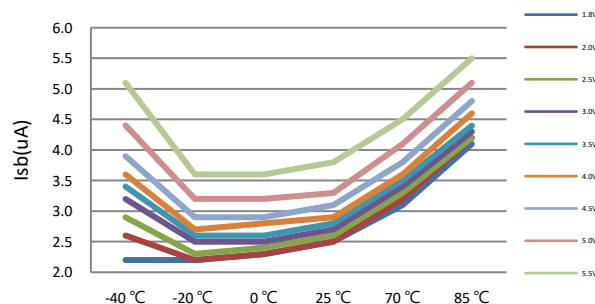
X'tal 16M GreenCurrent

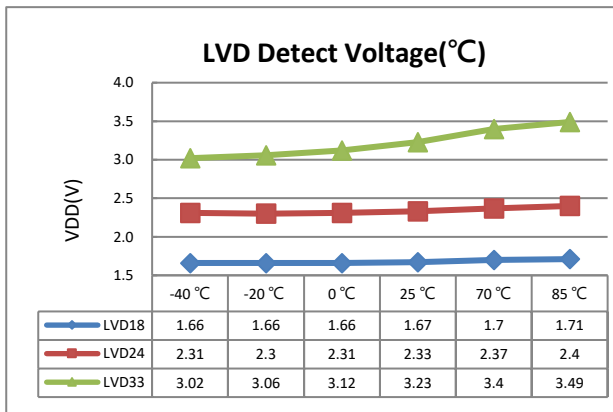


Standby Current



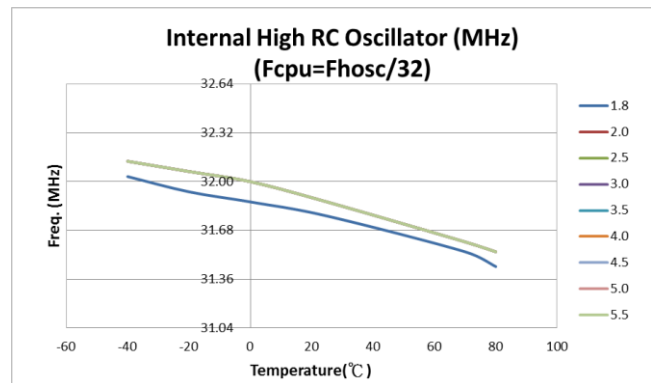
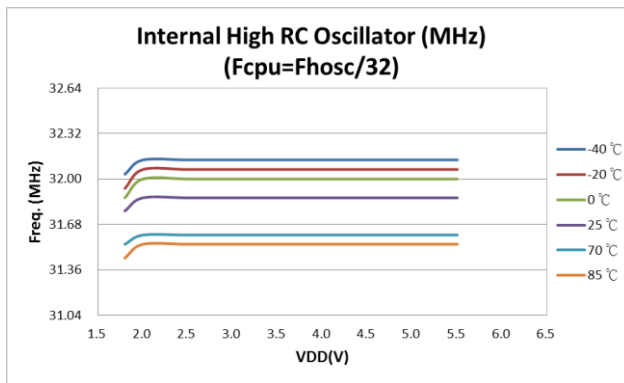
Standby Current





● IHRC Frequency - Temperature Graph

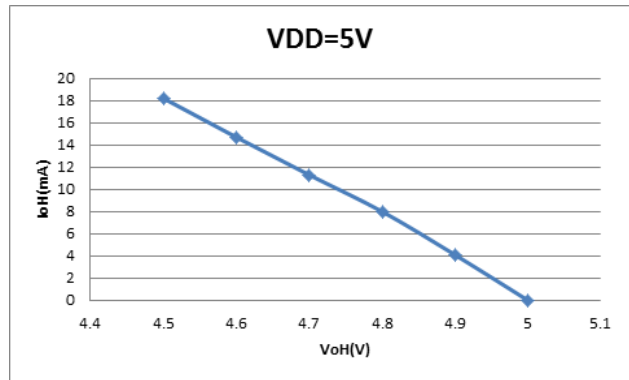
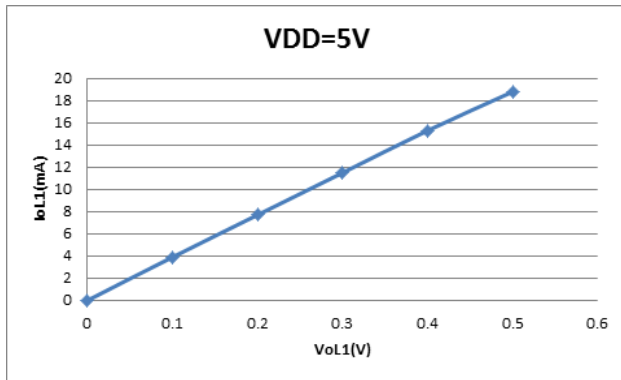
The IHRC Graphs are for design guidance, not tested or guaranteed. In some graphs, the data presented are outside specified operating range. This is for information only and devices are guaranteed to operate properly only within the specified range.



24.3 GPIO Characteristics

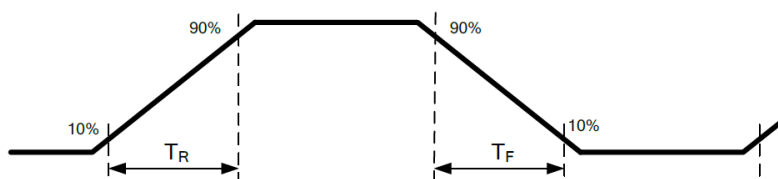
Parameter		Test Condition	MIN	TYP	MAX	UNIT
V _{IL}	Low-level input voltage		V _{SS}		0.3V _{DD}	V
V _{IH}	High-level input voltage		0.7V _{DD}		V _{DD}	V
I _{LEKG}	I/O port input leakage current	V _{IN} = V _{DD}			2	μA
R _{UP}	Pull-up resister	V _{DD} = 3V	100	200	300	kΩ
		V _{DD} = 5V	50	100	150	kΩ
I _{OH}	I/O output source current	V _{DD} = 5V, V _O = V _{DD} -0.5V	12	16		mA
I _{OL1}	I/O sink current (P14 – P17 , P2, P3, P4, P5)	V _{DD} = 5V, V _O = V _{SS} +0.5V	15	20		mA
I _{OL2}	I/O sink current (P0, P10 – P13)	V _{DD} = 5V, V _O = V _{SS} +1.5V	80	100		mA

* Ambient temperature is 25°C.



● I/O Slew Rate A.C. Electrical Characteristics

Typ. Value in Condition $C_L=8pF$		
VDD (V)	Rise Time, T_R (ns)	Fall Time, T_F (ns)
5.0	2.15	2.20
4.0	2.25	2.25
3.0	2.45	2.40
2.0	4.80	2.90
1.8	6.60	3.90



24.4 ADC Characteristics

Parameter		Test Condition	MIN	TYP	MAX	UNIT
V_{ADC}	Operating voltage		2.0		5.5	V
V_{AIN}	AIN0 channels input voltage	$VDD = 5V$	0		V_{REFH}	V
V_{REFH}	AVREFH pin input voltage	$VDD = 5V$	$V_{REFL}+2$		VDD	V
V_{REFL}	AVREFL pin input voltage	$VDD = 5V$	VSS		$V_{REFH}-2$	V
V_{EREF}	External reference voltage	$VDD = 5V$	2.0		VDD	V
V_{IREF}	Internal VDD reference voltage	$VDD = 5V$		VDD		V
	Internal 4V reference voltage	$VDD = 5V$	3.92	4	4.08	V
	Internal 3V reference voltage	$VDD = 5V$	2.94	3	3.06	V
	Internal 2V reference voltage	$VDD = 5V$	1.96	2	2.04	V
I_{AD}	ADC current consumption	$VDD = 3V$		0.65		mA
		$VDD = 5V$		0.70		mA
f_{ADCLK}	ADC clock	$VDD = 5V$			32	MHz

f_{ADSMP}	ADC sampling rate	$V_{\text{DD}} = 5\text{V}$			500	kHz
t_{ADEN}	ADC function enable period	$V_{\text{DD}} = 5\text{V}$	100			μs
DNL	Differential nonlinearity*	$f_{\text{ADSMP}} = 62.5\text{kHz}$		± 1		LSB
		$f_{\text{ADSMP}} = 250\text{kHz}$		± 1		LSB
		$f_{\text{ADSMP}} = 500\text{kHz}$		± 4.5		LSB
INL	Integral Nonlinearity*	$f_{\text{ADSMP}} = 62.5\text{kHz}$		± 2		LSB
		$f_{\text{ADSMP}} = 250\text{kHz}$		± 2.5		LSB
		$f_{\text{ADSMP}} = 500\text{kHz}$		± 4.5		LSB
NMC	No missing code*	$f_{\text{ADSMP}} = 62.5\text{kHz}$	10	11	12	Bit
		$f_{\text{ADSMP}} = 250\text{kHz}$		11		Bit
		$f_{\text{ADSMP}} = 500\text{kHz}$		9		Bit
V_{OFFSET}	Input offset voltage	Non-trimmed	-10	0	10	mV

* Parameters with star mark: $V_{\text{DD}} = 5\text{V}$, $V_{\text{REFH}} = 2.4\text{V}$, 25°C .

24.5 OPA Characteristics

Parameter	Test Condition	MIN	TYP	MAX	UNIT
V_{OPA}	Operating voltage	2.0		5.5	V
I_{OPA}	OPA current consumption	$V_{\text{DD}} = 3\text{V}$	90		μA
		$V_{\text{DD}} = 5\text{V}$	100		μA
V_{CM}	Common mode input range	$V_{\text{DD}} = 5\text{V}$	V_{SS}	V_{DD}	V
V_{OFFSET}	Input offset voltage	$V_{\text{DD}} = 5\text{V}$	0	10	mV
PSRR	Power supply Rejection Ratio*	$V_{\text{CM}} = V_{\text{SS}}$	50	70	dB
CMRR	Common mode Rejection Ratio*	$V_{\text{CM}} = -0.3\text{V to } 5\text{V}$, $V_{\text{DD}} = 5\text{V}$	50		dB
A_{OL}	Open loop gain*	$V_{\text{O}} = 0.2\text{V to } V_{\text{DD}} - 0.2\text{V}$, $V_{\text{CM}} = V_{\text{SS}}$	90		dB
V_{OS}	Output voltage swing	$V_{\text{OPP}} = 2.5\text{V}$	$V_{\text{SS}} + 15$	$V_{\text{DD}} - 15$	mV
I_{SC}	Output short current	$V_{\text{DD}} = 5\text{V}$ ($V_{\text{DD}} - 1/2 * V_{\text{DD}}$ and $V_{\text{SS}} + 1/2 * V_{\text{DD}}$)	25		mA
T_{OSR}	Output slew rate	$V_{\text{DD}} = 5\text{V}$, V_{O} rising	5		μs
		$V_{\text{DD}} = 5\text{V}$, V_{O} falling	5		μs

* Parameters with star mark are non-verified design reference.

24.6 Comparator Characteristics

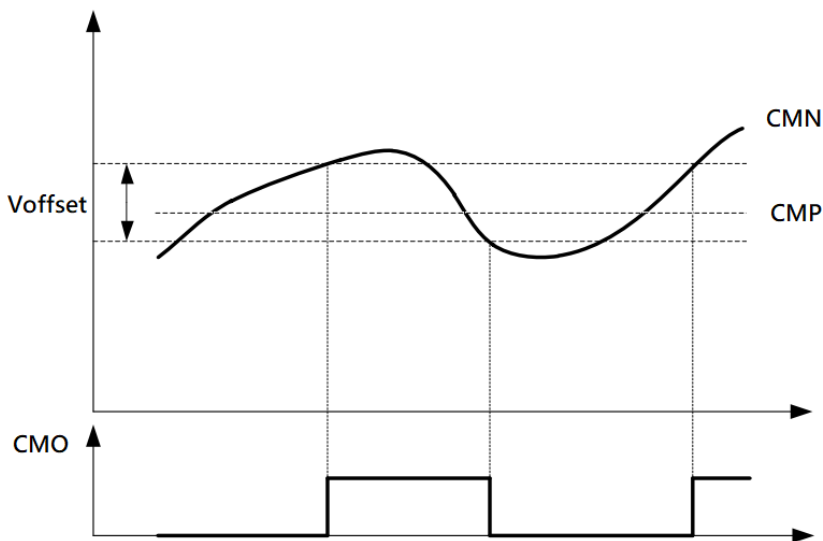
Parameter	Test Condition	MIN	TYP	MAX	UNIT
V_{CMP}	Operating Voltage	2.0		5.5	V
I_{CMP}	Current consumption*	$V_{\text{DD}} = 5\text{V}$	100		μA
V_{OFFSET}	Input offset voltage*	$V_{\text{DD}} = 5\text{V}$, $V_{\text{CM}} = 0.5V_{\text{DD}}$	-5	5	mV

T _{RS}	Response time	VDD = 5V, V _O rising	100			ns
		VDD = 5V, V _O falling	100			ns
T _{OSR}	Output slew rate	VDD = 5V, V _O rising	20			ns
		VDD = 5V, V _O falling	20			ns
V _{IREF}	Internal 4V reference voltage	VDD = 5V	3.92	4	4.08	V
	Internal 3V reference voltage	VDD = 5V	2.94	3	3.06	V
	Internal 2V reference voltage	VDD = 5V	1.96	2	2.04	V
V _{CMR}	Common mode input voltage	VDD = 5V	VSS+0.5		VDD-0.5	V

* Parameters with star mark are non-verified design reference.

VDD=5V	Voffset
Vp = Ext 2.5V	Vp-Vn = 4mV
	Vp-Vn = -3mV
VDD=3V	Voffset
Vp = Ext 1.5V	Vp-Vn = 7mV
	Vp-Vn = 0mV
VDD=2V	Voffset
Vp = Ext 1.0V	Vp-Vn = -1mV
	Vp-Vn = 3mV

VDD=5V	Voffset
Vp = Int 4.0V	Vp-Vn = 0mV
	Vp-Vn = -16mV
VDD=3V	Voffset
Vp = Int 3.0V	Vp-Vn = 9mV
	Vp-Vn = -5mV
VDD=2V	Voffset
Vp = Int 2.0V	Vp-Vn = -5mV
	Vp-Vn = -15mV



24.7 Flash Memory Characteristics

Parameter	Test Condition	MIN	TYP	MAX	UNIT
V_{dd}	Supply voltage	1.8		5.5	V
T_{en}	Endurance time	25°C	*100K		cycle

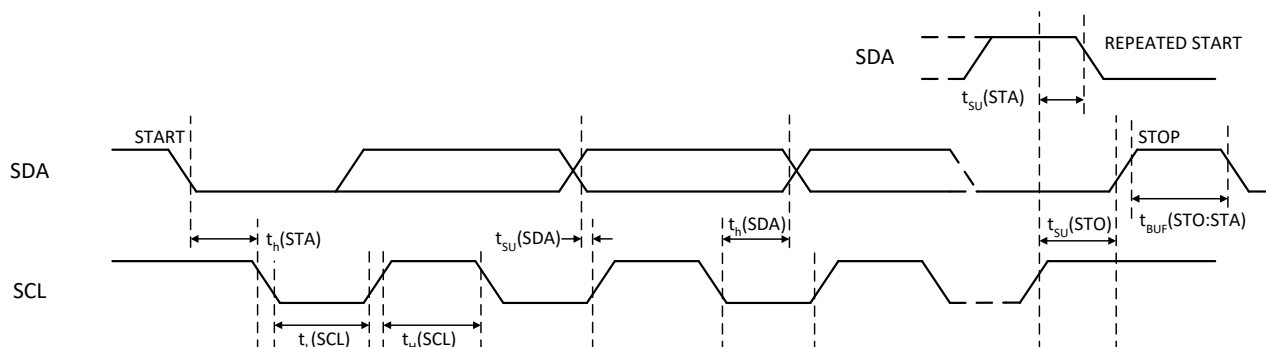
I_{wrt}	Write current	25°C	3	4	mA
T_{wrt}	Write time	Write 1 page=32 bytes, 25°C	6	8	ms

* Parameters with star mark are non-verified design reference.

24.8 I2C interface characteristics

Symbol	Parameter	Typical Formula	UNIT
$t_H(SCL)$	SCL clock high time	$2 * f_{sck}$	μs
$t_L(SCL)$	SCL clock low time	$2 * f_{sck}$	μs
$t_{SU}(SDA)$	SDA setup time	$2 * f_{sck} - 7 * f_{cpu}$	μs
$t_H(SDA)$	SDA data hold time	$7 * f_{cpu}$	μs
$t_H(STA)$	START condition hold time	$4 * f_{sck}$	μs
$t_{SU}(STA)$	Repeated START condition setup time	$2 * f_{sck}$	μs
$t_{SU}(STO)$	STOP condition setup time	$4 * f_{sck} + 8 * f_{cpu}$	μs
$T_{BUF}(STO:STA)$	STOP to START condition time	$4/3 * f_{sck} + 10 * f_{cpu}$	μs

*Standard mode I2C $f_{sck} = 100KHz$, Fast mode I2C $f_{sck} = 400KHz$



25 Instruction Set

This chapter categorizes the SN8F5708 microcontroller’s comprehensive assembly instructions. It includes five categories—arithmetic operation, logic operation, data transfer operation, Boolean manipulation, and program branch—which are fully compatible with standard 8051.

Symbol description

Symbol	Description
Rn	Working register R0 - R7
direct	One of 128 internal RAM locations or any Special Function Register
@Ri	Indirect internal or external RAM location addressed by register R0 or R1
#data	8-bit constant (immediate operand)
#data16	16-bit constant (immediate operand)
bit	One of 128 software flags located in internal RAM, or any flag of bit-addressable Special Function Registers
addr16	Destination address for LCALL or LJMP, can be anywhere within the 64-Kbyte page of program memory address space
addr11	Destination address for ACALL or AJMP, within the same 2-Kbyte page of program memory as the first byte of the following instruction
rel	SJMP and all conditional jumps include an 8-bit offset byte. Its range is +127/-128 bytes relative to the first byte of the following instruction
A	Accumulator

Arithmetic operations

Mnemonic	Description
ADD A, Rn	Add register to accumulator
ADD A, direct	Add directly addressed data to accumulator
ADD A, @Ri	Add indirectly addressed data to accumulator
ADD A, #data	Add immediate data to accumulator
ADDC A, Rn	Add register to accumulator with carry
ADDC A, direct	Add directly addressed data to accumulator with carry
ADDC A, @Ri	Add indirectly addressed data to accumulator with carry
ADDC A, #data	Add immediate data to accumulator with carry
SUBB A, Rn	Subtract register from accumulator with borrow
SUBB A, direct	Subtract directly addressed data from accumulator with borrow
SUBB A, @Ri	Subtract indirectly addressed data from accumulator with borrow
SUBB A, #data	Subtract immediate data from accumulator with borrow
INC A	Increment accumulator
INC Rn	Increment register
INC direct	Increment directly addressed location
INC @Ri	Increment indirectly addressed location
INC DPTR	Increment data pointer
DEC A	Decrement accumulator
DEC Rn	Decrement register
DEC direct	Decrement directly addressed location
DEC @Ri	Decrement indirectly addressed location
MUL AB	Multiply A and B
DIV	Divide A by B
DA A	Decimally adjust accumulator

Logic operations

Mnemonic	Description
ANL A, Rn	AND register to accumulator
ANL A, direct	AND directly addressed data to accumulator
ANL A, @Ri	AND indirectly addressed data to accumulator
ANL A, #data	AND immediate data to accumulator
ANL direct, A	AND accumulator to directly addressed location
ANL direct, #data	AND immediate data to directly addressed location
ORL A, Rn	OR register to accumulator

ORL A, direct	OR directly addressed data to accumulator
ORL A, @Ri	OR indirectly addressed data to accumulator
ORL A, #data	OR immediate data to accumulator
ORL direct, A	OR accumulator to directly addressed location
ORL direct, #data	OR immediate data to directly addressed location
XRL A, Rn	Exclusive OR (XOR) register to accumulator
XRL A, direct	XOR directly addressed data to accumulator
XRL A, @Ri	XOR indirectly addressed data to accumulator
XRL A, #data	XOR immediate data to accumulator
XRL direct, A	XOR accumulator to directly addressed location
XRL direct, #data	XOR immediate data to directly addressed location
CLR A	Clear accumulator
CPL A	Complement accumulator
RL A	Rotate accumulator left
RLC A	Rotate accumulator left through carry
RR A	Rotate accumulator right
RRC A	Rotate accumulator right through carry
SWAP A	Swap nibbles within the accumulator

Data transfer operations

Mnemonic	Description
MOV A, Rn	Move register to accumulator
MOV A, direct	Move directly addressed data to accumulator
MOV A, @Ri	Move indirectly addressed data to accumulator
MOV A, #data	Move immediate data to accumulator
MOV Rn, A	Move accumulator to register
MOV Rn, direct	Move directly addressed data to register
MOV Rn, #data	Move immediate data to register
MOV direct, A	Move accumulator to direct
MOV direct, Rn	Move register to direct
MOV direct1, direct2	Move directly addressed data to directly addressed location
MOV direct, @Ri	Move indirectly addressed data to directly addressed location
MOV direct, #data	Move immediate data to directly addressed location
MOV @Ri, A	Move accumulator to indirectly addressed location
MOV @Ri, direct	Move directly addressed data to indirectly addressed location

MOV @Ri, #data	Move immediate data to in directly addressed location
MOV DPTR, #data16	Load data pointer with a 16-bit immediate
MOVC A, @A+DPTR	Load accumulator with a code byte relative to DPTR
MOVC A, @A+PC	Load accumulator with a code byte relative to PC
MOVX A, @Ri	Move external RAM (8-bit address) to accumulator
MOVX A, @DPTR	Move external RAM (16-bit address) to accumulator
MOVX @Ri, A	Move accumulator to external RAM (8-bit address)
MOVX @DPTR, A	Move accumulator to external RAM (16-bit address)
PUSH direct	Push directly addressed data onto stack
POP direct	Pop directly addressed location from stack
XCH A, Rn	Exchange register with accumulator
XCH A, direct	Exchange directly addressed location with accumulator
XCH A, @Ri	Exchange indirect RAM with accumulator
XCHD A, @Ri	Exchange low-order nibbles of indirect and accumulator

Boolean manipulation

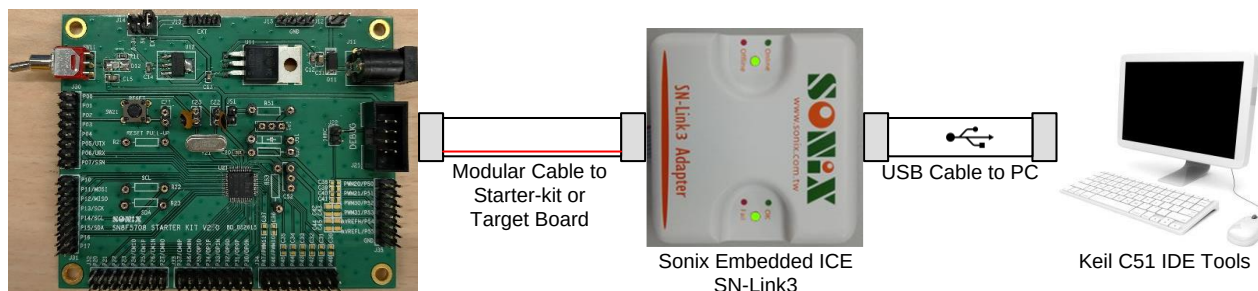
Mnemonic	Description
CLR C	Clear carry flag
CLR bit	Clear directly addressed bit
SETB C	Set carry flag
SETB bit	Set directly addressed bit
CPL C	Complement carry flag
CPL bit	Complement directly addressed bit
ANL C, bit	AND directly addressed bit to carry flag
ANL C, /bit	AND complement of directly addressed bit to carry
ORL C, bit	OR directly addressed bit to carry flag
ORL C, /bit	OR complement of directly addressed bit to carry
MOV C, bit	Move directly addressed bit to carry flag
MOV bit, C	Move carry flag to directly addressed bit

Program branches

Mnemonic	Description
ACALL addr11	Absolute subroutine call
LCALL addr16	Long subroutine call
RET	Return from subroutine
RETI	Return from interrupt
AJMP addr11	Absolute jump
LJMP addr16	Long jump
SJMP rel	Short jump (relative address)
JMP @A+DPTR	Jump indirect relative to the DPTR
JZ rel	Jump if accumulator is zero
JNZ rel	Jump if accumulator is not zero
JC rel	Jump if carry flag is set
JNC rel	Jump if carry flag is not set
JB bit, rel	Jump if directly addressed bit is set
JNB bit, rel	Jump if directly addressed bit is not set
JBC bit, rel	Jump if directly addressed bit is set and clear bit
CJNE A, direct, rel	Compare directly addressed data to accumulator and jump if not equal
CJNE A, #data, rel	Compare immediate data to accumulator and jump if not equal
CJNE Rn, #data, rel	Compare immediate data to register and jump if not equal
CJNE @Ri, #data, rel	Compare immediate to indirect and jump if not equal
DJNZ Rn, rel	Decrement register and jump if not zero
DJNZ direct, rel	Decrement directly addressed location and jump if not zero
NOP	No operation for one cycle

26 Development Environment

SONiX provides an Embedded ICE emulator system to offer SN8F5708 firmware development. The platform is an in-circuit debugger and controlled by Keil C51 IDE software on Microsoft Windows platform. The platform includes SN-Link3, SN8F5708 Starter-kit and Keil C51 IDE software to build a high-speed, low cost, powerful and multi-task development environment including emulator, debugger and programmer. To execute emulation is like run real chip because the emulator circuit integrated in SN8F5708 to offer a real development environment.



26.1 Minimum Requirement

The following items are essential to build up an appropriate development environment. The compatibility is verified on listed versions, and is expected to execute perfectly on later version. SN-Link related information is available to download on SONiX website (www.sonix.com.tw);

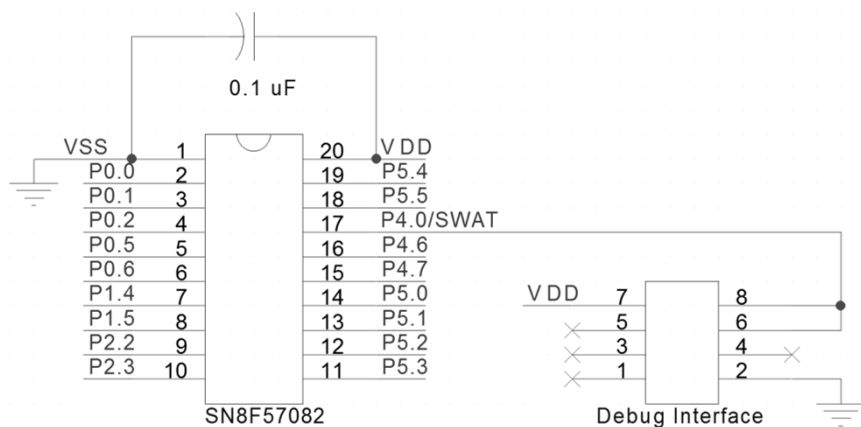
Keil C51 is downloadable on www.keil.com/c51.

- **SN-Link3 Adapter** with updated firmware version 1.02
- **SN-Link Driver for Keil C51** version 1.00.317
- **Keil C51** version 9.50a and 9.54a or greater.

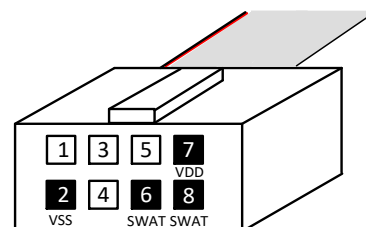
26.2 Debug Interface Hardware

The circuit below demonstrates the appropriate method to connect microcontroller's SWAT pin and SN-Link3 Adapter II.

Before starting debug, microcontroller's power (VDD) must be switched off. Connect the SWAT to both 6th and 8th pins of SN-Link, and respectively link VDD and VSS to 7th pin and 2nd pin. A handshake procedure would be automatically started by turn on the microcontroller, and SN-Link's green LED (Run) indicates the success of connection (refer *SN8F5000 Debug Tool Manual* for further detail).



example circuit



SN-Link header

26.3 Development Tool

SN-Link3 Adapter



Starter-Kit support SN8F5708 series



MP5 Writer



27 SN8F5708 Starter-Kit

SN8F5000 Starter-Kit provides easy-development platform. It includes SN8F5000 family real chip and I/O connectors to input signal or drive device of user's application. It is a simple platform to develop application as target board not ready. The Starter-Kit can be replaced by target board, because SN8F5000 family integrates embedded ICE in-circuit debugger circuitry.

27.1 Configurations of Circuit

These configurations must be setup completely before starting Starter-Kit developing.

1. Confirm to the circuit board whether elements are complete.
2. The power source of Starter-Kit circuit is chosen from 5.0V, 3.3V, external power or Micro USB via jumper.
3. The power source comes from 5.0V or 3.3V which must be connect to DC 7.5V power adapter.
4. If the power source is chosen from external power, then external power source connects to EXT pin.
5. The "RST" pin needs to connect pull high resister to VDD when external reset is chosen to use.
6. The "XIN" pin and the "XOUT" pin need to connect crystal/resonator oscillator components when system clock is setting crystal or RTC mode.
7. The "XIN" pin needs to connect external clock source when system clock is setting external clock input mode.
8. The Debug Port can connect SN-LINK Adapter for emulation or download code.
9. The MCU LED will light up and SN8F5000 family chip will be connected to power when power (VDD) is switched on.

27.4 Component Description

Number	Description
C30 – C41	12-ch ADC capacitors.
C42 – C45	AVREFH/AVREFL capacitors.
D12	MCU LED
J11	DC 7.5V power adapter
J13/J15	External power source.
SW21	External reset trigger source
J14	VDD power source is 5.0V, 3.3V or external power.
J21	Debug Port
J30 – J35	I/O connector.
R21, C21	External reset pull-high resister and capacitor.
R22, R23	I2C pull-high resisters.
SW11	Target power (VDD) switch
U21	SN8F5708F real chip (Sonix standard option).
Y21, C22, C23	External crystal/resonator oscillator components.

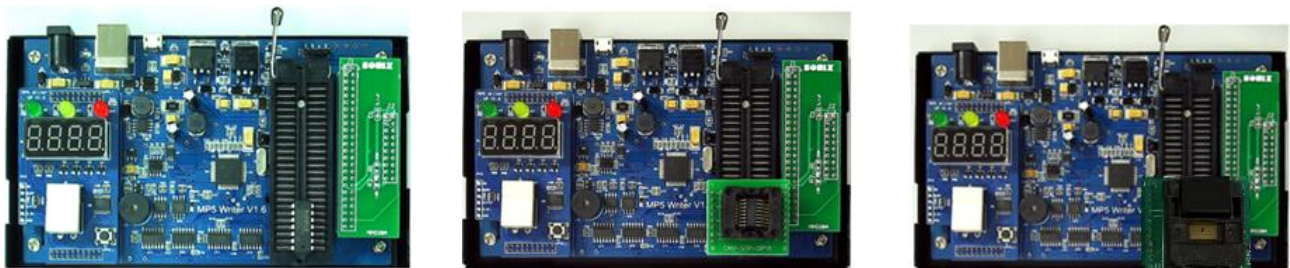
28 ROM Programming Pin

SN8F5708 Series Flash ROM erase/program/verify support SN-Link and MP5 Writer

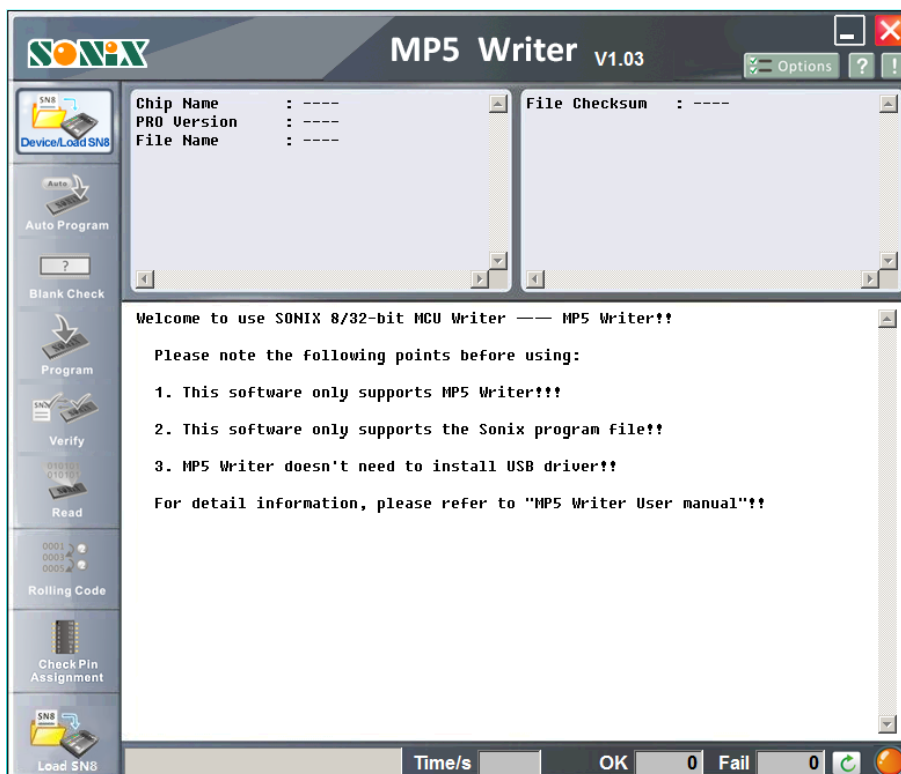
- SN-Link: Debug interface and on board programming.
- MP5 Writer: For SN8F5708 series version mass programming.

28.1 MP5 Hardware Connecting

Different package type with MCU programming connecting is as following, DIP, SOP, SSOP, TSSOP and QFN Illustration.

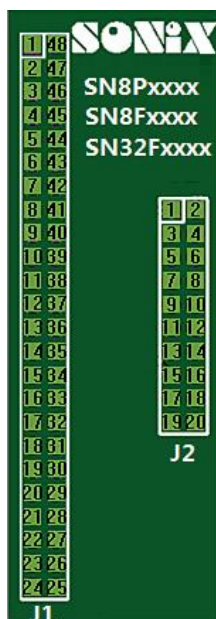


MP5 Software operation interface is as following.



28.2 MP5 Writer Transition Board Socket Pin Assignment

MP5 Writer Transition Board:



28.3 MP5 Writer Programming Pin Mapping

Writer Connector		MCU Pin Number	SN8F5708F/J		SN8F570870J		SN8F5707F		SN8F57086J	
J2 Pin Number	J2 Pin Name		MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number
1	VDD	VDD	42	42	39	40	39	41	35	39
2	GND	VSS	43	43	40	41	40	42	36	40
7	SWAT	P4.0	39	39	36	37	36	38	32	36
9	SWAT	P4.0	39	39	36	37	36	38	32	36
20	PDB	P1.4	8	8	7	8	7	9	7	11

Writer Connector		MCU Pin Number	SN8F5705F/J		SN8F57084K/S/T		SN8F57084J		SN8F57082S/T	
J2 Pin Number	J2 Pin Name		MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number
1	VDD	VDD	28	36	28	38	24	34	20	34
2	GND	VSS	29	37	1	11	25	35	1	15
7	SWAT	P4.0	25	33	25	35	21	31	17	31
9	SWAT	P4.0	25	33	25	35	21	31	17	31
20	PDB	P1.4	7	15	11	21	7	17	7	21

Writer Connector		MCU Pin Number	SN8F570822T		SN8F570812S		SN8F57085T		SN8F57087T	
J2 Pin Number	J2 Pin Name		MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number	MCU Pin Number	J1 Pin Number
1	VDD	VDD	1	15	13	29	16	32	15	31
2	GND	VSS	2	16	14	30	1	17	16	32
7	SWAT	P4.0	20	34	12	28	15	31	14	30
9	SWAT	P4.0	20	34	12	28	15	31	14	30
20	PDB	P1.4	7	21	1	17	4	20	9	25

28.4 SN-Link ISP Programming

SN-Link ISP programming hardware and software are as following.



28.5 SN-Link ISP Programming Pin Mapping

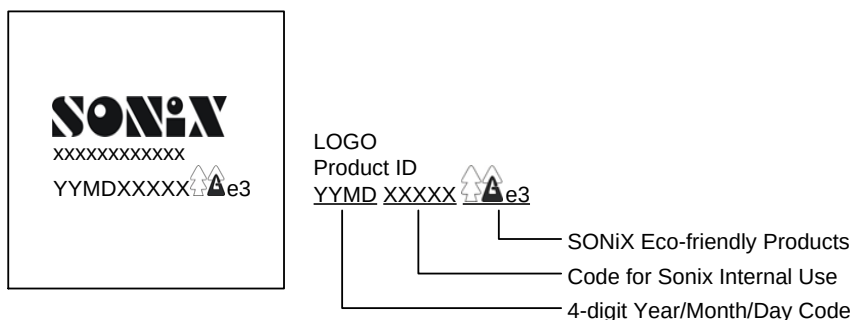
SN-Link Connector		MCU Pin Number	SN8F5708F/J	SN8F570870J	SN8F5707F	SN8F57086J
Pin Number	Pin Name		Pin Number	Pin Number	Pin Number	Pin Number
7	VDD	VDD	42	39	39	35
2	GND	VSS	43	40	40	36
6	SWAT	P4.0	39	36	36	32
8	SWAT	P4.0	39	36	36	32

SN-Link Connector		MCU Pin Number	SN8F5705F/J	SN8F57084K/S/T	SN8F57084J	SN8F57082S/T
Pin Number	Pin Name		Pin Number	Pin Number	Pin Number	Pin Number
7	VDD	VDD	28	28	24	20
2	GND	VSS	29	1	25	1
6	SWAT	P4.0	25	25	21	17
8	SWAT	P4.0	25	25	21	17

SN-Link Connector		MCU Pin Number	SN8F570822T	SN8F570812S	SN8F57085T	SN8F57087T
Pin Number	Pin Name		Pin Number	Pin Number	Pin Number	Pin Number
7	VDD	VDD	1	13	16	15
2	GND	VSS	2	14	1	16
6	SWAT	P4.0	20	12	15	14
8	SWAT	P4.0	20	12	15	14

29 Ordering Information

The figure below is an example of the marking. Contents such as the product ID or symbol may vary according to different packages.

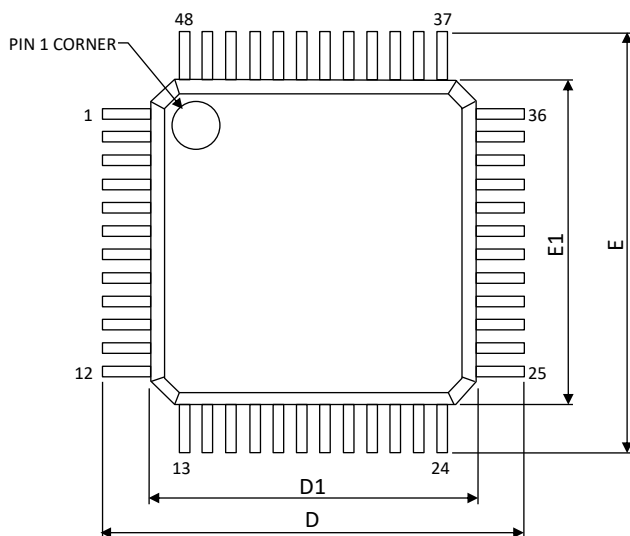


29.1 Device Nomenclature

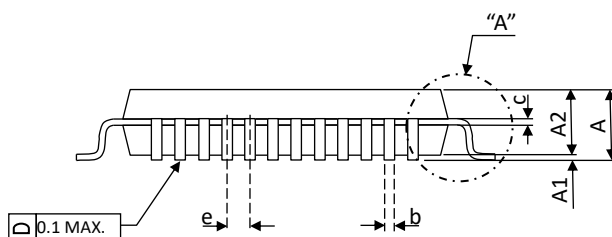
Full Name	Packing Type
S8F5708W	Wafer
SN8F5708H	Dice
SN8F5708FG	LQFP, 48 pins, Green package
SN8F5708JG	QFN, 48 pins, Green package
SN8F570870JG	QFN, 46 pins, Green package
SN8F5707FG	LQFP, 44 pins, Green package
SN8F57086JG	QFN, 40 pins, Green package
SN8F5705FG	LQFP, 32 pins, Green package
SN8F5705JG	QFN, 32 pins, Green package
SN8F57084KG	SKDIP, 28 pins, Green package
SN8F57084SG	SOP, 28 pins, Green package
SN8F57084TG	TSSOP, 28 pins, Green package
SN8F57084JG	QFN, 28 pins, Green package
SN8F57082SG	SOP, 20 pins, Green package
SN8F57082TG	TSSOP, 20 pins, Green package
SN8F570822TG	TSSOP, 20 pins, Green package
SN8F570812SG	SOP, 16 pins, Green package
SN8F57085TG	TSSOP, 16 pins, Green package
SN8F57087TG	TSSOP, 16 pins, Green package

30 Package Information

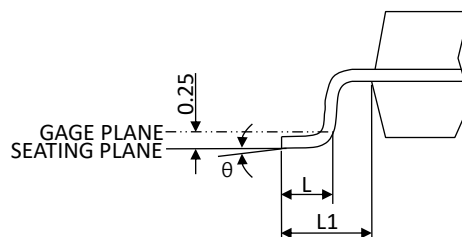
30.1 LQFP48



TOP VIEW



SIDE VIEW



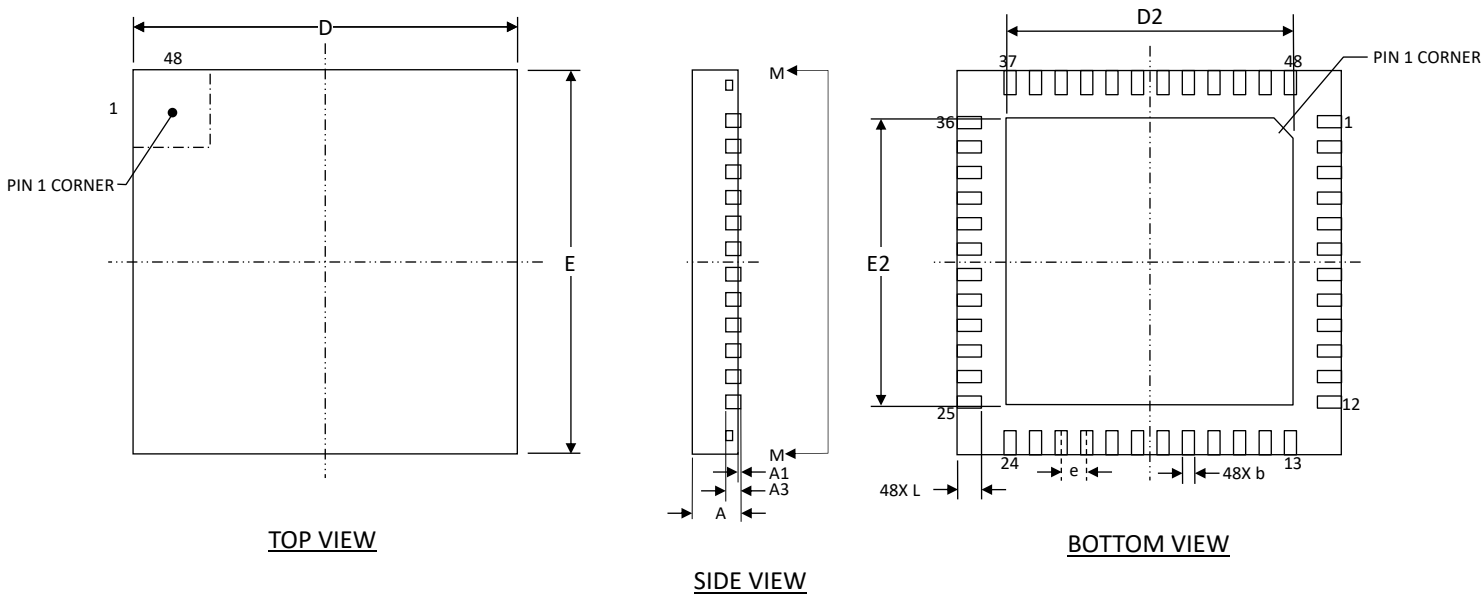
DETAIL "A"

SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.60	--	--	0.063
A1	0.05	--	0.15	0.002	--	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.17	0.22	0.27	0.007	0.009	0.011
c	0.09	--	0.20	0.004	--	0.008
D	9.00 BSC			0.354 BSC		
D1	7.00 BSC			0.276 BSC		
E	9.00 BSC			0.354 BSC		
E1	7.00 BSC			0.276 BSC		
e	0.50 BSC			0.020 BSC		
L	0.40	0.60	0.80	0.016	0.024	0.031
L1	1.00 REF			0.039 REF		
θ	0°	3.5°	7°	0°	3.5°	7°

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)
2. DIMENSIONS D1 AND E1 DO NOT INCLUDE MOLD PROTRUSION.
3. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION.

30.2 QFN48

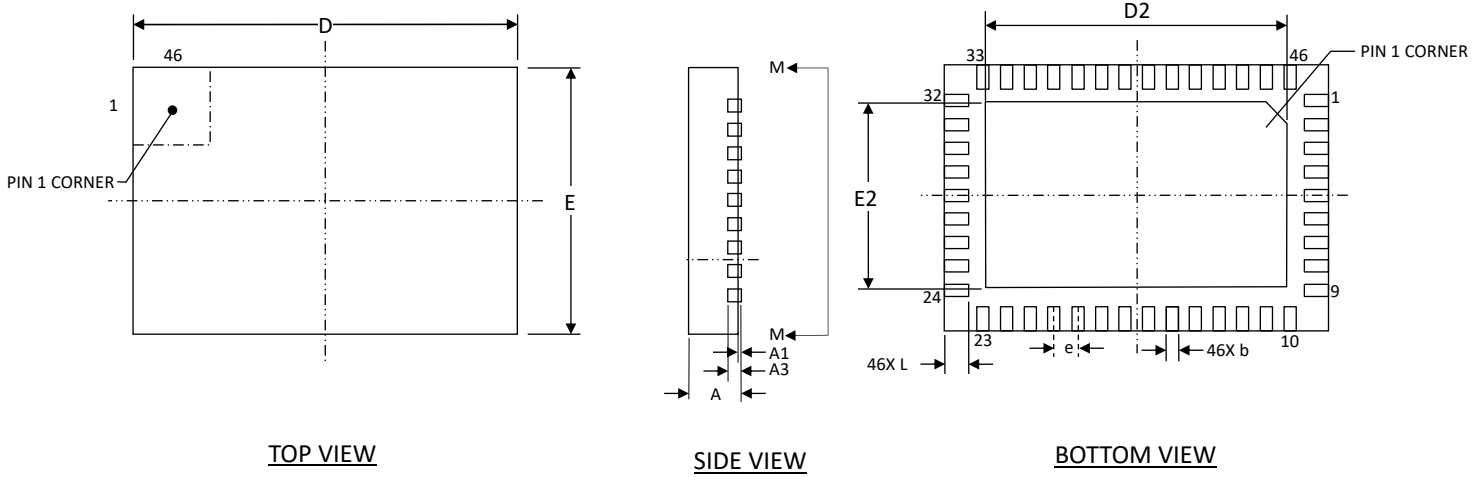


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.70	0.80	0.90	0.028	0.031	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.203 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	6.00 BSC			0.236 BSC		
E	6.00 BSC			0.236 BSC		
e	0.4 BSC			0.016 BSC		
D2	3.70	4.20	4.70	0.146	0.165	0.185
E2	3.70	4.20	4.70	0.146	0.165	0.185
L	0.30	0.40	0.50	0.012	0.016	0.020

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)

30.3 QFN46

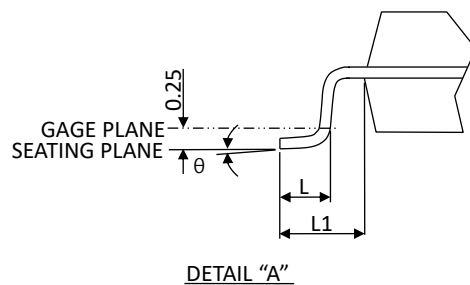
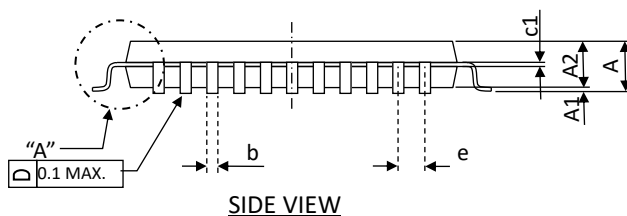
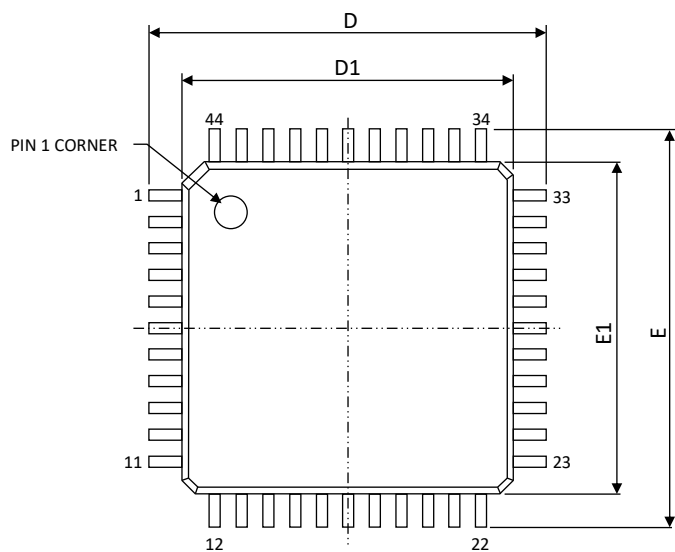


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.70	0.80	0.90	0.028	0.031	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.203 REF			0.008 REF		
b	0.20	0.25	0.30	0.008	0.010	0.012
D	6.5 BSC			0.256 BSC		
E	4.5 BSC			0.177 BSC		
e	0.4 BSC			0.016 BSC		
D2	5.00	5.10	5.20	0.197	0.201	0.205
E2	3.00	3.10	3.20	0.118	0.122	0.126
L	0.30	0.40	0.50	0.012	0.016	0.020

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)

30.4 LQFP44

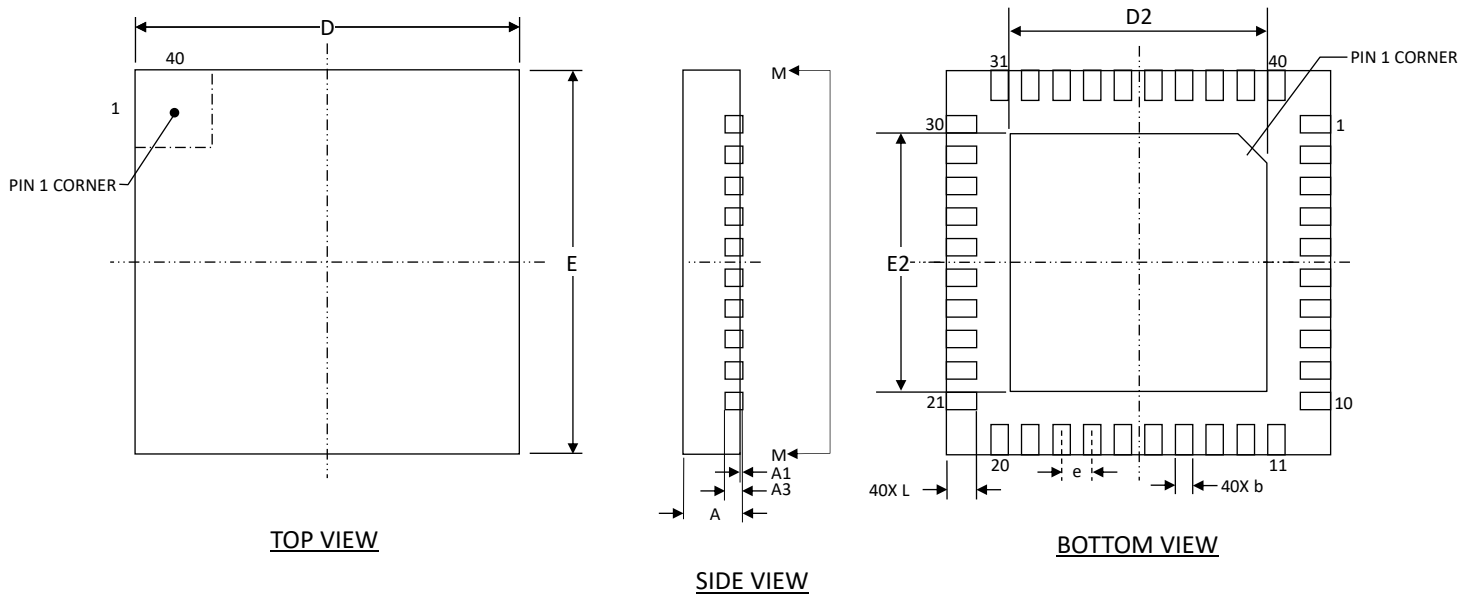


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.60	--	--	0.063
A1	0.05	--	0.15	0.002	--	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.28	0.37	0.45	0.011	0.015	0.018
c1	0.09	--	0.20	0.004	--	0.008
D	12.00 BSC			0.472 BSC		
D1	10.00 BSC			0.394 BSC		
E	12.00 BSC			0.472 BSC		
E1	10.00 BSC			0.394 BSC		
e	0.80 BSC			0.031 BSC		
L	0.45	0.60	0.75	0.018	0.0236	0.030
L1	1.00 REF			0.039 REF		
θ	0°	3.5°	7°	0°	3.5°	7°

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)
2. DIMENSIONS D1 AND E1 DO NOT INCLUDE MOLD PROTRUSION.
3. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION.

30.5 QFN40 5X5

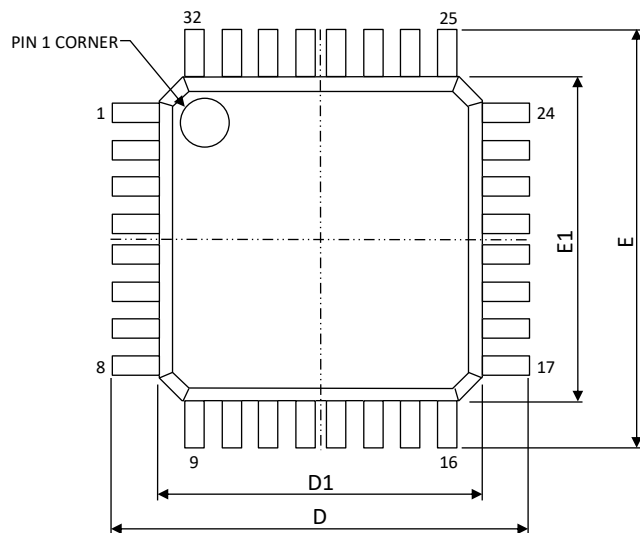


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.70	0.80	0.90	0.025	0.031	0.035
A1	--	0.02	0.05	--	0.001	0.002
A3	0.203 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	5.00 BSC			0.197 BSC		
E	5.00 BSC			0.197 BSC		
e	0.40 BSC			0.016 BSC		
D2	3.1	3.60	4.1	0.122	0.142	0.161
E2	3.1	3.60	4.1	0.122	0.142	0.161
L	0.25	0.35	0.45	0.010	0.014	0.018

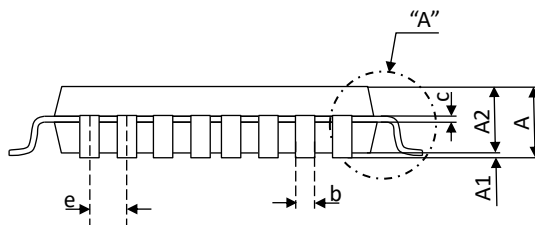
Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)

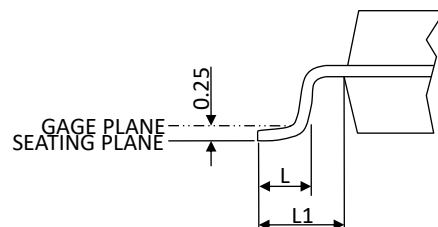
30.6 LQFP32



TOP VIEW



SIDE VIEW



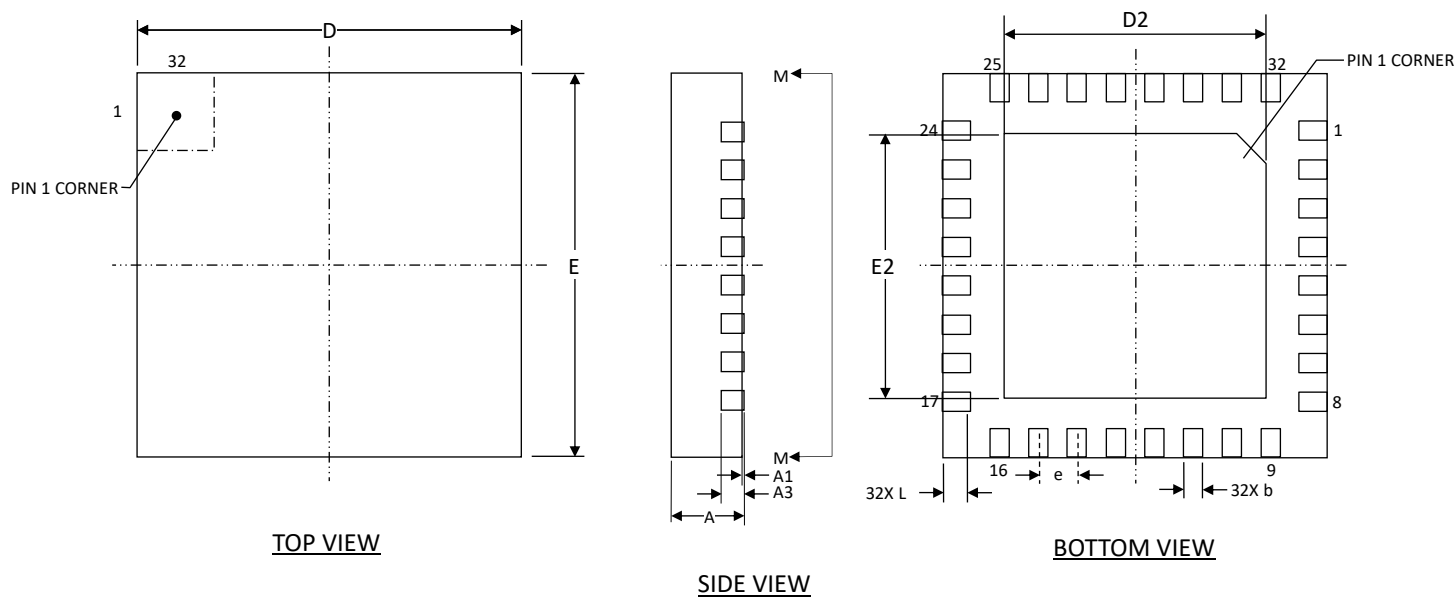
DETAIL "A"

SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.60	--	--	0.063
A1	0.05	--	0.25	0.002	--	0.01
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.30	--	0.45	0.012	--	0.018
c	0.09	--	0.20	0.004	--	0.008
D	9.00 BSC			0.354 BSC		
D1	7.00 BSC			0.276 BSC		
E	9.00 BSC			0.354 BSC		
E1	7.00 BSC			0.276 BSC		
e	0.80 BSC			0.031 BSC		
L	0.40	0.60	0.80	0.016	0.024	0.031
L1	1.00 REF			0.039 REF		

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)
2. DIMENSIONS D1 AND E1 DO NOT INCLUDE MOLD PROTRUSION.
3. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION.

30.7 QFN32 4X4

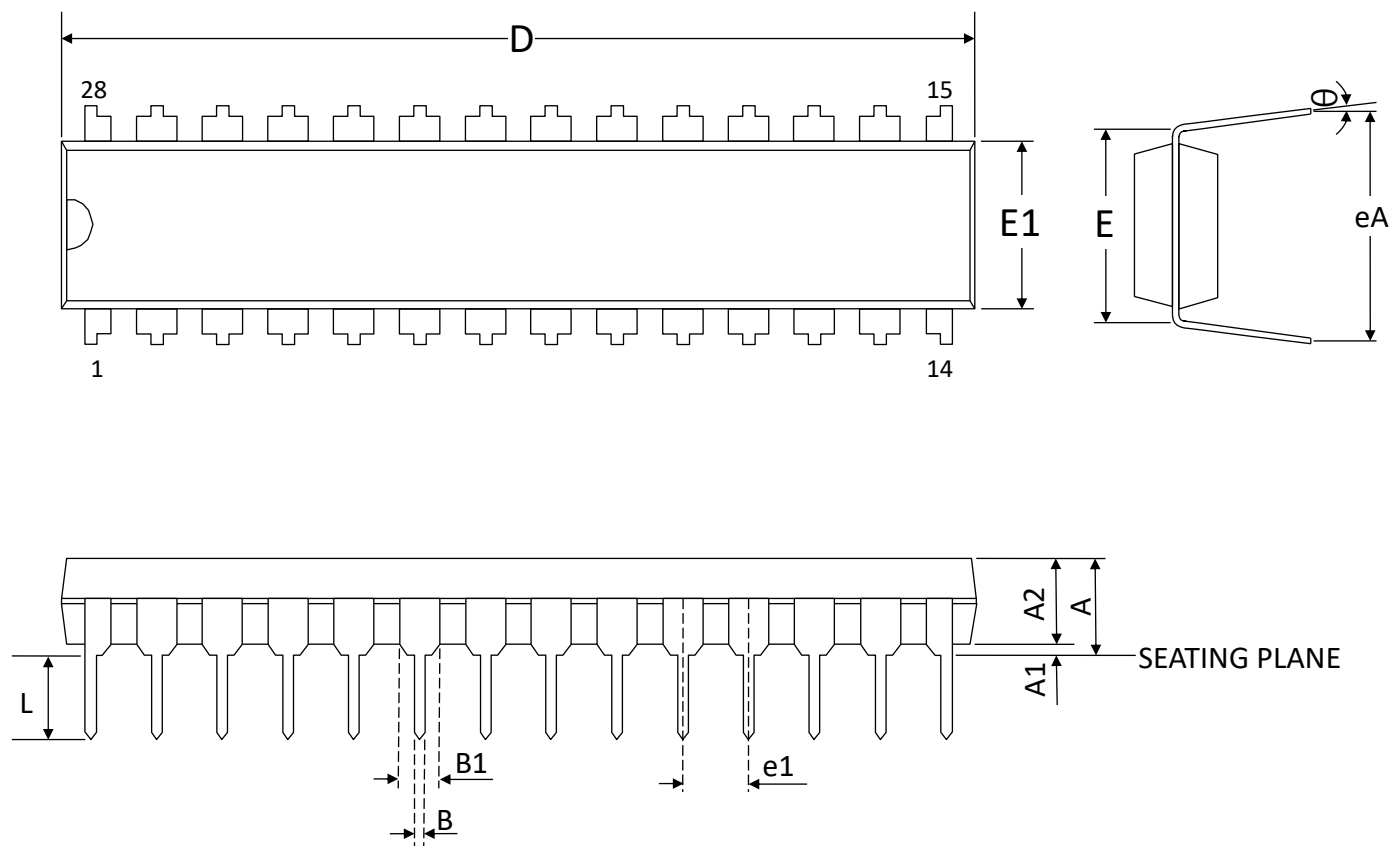


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.70	0.80	0.90	0.028	0.031	0.035
A1	0.00	0.02	0.05	0.000	0.000	0.002
A3	0.20 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	4.00 BSC			0.157 BSC		
E	4.00 BSC			0.157 BSC		
e	0.40 BSC			0.016 BSC		
D2	2.00	2.45	2.9	0.080	0.096	0.114
E2	2.00	2.45	2.9	0.080	0.096	0.114
L	0.25	0.35	0.45	0.010	0.013	0.017

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)

30.8 SKDIP28

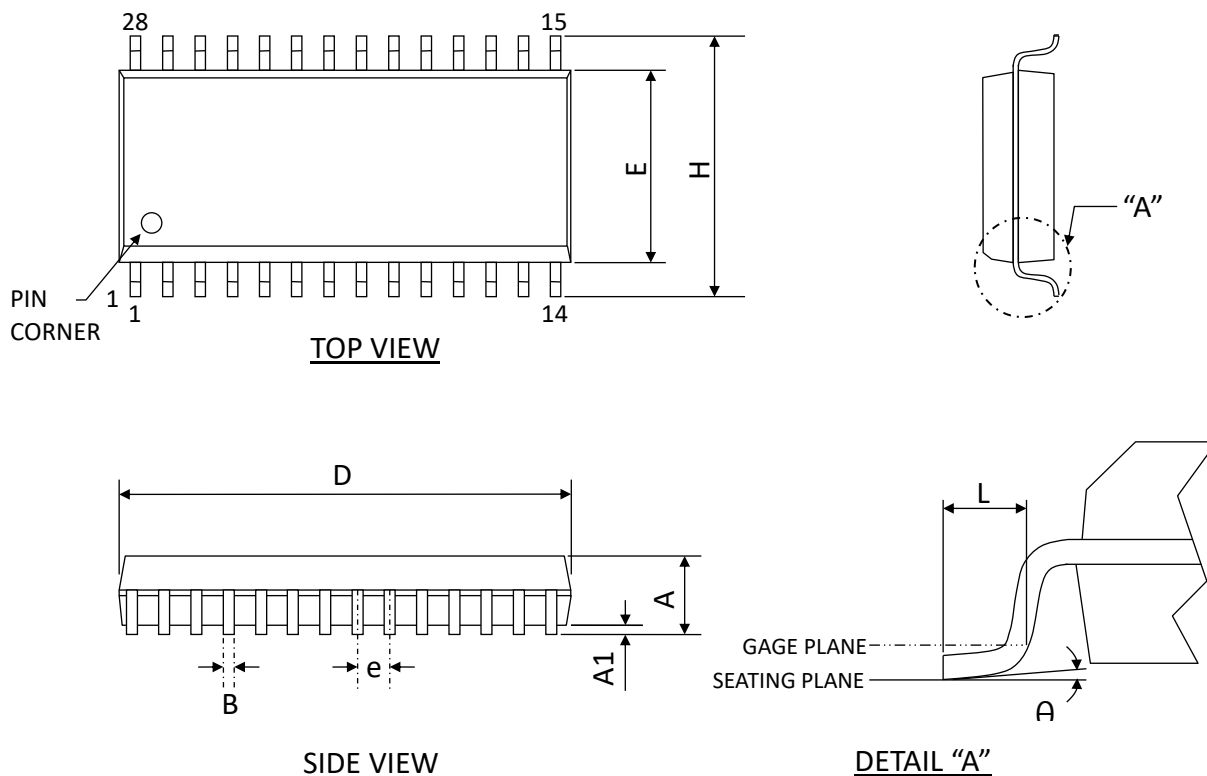


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	4.45	--	--	0.175
A1	0.38	--	--	0.015	--	--
A2	3.18	3.30	3.43	0.125	0.130	0.135
B	0.46 typ.			0.018 typ.		
B1	1.52 typ.			0.060 typ.		
D	35.18	35.31	35.56	1.385	1.390	1.400
E	7.87 BSC			0.310 BSC		
E1	7.19	7.32	7.44	0.283	0.288	0.293
e1	2.54 typ.			0.100 typ.		
L	3.05	3.30	3.56	0.120	0.130	0.140
eA	8.38	8.89	9.40	0.330	0.350	0.370
θ	0°	7°	15°	0°	7°	15°

Notes :

1. JEDEC OUTLINE : MS-015 AH
2. CONTROLLING DIMENSION : inch

30.9 SOP28

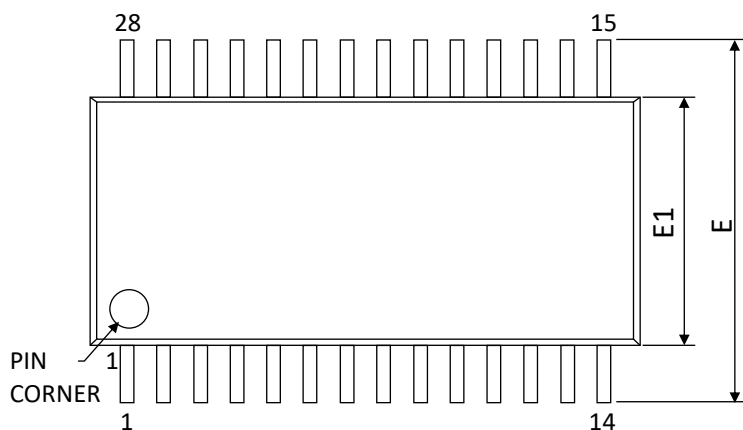


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	2.65	--	--	0.104
A1	0.10	--	0.30	0.004	--	0.011
B	0.31	0.41	0.51	0.012	0.016	0.020
D	17.70	18.20	18.70	0.697	0.716	0.736
E	7.50 BSC			0.295 BSC		
e	1.27 BSC			0.050 BSC		
H	10.30 BSC			405 BSC		
L	0.40	--	1.27	0.016	--	0.050
θ	0°	4°	8°	0°	4°	8°

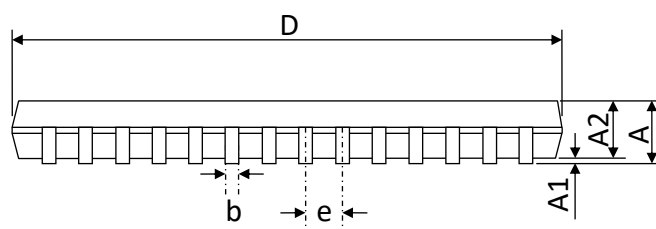
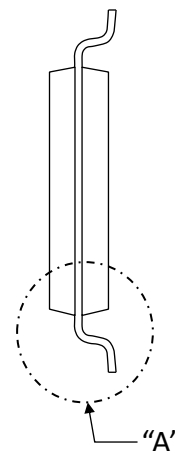
Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MO-119 AB

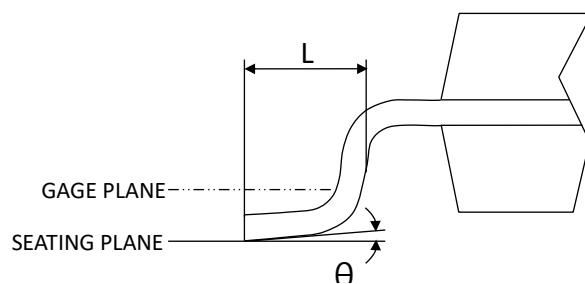
30.10 TSSOP28



TOP VIEW



SIDE VIEW



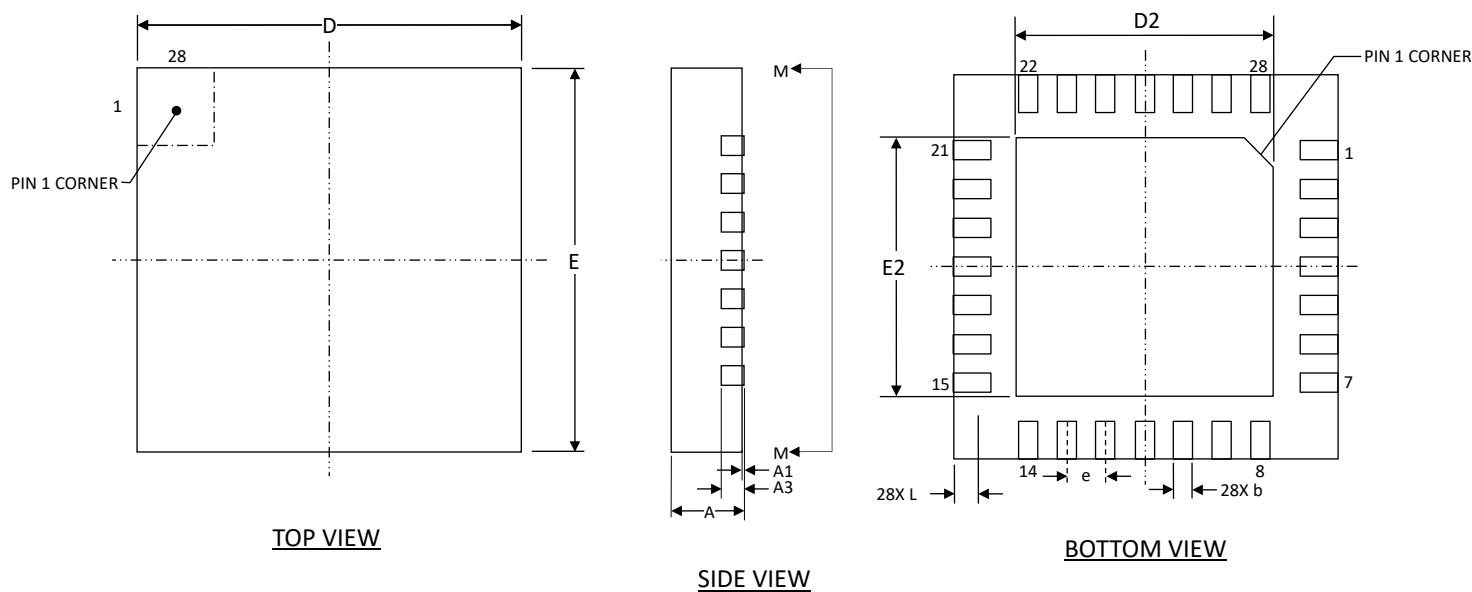
DETAIL "A"

SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.20	--	--	0.047
A1	0.00	--	0.15	0.000	--	0.006
A2	0.80	1.00	1.05	0.031	0.039	0.041
b	0.19	--	0.30	0.007	--	0.012
D	9.60	9.70	9.80	0.378	0.382	0.386
E	6.40 BSC.			0.252 BSC.		
E1	4.30	4.40	4.50	0.169	0.173	0.177
e	0.65 BSC.			0.026 BSC.		
L	0.45	0.60	0.75	0.018	0.024	0.030
θ	0°	--	8°	0°	--	8°

Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MO-153
3. DIMENSION 'D' DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BERRES.
4. DIMENSION 'E1' DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION.
5. DIMENSION 'b' DOES NOT INCLUDE DAMBAR PROTRUSION.

30.11 QFN28 4X4

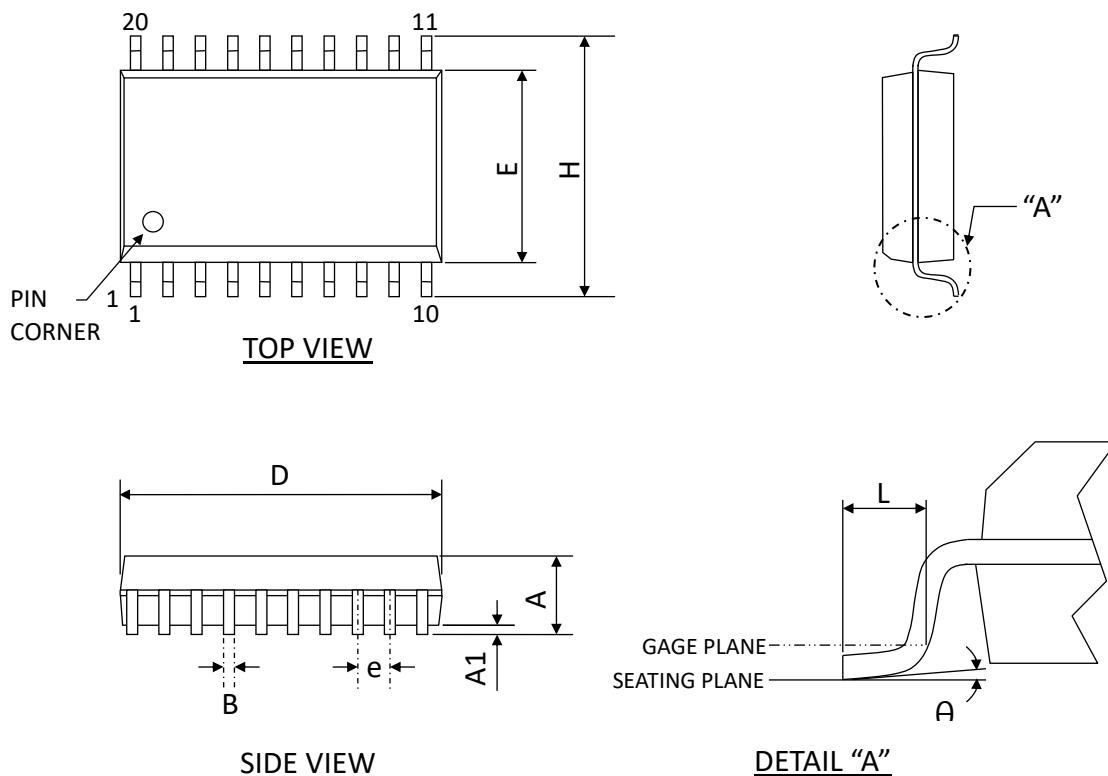


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.70	0.80	0.90	0.028	0.031	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.203 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	4.00 BSC			0.157 BSC		
E	4.00 BSC			0.157 BSC		
e	0.40 BSC			0.016 BSC		
D2	1.50	2.20	2.9	0.060	0.086	0.114
E2	1.50	2.20	2.9	0.060	0.086	0.114
L	0.30	0.40	0.50	0.012	0.016	0.020

Notes :

1. CONTROLLING DIMENSION : MILLIMETER (mm)

30.12 SOP20

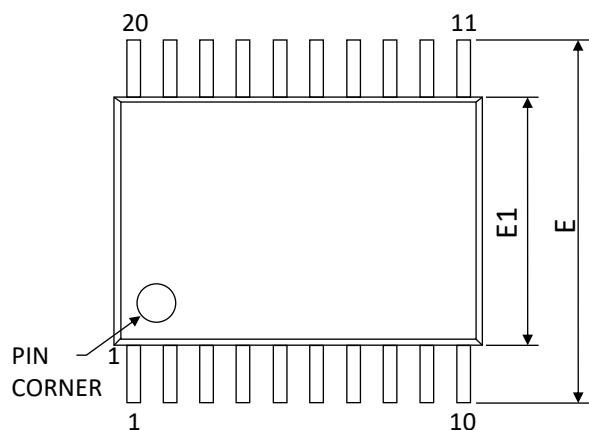


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	2.65	--	--	0.104
A1	0.10	--	0.30	0.004	--	0.012
B	0.31	0.41	0.51	0.012	0.016	0.020
D	12.80 BSC			0.503		
E	7.50 BSC			0.295		
e	1.27 BSC			0.050 BSC		
H	10.30 BSC			0.405		
L	0.40	--	1.27	0.016	--	0.050
θ	0°	4°	8°	0°	4°	8°

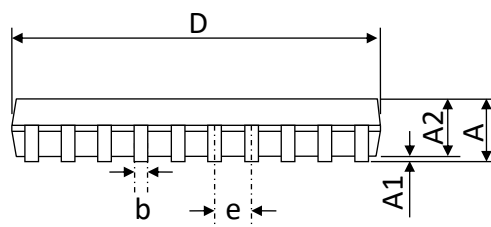
Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MO-013 AC

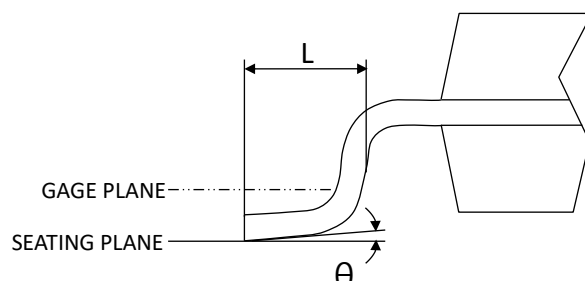
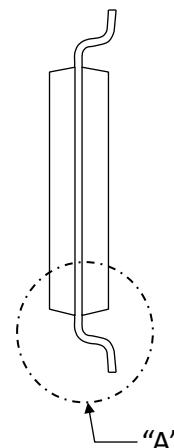
30.13 TSSOP20



TOP VIEW



SIDE VIEW



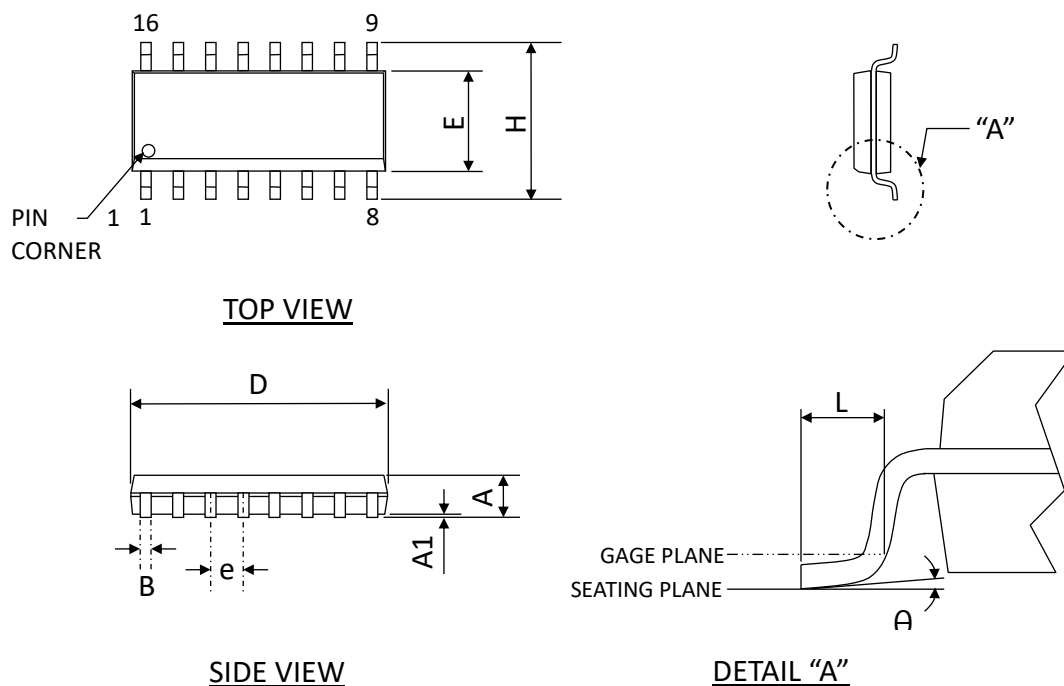
DETAIL "A"

SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.20	--	--	0.047
A1	0.05	--	0.15	0.002	--	0.006
A2	0.80	--	1.05	0.031	--	0.041
b	0.19	--	0.30	0.007	--	0.012
D	6.40	6.50	6.60	0.252	0.256	0.260
E	6.40 BSC.			0.252 BSC.		
E1	4.30	4.40	4.50	0.169	0.173	0.177
e	0.65 BSC.			0.026 BSC.		
L	0.45	0.60	0.75	0.018	0.024	0.030
θ	0°	--	8°	0°	--	8°

Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MO-153
3. DIMENSION 'D' DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BERRES.
4. DIMENSION 'E1' DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION.
5. DIMENSION 'b' DOES NOT INCLUDE DAMBAR PROTRUSION.

30.14 SOP16

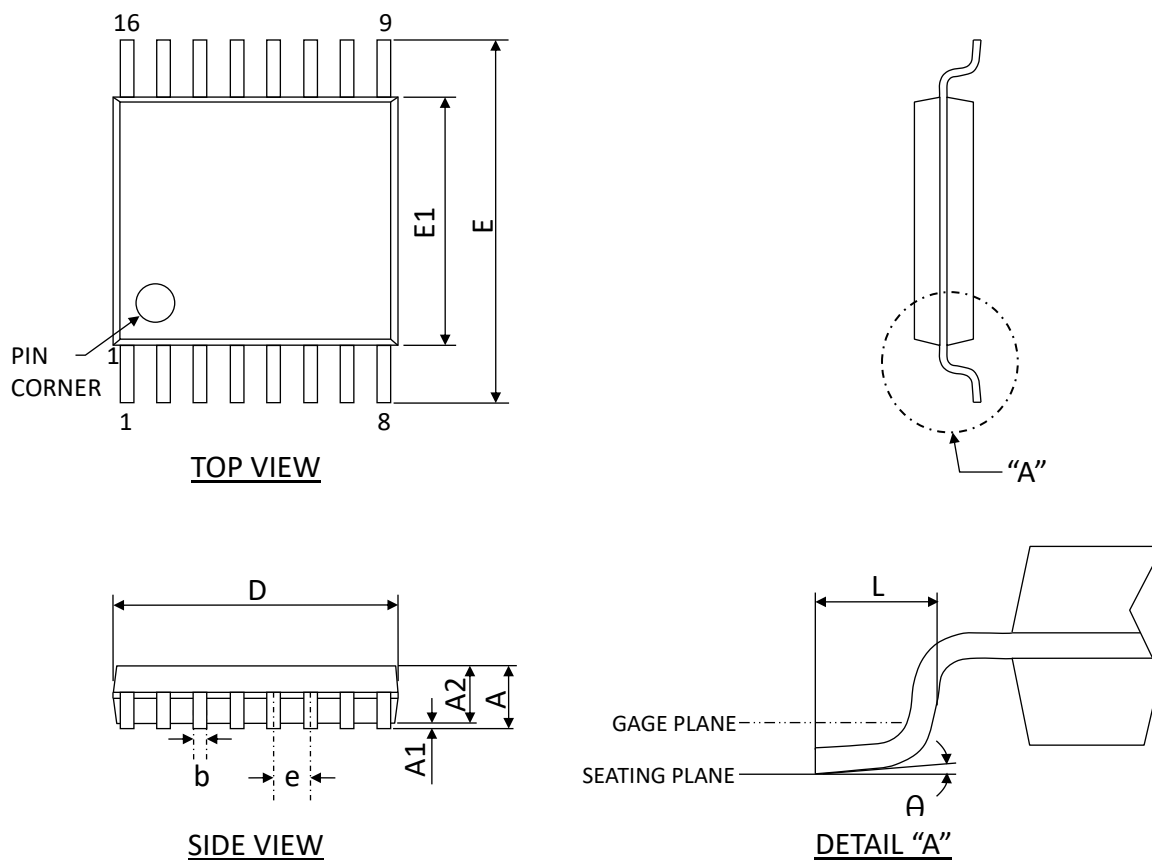


SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.75	--	--	0.069
A1	0.10	--	0.25	0.004	--	0.010
B	0.31	0.41	0.51	0.012	0.016	0.020
D	9.90 BSC			0.389 BSC		
E	3.90 BSC			0.153 BSC		
e	1.27 BSC			0.050 BSC		
H	6.00 BSC			0.236 BSC		
L	0.40	--	1.27	0.016	--	0.050
θ	0°	4°	8°	0°	4°	8°

Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MS-012 AC

30.15 TSSOP16



SYMBOLS	Dimension in mm			Dimension in inch		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	--	--	1.20	--	--	0.047
A1	0.05	--	0.15	0.002	--	0.006
A2	0.80	1.00	1.05	0.031	0.039	0.041
b	0.19	--	0.30	0.007	--	0.012
D	4.80	5.00	5.20	0.189	0.196	0.205
E	6.40 BSC.			0.252 BSC.		
E1	4.30	4.40	4.50	0.169	0.173	0.177
e	0.65 BSC.			0.026 BSC.		
L	0.45	0.60	0.75	0.018	0.024	0.030
θ	0°	--	8°	0°	--	8°

Notes :

1. CONTROLLING DIMENSION : mm
2. JEDEC OUTLINE : MO-153
3. DIMENSION 'D' DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BERRES.
4. DIMENSION 'E1' DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION.
5. DIMENSION 'b' DOES NOT INCLUDE DAMBAR PROTRUSION.

31 Appendix: Reference Document

Sonix provides reference document for users to help them quickly familiar SN8F5000 family (downloadable on cooperative website: www.sonix.com.tw).

Document Name	Description
SN8F5000 Starter-Kit User Manual	This documentation introduces SN8F5000 family all Starter-Kit, providing the user selects an appropriate starter-kit for development.
SN8F5000 Family Instruction Set	The document details the 8051 instruction set, and a simple example illustrates operation.
SN8F5000 Family Instruction Mapping Table	This document supplies the information about mapping assembly instructions from 8-Bit Flash/ OTP Type to 8051 Flash Type.
SN8F5000 Packaging Information	This documentation introduces SN8F5000 family microcontrollers' mechanical data, such as height, width and pitch information.
SN8F5000 Debug Tool Manual	This document teaches the user to install software Keil C51, and helped create a new project to be developed.

SN8F5708 Series Datasheet

8051-based Microcontroller

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